INTERNATIONAL MONETARY FUND

# SHAPING THE FUTURE OF ASIA

## Opportunities and Challenges



**Editors**

**RAHUL ANAND**
**THOMAS HELBLING**
**KRISHNA SRINIVASAN**

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# Foreword

Truly, the economic rise of Asia and the Pacific has been one of the world's most inspiring success stories. Hundreds of millions of people have been lifted out of abject poverty. Everyday life has improved on multiple fronts, from rising incomes to greater access to education, healthcare, and financial services to improvements in infrastructure and technology. This transformation has reshaped not only the region, but the world. Asia today generates two-thirds of global growth and accounts for close to 40 percent of world trade. The journey has not always been smooth, yet the region has put much truth in the saying that crisis can create opportunity: in the aftermath of the Asian Financial Crisis, country after country strengthened its policy frameworks, rebuilt its financial system, deepened trade integration, and invested in productivity. The results speak for themselves.

Nonetheless, further building on the region's achievements now poses new and formidable challenges. The very foundations of the region's growth and resilience are being tested and reshaped by geopolitical tensions, disruptions to world trade, concerns around energy security, demographic transitions, and technological change. Artificial intelligence promises substantial gains in productivity growth but also raises deep questions about labor markets, inequality, and the future of work. At the same time, Asia is aging faster than any other region in the world, even if demographic conditions continue to vary significantly across countries—some will still benefit from a growing labor force for a decade or more; others are seeing the opposite. These shifts are unfolding in a global economy defined by recurring shocks and increasing fragmentation.

*Shaping the Future of Asia* looks at these challenges and opportunities through a long-term lens. The chapters in this volume explore how Asia can maintain strong, inclusive, and resilient growth while adapting to a rapidly changing world. A central theme throughout the book is the need for agility and reform. Several chapters examine reforms that can cushion the growth, labor market, and fiscal impacts of the demographic transition. These include measures to strengthen labor force participation and mitigate the negative labor supply effects in aging economies, as well as measures to improve opportunities for young people.

Other chapters focus on the transformative potential of artificial intelligence and digitalization. The productivity gains these technologies can generate and the investments required to realize them can be a critical counterbalance to the demographic pressures noted above. The authors discuss the importance of digital infrastructure, financial development, education and skills training, and regulatory frameworks to encourage innovation while safeguarding stability and inclusion.

The volume also considers the future of trade and regional integration in a multipolar world. Asia's growth model has long benefited from openness, exports, and participation in global value chains. The authors show that deeper regional

integration, particularly through digital connectivity and cross-border financial and payment systems, can provide an important source of future growth and resilience.

*Shaping the Future of Asia* reflects a collaborative effort among IMF staff, academics, and policymakers across the region. It draws on extensive policy dialogue with Asian authorities, including most recently at the IMF's "Asia in 2050" conference, as well as the shared experiences of countries confronting common challenges under diverse circumstances.

Asia enters this period of global transition from a position of considerable strength, with dynamic economies, innovative private sectors, and a strong record of adaptation and reform. I have no doubt that Asia has the talent and spirit to remain a driving force in the global economy in the decades ahead.

Kristalina Georgieva
Managing Director
International Monetary Fund

# Preface

Since 1990, the Asia and Pacific region has undergone one of the most significant economic transformations in modern history, extending far beyond the earlier classic experiences of Japan and several other economies. What was once a diverse collection made up predominantly of developing and emerging markets has become a central driver of the global economy. Between 1990 and 2023, income per capita in the region more than doubled, lifting hundreds of millions of people out of poverty and dramatically improving living standards. Over the same period, the region's contribution to global growth also rose sharply, increasing from about one-quarter to nearly two-thirds of the world total. Profound social change accompanied this economic progress, including gains in life expectancy, expanding urbanization, and broader improvements in human well-being, especially in lower-income economies.

Yet the conditions and forces that made this remarkable ascent possible are changing. Economic growth is slowing from the high rates of the past, demographic trends are shifting, and the global economy has become more complex, uncertain, and fragmented. In some economies, investment and productivity have weakened, while the rapid expansion of global trade—one of the defining engines of the region's growth—has lost momentum. At the same time, rising geopolitical tensions are reshaping the global landscape in ways that could affect Asia and Pacific more than most regions.

The region itself is experiencing important structural changes. The populations in some economies are aging rapidly, creating new pressures on labor supply, public finances, and growth. Others continue to benefit from young and expanding workforces but face the challenge of creating sufficient economic opportunities for them. Technological advances, especially in digitalization and artificial intelligence, are opening new avenues for growth, productivity, and innovation even as they raise difficult questions about economic inequality and the future of work. Climate change has also become a more visible and urgent challenge, with more frequent and extreme weather events and natural disasters threatening lives, livelihoods, and development gains.

Change on this scale raises some fundamental questions. What explains the region's extraordinary transformation over the past 35 years? Can the same forces continue to sustain progress in the decades ahead? How should policymakers respond to a changing world that is challenging long-standing assumptions about demographics, technology, and globalization?

The answers are neither simple nor uniform across countries. Much of the region's success was built on structural transformation—the movement of labor and capital from less productive sectors and activities to more dynamic sectors, industries, and services—combined with rising levels of human and physical capital, gains in productivity, and favorable demographic trends. Yet several of

these tailwinds are now weakening. Population aging is expected to reduce labor's contribution to growth in many of the region's economies, and diminishing returns on investment have become a growing concern. These forces underscore the need for policymakers to maintain a strong focus on innovation, efficiency, and policy reform.

This book contributes to that ongoing discussion by bringing together a range of perspectives on key forces shaping the region's future, with structural transformation as a common thread throughout. The chapters examine how Asia and Pacific can sustain growth, promote inclusion, and remain a dominant force in the global economy.

As economies evolve, structural transformation will remain essential for sustaining growth and raising living standards. At the same time, demographic change will require policies that both harness the potential of younger populations and address the economic and social implications of aging.

Technology is another critical dimension of the region's future. Digitalization and artificial intelligence promise substantial productivity gains, broader access to services, and entirely new forms of economic activity. However, realizing these benefits will require governments to ensure that technology is widely accessible, and they must prepare societies to manage the disruptions that often accompany innovation.

The global environment will also be a decisive factor. The region's rise has been closely linked to its integration into global trade and financial systems. As those systems evolve and potentially become more fragmented, economies across the region can adapt by strengthening regional connections and diversifying economic partnerships. The global resurgence of industrial policy complicates this landscape, bringing both opportunities and risks for trade, investment, and economic structure.

Resilience—both economic and environmental—has become a defining priority. Whether confronting economic shocks, geopolitical uncertainty, or climate-related disruptions, the region's future prosperity will depend on both its capacity to grow and its ability to withstand and adapt to change. The frequency of climate-related events highlights the importance of policies that promote sustainable and inclusive development while protecting vulnerable populations and preserving hard-won gains.

Together, these issues point to a simple but powerful conclusion: the next chapter of Asia and Pacific's development will not be written by repeating the strategies of the past. It will be shaped by the region's ability to adapt to new realities, embrace technological and structural change, seize new opportunities, and strengthen its resilience in an increasingly uncertain world.

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CHAPTER 1

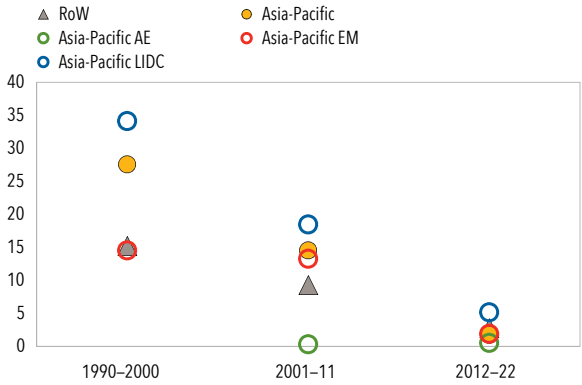
# Asia and Pacific at a Turning Point: Transformation, Megatrends, and the Next Growth Frontier

**Thomas Helbling, Mike Li, and Krishna Srinivasan**

The Asia and Pacific region has emerged as a pivotal force in the global economy after decades of remarkable and rapid economic development. Its GDP per capita more than doubled between 1990 and 2023, and quality of life has improved significantly. This transformation has elevated the region's contribution to global GDP growth from about a quarter to nearly two-thirds, underscoring its critical role in shaping global economic dynamics. The share of people living in extreme poverty fell dramatically from more than 25 percent to less than 5 percent (Figure 1.1), while life expectancy increased rapidly, especially in lower-income countries. Rapid economic growth and the related structural transformation drove these achievements, and that transformation has been central to the Asian economic miracle.

The remarkable progress bodes well for the region's prospects, but several megatrends present new challenges and opportunities. Rapid productivity gains that fueled earlier economic growth have slowed virtually everywhere since the global financial crisis. Meanwhile, population and labor force growth are decelerating in many countries while population aging accelerates. Still rising risks of geoeconomic fragmentation and indications of peak manufacturing, with the expected rise of the services sectors in production and trade, call into question the role of international trade and large manufacturing sectors as future engines of growth. New technologies, including digitalization and especially AI, could boost productivity and innovation, possibly even more than previous general-purpose technologies. The use of industrial policies has accelerated again in this context, raising old and new debates and issues. The transition to a sustainable future highlights the importance of sustainability for macroeconomic stability and the global energy system. These challenges and opportunities are crucial for policymakers, businesses, and individuals, including in several of the region's countries that aspire to achieve high-income status within the next 20 to 25 years (for example, Bangladesh by 2041, Vietnam by 2045, and India by 2047).

**Figure 1.1. Poverty Headcount Ratio**
*(Percent of population)*




Sources: World Bank World Development Indicators; and IMF staff calculations.
Note: The dots and triangles show the median of the population living on less than $2.15 a day at 2017 purchasing power-adjusted prices. Asia-Pacific EM includes CHN, IND, IDN, MYS, PHL, LKA, and THA. Asia-Pacific AE includes AUS, JPN, and KOR. Asia-Pacific LIDC includes BGD, LAO, MNG, and VNM. Data for JPN and KOR were unavailable for 1990–2000; hence, Asia-Pacific AE is not shown in that time period. RoW denotes rest of the world.

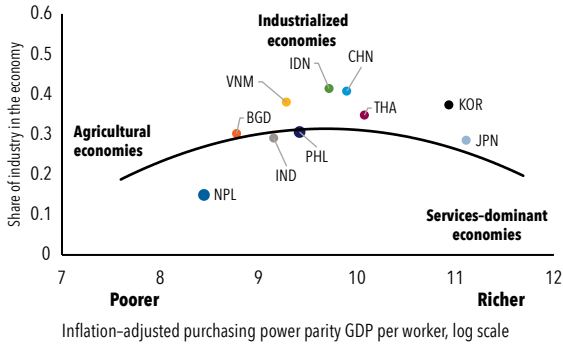
Several megatrends will be especially transformative for the region's economic future, affecting growth, development, and regional integration. Demographic dynamics will shape outcomes through aging and declining populations in most of the region's major economies and through the potential to harness a burgeoning young workforce in others. New technologies will boost productivity and innovation through digitalization and AI adoption. The evolving trade and financial landscape reflect geoeconomic fragmentation and the slowdown of merchandise trade growth. Industrial policies can also have a role in shaping economic outcomes and global dynamics. The transition to a sustainable future highlights the importance of sustainability for macroeconomic stability and is reshaping global energy demand.

This book explores the implications of these megatrends for the economic future of Asia and Pacific in the context of the region's growth experience since the 1990s. The targeted analyses of these megatrends can yield practical policy insights tailored to the region's unique challenges and opportunities and lay the groundwork for future research on other aspects of its evolving landscape.

## THE STRUCTURAL TRANSFORMATION AND INVESTMENT BEHIND THE ECONOMIC MIRACLE

The Asia and Pacific region has experienced a profound economic transformation over the past 30 years that lifted hundreds of millions out of poverty and improved living standards. Structural transformation has been central to the

**Figure 1.2. Structural Transformation in Asia**
*(Industry share of value added versus income level)*




Sources: GGDC/UNU-WIDER Economic Transformation Database, Penn World Table version 10.1; and IMF staff calculations.
Note: The curve represents global averages, estimated with 2018 data.

region's success, driving rapid growth and shifting labor and resources from lower-productivity agriculture into higher-productivity manufacturing and services (Figure 1.2 and Chapter 2). In China, the share of employment in agriculture dropped from about 60 percent in the late 1980s to less than 25 percent by 2020. This structural shift was vital to productivity growth because manufacturing jobs typically have higher output per worker than agricultural jobs.

Asia and Pacific countries industrialized more intensively than peer countries at similar income levels. Policies that promoted export-led industrial growth and openness to global trade contributed to transforming the region into "the world's factory," accelerating industrialization and supporting innovation spillovers. China built one of the world's largest manufacturing sectors, supported by pro-market reforms, rapid urbanization, significant infrastructure development, and deeper integration into global supply chains. Vietnam began industrializing in the 1990s with major reforms and liberalization that attracted significant foreign investment, which drove rapid growth in manufacturing sectors such as electronics, textiles, and footwear. This growth spurred rapid urbanization and the rise of export-oriented sectors. Also, targeted foreign investment in Malaysia's Penang electronics sector and Thailand's eastern seaboard automotive industry brought capital and technology, which supported regional supply chains.

The share of services in economies has begun rising as incomes rose and preferences changed. As manufacturing moved up the value chain and became more capital-intensive, labor-intensive service sectors absorbed a greater share of workers leaving agriculture—especially in retail, hospitality, government, and informal sectors—and offered more job opportunities. Technological advances transformed many services from local activities to internationally tradable activities. For example, a software developer in Vietnam could work for a company in California, and a call center agent in the Philippines could provide customer support for a

European bank. Countries in the region now export a wide range of IT and business services, and India and the Philippines became leaders in this field. Tourism remains a vital export sector. Even in China, service sectors (e-commerce, financial technology, and tourism) have grown rapidly and now account for a larger share of total value added and employment than industry. By 2040, services are expected to contribute 60–70 percent of GDP in many emerging market economies in the region, with growth concentrated in digital, finance, health care, logistics, creative, green, and tourism industries. Greater trade liberalization and the removal of regulatory barriers to increase competition in services could enhance productivity and unlock substantial additional growth potential.

Enabling continued structural transformation will be critical for sustaining growth in per capita incomes and improving economic well-being in the region. Shifting employment and resources toward more productive firms and sectors is expected to remain a force for growth. The service sector is poised to expand even more within the economy, but the ongoing rise of automation and supply chain reconfiguration will likely limit the manufacturing sector's ability to absorb labor from other sectors. Economic policies can support future transformation, including by equipping the workforce with the skills needed to adopt and use new technologies and production processes (Chapter 2).

Managing a smooth transition from traditional investment-led growth to new growth drivers will be another fundamental challenge for the region. High investment ratios and the related increase in capital intensity have been a hallmark of growth for many decades. The average investment-to-GDP ratio in the region's economies was approximately 27 percent as of 2024, a level markedly higher than that in Europe or the United States. Although these factors have spurred rapid income convergence, growing evidence of declining capital efficiency and diminishing returns has emerged. Increasing capital misallocation is a related concern, reflected in a widening dispersion of the marginal revenue product of capital across firms and a greater concentration of capital in less productive firms (Chapter 3).

Relatively underdeveloped capital markets that limit efficient capital allocation have contributed to the challenge. The issue is not one of overall availability of funding for investment—at 143 percent of GDP, credit to nonfinancial firms is well above global averages. The issue instead lies in the structure of the region's financial system, which remains highly bank-centric despite expanding capital markets. Financing through capital markets is still relatively limited and concentrated among a few large firms and state-owned enterprises, like the provision of bank credit. Listing costs, disclosure requirements, and limited secondary market activity discourage smaller firms from accessing capital markets, while low institutional investor participation and regulatory constraints limit market liquidity and size. Venture capital and private equity, though growing, remain small and concentrated. These constraints force many firms to rely on internal funds (especially small and medium-sized enterprises), which impedes their growth and innovation potential. Favoring larger firms and state-owned enterprises (a major part of the region's corporate landscape) has contributed to declining capital productivity and increasing misallocation, reducing opportunities for high-potential, innovation-driven young

firms (Chapter 3). Government interventions such as interest rate caps and directed lending also distort market incentives and hinder efficient capital allocation.

Improving financial systems to support more efficient capital allocation will be critical for strengthening growth and structural transformation. Achieving this outcome requires reforms that develop capital markets and diversify financial instruments to better match the scale and mix of financing that a broader range of firms demand. Deeper equity and bond markets, venture capital, and alternative nonbank channels are needed to support firms at different growth stages. Simplified listing and regulatory procedures, stronger corporate governance, increased institutional investor participation, and reduced financial repression can improve market liquidity and confidence. Enhancing market discipline for state-owned enterprises while broadening financing access for private, growth-oriented firms is also vital (Chapter 3).

Understanding the region's structural transformation and financial system provides insight into its past successes and what should be on the agenda for reform. The rapid shifts from agriculture to manufacturing and services and the ability to manage capital efficiency despite high investment and to develop capital markets to support private sector growth underpinned the region's impressive economic development.

## THE DEMOGRAPHIC TRANSITION IN THE NEXT QUARTER CENTURY

A sharp demographic transition resulting in aging populations and declines in the labor force is one of the megatrends influencing growth prospects in the region. The profound demographic shifts are driven primarily by two trends: declining fertility rates and increasing life expectancy (Chapter 4). Fertility has dropped from about six children per woman in 1965 to about 1.6 children per woman in 2026—well below the replacement rate of about 2.1 children per woman—while life expectancy has risen from 49 years to 74. As a result, working-age populations in China, Japan, and Korea are already shrinking, while the share of people ages 65 and older in the region is increasing rapidly to about 20 percent of the population by 2050 from 11 percent in 2025. The old-age dependency ratio, a key indicator of population aging, is set to double to more than 30 percent by mid-century, surpassing that of Africa or the Middle East regions. However, the ratio will remain below that of Europe and will be broadly similar to that of the Western Hemisphere.

Within these broad trends, the speed of aging varies across Asia and Pacific. The trend of rapid population aging at the regional level is shaped by China, Japan, Korea, and a few other countries where fertility rates declined earlier or faster. Yet some countries still benefit from a growing workforce. To illustrate the region's demographic diversity, it is helpful to group countries into three categories according to their position with respect to prospects for a demographic dividend—that is, the potential for economies to harness their growing workforces to boost economic growth. "Early-dividend" countries have working-age

populations that are growing faster than the overall population, and these are mostly lower-income countries, such as Bangladesh, Lao P.D.R., and many Pacific island countries. "Late-dividend" countries are mostly emerging market economies in the region, where the labor force grows at a relatively lower rate and accounts for a declining share of the overall population. "Post-dividend" countries face shrinking working-age populations in absolute terms and as a share of the population, and these are mostly advanced economies such as Japan and Korea, and some emerging market economies, including China and Thailand.

The demographic transition carries significant macroeconomic supply-side implications, but reforms that promote labor force participation and skills can alleviate them. As working-age populations shrink, aging reduces labor supply, risking labor scarcity and damping growth. This dynamic calls for concerted policy efforts to address the key barriers to increasing effective labor force participation, especially among demographic groups with lower participation rates caused by factors such as skills mismatches or limited access to flexible working arrangements. With advances in health care and technology extending healthy working lives, increasing older workers' participation can ease the labor supply impact. This can be achieved by raising statutory retirement ages, promoting reskilling (especially in digital skills), and enabling flexible, age-friendly work arrangements. Separately, a gender gap in labor force participation persists in the region, but it can be closed by enhancing education and training, reshaping workplace norms, expanding affordable childcare, and promoting family-friendly policies such as equal parental leave, thereby mitigating the labor supply effects of aging. Japan has succeeded in raising women's labor force participation rate over the past 10 years through enhanced childcare support from the government and improved maternity leave policies. Migration offers another potential avenue for mitigating the drag from aging. Together, these policy actions, supported by ongoing efforts in fiscal management and pension reforms, will support the region's countries in mitigating the demographic challenges and harnessing remaining demographic dividends.

## The Fiscal and Financial Implications of Population Aging

Population aging and the related rise in dependency ratios will increase government expenditure on pensions and health care. One reason for the increase is that retirement ages have not kept pace with rising longevity. Other factors determining the extent and speed of the increase are the coverage and benefit generosity of the public pension programs, and the mechanisms that fund them, which vary across the region and include noncontributory plans, pay-as-you-go defined-benefit systems, and defined-contribution plans (mostly in advanced economies such as Australia and Singapore). Public pension spending has reached about 7 percent of GDP in East Asia, where aging has been most prominent, and is well above the rest of the region. Meanwhile, the limited coverage and adequacy of pensions have led several countries to broaden mandatory and voluntary pension plans to include informal and part-time workers (for example, Indonesia) or raise contribution rates and replacement ratios (for example, Korea).

IMF staff analysis suggests a substantial increase in pension expenditure through 2050 without changes to the parameters of the system. In advanced economies, spending could rise by an average of 2 percent of GDP between 2024 and 2050, from about 6 percent to more than 8 percent of GDP. In emerging markets, spending is projected to increase by about 1 percent of GDP on average. As populations age, health spending is expected to increase and will grow even faster than spending on pensions, driven by rising unit costs amid increasing demand for health care services by older people (Chapter 5). Systemic inefficiencies compound these pressures, including overreliance on costly hospital-based care versus more efficient primary and preventive services, and spending needs to improve health services, especially in developing countries.

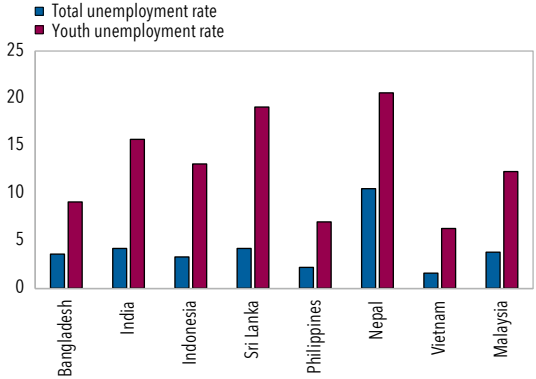
Governments will face challenging trade-offs as they strive to manage escalating age-related expenses, but timely reforms are essential to managing the rising pressures on fiscal positions while addressing other spending priorities and maintaining fairness in distributing the fiscal burden across generations is also crucial. This underscores the urgency of implementing reforms to improve the sustainability, coverage, and equity of pension and health systems. Potential policy measures include raising the retirement age, increasing contribution rates, better targeting of benefits, expanding health coverage with an emphasis on primary care and cost control, and finding the right balance between public and private roles in financing and service delivery.

Beyond public finances, population aging also affects financial systems mainly through changes in behavior. Older people typically lower their exposure to financial risks. They increase holdings of liquid and safer financial assets such as deposits and reduce holdings of riskier assets. Bank data also suggest that population aging tends to lead to reduced borrowing. To adapt, banks may offer more fee-based services and explore business areas beyond conventional lending. Although increased liquidity can support regulatory buffers, the search for returns could drive emerging risks into less transparent sectors, underscoring the need for regulatory adaptability. Data show that banks tend to allocate fewer assets to aging, slower-growing markets and redirect investments toward younger, faster-growing economies. This dynamic has significant implications for financial integration, competition, and stability and requires heightened supervisory vigilance. Aligning regulatory and supervisory systems with demographic trends is crucial to ensuring financial stability and enabling sustainable economic growth (Chapter 6).

## Demographic Dividend and Migration

Despite rapid aging in parts of the region, some countries will continue to benefit from demographic dividends for some time. The rising share of the working-age population in the total population has historically played a critical role in rapid growth episodes in the region, notably in the economic miracles in East and Southeast Asia in the 20th century. However, without reforms, the contribution of the growing share of the working-age population to the region's per capita income growth would be modest and peak by 2035 (Chapter 7). To maximize and sustain

**Figure 1.3. Unemployment Rate: Total and Youth**
*(Percent)*




Sources: ILO consolidated LFS database; and IMF staff calculations.
Note: All values are for 2024.

the benefits, reforms should target boosting education and skills training aligned with labor-market demands, while promoting job-rich private sector growth supported by stable macroeconomic and regulatory environments. Inclusive growth and the creation of high-quality jobs in the domestic economy are essential for avoiding socioeconomic tensions, which often intensify when interregional and intragenerational wealth and job inequalities widen (Figure 1.3).

Migration can be important to shaping the region's demographic transition. As countries and labor markets integrate, incentives for migration can increase with the demographic transition, changing labor supply and economic prospects across countries. Reducing labor-market friction and enhancing both domestic and cross-border mobility are vital to matching workers with available jobs and to structural changes induced by demographic shifts and technological progress. Aging post-dividend economies can benefit from labor inflows that can ease demographic pressures by slowing the decline in the working-age population. Emigration from dividend economies can help alleviate local labor-market pressures and support local consumption through remittances. But these benefits must be weighed against the risks from brain drain, especially in smaller nations, such as Pacific island countries. Effective migration and integration policies focusing on balancing labor supply losses and skill gains in origin economies, supporting productivity and inclusion in destination economies, and promoting coordinated regional mobility would maximize gains and minimize socioeconomic challenges in both sending and receiving economies (Chapter 7).

## NEW TECHNOLOGIES

New technologies and related changes to production structures constitute another megatrend affecting the region. They promise substantial productivity gains at a

time when the region's traditional growth engines—demographic dividends, export-driven manufacturing, and global value chain integration—are losing momentum. The region is already leading in digitalization and AI adoption through rapid innovation, substantial investment in digital infrastructure, and a vibrant startup ecosystem and is positioned to emerge as a leading consumer and innovator of AI. Singapore has leveraged AI in public services and is emerging as a regional AI innovation center, India has made significant strides in financial technology innovation and is becoming a global IT services hub, and Japan is pioneering AI integration in robotics and finance.

## Artificial Intelligence

AI stands out as providing a potentially powerful boost to productivity and economic growth. It is a general-purpose technology that drives sophisticated automation in the production of services and goods, including through large language models. Model-based simulations have found that AI adoption can boost growth by 0.2 to 0.8 percentage point annually in emerging and developing economies and 0.5 to 1 percentage point in advanced economies in the region, depending on the pace of AI technological advance (Chapter 8). It does so in part by transforming labor-intensive industries into more capital- and technology-intensive industries. This transformation can support productivity growth, particularly in aging economies with shrinking workforces (Chapter 4). Japan, for example, uses AI in robotics and manufacturing automation to sustain productivity amid rapid aging, with elder care robots improving service efficiency. Korea leads AI adoption in manufacturing and knowledge-intensive services, achieving notable productivity and profitability gains. Vietnam's expanding digital infrastructure and human capital are accelerating AI adoption, with potential for significant productivity and GDP growth.

However, the path of AI adoption can be uneven, leading to disparities in gains across the region. Large-scale AI adoption requires access to AI technology and investment in AI-enabling physical and digital infrastructure (such as data centers and specialized processors) to deploy AI widely and effectively. Advanced economies such as Japan, Korea, and Singapore are better prepared structurally for early AI adoption. Conversely, emerging market and developing economies such as Bangladesh, Thailand, and Vietnam could face delayed adoption because of lower skill levels, weaker productivity, and lower capital intensity, which risks widening economic divergence within the region. The AI-related investment surge could also raise global interest rates and contribute to widening economic divergence because it would dampen other investment in emerging markets and developing economies. Structural reforms focused on raising skills, boosting labor and capital productivity, and improving institutions can materially hasten AI adoption and help achieve sustained long-term economic gains (Chapter 8).

Uneven exposure, complementarity, and adoption within countries can also increase inequality in income and opportunity. AI's labor-market impact and

economic benefits will vary by sectors and occupations. High-skilled workers in AI-complementary roles are likely to benefit from wage gains through higher productivity, while low- and medium-skilled workers face displacement risks and wage declines during this transition. Women are especially overrepresented in high-exposure but low-complementarity jobs that are at risk of automation. Similarly, IMF staff analysis based on firm-level evidence in Korea points to greater productivity gains in larger and mature firms than in their smaller and younger peers, potentially widening firm-level disparities.

Economic policies can strengthen the benefits from AI while reducing dislocation and risks of widening inequalities. Active labor-market policies emphasizing retraining, upskilling, and realistic occupational mobility pathways can mitigate labor-market displacement risks. Improving access to finance for small and medium enterprises through expanded credit guarantees, tailored innovation funds, and blended finance with financial technology solutions can facilitate AI adoption and productivity gains beyond industry front-runners. Digital public infrastructure can be a vital foundation for broader AI access and have a crucial role in bridging the adoption gap. A nuanced and country-specific redistributive policy that strikes a delicate equity-efficiency balance is likely needed to support displaced workers as a safety net during the transition. A comprehensive policy package combining structural reforms and fiscal redistribution can help ensure that AI-driven efficiency gains translate into inclusive and sustainable development for the region (Chapter 8).

## Financial Technology and Digitalization

Innovations in financial technology, especially digital payments, can also potentially boost economic growth. Functioning as the essential "rails" (underlying infrastructure and networks) of the digital economy, digital payments form a crucial part of public infrastructure. Asia and Pacific leads the world in mobile broadband subscriptions and internet penetration and is supported by strong network readiness, which provides a solid foundation for innovation and adoption of digital payments. ASEAN countries have created one of the world's most innovative and interconnected cross-border mobile payment networks. However, its adoption has been uneven: economies at the technology frontier such as Japan, Korea, and Singapore show near-universal digital payment use, while many emerging and low-income economies face implementation challenges and notable urban-rural divides.

Digital payment innovation and adoption have the potential to improve government services significantly, lower transaction costs, and promote financial inclusion. India's digital payment infrastructure generated substantial fiscal savings by promoting formalization and enabled rapid household transfers during the COVID-19 pandemic. Sri Lanka's GovPay online payment platform shows how digital payments strengthen fiscal governance and improve government services by enhancing transparency, reducing leakages, and accelerating service delivery. Digital payments can also help unlock access to financing by leveraging transaction

data to reduce information asymmetries, enabling financial technology firms to extend credit increasingly to underserved borrowers, including small firms, with evidence already showing improved loan approval rates and terms.

Digital payments can also accelerate cross-border transactions and thereby facilitate stronger regional integration. Faster, less costly, and safer transactions and increased local currency usage in cross-border payments can reduce transaction costs in cross-border payments, lower foreign exchange risks, and strengthen financial stability. A project of the ASEAN, the Integrated QR Code Payment System, is expected to significantly enhance regional payment connectivity and lower transaction costs. Such multilateral efforts are also critical for mitigating payment challenges faced by the Pacific island countries because of geographic dispersion and loss of correspondent banking relationships.

A supportive regulatory and infrastructural environment is vital to fully realizing these benefits. Guaranteeing equitable access is essential to ensuring that digital payment systems promote inclusive economic growth. Mandating interoperability is vital to avoiding fragmentation and promoting competition and innovation, as the rapid growth of India's Unified Payments Interface illustrates. Governance frameworks that ensure consumer protection, cybersecurity, data privacy, and market fairness underpin trust and system resilience. Regulators must also address emerging risks related to market concentration, financial stability, and concerns about AML/CFT as digital payments expand (Chapter 9).

## GEOECONOMIC FRAGMENTATION AND THE FUTURE OF TRADE AND FINANCIAL INTEGRATION IN ASIA AND PACIFIC

Growing geoeconomic fragmentation is expected to continue and thus is becoming another megatrend that shapes the structure and gains from the region's continued integration into global markets. Trade and financial integration have been crucial for the region's rapid economic growth, including becoming part of global supply chains. The region's share of global capital flows—especially foreign direct investment—has also increased but relatively less than the rise in its export share. Emerging market economies in the region still have great potential for deeper trade and financial integration. However, with geoeconomic fragmentation, deeper integration can be more challenging to achieve, raising questions about its role as a future engine of growth in the region.

### Trade Integration

The region's rapid trade expansion in recent decades was powered by low-cost and abundant labor, declining transportation costs, and broad global trade liberalization. Deep integration into global supply chains has transformed the region into a major manufacturing hub, and it now accounts for more than one-third of global trade,

about half of which takes place within the region. China has emerged as the largest trading partner for several major economies and the principal import source for many others.

Large shifts in global trade and finance policies present new challenges. Geoeconomic fragmentation has already reshaped trade patterns and supply chains in the region, as demonstrated by the development during the US-China trade tensions in 2018–19 and the changing geography of foreign direct investment. The 2025 US tariff shocks have affected the region disproportionally, as have other increases in protectionism, given its high trade openness, and contributed to further reconfiguration of trade and supply chains. Many Asia and Pacific economies also experienced a structural shift from manufacturing toward services, sectors that have historically been more protected in the region, which highlights the need to focus more on services trade.

Asia and Pacific has scope to strengthen intraregional integration. Despite robust trade integration in intermediate goods, the share of intraregional trade in final goods remains comparatively low at less than 30 percent versus approximately 60 percent in European and North American trading blocs, highlighting an opportunity for growth through deeper integration. Regional trade agreements tend to be fragmented, largely bilateral, or bloc-based, lacking the wide, legally binding plurilateral structures seen in those of Europe or North America (Chapter 10). These fragmented accords often result in overlapping rules and inconsistent standards, adding friction especially to intraregional trade. Compared with other regions, Asia and Pacific trade arrangements are often shallower and have less comprehensive coverage and enforcement, particularly regarding issues outside World Trade Organization mandates such as competition policy, environmental standards, and e-commerce.

Deeper regional integration can be an engine of growth. Trade agreements that cover a broader range of enforceable provisions could raise the region's real GDP by about 1.8 percent in the long term, primarily through unilateral reductions in nontariff barriers that lower trade costs, improve access to intermediate inputs and resource allocation in line with comparative advantage, thereby boosting productivity and capital accumulation (Chapter 10). Gains are even larger when many countries liberalize concurrently because enhanced regional demand reinforces growth prospects. To ensure inclusive growth, complementary social protections and labor-market policies are crucial to mitigating the adverse distributional impacts of trade reforms.

## ASEAN Integration

The scope for stronger regional integration is also relevant for ASEAN member countries, despite already significant achievements. ASEAN member countries currently constitute the fifth largest economy in the world and are deeply integrated into global supply chains. The group's framework for economic and financial integration strengthened significantly in the past three decades, during which it established the ASEAN Free Trade Area (1992), launched the ASEAN Economic

Community (2015) that enabled deeper integration in services, investments, and labor; and upgraded the ASEAN Trade in Goods Agreement (2025). With intra-ASEAN trade at only about 20–25 percent of total exports—far below levels in other major trading blocs such as the European Union or the United States–Mexico–Canada Agreement, the potential for greater trade integration among the member countries is significant, even if export complementarity among ASEAN member countries is lower than that with key external partners such as China and the United States.

To realize these opportunities, ASEAN member countries should undertake targeted reforms. Evidence points to significant potential to boost regional trade and economic integration by advancing reforms in digital government, logistics, and human capital, especially in lower-income ASEAN member countries such as Cambodia and Lao P.D.R. (Chapter 11). Promoting resilient global value chain integration through supplier upgrading, cross-border industrial clusters, and diversified input sources can deepen production ties and buffer against shocks. Regulatory alignment and investor protections can catalyze regional capital flows, reinforce supply chains, and encourage intra-ASEAN investment. Reducing non-tariff barriers—which remain significant in ASEAN—and advancing regulatory harmonization will help reduce fragmentation and boost trade predictability.

## Policies to Navigate Fragmentation

How should policymakers respond to the new constraints of geoeconomic fragmentation and a shifting international landscape? In a guest contribution (Chapter 12), Danny Quah argues that the economic progress of smaller states—which has underpinned much of the progress in the Asia and Pacific region—is now at risk with more geoeconomic fragmentation and strategic interdependence. This transition can challenge traditional views on the benefits of trade openness, economic efficiency, and accepting the multilateral rules of the global trading system.

Yet for smaller states, international cooperation remains critical as they seek to sustain trade and financial integration as engines of growth and development. A stronger focus on regional integration, as discussed above, will be important. However, Quah contends that the region's smaller states must also reconsider strategies that rely on passively accepting international prices and norms. Instead, they could form loosely structured, open alliances, whose members cooperate when interests align. Such alliances do not require hegemonic leadership but must be sufficiently multilateral to be effective.

## Financial Integration

Financial integration in Asia and Pacific has advanced more slowly than trade integration. The region's financial integration remains uneven, with advanced economies becoming well integrated globally while emerging markets exhibit more limited integration. Across financial instruments, foreign direct investment and cross-border banking are increasingly linked internally in the region,

primarily reflecting strong trade integration. However, as a percentage of GDP, related asset positions remain modest, compared with those in Europe and the Americas. By contrast, foreign portfolio investment positions show little correlation with trade, remain limited in scale, and are concentrated in a few advanced economies. Policies to strengthen financial integration are increasingly relevant for boosting domestic demand and cultivating new sources of growth (Chapter 13).

Regional financial integration is likely to strengthen in the next decade, given ongoing integration in cross-border payment systems, including among ASEAN members, which boosts connectivity and lowers financial transaction costs. Strong legal frameworks and smaller government footprints in destination countries are important pull factors that attract financial inflows, while economic instability and weaker institutions are deterrents (Chapter 13). Capital flows also pose challenges by increasing exposure to global shocks and financial contagion. Reforms to manage such challenges should include strengthening regulatory frameworks, improving supervisory capacity, enhancing cross-border payment infrastructure, managing financial risks through macroprudential tools, and promoting regional cooperation.

## THE NEW INDUSTRIAL POLICY WAVE

Asia and Pacific countries have increasingly used industrial policy tools to strengthen or maintain their positions in some sectors or product markets and to advance national security objectives, in line with global trends. Such vertical policy interventions can take many forms, such as subsidies, export promotion, import restrictions, protectionism in procurement, local content requirements, and investment restrictions, among others. The region has long relied on industrial policies but has increased their use significantly since 2009, especially in emerging markets and advanced economies. Domestic subsidies are the main instrument. Approximately 80 percent of industrial policy interventions have focused on industrial and manufacturing products, while interventions in the service sectors were limited. Most interventions focused on either strengthening the competitiveness of new products closely related to a country's current export basket or promoting innovation and technological leapfrogging in new products. By contrast, interventions aiming to correct market failures seem less prominent (Chapter 14).

Despite the resurgence, long-standing concerns about the effectiveness of industrial policies remain relevant. The positive effects of industrial policies generally dissipate within a few years, implying a limited structural impact. Notably, domestic subsidies and export incentives are found to enhance comparative advantage only for products for which countries already have an existing edge. The seemingly successful cases of industrial policy interventions boosting exports could mean that those products are already competitive in global markets. Import-limiting measures are associated with transient import substitution effects but

rarely translate into sustained industrial competitiveness. At the firm level, industrial policy interventions are often associated with industries exhibiting lower productivity, with no systemic or significant productivity catch-up observed. These findings raise questions about whether industrial policy is targeted and selective and whether the benefits outweigh the costs, including fiscal costs.

Ultimately, achieving sustainable economic transformation will require parsimonious, balanced, well-targeted industrial policies complemented with robust structural reforms. Caution is suggested in expanding the use of industrial policy without considering institutional capacity and policy design. Although some evidence of localized successes exists, mostly in advanced economies in East Asia with established institutions, the broader findings call for a prudent and carefully designed approach to industrial policy. Effective industrial policies should focus on addressing well-identified market failures, be time-bound and cost-effective, and be supported by strong governance to minimize rent-seeking and inefficiencies. Crucially, horizontal structural reforms remain indispensable for widespread and sustained economic benefits (Chapter 14).

## ASIA AND PACIFIC'S TRANSITION TO A SUSTAINABLE FUTURE

As the region's industrial landscape evolves in response to shifting policies, a transition to a sustainable future is imperative. Asia and Pacific is a powerhouse for global growth but also a large greenhouse gas emitter because of its heavy reliance on coal for generating electricity and for other energy, especially in China, India, and Indonesia. Yet the region also faces acute climate vulnerabilities and has been a leader in developing and installing renewable energy capacity. Rising sea levels threaten densely populated coastal cities, such as Bangkok and Ho Chi Minh City, and low-lying archipelagos, such as Maldives and parts of the Philippines. Changing monsoon patterns and more frequent droughts jeopardize agriculture across the region, and increasingly extreme heat risks reducing labor productivity and human health. Extreme weather events strain infrastructure and disrupt supply chains, highlighting the urgent need for climate-resilient development and energy diversification.

The green transition is essential for both the economy and the well-being of the population, yet its economic and social impacts must be managed effectively (Chapter 15). Only then can the transition provide substantial economic opportunities from the investment in renewable energy, energy-efficient retrofitting, electric vehicle manufacturing and charging infrastructure, and climate-resilient development. These investments will create new industries and jobs and help modernize and strengthen economies.

The Asia and Pacific region is already emerging as a global leader in green technology. China leads the world in producing solar panels, wind turbines, and electric buses; India is pursuing one of the most ambitious renewable energy expansion plans; and Japan and Korea are making significant investments in

battery and hydrogen technologies. Their leadership in green innovation positions the region to drive a sustainable and resilient future.

Climate adaptation and the green transition will also demand increased access to finance, a critical challenge in the region. Mobilizing funds from both public and private sources is vital to supporting renewable energy deployment, climate-resilient infrastructure, and the overall green transition. Yet obstacles such as limited fiscal space, high borrowing costs, and inadequate risk-sharing mechanisms often hinder investment. Limited scale and a thin pipeline of investable climate adaptation projects constrain private sector participation; a challenge that cross-border projects can help overcome. Effective climate finance empowers countries to achieve their emissions targets and adapt to climate impacts, and it also attracts private investment, accelerates technology adoption, and promotes inclusive, sustainable growth.

## HOW THE MEGATRENDS WILL SHAPE ECONOMIC PROSPECTS IN ASIA AND PACIFIC

After remarkably rapid economic growth and development in recent decades, the Asia and Pacific region faces several global megatrends that could challenge future achievements: demographic transition, new technologies, a shifting global trade and financial landscape, greater use of industrial policies, and the urgent transition to sustainability.

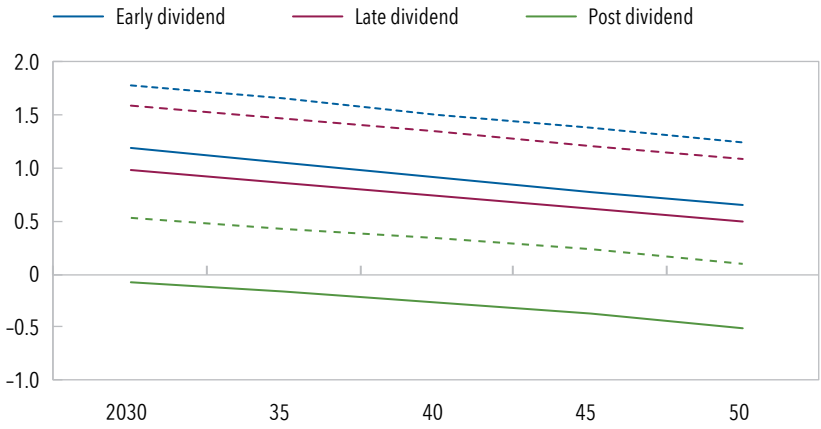
Although megatrends pose challenges, they also offer opportunities for the region's economies because the trends pull in different directions, and the overall economic impact depends crucially on policy responses. The demographic transition on its own will, for example, lower overall economic growth in rapidly aging economies through the labor supply impact but not necessarily in per worker or even per capita terms, depending on investment and productivity. At the regional level, countries still benefiting from a demographic dividend could become a new vortex of growth.

A simple growth accounting framework illustrates the big picture implications of the megatrends for the region's economic prospects in the next 25 years, between 2025 and 2050. Overall, the analysis in this book underscores the critical role that the balance between the negative labor supply effects of the demographic transition and the potential productivity gains from AI and other factors will play in this regard. As illustrated in Figure 1.4, if the increase in the latter is sizeable, the negative labor supply effects could be offset even in post-dividend countries as a group. However, if the productivity gains are small, the negative labor supply effects progressively subtract from economic growth in the post-dividend countries, while growth will decrease in the early-and late -dividend economies.

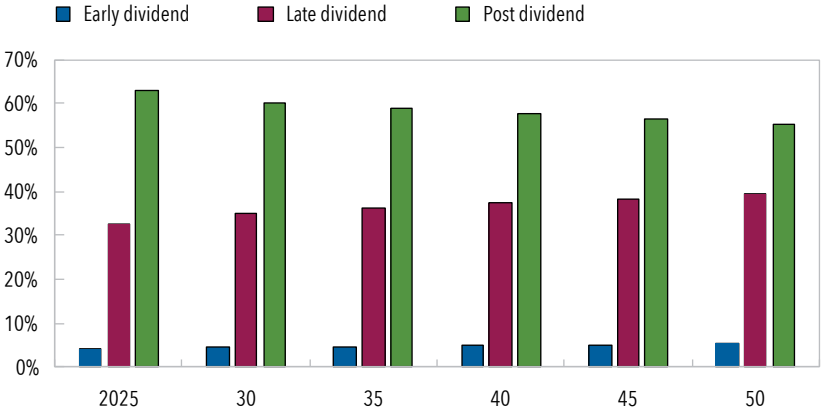
In terms of regional growth, the combined impact of the two principal megatrends—the demographic transition and new technologies—could thus be relatively small, that is, less than one percentage point relative to 2025. Nevertheless, the negative labor supply effects also mean that the share of the post-dividend

## Figure 1.4. Asia and Pacific: Growth Impact of Demographic Transition and Productivity Increases

**1. Growth Impact**
*(In percentage points)*




**2. Weight in Regional GDP**
*(In percent)*




Sources: UN (2024), UN Population Projections, and IMF staff calculations.
Note: The top chart shows the growth impact of the projected change in the statutory labor force and the assumed factor productivity effects of AI. The solid lines show combined effects with an assumed TFP growth rate of 0.2 percent. The dotted lines show the combined effects with an assumed TFP growth rate of 0.8 percent.

countries in the regional output will decline, while that of the late-dividend countries in particular will increase. The relatively small change in the regional distribution of output reflects the high initial share of post-dividend countries in regional output at the outset, in 2025.

In short, Figure 1.4 highlights the critical importance of future increases in productivity growth, given the drag from the demographic transition in the region. The increase can be because of AI, new technologies, or structural reforms more broadly.

As for trade and financial integration, analysis in the IMF's Asia and Pacific *Regional Economic Outlook* in 2022 and 2023 highlighted the efficiency losses from increased geoeconomic fragmentation, resulting in lower GDP and overall incomes. The gains from regional trade integration can lift annual growth by about 0.1 to 0.3 percentage points depending on the country, yet those gains need to be seen against recent and potential further losses from fragmentation.

Sustained gains from industrial policies that could offset losses from other sources seem unlikely, given the evidence on their impact on productivity, including at the firm level. Climate vulnerabilities will be a drag on growth, but green transitions can support structural transformation and become a new source of growth through capital accumulation, employment, and productivity gains from increased energy efficiency.

Recognizing the megatrends and their interplay is critical for policymakers formulating strategies and for businesses and individuals adapting to the evolving economic environments. The impressive achievements in lifting hundreds of millions out of poverty and boosting living standards provide strong foundations, but sustaining growth momentum will require policy visions that embrace these trends and the opportunities they present. The policy agenda should include measures to prepare workers for the future economy, strengthen financial systems to improve resource allocation, and support innovation systems that spread benefits broadly across society. The need to enhance social protection systems so that they balance productivity gains with equity and promote labor-market inclusivity by enabling greater participation of women and older workers is equally important, as is implementing sustainable urban and rural development strategies that mitigate climate risks. Regional cooperation on trade, finance, and technology and knowledge can be a powerful multiplier.

Asia and Pacific's resilience and dynamism will be crucial for global economic prosperity through to 2050. The region's continued ascent as an economic powerhouse depends on collective commitment and proactive engagement from governments, the private sector, and civil society. The path forward may be complex and uncertain, but with vision, collaboration, and determination, the region remains well positioned to sustain its extraordinary growth trajectory and reinforce its vital role in the global economy.

CHAPTER 2

# Structural Transformation in Asia and Pacific

**Chikako Baba, Rahul Giri, Haibo Li, Chris Papageorgiou, and Alasdair Scott**

The Asia and Pacific region has achieved remarkable economic progress over the past 30 years. Favorable demographics contributed to it, but strong investment and sustained productivity growth drove most of the gains. Structural transformation was also significant but often overlooked, especially the shift of workers from agriculture to higher-productivity sectors. Manufacturing, supported by rapid integration into global trade, has long been central to this process, but more recently, services have grown in importance across the region. Sustaining growth will require continued transformation. As countries grow wealthier, services will become even more prominent. Shifting workers into more productive service activities will require greater investment in education and training to build the skills needed to adopt and use emerging technologies, including artificial intelligence (AI). Improving tradability and encouraging competition will also be essential to deepen international integration in services. However, many economies will still need to boost agricultural productivity to support broader structural shifts and overall growth. Lowering barriers that prevent labor and resources from moving across sectors will remain important. Because population aging and decline affect much of the region, policies that raise labor force participation—especially among older workers and women—will become more critical.

## INTRODUCTION

The Asia and Pacific region has experienced a significant economic transformation over the past 30 years that has lifted hundreds of millions out of poverty and greatly improved living standards. Since 1990, average incomes (GDP per person) in the region have more than doubled, and the region's contribution to global growth has risen from about one-quarter to nearly two-thirds. The share of people living in extreme poverty has plummeted from more than 25 percent to less than 5 percent, and life expectancy has climbed rapidly, especially in lower-income countries. These gains reflect rapid economic growth and structural transformation (also referred to as structural change)—a fundamental shift in the

economy's structure as workers move from farms into factories and service jobs. This transformation has been critical to the Asian "economic miracle."

Several emerging challenges raise doubts about whether the rapid progress so far can continue. Population growth and labor force expansion are slowing, and many societies are aging. The rapid productivity improvements that drove previous growth have moderated in many countries since the global financial crisis. Meanwhile, the rapid global trade expansion that fueled Asia's export-led growth has slowed, and world trade growth has plateaued in recent years. Geopolitical tensions and discussions of decoupling[1] or reshoring supply chains raise concerns that the highly interconnected global value chains that have benefited the region could fracture, and climate change looms over all those challenges. The tailwinds that propelled Asia's rise might not be as strong in the future.

It is timely to reflect on what drove Asia and Pacific's rapid growth and structural transformation and how the region can sustain growth going forward. In the region's development since 1990, labor and resources shifted from traditional agriculture into industry and services, a reallocation that boosted productivity. But as countries become wealthier, Asia's economic structure will evolve, likely becoming increasingly service oriented. It is important to understand the implications of this change for productivity growth to understand the region's growth prospects. Put differently, could services emerge as the new growth engine or will manufacturing still have a critical role to play? How countries can continue to raise productivity amid demographic changes, technological disruptions, and a changing global trade environment is an open question.

## GROWTH AND PRODUCTIVITY

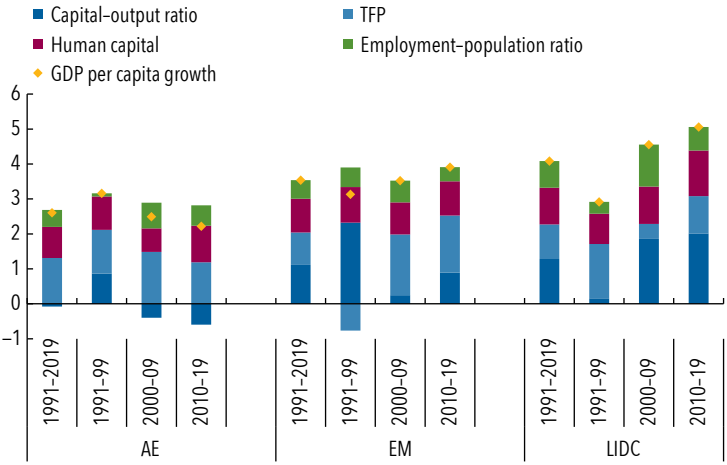
There are two broad drivers of growth: more inputs and more efficient use of inputs. Labor force growth, increases in human capital (skills and expertise of the labor force), and investments in physical capital (machines, factories, structures) contribute to increasing the amount of inputs. Productivity gains, measured by total factor productivity, reflect more efficient use of those inputs, meaning higher output even with unchanged inputs.

Figure 2.1 shows that capital investment had a significant role in the region's emerging markets and low-income developing countries, supported by a steady increase in human capital and the labor force, especially in earlier years. Many Asian countries enjoyed a demographic dividend—the potential for young economies to harness their growing workforces to boost economic growth. A rising share of working-age population accelerates economic growth.[2] Countries such as

[1] Decoupling means reducing economic interdependence between countries (or blocs) in production and supply chains.

[2] Economic growth acceleration caused by an increase in the share of working-age population commonly occurs during the demographic transition, when both fertility and mortality rates are reduced. Mortality rates fall faster than fertility rates at first, creating a population boom and increasing the share of young dependents, which typically slows economic growth. As fertility declines and that

**Figure 2.1. Decomposition of GDP per Capita Growth**
*(Annual averages, percent)*




Sources: International Labour Organization; Penn World Table version 10.01; United Nations; and IMF staff calculations.
Note: Aggregate groups are based on simple averages and current income group classification. AE = advanced economy; EM = emerging market; LIDC = low-income developing country; TFP = total factor productivity.

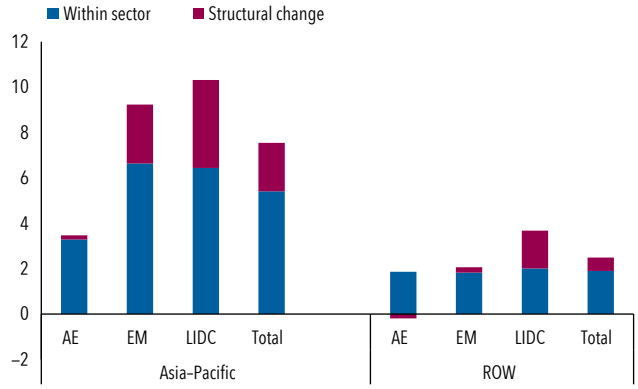
Bangladesh, India, and the Philippines had young populations that increased the labor force faster than the overall population and added to per capita output. Productivity became more important in emerging markets over time, whereas its contribution to growth in low-income developing countries was strongest in the 1990s. In the advanced economies, productivity gains and human capital growth contributed most to income growth, whereas physical capital contributed little over the entire period. Falling birth rates reduced working-age populations, particularly in Japan and Korea, but higher labor force participation—especially more women entering paid work—and workers delaying retirement helped offset the drag on growth by preventing the employment-to-population ratio from falling.

As income levels rise, productivity gains become increasingly critical for continued convergence. Productivity gains have two components: within-sector productivity gains (a factory produces more per worker because of better technology or management) and gains from between-sector shifts, or structural change (capital and workers moving from low-productivity to high-productivity sectors). Both components have contributed to productivity growth in the region (Figure 2.2).[3]

cohort enters the workforce, per capita income can rise through an expanded labor supply, higher savings, and greater investment in human and physical capital. The growth potential from a rising share of the working-age population relative to dependents (children and elderly adults) is the demographic dividend (Bloom, Canning, and Sevilla 2003).

[3] A lack of sector-level capital stock data prevents total factor productivity accounting of within-sector versus across-sector contributions and limits the analysis to labor reallocation and labor productivity.

**Figure 2.2. Contribution of Structural Change to Productivity Growth, 1990–2018**
*(Change between 1990 and 2018 relative to 1990)*




Sources: GGDC/UNU-WIDER Economic Transformation Database; and IMF staff calculations.
Note: Aggregate groups are based on simple averages. AE = advanced economy; EM = emerging market; LIDC = low-income developing country; ROW = rest of the world.

Structural change has been most important for productivity growth in low-income developing countries in Asia, accounting for about 40 percent of labor productivity growth. In emerging markets, this share drops to about 25 percent, and in advanced economies, structural transformation contributes only marginally to productivity growth.

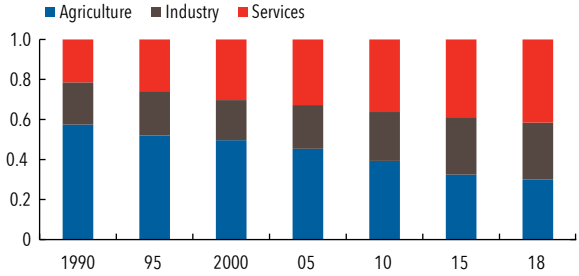
The large contribution of structural transformation to productivity growth reflects a significant reallocation of inputs to more productive sectors in Asian economies along the convergence path. The next section examines the nature of this reallocation.

## DEVELOPMENT AND STRUCTURAL TRANSFORMATION IN ASIA AND PACIFIC

### Industrialization since the 1990s

In 1990, many Asia and Pacific countries were still largely agrarian. Large shares of the workforce lived in rural areas, and agriculture generated most output and employment. Over the next 30 years, a massive shift occurred as workers left agriculture and moved to cities, where they found jobs in construction, offices, shops, and a wide range of business services, and in factories producing textiles, toys, electronics, cars, and heavy machinery. On average, the share of workers in agriculture fell by nearly half in the region between 1990 and 2018 (Figure 2.3). This shift was especially pronounced in fast-growing low-income countries and emerging economies. In 1990, about two-thirds of the workforce in India and Vietnam were in agriculture, but by 2020, Vietnam's farming share dropped below 40 percent and India's below 45 percent, driven by growth in

**Figure 2.3. Share of Employment**
*(Weighted average across countries)*




Sources: GGDC/UNU-WIDER Economic Transformation Database; and IMF staff calculations.

manufacturing and services. Vietnam industrialized and urbanized rapidly, and India expanded in both industry and services but mostly services. China experienced one of the most dramatic shifts: about 60 percent of workers were in agriculture in the late 1980s, but that fell to less than 25 percent by 2020 because tens of millions moved to cities to work in factories and construction, powering China's economic boom.

Crucially, people left the farms but found more productive work in cities. When labor shifted from low-productivity agriculture into higher-productivity industry or services, overall productivity increased. This reallocation, or structural change, accounts for the large contribution of structural transformation to labor productivity growth in low-income developing countries and emerging markets.

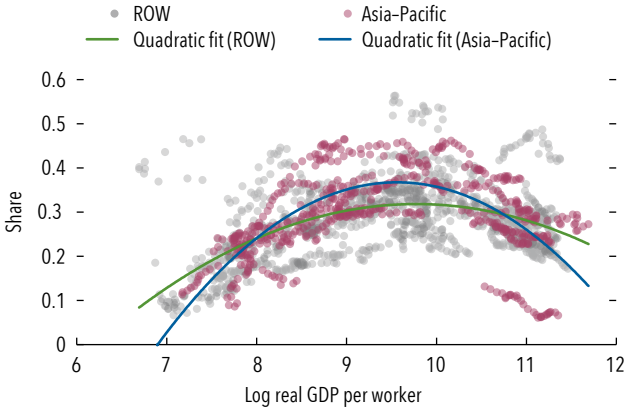
Economic development has long been associated with structural change. As countries develop, agriculture's role shrinks while modern sectors grow, bringing higher overall productivity and incomes (Kuznets 1966). Nobel laureate W. Arthur Lewis in the 1950s (Lewis 1954, 1958) described a "dual economy" model: developing economies begin with a large traditional farming sector with surplus labor, and as modern industries grow, they draw workers out of agriculture, fueling rapid growth until the surplus labor is absorbed (the "Lewis turning point"). This process characterized the historical development of today's advanced economies and has been evident in many Asian countries' trajectories.

The reallocation out of agriculture led Asia and Pacific economies, especially in East and Southeast Asia, to become industrialized, more so than other countries. At similar income levels, industry tended to account for larger shares of GDP and employment in a typical Asian country. Figure 2.4 shows that Asia and Pacific countries mostly lie above the typical hump-shaped curve that links per capita income with the industrial share of value added. Manufacturing largely explains this pattern.

It is not straightforward to determine whether Asia's rapid industrialization was inevitable, orchestrated by state design, or driven by a confluence of factors. Forces that usually accompany growth also drive structural change. Herrendorf, Rogerson, and Valentinyi (2014, pp. 855–941) provide a comprehensive review of how demand-side forces (such as rising incomes that change consumption patterns) and supply-side improvements (such as technological progress in certain

**Figure 2.4. Industry Share of Real Value Added versus Income**

*(Share)*




Sources: GGDC/UNU-WIDER Economic Transformation Database; Penn World Table version 10.01; and IMF staff calculations.

Note: "Quadratic fit" refers to second-order polynomial fit. ROW = rest of the world.

sectors) drive this process. As incomes rise, people spend less on food and more on manufactured goods and services, which shifts the economy's expenditure composition. Concurrently, technological advances in agriculture raise farm productivity and allow workers to move to other sectors without reducing the food supply. Thus, agriculture's share of employment falls. In the initial stages, many workers move into industry (manufacturing, mining, and construction), which can absorb unskilled labor and has potential to achieve rapid productivity gains. At later stages, services become dominant because demand for services rises more with income than demand for agricultural and industrial goods. Productivity growth also tends to be slower in services than in industry and therefore services need more resources to meet rising demand. These forces explain the hump-shaped pattern of industrialization, which has been the canonical pattern observed in the development of the United States and Western Europe, and even Japan and Korea.

Deliberate strategies for rapid industrialization likely contributed: Many Asian nations prioritized manufacturing growth and invested in factories, infrastructure, and skills, often supported by policies that promoted exports and maintained macroeconomic stability.[4] Part of this strategy took advantage of globalization. Sharp

[4] Another view credited active government industrial policy—strategic support for selected industries through subsidies, trade protection for infant industries, and state-directed credit. In the 1980s and 1990s, Krueger (1985) and other economists criticized protectionist import substitution strategies as often unsuccessful. The East Asian Miracle led some to argue that government interventions, if done judiciously, helped jump-start growth (for example, Rodrik, Grossman, and Norman 1995; Stiglitz 1996). Others, such as Krugman (1994), argued that East Asia's growth was mostly because of mobilizing inputs (labor and capital) instead of miraculous efficiency gains, cautioning that growth

declines in shipping costs (with the introduction of container ships) and improvements in communications (the internet) allowed production processes to be unbundled across countries (Baldwin and Forslid 2020). Asia became the world's factory by taking advantage of these trends. Multinational companies established supply chains that allow them to produce components in many countries and ship them to one country for final assembly. Asia and Pacific economies readily integrated into these global value chains. Global electronics manufacturing shifted to Asia on a large scale. Japanese and Western firms moved labor-intensive assembly to countries such as Malaysia and Thailand and later China and Vietnam. This shift boosted industry in host countries. Foreign direct investment poured into places such as Malaysia's Penang (electronics hub), Thailand's eastern seaboard (auto plants), and China's coastal provinces, bringing capital, technology, and expertise. Governments in the region supported the shift by building export processing zones, maintaining political stability, and often keeping wages competitive.

Openness was crucial to enabling the shift: Asia and Pacific countries reduced import tariffs on needed inputs (such as machinery) and kept export channels open. Effective tariffs on manufactured imports in Asia fell steadily and reached low levels by the 2000s. With access to foreign markets, even relatively small countries could industrialize beyond what domestic demand alone could support. Empirical analysis confirms this mechanism: historically in Asia, a 10 percentage point increase in manufacturing exports (as a share of GDP) is associated with about a 6 percentage point increase in the share of manufacturing in domestic value added.

## Services and Structural Transformation

What has this transformation meant for the importance of services in the region's economies? Services include retail shops, restaurants, banking, education, health care, transportation, telecommunications, and public administration. Because of the concentration in industry in many Asian economies, the shares of value added and employment in services have been low compared with peers in the rest of the world at similar income levels. Even so, since 1990, Asia's service sector has absorbed more workers leaving agriculture than industry has. The share of employment in services doubled in Asia and Pacific between 1990 and 2018. In some Asian countries, services now account for the largest portion of employment and a growing share of GDP. Domestic consumption shifting toward services as incomes rise is an important part of the development story. Once households meet basic material needs such as food, clothing, and shelter, they spend more on services such as entertainment, travel, education, health care, and dining out. If income doubles, demand for manufactured goods might not double (because of limited needs for appliances or cars, for example), but demand for services such

could slow once easy gains were exhausted. Although certain governments were successful at guiding resources into export industries, how much industrial policy mattered is uncertain, considering many other factors. Clearly, global trade integration was crucial.

as tourism or health care often increases more than proportionally. As Asia's middle class grew, demand also grew for services that cater to it. Some service industries are labor intensive and create many jobs quickly, such as retail and hospitality (shops, street vendors, hotels, and restaurants), which employ many people and often require few advanced skills. As cities grew and incomes rose, these sectors absorbed workers. Governments also increased public and social services (schools, clinics, police, and government offices), which created professional jobs. In developing Asia, many people who left rural areas but did not find factory jobs worked in urban services in both formal and informal roles such as driving taxis or selling goods in markets.

IMF (2024a) found that tradable services in the region—including finance, business services, information technology (IT), transport, and communications—showed especially strong employment growth, driven by emerging markets. In low-income developing countries, industry's share of employment has risen at a pace comparable with that of services, driven both by manufacturing and nontradable industry (construction and utilities). Advanced economies were already service dominant, and that shift has continued.

To sum up this journey: from 1990 to 2020, Asia and Pacific economies underwent a historic transformation. Typically, countries first raised agricultural productivity, which freed up labor. In most cases, workers moved to industry—especially manufacturing—which grew rapidly in many economies thanks to exports and investment. Over time, as incomes rose further, services became increasingly important.

## Different Paths of Structural Transformation

Important departures from this broad pattern occurred, however. South Asia, especially India, did not experience as large an increase in manufacturing as east Asian peers at similar stages. India's development path has been distinctive: it experienced some industrial growth but had an even bigger surge in services. Unlike east Asian peers, India did not embrace policies to increase openness. The Philippines also largely missed the manufacturing expansion. Instead, services increased, especially with a boom in call centers and business process outsourcing in the 2000s. In recent years, both India and the Philippines have tried to reinvigorate manufacturing and attract foreign direct investment.

Some advanced or resource-rich economies in the region followed different paths. Australia and New Zealand shifted from agriculture and mining to service-dominated economies without building large manufacturing sectors. In the Pacific island countries, manufacturing never took hold because of their small market size and remote location, and structural change often meant moving from subsistence agriculture and fishing into services such as tourism and public sector jobs. Fiji and Samoa have relatively large service sectors (such as tourism and banking) and a smaller industry sector. Papua New Guinea shifted from traditional agriculture to mining, which does not employ many people, and informal services, with a limited manufacturing expansion.

## Relative Importance of Key Forces Driving Structural Transformation

An assessment of the relative importance of three key forces highlighted in the literature for structural transformation in the region—productivity, trade, and income—was conducted in a global context.

The assessment tool is a global general equilibrium model of structural transformation, building on Sposi and others (2024) and applied to a sample of countries, with a particular focus on Asia and Pacific economies for 2007–23.[5] The model features multiple countries, three sectors (agriculture, industry, and services), and three well-known mechanisms driving structural transformation: different income elasticities of demand that change the allocation of consumption expenditure as income increases; biased sectoral productivity growth that affects relative prices of goods in the three sectors; and global trade that works through increased specialization (and the resulting increase in demand for production inputs) in the sector with comparative advantage. Consistent with the literature, the measure of structural transformation is the change in sectoral value-added shares.

Overall, different income elasticities of demand predominantly explain structural transformation between 2007 and 2023, followed by trade and then sector-biased productivity growth. This complements the findings of Sposi and others (2024). The model also confirmed that global trade had a crucial role in structural transformation across the region. Between 2007 and 2023, global trade accounted for 22 percent of structural transformation in the agriculture sector, 39 percent in industry, and 10 percent in services across the region.

For economies with a comparative advantage in industry, global trade would have boosted industrial activity through stronger foreign demand. This channel would have strengthened even more because of the decline in trade costs over that period.[6] Model simulations indicated that if trade costs remained at their higher 2007 levels, most of these economies would have experienced weaker industrial activity. Emerging markets that benefited the most from the decline in manufacturing trade costs would have experienced the largest declines in their industrial shares: 8 percentage points in Mongolia, followed by Lao P.D.R. (7 percentage points), Bhutan (4 percentage points), and Vietnam (3 percentage points). Large industrial economies also would have faced declines, though smaller—0.5 percentage point in Korea and 0.3 in China. Meanwhile, industrial shares would have increased by about 2 percentage points in Bangladesh, 1 in the Philippines, and 0.6 in Japan, largely because these economies experienced smaller declines in trade costs over the

[5] The sample used to calibrate and simulate the model includes 19 Asia and Pacific economies, 33 economies in the other regions, and an aggregation representing the rest of the world, covering 2007–23.

[6] Trade costs were modeled as iceberg costs, as is common in quantitative trade models. Just like an iceberg loses mass as it travels, goods effectively shrink in transit because of trade frictions. Although trade costs have risen in recent years, they have declined overall for most countries compared with their 2007 levels.

past two decades than many regional peers and therefore would have faced a smaller increase in trade costs when anchored at their 2007 levels.

Similarly, higher services trade costs would have implied lower services shares in service-dominated economies. The decline would have been the largest in Cambodia at about 4 percentage points, followed by Maldives at about 1.5 percentage points, and Japan, Malaysia, and Nepal at about 1 percentage point each. India's services share would have fallen only marginally—about 0.1 percentage point, reflecting the moderate decline in its services trade costs over the past two decades.

## LOOKING BACK TO LOOK AHEAD

The analysis so far shows that the historical pattern of transformation moves from agriculture to industry to services. Multiple pathways exist within this pattern. Besides the key forces driving structural transformation, policies, geography, and resources also influence how transformation unfolds. Understanding such interactions sets the stage for examining the next phase of transformation: the rise of the service economy and its implications for future growth.

For the region, tradable services have the highest value-added share across sectors and will likely surpass agriculture to become the sector with the highest employment share. By contrast, manufacturing's share of value added in the region has declined since the global financial crisis, driven by emerging markets, whereas the employment share of manufacturing has remained largely stagnant. Industry's share of employment has increased because of construction and utilities.

These trends suggest that the region could be at an inflection point. Manufacturing might have peaked, and services could continue to dominate the region. The shift toward services raises concerns about its implications for long-term growth potential.

### The Role of Manufacturing in Economic Convergence

Historically, manufacturing has been considered the engine of growth for developing economies for good reasons: it can employ many moderately skilled workers, it is tradable (so industries can scale up by selling globally), and factories often act as conduits for innovative technologies and methods. Manufacturing often generates spillovers: a successful manufacturing sector can spur improvements in infrastructure and education and the growth of supporting industries such as logistics, design, and maintenance services. Korea's development illustrates this pattern: light manufacturing (textiles) led to heavy industry (steel and shipbuilding) and then to high-tech (electronics and cars). Some economists have argued that manufacturing acts as an "escalator" for development. Industries such as textiles, steel, autos, and electronics helped economies such as Japan, the United Kingdom, the United States, and more recently China raise incomes. Rodrik (2013) provided evidence that manufacturing industries exhibit unconditional

productivity convergence, meaning that manufacturing sectors in low-income countries tend to catch up with productivity levels in wealthier countries, even if the overall economy lags. These observations reinforce the view that industrialization is a reliable path to development.

Services have been considered less productive than manufacturing, a phenomenon often called the Baumol effect or Baumol's cost disease. Baumol (1967) argued that an economy dominated by services will inevitably face slower growth because many services cannot easily become more efficient. His classic example is a string quartet, which needs the same number of musicians and time to perform the same piece of music today as it did 200 years ago. However, recent data and research (including findings in IMF 2024a) suggest that services can also be a driver of growth, not just manufacturing.[7] In Asia, average productivity in modern tradable services (sectors such as business services and finance) can rival that in manufacturing. IMF (2024a) found that productivity in these modern services in Asia has been catching up with the global frontier faster than manufacturing productivity.[8] This evidence challenges the view that manufacturing is essential for continued convergence. Thus, even though Baumol's insight holds for some traditional services, technology is changing outcomes in many others. Education, for example, can be delivered online to millions, software can be replicated at near-zero cost, and automation can increase a call center worker's productivity.

Premature deindustrialization is another concern. Rodrik (2015) observed that many developing countries have begun to lose industrial share (in employment and output) at much lower income levels than early industrializers did. Some countries peak in manufacturing and then shift to services at middle or low income levels, whereas earlier industrialized economies shifted to services at high income levels. Globalization and automation contribute to this pattern by making manufacturing more efficient globally, reducing labor demand and increasing competition that makes it harder for latecomers to break into certain industries. Premature deindustrialization raises concerns because countries that shift to services too early, without an industrial boom, might miss a traditional engine of growth. The Asia and Pacific region shows mixed patterns, with parts of East Asia deeply industrialized and some South and Southeast Asian countries having smaller manufacturing bases.

Less-developed economies might struggle to harness the gains from manufacturing-driven growth and job creation as successfully as their predecessors in the region. The increasing skill and capital bias of manufacturing, together with the

[7] The data used in IMF (2024a) allow calculation of sectoral purchasing-power-parity prices rather than applying the same purchasing-power-parity price across all sectors, and they distinguish among different services subsectors. This distinction matters for comparisons of productivity levels between manufacturing and services. Rodrik (2013) used nominal exchange rates to compare productivity across countries.

[8] The evidence aligns with similar recent findings in Herrendorf, Rogerson, and Valentinyi (2022), which compared goods-producing sectors with those that do not produce goods, but because of a lack of data, they do not analyze the importance of different services subsectors for convergence.

increasingly internationally integrated nature of production, even as fragmentation reorganizes it, could limit the scope for labor-intensive manufacturing. Baldwin (2019) and Rodrik and Stiglitz (2024) argue that manufacturing might not absorb labor as it once did because of automation and changing global demand. They also suggest that the traditional east Asian model of export-led industrialization is becoming less viable and effective.

## Services and the Digital Era

Historically, many services (retail, health, education, and personal services) were considered local and nontradable, but the internet and digitalization have changed this pattern for many services. Baldwin (2019), among others, underscores how telecommunications technologies allow trading services internationally to a much greater extent. Workers in countries such as the Philippines and Vietnam can provide software development, design, and customer service to firms abroad, for example. Since about 2010, global trade in services has grown faster than trade in goods. For economies in the Asia and Pacific region, this expansion enables services exports directly over digital networks, not just through tourism or labor mobility.

However, many developing countries in Asia still lag in modern services exports compared with advanced economies. Barriers to trade in services are one reason. Many countries impose stricter regulations on foreign entry or data flows in services than in goods trade. Restrictions on foreign ownership of banks or telecommunications companies could slow innovation in those sectors, and heavy licensing requirements for professionals can limit the growth of business process outsourcing. IMF (2024a) highlights that heavily industrialized Asian countries have often maintained more restrictive services policies, even as trade in goods has been open. As Asia evolves, liberalizing trade and encouraging competition in services could support stronger growth. Greater regional cooperation in services could also emerge. Trade agreements, such as the Regional Comprehensive Economic Partnership enacted in 2022 and involving many Asia and Pacific economies, include provisions on services and investment. The partnership and similar agreements could eventually harmonize regulations and reduce barriers and thus support a larger regional market for services.

So far, India and the Philippines stand out as leaders in exportable IT and business services. Other countries, such as Malaysia and Vietnam, are also increasing IT service exports. Malaysia has developed the Multimedia Super Corridor technology park, and Vietnam has a budding software industry. Small states such as Fiji have experienced promising growth in niche outsourcing markets, drawing on their educated workforces and time zone advantages.

India has experienced service-led development since the 1990s. The country has a large manufacturing sector, but its growth has tilted heavily toward services, especially IT and business process services. Starting in the 1990s, India became a global hub for software development and outsourcing. Cities such as Bangalore, Hyderabad, and Pune, with technology parks and call centers, developed into

major centers for IT-enabled services. By the 2000s and 2010s, India's IT and business process outsourcing exports ranked among the largest in the world, with firms in the sector providing IT services globally. As a result, services account for a significant share of India's GDP—more than 40 percent—and an increasing share of employment in services, although a large, informal agricultural workforce remains.

The Philippines also found a niche in service exports in the 2000s. It became a leading destination for voice-based business process outsourcing and other back office services for foreign companies. This development offset a lackluster manufacturing sector and supported the growth of an urban middle class employed in customer service and technical support roles in Cebu, Manila, and other cities. As a result, services grew to about 60 percent of GDP and more than half of employment.

Services have surged even in China, "the world's factory." China's service sector—e-commerce, financial technology (or fintech), tourism, and real estate—has grown larger than its industrial sector in GDP value-added terms. As incomes rose, consumption shifted toward services such as travel and hospitality. Technology-driven services also grew rapidly and generated substantial employment in sectors that were small or nonexistent 15 years ago.

Not all growing services are high tech. Tourism has boomed in many Asia and Pacific countries and provides substantial employment, especially in small Pacific island countries such as Fiji and Maldives, where it is often the largest source of jobs and foreign exchange. Tourism is tradable, benefiting from both domestic and foreign demand (foreign visitors spend within the destination economy, effectively importing their consumption), and it is labor intensive, absorbing many lower-skilled workers. Tourism contributes significantly to structural change in countries with natural or cultural attractions, but it can be volatile, as the COVID-19 pandemic proved when the collapse in travel caused severe recessions in tourism-dependent economies. The volatility highlights the importance of diversification even within services.

## The Continued Importance of Agriculture

Agriculture still matters in the region's countries, and increases in agricultural productivity are crucial for structural transformation. In many developing countries in Asia, increasing agricultural productivity remains a priority even as workers leave farms, given the large gaps in their farm productivity compared with the most productive economies. These gaps have remained at their 1990 levels (IMF 2024a). As a result, yields per worker in parts of South Asia or Southeast Asia are a fraction of those in economies such as Korea or Japan. Many factors contribute to these gaps, including small landholdings, limited mechanization, and a lack of rural finance. Policy distortions also persist. Asia as a region has maintained relatively high protection for agriculture (tariffs and subsidies), which can sometimes lead to inefficiencies despite aims to support farmers.

## OUTLOOK FOR STRUCTURAL TRANSFORMATION IN ASIA AND PACIFIC

Historical trends and emerging developments provide a basis for assessing how the region's economic structure and growth drivers could evolve in the coming 25 years or so. Although precise outcomes are uncertain, current patterns and new developments point to several probable trajectories and challenges for structural transformation.

### The Continued Rise of Services

Agriculture's share of both output and employment will likely continue to decline in most Asian economies. In countries where many people still work in agriculture (Cambodia, Myanmar, and Nepal and, to a lesser extent, India, Indonesia, and Thailand), its share of output and employment should continue declining as industry and services grow. Managing rural-to-urban migration is crucial to ensuring that cities can absorb new workers with jobs, housing, and services, and that rural livelihoods become more lucrative (farm consolidation and higher-value crops are examples). The social dimension also matters, including support for aging rural populations and making sure that rural youth have opportunities locally.

Manufacturing's share of employment has peaked or will peak soon in many emerging Asian economies. Although manufacturing continues to grow in absolute terms, its share of employment is no longer increasing and, in some cases, is declining relative to services. Automation contributes to this trend, because factories can produce more with fewer workers through robotics and improved processes (Jacob 2013; ILO 2022). Competition and shifting comparative advantage also contribute, with some economies losing manufacturing activity to others. However, low-income Asian economies still have scope to expand manufacturing in absolute terms. In the 2010s, Bangladesh and Vietnam increased manufacturing employment, including growth in garments in Bangladesh and in electronics and apparel in Vietnam. Cambodia and Myanmar also attracted garment production as China moved up the value chain and wages rose, creating tens of thousands of jobs, although political turmoil has disrupted Myanmar's progress. Premature deindustrialization could become a policy concern, especially in the context of increased geoeconomic fragmentation over the past decade. Global firms are pursuing China+1 strategies that diversify production across locations beyond China to improve resilience—Vietnam, for example—and because of policy efforts.

Services are expected to account for a larger share of the region's economies by 2040. In many emerging markets, services could rise to 60–70 percent of GDP, from about 50 percent. The composition of services will also change, with growth in modern services such as digital and IT services, finance, outsourced corporate services, creative industries, health care, education, and environmental services—sectors that grow with rising income and technological advancement.

Digital and IT services are likely to expand, including software development, IT consulting, and emerging fields such as AI and data analytics. India aims to move

up the value chain from basic IT outsourcing to product development and research and development services. China is a global leader in some digital sectors, such as mobile apps and fintech, and could export those services or set standards abroad.

Financial sectors will deepen as Asian economies mature. Emerging market economies such as India, Malaysia, and Thailand are expanding banking, insurance, and fintech. Mobile payments in India have grown since the launch of the Unified Payments Interface system. Fintech especially can create high-tech jobs and improve access for small businesses, which could support structural change by funding new enterprises. New initiatives are establishing the infrastructure needed for cross-border payments, such as interoperability of domestic QR-based payments systems in Cambodia, Indonesia, Malaysia, the Philippines, and Thailand to enable small-value cross-border transactions.

Health care and education will expand with demographic and income trends. Aging populations in East Asia are increasing demand for health care services, and countries with younger populations are investing more in health care as incomes rise. Medical services can also become an export: medical tourism is well established in India, Malaysia, and Thailand and attracts patients from across Africa and Asia who seek affordable, quality treatment. Education, especially higher education, could also become a tradable service. Universities in Australia and Singapore already attract Asian students, and potential exists for online education exports.

Logistics and transport services will grow with intra-Asian trade, increasing demand for ports, warehousing, and freight services. Countries such as Malaysia and Vietnam are emerging as regional logistics hubs. The region is also poised to become the largest air travel market, which will create more jobs in related services.

Creative industries and media will continue to expand, supported by digital platforms that allow content created anywhere to reach global audiences. South Korea exports popular music and television content globally and has become a cultural export powerhouse, whereas India's film industry reaches a large domestic audience and a broad diaspora. These industries generate export revenue and create service jobs in production and marketing.

Environmental services related to renewable energy, energy efficiency, and environmental management will grow as the world and Asia confront climate change. Activities such as installing and servicing solar panels, managing smart grids, and providing sustainability consulting could employ many workers.

Tourism is likely to rebound after the pandemic and grow as the Asian middle class travels more within the region. New tourism markets could emerge, such as Silk Road tourism in central Asia and ecotourism in Borneo.

## The Challenge of Job Creation

Rodrik and Sandhu (2024) argue that the future of developing countries lies in services because manufacturing has a limited role in job creation. Raising productivity in labor-absorbing services such as retail, hospitality, and health care must

therefore become a policy priority. They note that manufacturing often functions as an enclave in many countries, with high-tech factories that employ relatively few workers. Recent research suggests that even though manufacturing remains important, especially for exports and innovation, countries cannot rely on it alone to employ growing labor forces. Instead, they must raise productivity in service sector jobs, where most people are concentrated. Ghani and Kharas (2010), and more recently Rodrik and Stiglitz (2024), discuss how the traditional growth model might be changing to emphasize services and green industries. They propose a strategy centered on clean energy investments and higher productivity in labor-intensive local services.

India's modern service sectors produce a high output with few workers, indicating high productivity. IT services employ only a few million workers in India, but they generate tens of billions in revenue. This highlights a challenge: these sectors boost GDP and exports but do not directly employ large numbers of semiskilled workers. However, they have indirect effects. The well-paid IT workforce supports many other jobs in construction, retail, and domestic services in the cities. India's experience shows that services can be internationally competitive even in a lower-middle-income country if the workforce has the right skills (such as English-speaking engineers) and connectivity. It also highlights a duality between high-end and low-end services, a pattern seen across Asia.

However, many service jobs in the Philippines are still in low-productivity areas, and many Filipinos continue to work abroad. Although services have boomed, creating enough good jobs at home remains a challenge.

Service sectors vary widely in productivity. Some services are highly productive and often pay high wages (finance, IT, telecommunications, high-end consulting, and engineering), whereas others have much lower productivity, often only slightly above traditional farming (small retail shops, personal care, and driving a motorized rickshaw). When employment shifts to services, the effect on overall productivity depends on which activities grow. The productivity of a rural worker who becomes a construction laborer or a security guard in the city could increase only modestly compared with farm work, whereas it could increase more for a worker who becomes a bank clerk or a call center agent. Many Asian countries have experienced growth in productive service sectors, which is encouraging. IMF (2024a) shows that in Asia, modern services (business, financial, and information and communication technology) are on average more productive than manufacturing when measured by value added per worker. Thus, shifting from manufacturing into these high-value services can raise overall productivity. Singapore's banking and high-tech services have productivity levels comparable to advanced manufacturing. However, many traditional services in Asia, such as petty trade in India or informal transport in Indonesia, absorb labor but do not increase output per worker significantly.

Policymakers face the challenge of supporting more productive, higher-value services. Meeting that challenge often requires investing in education to supply skilled labor, technology such as broadband and digital payment systems, and openness to competition. Such competition allows many service industries to

benefit from scale and innovation that competitive markets or foreign investment often provide. Telecommunications and internet services have transformed many Asian economies in the past 20 years. Countries that liberalized telecommunications early, such as Malaysia or Thailand in the 1990s, achieved faster internet adoption than countries that maintained state monopolies. With the rise of fintech, e-commerce, and digital platforms, the regulatory environment will shape how these service sectors evolve and how many jobs they create.

## Megatrends and Their Impact

Asia's demographic landscape will change significantly in the next 25 years (Chapter 4). East Asia is aging rapidly: Japan already has a highly aged population, and China, Korea, and Thailand are aging at an unprecedented pace because of low birth rates. China's population numbers have begun to decline, and its workforce is shrinking. Some Southeast Asian countries (Thailand and Vietnam) risk growing old before becoming rich. However, South Asia and parts of Southeast Asia (including India, Indonesia, Malaysia, and the Philippines) still have youthful populations and will continue adding workers. So, the source of demographic growth will shift, with South Asia overtaking East Asia as the main source of new workers in the region. In countries with rapid aging, the workforce will grow at a slower rate or will decline, and demographics will be a drag on growth. In these countries, productivity increases will be crucial to maintaining GDP growth.

Demographics will also shape structural change. Demand patterns in older societies are different from those in younger societies. Older societies can require more health care services and fewer education services. Aging might also lead to labor shortages in labor-intensive sectors, which could encourage greater use of automation in manufacturing and some services, such as Japan's use of robots for elder care tasks such as lifting patients.

Emerging technologies, especially AI and advanced automation, introduce significant uncertainty about jobs and economic structure. AI can automate certain tasks, especially in services (such as basic accounting, customer service using chatbots, and some aspects of medical diagnosis or legal work) and also in manufacturing. It can also raise productivity and create new service industries such as data science and AI model training. Historically, technological change has not only disrupted labor markets in some sectors but also created new types of jobs and raised real incomes. Large Asian emerging markets such as China, India, and Indonesia are well positioned for AI adoption, and advanced Asian economies are among the leaders (Chapter 8). Asia not only can harness AI to boost productivity but also must invest in education, including better aligning it with new skills being demanded, as well as reskilling to prepare workers for changing skill needs, and adopt policies to address the risks of widening within-country and cross-country inequality.

Climate change could affect regional growth prospects significantly. Asia and Pacific region is highly vulnerable to rising sea levels that could threaten coastal megacities and low-lying countries. Changing monsoon patterns threaten agriculture in South Asia, and extreme heat could reduce labor productivity because of

the dangers it poses for outdoor workers. The global push to reduce carbon emissions will also transform coal and other high-carbon industries, which will face pressure to contract. Asia relies heavily on coal for power and will need to shift to renewable energy, creating both risks and opportunities.

Green policies could create new sectors and jobs. Renewable energy installation and maintenance already employs a growing number of workers. Asia is also becoming a leader in certain green technologies, especially China, which is leading in development and deployment of renewable energy. India has ambitious renewable capacity expansion plans, whereas Japan and South Korea invest heavily in battery technology and hydrogen. The climate challenge could also spur a new wave of industrial and service growth in clean technology—a new structural transformation from fossil fuels and nonrenewable resources to green economy sectors. Regions that depend on fossil fuel extraction will need to reskill workers and diversify local economies as those industries decline, which will affect coal mining areas in India and oil and gas in Indonesia and Malaysia.

Climate-induced displacement poses risks. If certain areas become uninhabitable (coastal areas or small islands exposed to frequent cyclones and floods), migration and sudden structural shifts can follow. Governments will need long-term planning for risk mitigation, including building resilient infrastructure and potentially guiding migration inland in a structured way.

The future of structural transformation in Asia and Pacific will involve navigating a more service-oriented landscape, leveraging technology, managing demographic shifts, and aligning growth with sustainability. Outcomes will vary by country and reflect the region's diversity, but all countries are expected to continue moving up the value chain.

## CONCLUSION AND POLICY RECOMMENDATIONS

The region's economic rise since the 1990s has been driven by a dynamic process of structural transformation, which has shifted labor and resources from low-productivity agriculture to higher-productivity industry and services. Sustaining this momentum will require navigating the next phase of transformation while confronting demographic change, technological disruption from AI, and the growing urgency of climate change.

Structural transformation remains a key driver of productivity growth across the region. Countries with large agricultural workforces must close productivity gaps in farming through land reforms, infrastructure investment, access to finance, and climate-resilient practices. Facilitating the reallocation of labor from low-productivity agriculture to higher-productivity sectors in manufacturing or services can raise aggregate output significantly. However, the traditional path of labor-intensive industrialization is becoming less accessible because global manufacturing has become more skill and capital intensive, driven partly by automation, AI, and worsening fragmentation.

As economies become more service oriented, addressing inefficiencies and misallocation within service sectors becomes more important. IMF (2024b) found that two-thirds of misallocation losses stem from structural barriers such as limited market entry, weak competition, and constrained financial access, which are especially acute in services. Reforms that promote competition, reduce regulatory burdens, and enhance access to finance can unlock significant productivity gains. Policymakers should also upgrade labor-intensive services to ensure that growth remains inclusive and generates jobs.

Demographic trends present a dual challenge. Aging economies such as China, Japan, and Korea will need to boost labor force participation, especially among women and older workers, and reform pension and health care systems. They will also need to consider targeted immigration strategies. Younger economies such as India and Philippines face the challenge of creating productive employment for expanding workforces. Investing in education, vocational training, and digital infrastructure is essential to harnessing the demographic dividend and preparing for a more complex labor market.

AI and automation are reshaping work and labor markets across sectors. Although these technologies can raise productivity and create new jobs, they also risk displacing workers. To mitigate these risks, governments must invest in digital infrastructure, promote inclusive access to technology, invest in education and reskilling, and strengthen social safety nets. Regulatory frameworks must evolve to ensure ethical AI deployment, protect data privacy, and foster innovation without increasing inequality.

Climate change adds complexity, and Asia and Pacific is highly vulnerable to its impacts. The global shift toward low-carbon growth also creates risks and opportunities. Countries must accelerate the transition to green industries, invest in climate-resilient infrastructure, and support sustainable agriculture. Integrating environmental sustainability into structural transformation strategies will be critical for long-term resilience and inclusive development.

Asia and Pacific's rapid growth and structural transformation show the power of adaptation and strategic policymaking. By embracing the next wave of structural transformation with foresight and inclusivity, and proactively addressing demographic, technological, and environmental challenges, the region can chart a path toward sustainable and equitable growth in the decades ahead. To thrive in this evolving landscape, Asia and Pacific economies must continue to focus on productivity-enhancing reforms.

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CHAPTER 3

# Investment Inefficiency and Capital Misallocation: The Role of Financial Structure

**Rahul Giri, Anne Oeking, Akshat Singh, Jeongwon Son, Ying Xu, and Li Cui**

Asia and Pacific economies have experienced a steady decline in growth since the global financial crisis. Compared with other regions, Asian countries still have higher investment rates, and capital deepening has a larger role in the region's economic development and income convergence than labor input or total factor productivity. However, falling return on capital investments, reflecting lower capital efficiency, have raised questions about drivers of future growth. Openness to global trade and participation in global value chains remain central to the region's growth, but rapid shifts in global trade policies are increasing pressure on the trade-oriented, investment-intensive Asian model that has long underpinned the region's growth aspirations. Economic prospects in Asia now depend more on critical domestic reforms, and measures to enhance capital efficiency are essential for lifting growth potential.

Greater capital efficiency is also important for economic rebalancing in the region. Improving returns to capital would allow more resources to shift toward consumption. Capital misallocation in Asia is especially acute in the nontradable and service sectors. Addressing these inefficiencies is critical for structural transformation in the region, strengthening the services sectors, boosting job growth, and increasing household income and spending.

The financial system channels savings into investment, and an effective intermediation structure should support productive investment that drives growth. However, economic growth suffers when funds are directed to unproductive activities or exclude high-potential borrowers. Factors beyond the financial system, such as industrial policies and ownership structures unique to Asia, can also contribute to growth inefficiency. However, these policies often affect the investment process by influencing the price and direction of capital allocation, becoming intertwined with and reinforced by the structure of financial intermediation. Improving that structure remains essential for reducing inefficiency and enhancing resource allocation.

## THE DOMINANT ROLE OF CAPITAL IN GROWTH AND ITS DECLINING EFFICIENCY

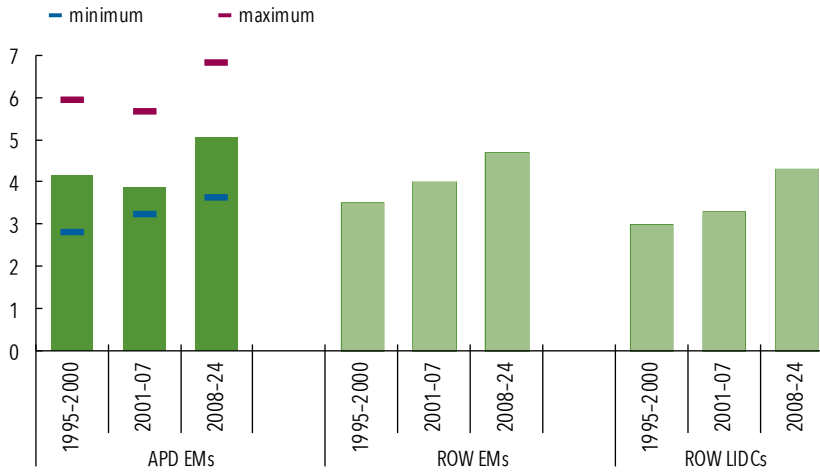
Economic growth in the region's countries has generally been more capital intensive than elsewhere. The investment-to-GDP ratio averaged 27 percent of GDP in 1995–2024, well above the levels in Europe and the United States. Many economies in the region—from rapidly growing low-income countries to middle-income countries in the Association of Southeast Asian Nations, China, and India, to more advanced Asian economies—have consistently invested a greater share of GDP than countries at similar income levels outside the region (IMF 2025a). A high savings rate and pro-growth policies have enabled this strong investment. The region's relatively capital-intensive growth model has delivered decades of impressive income convergence and has continued to do so across the region in the past three decades.

However, a capital-intensive growth model is sustainable only if additional investment continues to generate output growth, offsetting declines in the marginal return to capital through technological gains and higher productivity, and it is along this dimension that Asia's record has become more mixed. The incremental capital-to-output ratio—the additional units of capital needed to produce one more unit of output—has risen more sharply in most Asian economies than in the rest of the world, indicating a marked deterioration after the global financial crisis (Figure 3.1). The rise in the incremental capital-to-output ratio has not been limited to Asia; returns on investment have declined in other emerging markets and low-income countries since the global financial crisis. However, ratios in large Asian countries tend to be higher and have increased faster than in other countries. Ratios in several major Asian economies exceeded 4, significantly above the levels generally considered efficient (about 3). Many Asian economies continued to invest heavily but generated lower investment returns.

Consistent with the macro evidence, firm-level data reveal a decline in asset efficiency, reflected in a significant drop in the corporate return on assets in the period after the global financial crisis—a decline in average return on assets that is sharper in most regions of Asia and Pacific than in the rest of the world (IMF 2025a). A closer look at the drivers of return on assets reveals that the decline stems primarily from weaker asset efficiency in revenue generation, not from compressed profit margins, which underscores the importance of capital efficiency in sustaining returns to capital.

A formal measure of the allocative efficiency of capital, captured by the dispersion of the marginal revenue product of capital (MRPK) across firms (following Hsieh and Klenow 2009), indicates rising capital misallocation. Greater MRPK dispersion implies that capital is not being used where it generates the most value, which reduces capital efficiency and overall productivity (total factor productivity). The measured index of misallocation increased by about 25 percent in Asia and Pacific between the period before the global financial crisis and the period after (Figure 3.2), suggesting that capital is increasingly concentrated in less-productive firms. The rise in misallocation exceeds that in the United States, where MRPK

**Figure 3.1. Incremental Capital-to-Output Ratio, 1995–2024**
*(Average for each period)*




Sources: IMF *World Economic Outlook*; and IMF staff calculations.
Note: Incremental capital-to-output ratio is investment rate divided by real GDP growth rate. EM = emerging market; LIDC = low-income developing country; ROW = rest of the world.

dispersion declined in both manufacturing and services, and matches the increase observed in Europe.

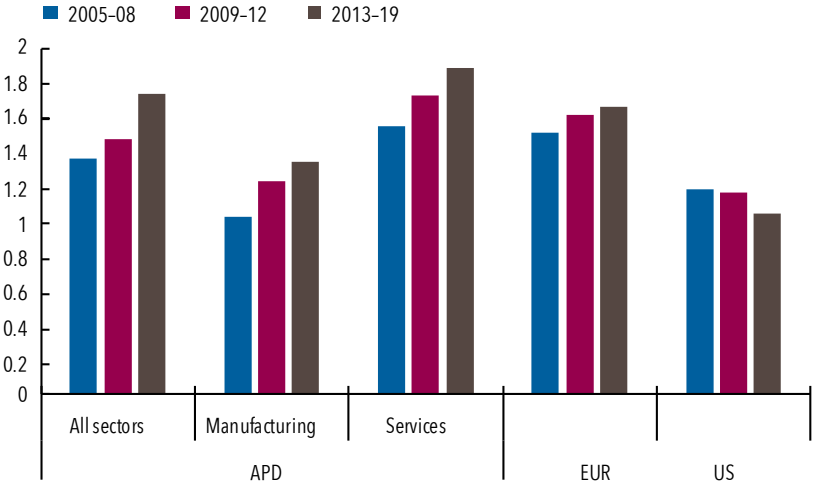
Firms in less tradable sectors—those with lower levels of foreign trade—account for a larger share of allocative inefficiency. MRPK dispersion in Asia's service sector exceeds that in Europe and the United States and has increased the fastest among the three. Within services, capital misallocation is lower in tradable services (trade, transport, business services, and financial services) than in nontradable services and has increased more slowly. The same pattern holds for tradable industry (manufacturing and mining) compared with nontradable industry (construction and utilities).[1]

These symptoms—higher incremental capital-to-output ratio, lower returns, and increased misallocation—point to a decline in capital efficiency in much of the region after the global financial crisis.[2] The lower level and slower increase in misallocation of more traded sectors indicate that productive gains from trade have offset domestic inefficiency. It also underscores the need for stronger domestic reforms to unlock productivity potential amid global uncertainty and rising trade protection.

[1] The sectoral nomenclature follows that of IMF (2024).

[2] Investment measures include real estate sector investment. Capital misallocation resulting from financial constraints is not sector specific, though varying degrees of capital misallocation can occur across sectors (see the section "From Financial Intermediation to Capital Misallocation").

**Figure 3.2. Capital Misallocation**
*(Standard deviation of MRPK)*




Sources: Orbis; and IMF staff calculations.
Note: See Annex 1.2 for methodology and data. MRPK = marginal revenue product of capital.

## FROM FINANCIAL INTERMEDIATION TO CAPITAL MISALLOCATION

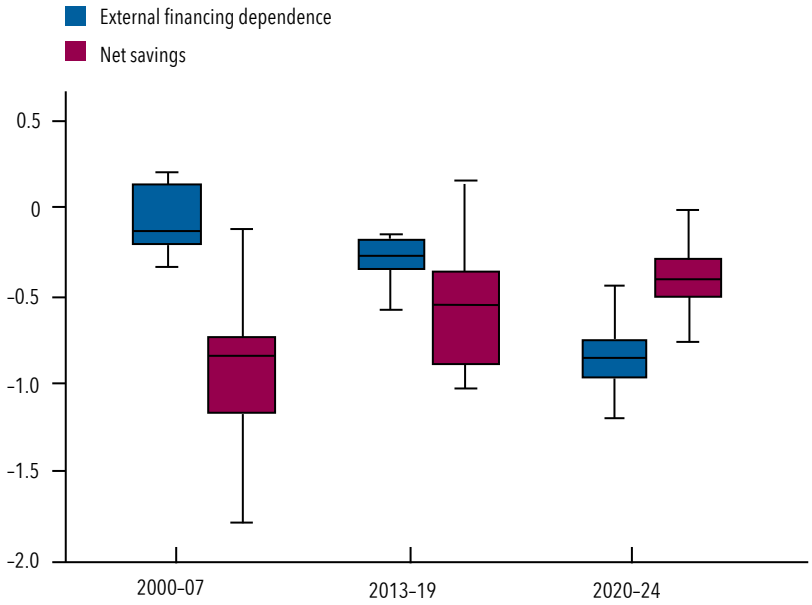
Financial systems have expanded steadily in the region, supporting economic development except during market stress episodes such as the 2013 taper tantrum.[3] However, several stylized facts and subsequent firm-level analysis suggest that inadequacy in financial intermediation has contributed to declining capital efficiency.

First, in many Asian countries, financial controls such as interest rate limits and government-directed lending are still more common than in the rest of the world. The region moved steadily toward more flexible interest rate policies from the 1980s through the late 2000s. However, the trend stalled and reversed after the global financial crisis and up to 2017, when the data on financial controls end, especially in emerging market and low-income economies and relative to peers in other regions (IMF 2025a). Although such measures often aim to support specific sectors, protect borrowers, and potentially accommodate fiscal dominance, they tend to interfere with market functioning. By distorting market incentives and signals, the measures could reinforce sectoral policies and result in rent seeking and efficiency losses that are a drag on growth (Jafarov, Maino, and Pani 2019).

[3] In 2013, a shift in expectations of market participants for the timing of the tapering of the Federal Reserve's asset purchases, and its ramifications for normalization of US monetary policy, led to sharp increases in longer-term US Treasury yields and volatility in global financial markets.

Second, a sample of listed firms in Asia shows that reliance on external financing to fund operations has declined since the global financial crisis, despite expanding financial systems and accommodative conditions, while net savings have increased (Figure 3.3). In the early 2000s, a more cautious response after the 1997 Asian financial crisis and greater internal financing within conglomerates likely contributed to this trend. Tighter financial regulations and corporate strategies to reduce exposure to volatility also reflect that shift. However, the trend's persistence suggests that reliance on internal funding is less likely caused by weak demand for external financing and more likely reflects inadequacy in financial intermediation. Conceptually, the inadequacies extend beyond the availability of funding. Their prevalence during generally accommodative financial conditions since the global financial crisis indicates a mismatch between firms' needs and the financial instruments available to them. IMF (2009) finds that inefficiencies arising from underdeveloped financial systems contributed to a surge in corporate savings after the Asian financial crisis. Survey data indicate that collateral requirements are more restrictive in Asia than in other regions (IMF 2025a, Online Annex 2), and limited access to equity markets creates specific challenges for startups.

**Figure 3.3. External Financing Dependence**
*(Ratios)*




Sources: Capital IQ; and IMF staff calculations.
Note: External financing dependence is calculated as ([capex – operational cash flow] / capex). Net savings is calculated as ([net income – dividends paid – capex] / total assets). The boxes show the interquartile range, the horizontal line denotes the median, and the whiskers indicate the range of observations, which are at the sector level. Capex = capital expenditure.

Third, financial resources have also shifted from smaller firms to larger ones. Although the change in leverage at an aggregate level presents a mixed picture across countries, leverage within an industry and country has increased more for larger firms, indicating a shift in financial resources toward them. These firms also tend to experience larger declines in the productivity of capital (MRPK), contributing to greater capital misallocation through wider MRPK dispersion.

Lastly, evergreening has increased across the region, reflected in the rise of number of "zombie" firms. Zombie firms are firms that cannot cover their interest expenses with operating income over extended periods, but they continue operating. The share of corporate debt held by zombie firms has nearly tripled in the region since the global financial crisis, compared with a flat or declining trend in other regions (Figure 3.4). The zombie debt share rose from an average of about 2 percent in 2010–12 to approximately 7 percent in 2020–23.

Taken together, these developments suggest that financial resource misallocation might contribute to broader capital misallocation in the region and widening divergence in access to finance across firms. Accommodative financing conditions have lowered borrowing costs, but large and less-productive firms have benefited disproportionately, while smaller, younger, and more productive firms face tighter financing constraints because of collateral requirements, a limited set of financing instruments, and weak restructuring incentives.

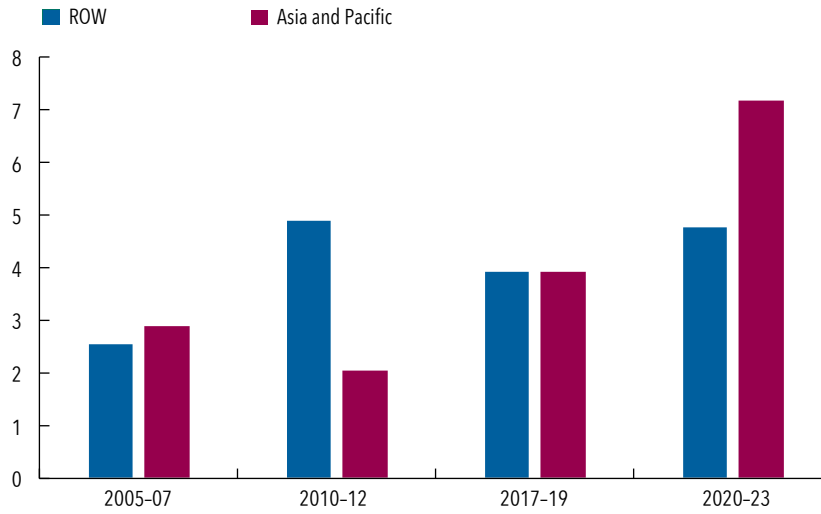
Empirical tests establish how financial intermediation relates to capital misallocation. Financial inadequacy can coexist with other policies or institutional features. For example, preferential policies for certain sectors or firms could lead to capital misallocation. The empirical strategies isolate the impact of financial factors and identify specific channels through which financial misallocation might have affected capital allocation. Using firm-level data, the analysis first examines how financing constraints affect company investment and growth and identifies the firms most affected. Next, it examines recent increases in evergreening among firms and the effects of easing financing conditions. Finally, the analysis quantifies the loss in real value added arising from financial misallocation, separating it into that attributable to funding availability and mismatches in financing instruments, and examines how these effects differ across countries at different stages of development.[4]

## Financing Constraints and Misallocation of Capital

To demonstrate and assess the impact of financing constraints on firm growth, the analysis followed Rajan and Zingales (1998) and tested whether firms in sectors that rely heavily on external financing grow faster in countries with more developed financial systems. The hypothesis was that if financing

[4] For details of methodologies and data, see IMF (2025a, Online Annexes 1.3–1.6).

**Figure 3.4. Share of "Zombie" Firms in Corporate Debt**
*(Percent, average for each period)*




Sources: Capital IQ; and IMF staff calculations.
Note: "Zombie" companies are firms older than 10 years that could not cover interest payments with operating income for three consecutive years. ROW = rest of the world.

constraints matter for growth, firms in industries that depend more on external finance will have higher growth rates when financial development relaxes those constraints.

Following the methodology of Li (2020), the analysis found strong evidence of financing constraints in Asia (Annex Table 3.1.1, panel 1, column 2): a statistically significant positive interaction between external finance dependence and level of financial development, captured by the financial development index. The results confirm that easing financial constraints through financial development helps firms grow. Annex Table 3.1.1 also shows that the effect of easing financing constraints on growth is more pronounced in Asia than for the global sample (column 1).

Financing constraints also affect firms differently, with the strongest effects on small firms (those with low assets) and young firms (columns 3 and 5). Firm investment follows a similar pattern, measured by capital expenditure (Annex Table 3.1.1, panel 2). Financing ability constrains Asian firms' investments, and the effect is stronger than observed globally and stronger for smaller and younger firms.

Industry differences in firm composition can cause financing constraints to affect industries differently. Service firms tend to be smaller and younger than manufacturing firms, consistent with the more dynamic nature of service

industries, in which specialization and customized needs typically drive businesses instead of scale.

Financing constraints on the expansion of smaller firms would lead to capital misallocation because smaller firms tend to have higher capital productivity than larger firms. An alternative analysis supports that assessment. Gopinath and others (2017) show that smaller firms that cannot borrow must rely on internal savings for investment. The constraint limits their investment and slows their growth even though the firms have higher MRPK than larger, unconstrained firms. Thus, in financing-constrained environments, leverage and firm size are positively correlated, whereas MRPK and firm size are inversely correlated. The data confirm those correlations for the Asia and Pacific region (Annex Table 3.1.1, panel 3). Firms classified as large based on precrisis size (assets) could borrow more and invest more than firms classified as small—2.0 percent higher investment and 0.1 percent higher leverage during the period after the global financial crisis for a 1 percent larger precrisis size. The rise in capital misallocation after the crisis mainly reflects less efficient capital use among larger firms (lower MRPK).

## Evergreening Aided by External Financing

A rising share of zombie firms is a symptom of capital misallocation. More evergreening allows unproductive zombie firms to survive and absorb resources that could be used more productively elsewhere. Although policy support during the pandemic to cushion economic shocks likely increased evergreening in other regions, the increase in Asia began well before the pandemic, underscoring a sustained upward trend since the global financial crisis. Asian zombie firms tend to be small and middle-aged (10–14 years old) and are concentrated in sectors such as real estate, utilities, and manufacturing, which tend to rely more heavily on external financing. They also generally have weaker balance sheet positions and lower savings than non-zombie firms. Zombie firms are present across Asian countries, ranging from economies with a very small zombie share to those with a share exceeding 10 percent, though data coverage varies.

Formal statistical tests established links between the increase in zombie firms and their access to external financing. Other factors also contribute to the rise in zombie firms, including subsidies, industrial policies favoring state-owned enterprises (SOEs), pandemic-related fiscal support, and weak bankruptcy laws. The empirical strategy isolated the effects of external financing through sectoral variation in firms' sensitivity to external financing. The results show strong evidence that the prevalence of zombie firms has grown faster in industries that rely heavily on external financing because the decline in financing costs—proxied by policy rates—reduced financial pressure on firms. Quantitatively, lower interest rates after the global financial crisis and during the COVID-19 pandemic are associated with an increase in zombie firms' share in total assets of about 0.4 percent for every 1 percent decline in interest rates (Annex Table 3.1.1, panel 4). Other studies report similar findings (Banerjee and Hofmann 2018; Albuquerque and

Iyer 2024). For firms that require external financing, easier financing conditions account for nearly two-thirds of the increase in the share of zombie firms since the global financial crisis. The relationship also holds among relatively larger zombie firms (those with more assets than the median zombie firm) and firms in industry and tradable sectors. No such evidence was found for service firms, which suggests that manufacturing firms benefit more from easier financing conditions, consistent with better financing access in those sectors. Cheaper financing reduced pressure on lenders to clean up their balance sheets and allowed financing to flow to weaker firms, contributing to distorted capital allocation toward unproductive firms. In this connection, interest rate controls (which became more common in the region after the global financial crisis) likely eased financial pressure on nonviable firms even more.

The increase in the share of zombie firms captures only the direct impact on misallocation. Albuquerque and Iyer (2024) and other studies also show that zombie firms can harm healthy firms within the same industries. Zombie firms depress investment, productivity, and job creation because of negative macrofinancial spillovers, such as increased competition for input goods, excessive supply of goods, lower market prices, higher labor costs, and lower profits for productive, healthy firms. The analysis did not explore potential spillovers, but a persistent rise in zombie firms can lead to even greater capital misallocation, and strategies to restructure bad debt are essential for improving capital allocation.

## Financial Misallocation: Scale versus Mix of Financing

Firm-level data confirm that financial misallocation has influenced capital allocation in Asia. An analysis examined the extent to which this effect stems from the scale of financing and the mix of instruments and what that implies about potential gains from reducing financial misallocation. The analysis used a tractable framework that quantified financial misallocation using observed firm-level stocks of two broad categories of financing instruments: debt and equity.

The analysis adopted the framework of Whited and Zhao (2021), in which debt and equity are imperfectly substitutable inputs into the production process. Firms' use of financial liability is therefore linked directly to their value added (see Annex 3.1 for more details). Firms differ in the financial productivity or efficiency with which they transform a given amount of such liabilities into value added. Rearranging these liabilities across firms by moving gross resource flows from less efficient to more efficient firms can therefore generate productivity gains and raise aggregate value added. Without distortions, financial intermediation should enable assortative matching between the total amount of liabilities, or scale of financing, and firm efficiency. Diminishing marginal returns imply that the composition of finance also affects aggregate real value added, and additional gains can arise from optimizing the mix of debt and equity.

An efficient scale of total financing, implied by firm-level productivity, and an efficient mix of debt and equity, determined by the sector-specific optimal debt-equity ratio, exist within narrowly defined sectors. Financial misallocation

therefore is captured by extent to which the observed distribution of total liabilities—debt and equity—across firms within narrowly defined sectors differs from the observed distribution of firm-level productivities, and the extent to which observed firm-level debt-to-equity ratios deviate from the sector-specific optimal ratio.

The estimation of gains from removing financial misallocation was based on two counterfactual scenarios. In the first, the analysis considered debt and equity to be perfect substitutes. Under perfect substitution, the composition of liabilities is irrelevant for real value added, and aggregate productivity gains reflect only the benefits of reallocating the total amount of financing across firms. Optimal allocation in this case entails distributing total liabilities according to firm-level productivity, capturing the effects of the scale of financing.

In the second counterfactual scenario, the analysis used the data to infer the elasticity of substitution between debt and equity for each narrowly defined sector. Optimal allocation in this setting involves allocating total liabilities based on the firm-level productivity calculated in the first step and an efficient sector-specific mix of debt and equity consistent with the estimated elasticities.

The first counterfactual scenario estimated the gains from correcting distortions in the scale of financing, while the second captured the gains from correcting distortions in both the scale and the mix of financing.[5] Comparing the two estimates allowed the importance of scale and mix distortions to be quantified.

The results show that although the amount of financing is the main issue in Asia overall, the importance of the financing mix generally rises with income levels. Therefore, the need for diverse financing options becomes more important as countries develop (Figure 3.5).

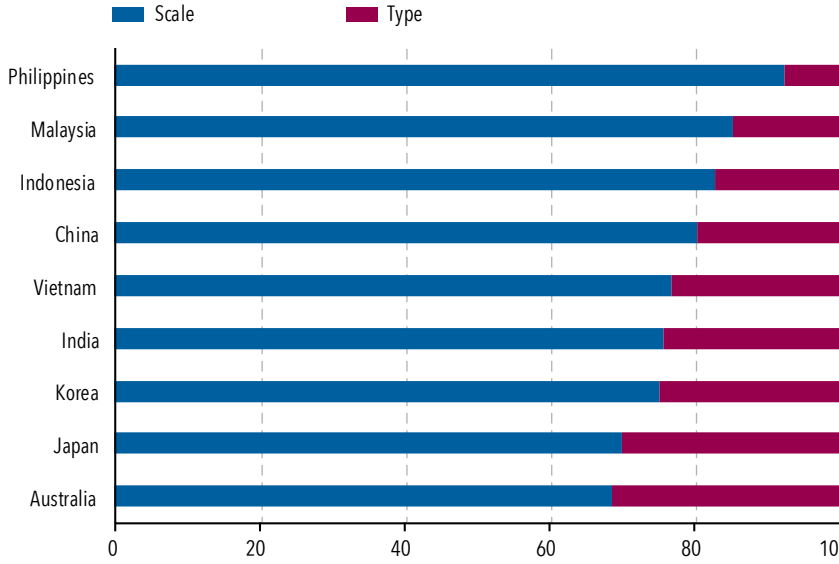
Financial misallocation is more acute in services than in goods sectors, likely because goods sectors benefit from greater availability of tangible capital to secure financing. Scale is also more constraining for services, indicating that insufficient financing is a particularly binding constraint for service-oriented activities in Asia. The analysis also suggests a substantially higher degree of financial misallocation in Asia compared with the United States, with productivity gains from eliminating financial misallocation in Asia about three times higher in Asia than in the United States.

## Financing Constraints on Fast-Growing Small Firms

Small, young, rapidly growing firms—often called "gazelles"—are likely to face especially severe financing constraints. Such firms have contributed significantly to aggregate productivity, employment, and output across global economies (Decker and others 2014; Haltiwanger and others 2016; Goswami, Medvedev,

[5] Firm-level sales were used as a proxy for value added because of data availability constraints for a subset of countries, which might have introduced a potential bias related to differences in the intensity of intermediate input use across firms. To assess the robustness of the results, the analysis was reestimated using value added for the countries where such data are available. The results are similar to those obtained using sales.

**Figure 3.5. Importance of Scale versus Mix of Financing in Explaining TFP Gains**
*(Percent)*




Sources: Orbis database; and IMF staff calculations.
Note: The figure shows the counterfactual gains in real value added for each country if finance misallocation were reversed under empirically estimated elasticity of substitution between debt and equity and under the alternative scenario of perfect substitutability. The gains for each country are averaged over 2010–22. TFP = total factor productivity.

and Olafsen 2019). However, in most countries, including Asian advanced economies, gazelles have a small footprint and account for less than 2 percent of new entrants. In those economies, gazelles carry higher leverage than other firms do from a young age, consistent with the fact that early access to finance is crucial for gazelle formation. However, firm-level data suggest that gazelles also face higher borrowing costs than other firms, even after accounting for higher leverage—on average by 63 basis points in Asian advanced economies—despite having higher average productivity of capital. This could reflect gazelles' risk profile but could also signal resource misallocation or an inability to access adequate financing. Borrowing costs are especially elevated for gazelles with a high level of intangible capital (up to 80 basis points more)—which is more difficult to use as collateral for standard bank loans.

By contrast, data show that the share of gazelles among new entrants is much lower in emerging Asia, and gazelles in these economies do not carry excess leverage or face excess interest costs, in contrast to the pattern observed in peer emerging markets. These observations suggest that financing constraints on gazelles are even more severe in emerging Asia than elsewhere, which has suppressed their

emergence. Traditional bank and equity financing might not provide financing that is sufficiently tailored or accessible to gazelles.

Across Asia, greater mismatches in financial instruments would constrain the growth of innovative and young firms. Broadening the range of tailored financing options, including deeper private capital markets, can support the emergence and growth of young, high-growth, high-potential firms. For example, venture capital has been shown to be especially suitable for supporting gazelle growth (Ando 2025).

## STRUCTURE OF CORPORATE FINANCE IN ASIA

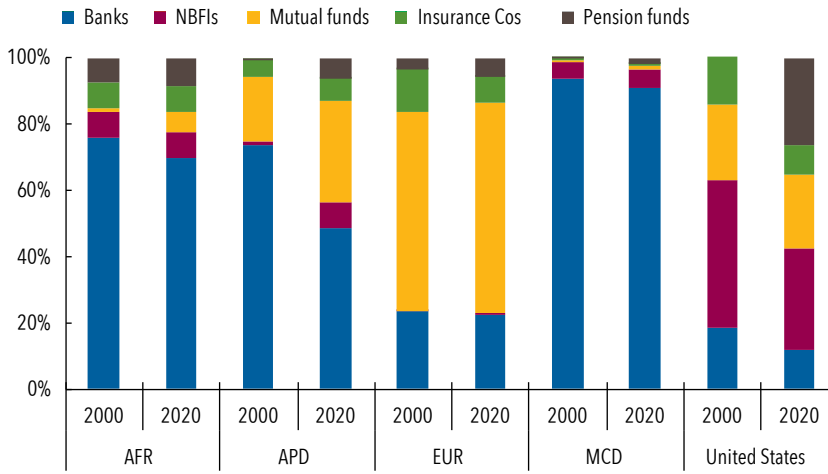
### Asian Financial Systems: Key Features and Constraints

The structure of financial systems in the region has affected how financial resources are allocated across firms. Overall financial development in Asia is high and comparable to other regions, though differences within the region are large. Measured by financial access, depth, and institutions, the level of financial development appears higher in Asian economies than in peer economies outside the region with similar GDP per capita, based on the IMF Financial Development Index (IMF 2025a, Online Annex 2). Financial systems have expanded steadily across the region, and except for episodes such as the 2013 taper tantrum, financial conditions have supported investment since the global financial crisis. Financing levels are generally higher in East Asia and more developed economies and remain relatively low in several South Asian economies such as India and Indonesia.

Bank dominance remains a defining feature of the region's financial systems (Figure 3.6). In Asia, bank credit extended to nonfinancial companies equals 143 percent of GDP, far above the global average of 96 percent (OECD 2025). Banks dominate financial intermediation in the region's emerging markets, low-income developing economies, and advanced economies (IMF 2025a, Online Annex 2). Market-based financing exceeds bank financing slightly in only a few countries, including India and Indonesia. Banks have been crucial for mobilizing savings and supporting Asia's rapid economic growth and industrialization by pooling deposits, reducing informational asymmetries through long-term relationships with firms, and benefiting from economies of scale in information gathering.

However, the bank-dominated systems also face limitations in supporting broad-based growth, and financial inclusion gaps remain large in lower-income countries. In 2024, nearly 40 percent of adults in low- and middle-income Asia and Pacific economies lacked a bank account. Small and medium enterprises often face even tighter financing constraints. As countries develop, the structure of bank lending itself can become restrictive because banks generally prefer tangible collateral and predictable cash flows, favoring established firms over small and young firms. Sectors with fewer tangible assets, such as technology and services, or those with novel business models, can struggle to obtain credit. The structure can bias capital allocation toward safe, traditional industries and away

**Figure 3.6. Structure of Financial Sector**
*(Percent of financial sector total assets)*




Sources: Global Financial Development Database; and IMF staff calculations.
Note: NBFI = nonbank financial institution.

from emerging industries, slowing innovation and reducing overall investment efficiency. In addition, financial repression has historically been more common in bank-dominated systems. Governments have sometimes capped interest rates or directed banks to lend to favored sectors, which can distort credit allocation even more. Weak regulation can worsen these problems because banks might evergreen loans to keep troubled zombie firms afloat, undermining productivity and long-term growth.[6]

Many Asian countries have aimed to develop capital markets to establish a more diversified financial structure and impose stronger financial discipline. Capital markets could complement bank financing in several important ways. Deep stock and bond markets can mobilize funding from a broad investor base to support high-potential ventures, unlike banks, which rely on relationship-based finance and favor stability and incumbents, potentially suppressing high-growth opportunities among riskier, smaller, and younger firms.[7] Market-based financing

[6] See, for example, Caballero, Hoshi, and Kashyap (2008), Acharya and others (2019), and Blattner, Farinha, and Rebelo (2023) for Japan in the early 1990s and Europe in the early 2010s. More recently, Albuquerque and Iyer (2024) find that zombie shares can increase even with a strong banking sector and tight macroprudential policies if insolvency frameworks are not well prepared to manage restructuring.

[7] Levine (1991) and others note that large liquid markets provide investors with strong incentives to invest in longer-term and higher-return projects because they can sell their shares before the project matures. For example, venture capital and equity markets strongly influenced the rise of the technology sector in the United States.

can also provide stable funding to companies during economic downturns or periods of tighter lending, substituting for bank credit, which is procyclical. Such financing is especially crucial for enabling smaller companies to borrow and invest.[8] Market-based systems can also strengthen corporate governance through mechanisms such as equity ownership, takeovers, and managerial compensation tied to stock performance (Jensen and Murphy 1990). However, market-based systems require robust legal and informational infrastructure, including well-defined property rights, transparency, disclosure, and investor protections (for example, Demirgüç-Kunt and Levine 2001). Without such foundations, market-based finance cannot protect minority shareholders adequately and becomes more vulnerable to bubbles and volatility.

Asia's capital markets have expanded substantially since 2000. By 2020, stock market capitalization as a share of GDP reached 133 percent in Korea, about 98 percent in India, and 83 percent in China. Several Southeast Asian markets, including Malaysia and Thailand, now have stock market sizes on par with the size of their economies, growth that reflects both rising equity valuations during economic growth and an increase in listed companies and equity issuance over time. Corporate bond markets in the region have multiplied in size since 2000. OECD (2025) finds that annual corporate bond issuance in Asia increased more than any other region, driven mainly by China's bond market expansion. Outstanding corporate bonds increased by $9 trillion from 2000 to 2024, about twice the increase in Europe and the United States. Asia's share of global corporate bond issuance also surged, with the region's share of the global corporate bond market up from 10 percent in 2000 to 29 percent in 2024.

However, Asian financial markets face several structural challenges. A few large companies with high concentration of ownership often dominate stock markets except in financial hubs such as Hong Kong Special Administrative Region. In Indonesia, the Philippines, and Thailand, the top 10 listed companies account for about two-thirds of total market capitalization, compared with 36 percent in the United States, reflecting the dominance of conglomerates and company groups. In addition, SOEs account for about 26 percent of total market capitalization in Asia, with even higher shares in China, Malaysia, Singapore, and Vietnam, far above the 5 percent share in the rest of the world. Corporate owners hold nearly one-quarter of shares in Asia, compared with the global average of less than 10 percent. In several countries, the top three shareholders control more than half of all shares (OECD 2025). The market and ownership structure increases the importance of corporate governance, comprehensive information disclosure, and strong protection of minority shareholders, which are essential for strengthening investor confidence and participation.

The low level of institutional investor participation in capital markets reflects both market structure and investment restrictions. Domestic institutional

[8] See Becker and Ivashina (2014) on firm-level evidence and cyclicality of credit supply.

investors in Asia (pensions, insurance, and asset managers) are expanding but remain less prominent than their counterparts in Europe or the United States. Institutional investors hold 69 percent of company shares in the United States and 38 percent in Europe, compared with only 18 percent in Asia, where corporations and the public sector hold more shares (OECD 2025). Many countries rely on banks or government-linked institutions to buy corporate bonds. Regulatory constraints on institutional investors' mandates also limit participation. For example, local pension funds in Asia tend to follow conservative mandates and invest mainly in government bonds and deposits, with limited allocations to domestic equities or corporate bonds. In some countries, including Bangladesh, Lao P.D.R., and Mongolia, pension funds cannot invest in equities (OECD 2025), limiting demand for new issuance, especially from smaller private companies.

Several key markets experience low liquidity, especially corporate bond and growth equity markets. Except for a few large markets, Asia's corporate bond markets face low investor participation, inadequate hedging tools, and challenges to credit rating coverage. Most jurisdictions report little secondary market activity, and large companies drive most trading in more advanced markets such as Japan, Korea, and Singapore (OECD 2025). Low liquidity limits investors' ability to evaluate information effectively and reduces incentives to participate. Participation of institutional investors could improve market liquidity. However, pension funds and insurance companies in Asia typically invest in government bonds to manage risks or require minimum credit ratings for investments, restricting allocations to corporate bonds. The limited opening of financial markets also constrains foreign investor participation. Similarly, growth stocks tend to have low trading volumes, partly because controlling shareholders hold large stakes that reduce the free float of shares.

Growth firms face additional hurdles from regulatory requirements, listing fees, and underdeveloped markets. For midsize firms, listing procedures, underwriting fees, and disclosure requirements can create significant barriers, given their modest financing needs. Thus, equity financing is an option mainly for larger companies or firms in popular sectors that can attract high valuations. Some markets have introduced small and medium enterprise boards or reduced listing costs in growth markets by replacing flat fees with proportional charges or sliding fee schedules based on offering size (OECD 2025). Similarly, corporate bond issuance usually requires a credit rating and compliance with disclosure standards, which pose challenges for smaller firms. Although some countries have relaxed rules to allow unrated bond issuance or private placements for smaller firms, investor interest remains limited. Many Asian countries such as China, India, Indonesia, and Japan introduced policies to reduce the tax incentives of debt financing and to support a more balanced corporate capital structure, but enforcement varies, and coordination problems with other tax benefits could weaken their effectiveness.

Private equity and venture capital have expanded rapidly as complementary sources of corporate financing, especially for growth companies. Venture

investment in Asia reached record levels in the late 2010s, with unicorn startups emerging in China, India, and Southeast Asia.[9] The expansion improved access to equity-like financing for the most dynamic startups. However, venture capital is concentrated in a few technology hubs and is unavailable to the broader small and medium enterprise sector. Private equity remains much smaller in Asia than in Europe and the United States, and limited exit opportunities caused by underdeveloped private markets are a major obstacle. Regulatory policies such as capital controls, antitrust policies, and government approval requirements for certain investments have also affected private equity growth (OECD 2025).

The concentration of large firms, lack of investor diversification, high listing requirements for smaller companies, and low liquidity in important market segments have constrained the potential of market development and restricted companies' funding opportunities. As a result, their ability to serve the corporate sector—especially small and young firms that lack the scale to attract bank financing—remains constrained, despite Asia's expansion of capital markets. Reform priorities vary across countries, but crucial areas include strengthening minority shareholder protections to attract investors, improving and simplifying listing procedures for small companies, allowing more flexible mandates for institutional investors, and reducing financial repression to deepen markets and improve liquidity.

## Financial Intermediation, Capital Misallocation, and SOEs

Intermediation of financial resources is linked closely to other institutional features (such as state ownership of companies and financial intermediaries) and to policy choices (such as industrial policies). Together, these factors drive capital misallocation and compound each other's effects. Understanding financial intermediation in Asia requires considering its interaction with other potential sources of distortion, particularly extensive state ownership of companies.

SOEs are an important part of corporate Asia, where about 70 percent of the world's listed SOEs are located (OECD 2025). Their presence ranges from a few dozen entities in smaller economies (such as Bhutan, Nepal, and Samoa) to several thousand in the largest markets (including China, India, Indonesia, and Vietnam), reflecting varying degrees of state involvement (Dall'Olio and others 2022; World Bank 2023).[10] In several regional economies, SOE revenues equal

[9] A unicorn is a startup company valued at over US$1 billion which is privately owned and not publicly listed.

[10] Traditionally, SOEs have been defined as companies in which the central government hold a majority or full ownership stake. However, this chapter uses a broader definition, based on the World Bank's Business of the State database, that includes firms with minority, indirect, or subnational ownership, comprising all companies with more than 10 percent government ownership (World Bank 2023). The data set excludes data on regional advanced economies and China. Jurzyk and Ruane (2021) show that China's publicly listed SOEs accounted for about 30 percent of revenues in all industrial firms.

40–50 percent of GDP and together account for 25 percent of regional market capitalization. However, in Vietnam, the number of SOEs has declined, and they account for only 0.5 percent of all firms and employ 9 percent of the labor force yet still hold 29 percent of total assets in the economy.[11] Across the region, these enterprises maintain strong positions in competitive sectors and often hold majority stakes in strategic industries, underscoring their continued influence on market structures and policy priorities. Rationales for such policies range from supporting developmental goals in sectors where SOEs perform core public functions to managing fiscal costs and using SOEs as instruments of industrial policy.

Preferential credit policies that distort credit allocation toward SOEs could constrain productive firms from expanding and gaining market share (for example, Chen and Irarrazabal 2015; Meza, Pratap, and Urrutia 2019; Jurzyk and Ruane 2021). In Vietnam, Le (2022) found that preferential treatment for SOEs was the main contributor to capital misallocation. Examples include preferential access to credit from development and state-owned banks and allocation of prime land at below-market prices, which SOEs could then use as collateral to obtain bank loans. Reducing such misallocation could generate total factor productivity gains exceeding 50 percent. IMF (2017) also found evidence of credit misallocation between SOEs and non-SOEs among listed firms. In addition, borrowing costs for SOEs have been significantly lower than for domestic private firms, which could allow less-productive SOEs to remain competitive and gain market share while reducing overall productivity. Similar patterns—lower productivity and lower borrowing costs for SOEs—appear globally (IMF 2020).

Preferential credit policies toward SOEs could coincide with broader financial distortions, especially when support for SOEs shifts resources more toward large firms at the expense of smaller, more productive firms, or when they encourage loan evergreening through easy financing for SOEs. Strengthening internal controls and governance and distinguishing commercial objectives from other objectives are critical for improving capital allocation associated with SOEs. Strengthening financial intermediation remains an important part of policy efforts, especially as governments pursue SOE reforms, including public equity listings in countries such as China and Vietnam to improve transparency and efficiency. Better financial intermediation can strengthen market discipline for SOEs while broadening financial access for private and growth-oriented firms and leveling the playing field.

Given diversity across SOEs in size, sector, and state involvement (also discussed in IMF 2020), a tailored approach is essential. In strategic sectors where SOEs perform core public functions, such as utilities, the emphasis should be on improving performance and establishing sustainable business models. However,

[11] Tu Anh, "Hiệu quả chưa tương xứng với nguồn lực (Performance Incommensurate with Productive Resources)." PetroTimes, January 5, 2019, https://petrotimes.vn/hieu-qua-chua-tuong-xung-voi-nguon-luc-525353.html.

in competitive sectors where private firms operate efficiently, policy should prioritize efficient use of public resources and avoid preferential access that crowds out private credit.

## CONCLUSION

The growth slowdown in the Asia and Pacific region since the global financial crisis has coincided with lower investment efficiency and greater capital misallocation. Inadequacy in financial resource allocation has contributed to these problems. Despite the rapid expansion of financial systems, financing constraints for the corporate sector—particularly smaller, innovative, and service sector firms—have increased. Evergreening of corporate loans has also increased, which ties up financial resources in unproductive projects and risks crowding out more productive projects. The empirical evidence shows a strong link between capital misallocation and financial distortions.

Improving financial structures has become a key policy priority in Asia, yet important gaps remain. Financial inclusion is still limited in lower-income countries, and many people lack access to even basic financial services. In middle- and higher-income economies, access to credit is often constrained for small and innovative firms because banks tend to favor large, asset-rich borrowers. Corporate governance concerns, low institutional investor participation, incomplete equity financing options, and financial restrictions and repression also hinder the development of a more diversified financial structure. In addition, easier financing conditions after the global financial crisis increased inefficiency by prolonging the survival of poorly performing firms with unviable debt. Structural reforms to Asia's financial systems are needed to address these challenges.

Improving financial intermediation by expanding financial access and providing a better mix of instruments would boost capital efficiency and would help to boost productivity growth. The productivity gains from eliminating financial misallocation are three times as large as those for the United States. Service firms would account for much of the increase because they tend to face greater financing constraints and thus benefit more from removing the distortion. In lower-income countries, most gains stem from providing adequate funding to enable corporate growth. As countries develop, a better financing mix becomes increasingly important. Stronger equity financing would provide much-needed risk capital and long-term investment that would support innovation and productivity growth.

Policy measures to ease financing constraints across the corporate sector would improve capital allocation. Priorities should reflect each country's level of development. In lower-income countries, the main challenge is ensuring financial inclusion and access to finance. Higher-income countries should offer a broader mix of financing options. Deeper capital markets could complement banking services and provide more effective financial intermediation, improving

capital allocation. Critical steps include developing deeper equity and bond markets by streamlining listing procedures, diversifying the investor base, and prioritizing regulatory reforms that lower barriers to nonbank financing while strengthening financial regulation and insolvency frameworks. Given the concentrated market and ownership structures, stronger corporate governance, better disclosure, and robust shareholder protections are essential for increasing investor participation. Reducing financial repression and restrictions would also support market development. At the same time, regulators will need to monitor new emerging risks (including those related to the participation of new investors), strengthen supervisory oversight, and close data gaps (IMF 2023; Zhu, Zhang, and Zhang 2023).

Restructuring bad debt can improve capital allocation and lower the likelihood that governments will need to absorb losses through bank recapitalization or other support measures. Although low interest rates could help manage debt burdens, they can also lead to misallocation and obscure risks through mispricing. In addition, failure to address the costs of nonviable debt transparently could threaten fiscal sustainability. Stronger financial oversight and risk management are essential for preventing evergreening.

Improving financial intermediation will benefit smaller and service sector firms that face the greatest financing challenges and thus support job creation and income growth. Better financial systems also reduce firms' incentive to hoard cash and encourage more dividend payouts, which can boost consumer spending. Together, these changes will help Asia and Pacific move toward more balanced growth, rely less on external demand and more on domestic drivers, and strengthen economic resilience.

## ANNEX 3.1.

**ANNEX TABLE 3.1.1.**

Role of Financing Constraints for Firm Growth and Capital Misallocation

| 1. Financing Constraints and Firm Growth | | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | Global | Asia and Pacific | Small | Large | Young | Old |
| External finance dependence × financial development index | −0.0335 | 0.136*** | 0.257*** | 0.0208 | 0.277*** | 0.0177 |
| Number of observations | 284,063 | 185,305 | 51,869 | 133,436 | 72,058 | 113,247 |

Sources: Capital IQ; Financial Development Index Database; IMF 2025a, Online Annex 1.3; July 2025 *World Economic Outlook*; and IMF staff calculations.

Note: The dependent variable is firm growth (change in log assets). "Global" refers to the sample with all economies, and "Asia and Pacific" refers to the sample with Asia and Pacific economies. "Small" refers to the sample with firms with assets less than or equal to the median, and "large" refers to the sample with firms with assets greater than the median. "Young" refers to the sample with firm age less than or equal to the median, and "large" refers to the sample with firm age greater than the median.

*$p$ < .10; **$p$ < .05; ***$p$ < .01.

| 2. Financing Constraints and Capital Expenditure | | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | Global | Asia and Pacific | Small | Large | Young | Old |
| External finance dependence × financial development index | 0.259*** | 0.365*** | 0.604*** | 0.147*** | 0.571*** | 0.214*** |
| Number of observations | 271,171 | 165,527 | 47,899 | 117,628 | 77,553 | 87,974 |
| Fixed effects | Country, sector, year | Country, sector, year | Country, sector, year | Country, sector, year | Country, sector, year | Country, sector, year |

Sources: Capital IQ; IMF Financial Development Index database; IMF 2025a, Online Annex 1.3; July 2025 World Economic Outlook; and IMF staff calculations.
Note: The dependent variable is capital expenditure (log capital expenditure). "Global" refers to the sample with all economies, and "Asia and Pacific" refers to the sample with Asia and Pacific economies. "Small" refers to the sample with firms with assets less than or equal to the median, and "large" refers to the sample with firms with assets greater than the median. "Young" refers to the sample with firm age less than or equal to the median, and "large" refers to the sample with firm age greater than the median.
*$p$ < .10; **$p$ < .05; ***$p$ < .01.

| 3. Growth in Capital, Change in MRPK, and Change in Leverage and Firm Size | | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| | Dependent Variable: Capital Growth | Dependent Variable: MRPK Change | Dependent Variable: Leverage Change |
| Log net worth | 1.980*** | −0.444** | 0.056*** |
| Log capital | −4.499*** | 1.563*** | −0.033*** |
| Log productivity | 1.835*** | −4.636*** | −0.024*** |
| Fixed effects | Sector X country | Sector X country | Sector X country |
| Number of observations | 46,687 | 46,687 | 46,687 |

Sources: Orbis; IMF 2025a, Online Annex 1.4; and IMF staff calculations.
Note: MRPK = marginal revenue product of capital.
*$p$ < .10; **$p$ < .05; ***$p$ < .01.

| 4. Zombie Firm's Rise and Financing Costs | | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| External finance dependence × policy rate | −0.04** | −0.04** | −0.04** | −0.03* |
| Inflation | | | | −0.2 |
| Fixed effects | Country, year, sector | Country X year, sector | Sector X year, country | Sector X year, country |
| Number of observations | 26,243 | 26,233 | 26,026 | 24,595 |

Sources: Bloomberg Finance L.P.; Bank for International Settlements; Capital IQ; CEIC Data; IMF 2025a, Online Annex 1.6; July 2025 *World Economic Outlook*; and IMF staff calculations.
Note: The dependent variable is zombie firms' share of assets.
*$p$ < .10; **$p$ < .05; ***$p$ < .01.
Notes: See online Annex of Chapter 3 of the October 2025 *Regional Economic Outlook for Asia and Pacific* - 1.3 for panels 1 and 2, 1.4 for panel 3, and 1.6 for panel 4.

## Importance of Scale versus Mix of Financing

Firms need to issue debt and raise equity to finance the acquisition of physical capital, labor input, and other inputs of production. Rather than imposing a specific theory for why debt and equity are not perfectly substitutable and for how the total amount of financing is distributed into its various applications, the analysis postulated a direct mapping from financial liabilities into real output that captured these unmodeled elements in reduced form, following Whited and Zhao (2021). The distribution of finance into capital, labor, and innovation was subsumed in a finance-based measure of productivity, labeled as total finance benefit. Formally,

$$F_{si} = Z_{si}\left[\alpha_s D_{si}^{\frac{\gamma_s-1}{\gamma_s}} + \left(1-\alpha_s\right)E_{si}^{\frac{\gamma_s-1}{\gamma_s}}\right]^{\frac{\gamma_s}{(\gamma_s-1)}}, \tag{3.1.1}$$

where $s$ refers to the sector, $i$ denotes the firm, $Z_{si}$ denotes the total finance benefit backed out from the data based on the methodology in Whited and Zhao (2021), $\gamma_s$ is the industry-specific elasticity of substitution between debt $D$ and equity $E$ (estimated from the data), and $\alpha_s$ is the industry-specific weight of debt in real output. The weight was estimated as the undistorted share of debt cost relative to the total cost of debt and equity, with each component adjusted according to the elasticity of substitution. The real output is denoted by $F_{si}$ Empirically, the analysis used a firm's sales as the proxy for output and followed Kmenta (1967) to estimate the elasticity of substitution between debt and equity from the constant elasticity of substitution production function at the country-sector level.

Wedges, idiosyncratic to the firms, distort the aggregate scale of the firm and the relative price between debt and equity. These are the debt wedge $\tau_{Dsi}$ and the equity wedge $\tau_{Esi}$ Every period, firms choose debt and equity, taking their prices as given. Therefore,

$$\alpha_s\frac{(\sigma-1)}{\sigma}\frac{P_{si}F_{si}}{\alpha_s D_{si} + \left(1-\alpha_s\right)D_{si}^{\frac{1}{\gamma_s}}E_{si}^{\frac{\gamma_s-1}{\gamma_s}}} = r\left(1+\tau_{D_{si}}\right), \tag{3.1.2}$$

$$\alpha_s\frac{(\sigma-1)}{\sigma}\frac{P_{si}F_{si}}{\alpha_s E_{si} + \left(1-\alpha_s\right)E_{si}^{\frac{1}{\gamma_s}}D_{si}^{\frac{\gamma_s-1}{\gamma_s}}} = \lambda\left(1+\tau_{E_{si}}\right), \tag{3.1.3}$$

where $r$ denotes the undistorted cost of debt, $\lambda$ denotes the undistorted cost of equity, and $\sigma$ is the elasticity of substitution for the real benefit of finance

between firms in an industry and was set to 1.77. The values of these parameters were set based on Whited and Zhao (2021). Under no distortions, these marginal returns are equalized across firms. Therefore, any dispersion in marginal returns constitutes evidence of misallocation of financial liabilities. Solving a benevolent social planner's problem of maximizing aggregate real output subject to a given aggregate amount of debt and equity in the industry yields the following solution to the optimal debt and equity allocations under no distortions:

$$\widehat{D}_{si} = \frac{\left(Z_{si}^{\sigma-1}\right)}{\left(\sum_{i}^{M} Z_{si}^{\sigma-1}\right)} D_{s}, \tag{3.1.4}$$

$$\widehat{E}_{si} = \frac{\left(Z_{si}^{\sigma-1}\right)}{\left(\sum_{i}^{M} Z_{si}^{\sigma-1}\right)} E_{s}, \tag{3.1.5}$$

where $\hat{D}_{Si}$ and $\hat{E}_{Si}$ stand for the aggregate debt and equity holdings, respectively, allocated to industry $s$. Given the definitions of efficient debt and equity holdings, the efficient real output at the firm level is given by the following equation:

$$\widehat{F}_{si} = Z_{si}\left[\alpha_{s}\widehat{D}_{si}^{\frac{\gamma_{s}-1}{\gamma_{s}}} + \left(1-\alpha_{s}\right)\widehat{E}_{si}^{\frac{\gamma_{s}-1}{\gamma_{s}}}\right]^{\frac{\gamma_{s}}{(\gamma_{s}-1)}} \tag{3.1.6}$$

Aggregate gains from resolving financial misallocation can then be derived as the ratio between the aggregate real output in the undistorted economy and the one observed in the data.

The analysis decomposed the contribution of the level of financial liabilities versus the composition (that is, the debt-to-equity ratio) in explaining the overall degree of financial misallocation by comparing the baseline estimates—derived under sector-specific elasticities of substitution between debt and equity—to those obtained under an alternative scenario that assumed perfect substitutability. Under perfect substitution, the composition of liabilities is irrelevant for real output; thus, the productivity gains in this scenario reflect only the benefits from reallocating the total amount of finance across firms. The difference between the gains under the baseline and under perfect substitution quantifies the contribution of debt-to-equity ratios in accounting for the aggregate effects of financial misallocation.

The analysis covered 2010 to 2022, using data from Orbis for Australia, China, India, Indonesia, Japan, Korea, Malaysia, the Philippines, and Vietnam.

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CHAPTER 4

# Adapting to an Aging Asia: Challenges, Opportunities, and Policy Responses

Hoda Selim, Michael Green, and Mike Xin Li[1]

## INTRODUCTION

Population aging is advancing faster in Asia and the Pacific than in any other region of the world, propelled by longer lives and lower fertility. The region's population aged 65 and older is expected to nearly double by 2050 (to 880 million from 450 million in 2024) and account for one in five people. However, demographic patterns vary widely across the region, with some economies with younger population still expected to enjoy future growth tailwinds, while others will face drags from shrinking working-age populations. This diversity creates an opportunity to draw lessons from other regions and from the few Asian economies that have advanced the most in the demographic transition.

Demographics are shaping the region's economic outlook. Population aging affects labor supply, productivity, and long-term growth, while also constraining fiscal policy. However, policy choices can mitigate the economic consequences.

The literature has initially emphasized that aging reduces the working-age population and depresses labor supply and output, everything else equal (Gagnon, Johannsen, and López-Salido 2021; Kotschy and Bloom 2023). As a larger share of the population reaches retirement age and low fertility rates reduce the number of young workers, an aging workforce supplies less labor. A declining working-age population can lead to labor shortages and lower growth. The effects become more acute as the demographic transition accelerates. Other studies have argued that deteriorating physical and cognitive capacities reduce productivity as workers age (Truxillo, Cadiz, and Hammer 2015).

However, aging is not inherently a negative development. Aging partly reflects progress in public safety, public health, and material living standards. IMF (2025) finds that in many countries, people are living longer and remaining healthier as they age, which is associated with longer and more productive lives. Improvements in health are linked to advances in health care, technology, and education, which

[1] We thank Thomas Helbling and other colleagues at the IMF for comments on an earlier draft.

have strengthened the productivity profile of older workers and improved physical and cognitive health at older ages. Longer periods of good health enable workers to remain economically active for longer and moderate the impact of demographic aging on labor supply and economic growth. Shifts in production toward innovative goods and services tailored to older populations can stimulate new sources of demand and growth.

Aging societies can also adapt by innovating. Acemoglu and Restrepo (2017) argue that the drag of falling labor supply on productivity can be offset at least partly by the endogenous response of technology, through which countries undergoing pronounced demographic change adopt labor-saving technologies more quickly. In Asia, this process is already playing out in Japan and Korea. Higher real wages, the result of the shrinking of the working-age population relative to demand, give firms the incentive to invest in technologies that make labor more productive. The increase in productivity can partially of even more than offset the negative impact on economic growth from an aging population.

Beyond the labor market, population aging can affect consumption significantly, including through higher savings rates and shifts in consumption patterns across goods and services. Ando, Hidano, and Son (2025) find that real consumption is projected to decline in aging populations even as real income continues to increase. Moreover, consumption is likely to experience some shifts with spending on education and transportation shrinking faster than spending on health care. Aging also increases demand for health care services and thus raises demand for services relative to goods. Understanding these trends helps policymakers anticipate future expenditure pressures and develop long-term budget strategies.

This chapter provides an overview of the demographic transition in Asia and Pacific, which is unfolding in a global context but also varies substantially within the region. It also explores the extent to which population aging and rising old-age dependency ratios are creating pressures in labor markets. The chapter then examines the scope for policies to increase labor force participation among women and older workers as well as the role of healthy aging in reducing the adverse effects of aging on labor markets.

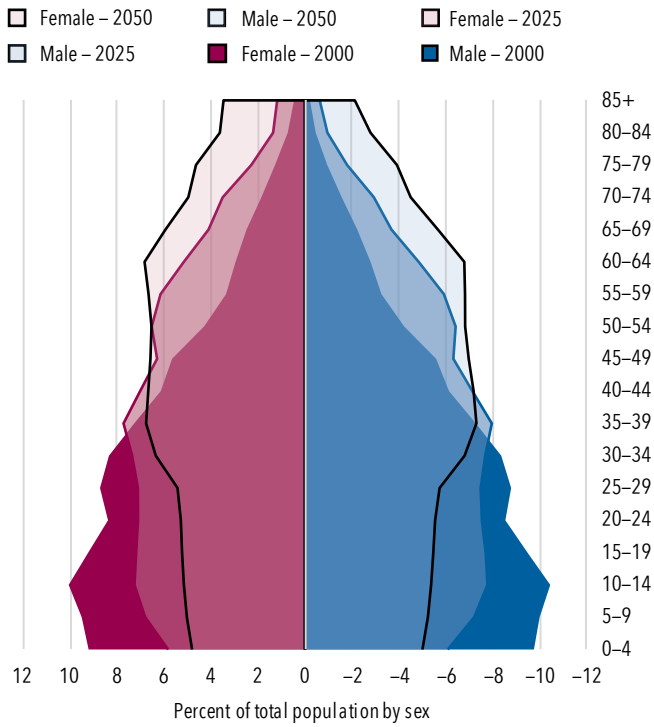
## THE PACE AND PATTERN OF AGING IN ASIA

Asia's population pyramids show that the region has aged between 2000 and 2025 amid rapid economic development and will age even more quickly in 2025–50 according to the United Nations' population projections (Figure 4.1). The region's working-age population (WAP)—defined as people aged 15–64—has increased on average by about 1.2 percent annually between 2000 and 2024. It is expected to continue to increase in 2026, accounting for over two-thirds of the population. Meanwhile, the share of individuals aged 65 or older has also expanded, to about 11 percent in 2025 from just over 6 percent in 2000.

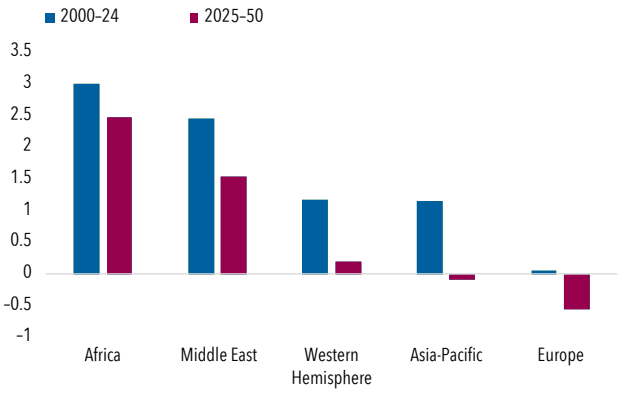
Going forward, the population structure of the Asia and Pacific region is projected to be significantly older by 2050. The WAP is expected to decline by 0.1 percent per year until then, making Asia the region that will experience the

## Figure 4.1. Demographics in the Asia-Pacific Region, 2000–50

**1. Population Pyramids, 2000–50**

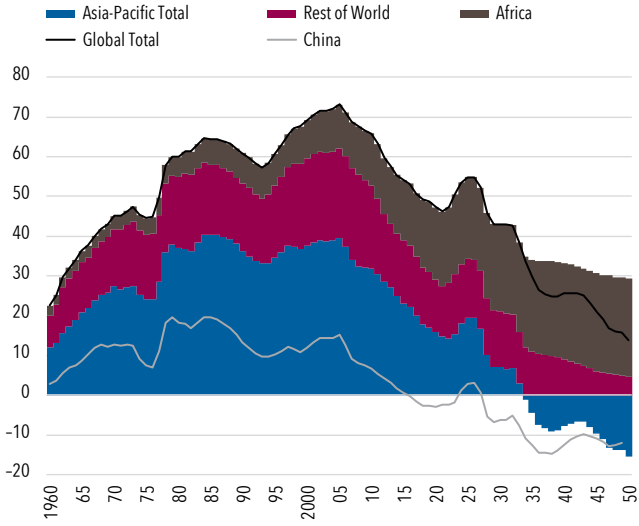



**2. Average Annual Change in Working-Age Share of**
*(Percentage)*

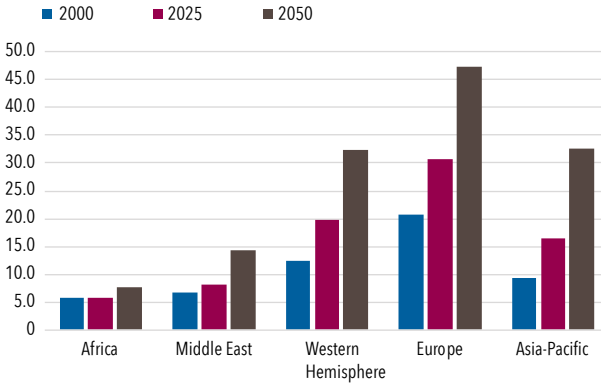


## Figure 4.1. Demographics in the Asia-Pacific Region, 2000–50 (Continued)

**3. Annual change in Working Change Population by region, 1960–2050**
*(in millions)*




**4. Old-Age Dependency Ratios, 2025 and 2050**




Sources: UN Population Prospects and IMF Staff calculations.

sharpest decline compared with the previous 25 years. Within the region, the decline in WAP is expected to be driven mainly by China, which is expected to account for about three-fourths of the projected fall. Between 2025 and 2050, the expected decline in China's WAP—estimated at 240 million—would be equivalent to a loss of 8.6 percent of Asia-Pacific's current working-age population. Meanwhile, the old-age dependency ratio (the population aged 65+ as a percentage of the working-age population) is expected to double by 2050, representing

the largest increase of all regions. By this metric, the Asia and Pacific region will be the second oldest region globally.

The region's rapid aging reflects both declining fertility rates and rising life expectancy. Fertility rates fell from six children per woman in 1965 to the replacement rate of 2.1 in 2013 and to 1.6 in 2025. Life expectancy rose from 49 years in 1965 to 74 in 2025, an increase of more than 50 percent across two generations.

A Shapley decomposition estimated the relative contributions of changes in fertility, life expectancy and demographic momentum to population aging in the region, as measured by the old-age population share. The demographic momentum is the effect of the population's existing age structure on future aging dynamics, independent of changes in fertility or life expectancy. In this approach, each factor's contribution reflects its direct effect and an equal share of the joint effect resulting from the interaction between fertility and life expectancy and momentum, ensuring that the contributions sum exactly to the total change in the old-age population share.

The estimates show that declining fertility during since the sixties has driven population aging in Asia, accounting for nearly two-thirds (63 percent) of the increase in the old-age population share. Rising life expectancy explains most of the remaining increase (34 percent). Looking ahead, however, the vast majority in the change in the old-age population share over 2025—50 is projected to come from past, rather than contemporaneous or future changes in fertility and life expectancy (Figure 4.2, panel 2). This 'Momentum' part represents the aging associated by generations born in the past that would take place if fertility and mortality remained constant at 2025 levels. In particular, the existing age structure will continue to exert a strong mechanical effect on the population's evolution, as cohorts move through the age distribution. As a result, even in the absence of further declines in fertility or gains in life expectancy, aging pressures would continue to intensify. Momentum is projected to account for 74 percent of population aging in Asia between 2025 and 2050. In terms of marginal changes, as life expectancy is expected to rise by much more than fertility is expected to fall, gains in the former are projected to become the main marginal contributor to aging, accounting for 23 percent of the total expected aging during 2025–50.

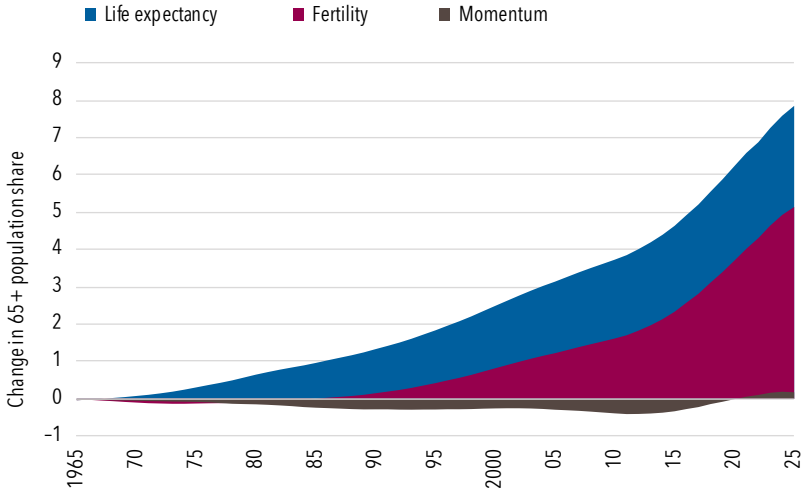
In absolute terms, the region remains younger than other regions. In 2025, the share of the population age 65 and older (16 percent) remained lower than in Europe (30 percent) and the Western Hemisphere (20 percent). The working-age population, estimated at about 68 percent of the total population, remains larger than in any other region but is expected to peak in 2027. Asia and Pacific is not the first region to reach that point: the working-age population peaked in Europe in 2009 and in the Western Hemisphere in 2022, but Africa and Middle East and Central Asia are not expected to peak until the 2070s.

Asia and Pacific stands out from the other regions in two respects. First, the pace of aging is exceptionally fast (Figure 4.1, panel 3). The share of the population age 65 and older doubled from 7 percent to 14 percent in about 20 years,

## Figure 4.2. Past and Future Drivers of Aging in the Asia Pacific Region[1]

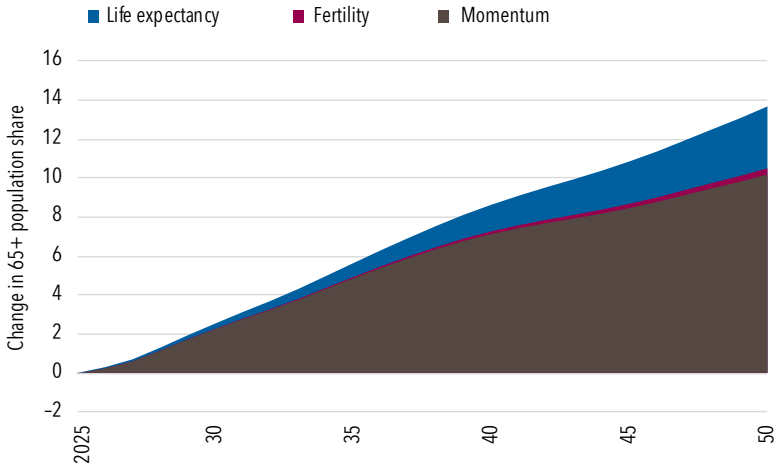
**1. Marginal Drivers of Change in Old-Age Share, 1965–2025**

*(Cumulative effects in percentage points)*




**2. Marginal Drivers of Projected Old-Age Share 2025–50**

*(Cumulative effects in percentage points)*




Sources: UN Population Projections and IMF Staff calculations.

[1]This approach uses historical fertility and mortality rates and only compares the two factors and momentum, leaving out net migration (Shorrocks, 2013). It assigns to each factor its average marginal contribution in each scenario, allocating the interaction equally between the two factors. This entails running simulations of three separate counterfactual scenarios for the years 1965–2025: one in which fertility rates remained constant at their 1965 levels ($S_{CF}$); one in which mortality rates remained constant at their 1965 levels ($S_{CM}$), and one where both remained constant (SB). The pure fertility effect is calculated as $S_{CM} - S_B$. The pure mortality effect is calculated as $S_{CF} - S_B$. If we let actually observed old-age population share = $S_A$, then the total change is $S_A - S_B$. Therefore, the interaction effect is $(S_A - S_B) - (S_{CM} - S_B) - (S_{CF} - S_B)$. This interaction effect is split evenly between fertility and mortality. Momentum is calculated as, e.g., $S_B(2025) - S_A(1965)$.

less than half the time that Europe or the Western Hemisphere took to double. As discussed earlier, aging is likely to continue and could even accelerate in the coming decade or so.

Second, aging within Asia also varies substantially because countries age at different rates. To examine this heterogeneity, an analysis grouped countries according to the effects of their expected demographic developments on labor supply. The demographic dividend—the potential for young economies to harness their growing workforces to boost economic growth—was crucial to the Asian economic development experience in the 1990's, during which many Asian emerging markets experienced rising working-age population shares between the 1960s and 1990s. Conversely, countries that underwent demographic transition earlier, such as Japan, have experienced slower output growth because of a shrinking working-age population since the mid-1990s.

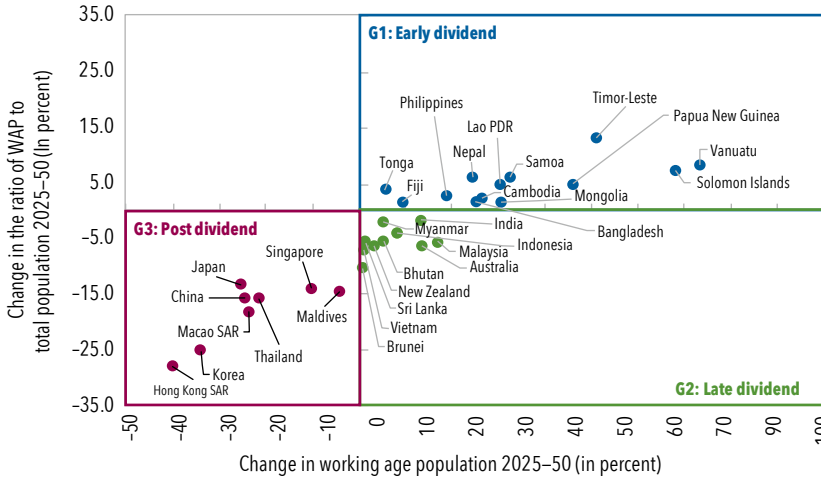
Figure 4.3 shows the three groups of countries in the context of the demographic dividend, based on the projected evolution of two variables between 2025 and 2050: change in the working-age share of the population versus change in the total working-age population.

- The *early dividend group* includes countries where the working-age population is expected to grow faster than the total population until 2050, suggesting a favorable impact of demographics on aggregate GDP and consumption. Of the 17 Asia-Pacific economies in this group, 15 are PICs and LICs, while two are emerging market and developing economies (EMDEs) (Mongolia and the Philippines).
- In the *late dividend group,* the WAP continues to grow but at slower rate than the general population. This will lead to a decline in the share of the WAP and, thus, a rising dependency ratio. Most countries in this group are EMDEs, along with two advanced economies (AEs) (Australia and New Zealand).
- *In the 'Post-Dividend'* group, the working-age population is declining, both in absolute terms and as a share of the overall population. This group includes advanced economies—some of which have seen their populations falling already, such as Japan and Korea—and EMDEs, such as China and Thailand. This group is projected to lose about a quarter of their WAP by 2050, driven by declines in China, Japan, and Korea, which is driving the expected decline in Asia's WAP until 2050 (Figure 4.4). The group will account for more than a third of the decline in the global WAP over the same period.

As a caveat, it should be noted that the categories of demographic groups are based on long-term population forecasts that are highly uncertain and could prove inaccurate. For example, United Nations forecasts of Asia's total fertility rate have projected continued declines since at least 1994, yet successive forecast vintages have proved overoptimistic and required further downward revision as fertility rates have declined faster than expected (Figure 4.4). Current projections suggest broadly stable fertility rates, but authorities should prepare for even faster-than-expected population aging if fertility continues to decline.

## Figure 4.3. Asia: Aging across the Region

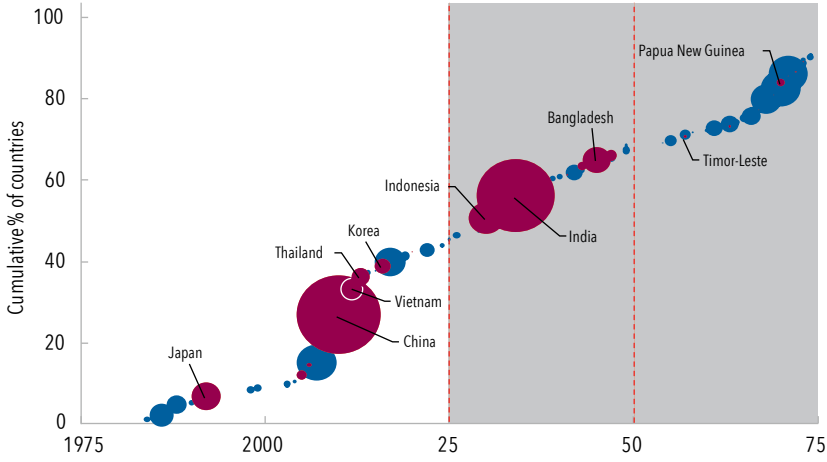
### 1. Demographic Shifts across Asia and Pacific, 2025–50




Sources: UN Population Prospects database 2024; and IMF staff calculations.
Note: Projections are based on the United Nations' World Population Prospects 2024 (which uses data publicly available up to the end of 2023). While using the UN's population projections ensures cross-country consistency, the UN's projections may differ from those of the authorities because of different underlying assumptions, notably on migration. For instance, in Hong Kong SAR, the authorities project much larger immigration than implied by the UN Projections

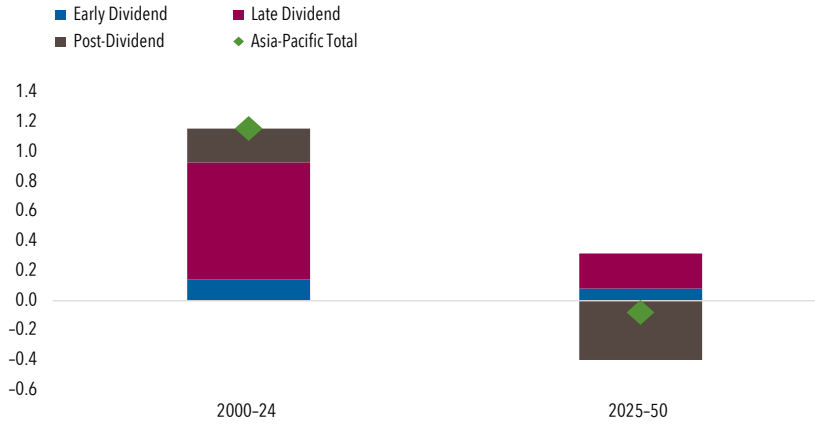
### 2. Countries with Declining Working-Age Populations

*(Percent cumulative share; bubble size proportional to population)*




Sources: United Nations Population Prospects database; and IMF staff calculations.
Note: Orange bubbles indicate Asia-Pacific countries; blue bubbles indicate other countries

**Figure 4.4. Contribution to the decline in WAP in APD by dividend group**




Sources: UN Population Projections and IMF Staff calculations

## WHAT AGING MEANS FOR ASIA AND PACIFIC IN THE NEXT 25 YEARS

The rapid demographic transition in Asia and Pacific will affect the region's economic prospects, including through its impact on labor supply, productivity, and potential growth. For the region as a whole, the decline in the working-age population is projected to begin around 2027, leading to a decline in its share of the total population from 68 percent in 2024 to about 64 percent by 2050 (Figure 4.6).

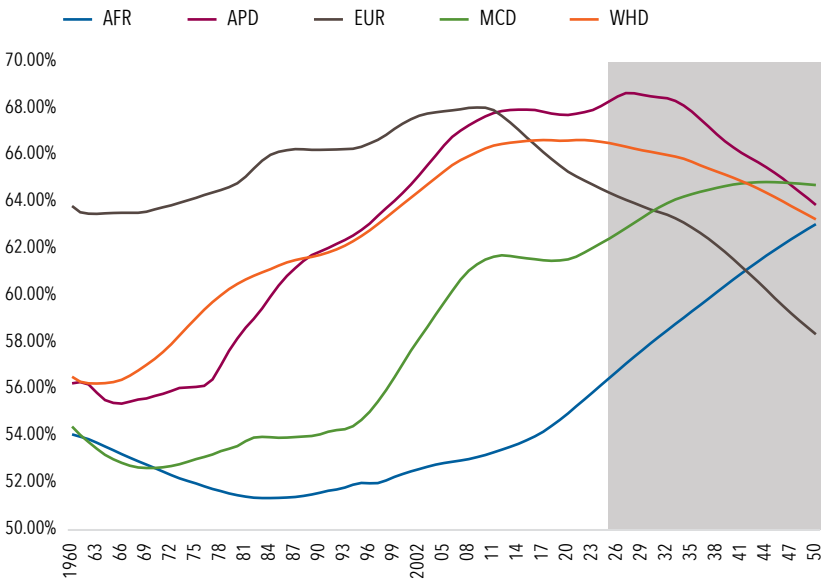
Population aging will undoubtedly exert some macroeconomic pressures. Nevertheless, it should be a manageable economic challenge if countries take steps to adapt to their aging populations. The reasons are the following.

- In 2050, Asia-Pacific's working-age population share is projected to remain the second highest globally, after the Middle East (Figure 4.5). Moreover, as discussed below, labor force participation rates in the region are currently relatively high (at around 63 percent), reflecting the increasing participation of women and older workers in recent decades. As a result, it seems likely that labor force participation will increase further, including in some post-dividend countries, which can offset the labor supply effects from a declining WAP.
- Healthy aging can partly mitigate the adverse economic effects of population aging in Asia. Gruss and others (2025) provide evidence that many countries in Asia—including China, India, Japan, Korea, and Thailand—have experienced healthy aging with positive spillovers on long-term growth and labor supply because of the increase in labor force participation by

## Figure 4.5. Asia and Pacific: Labor Market Indicators

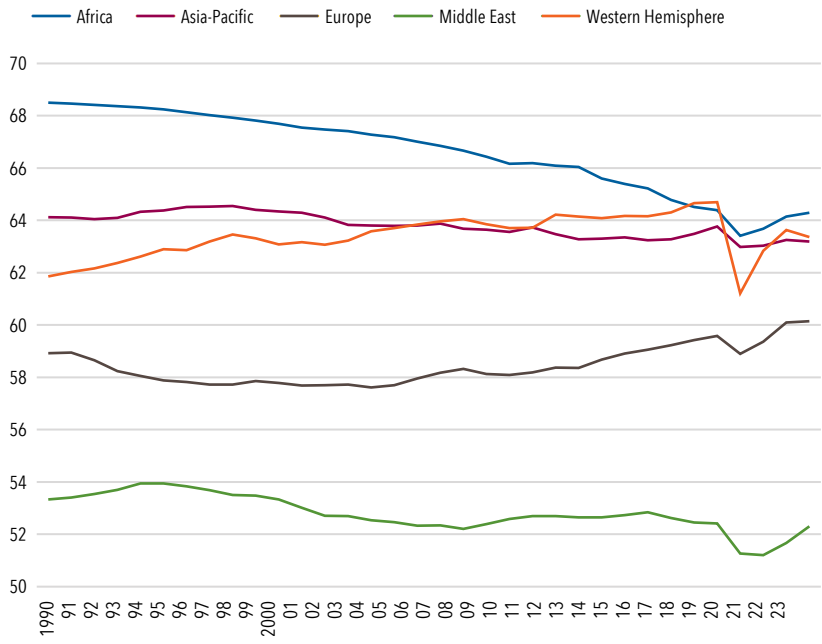
**1. Working-Age Share of Population by Region**

*(Population ages 15–64 as a share of total population)*




Sources: United Nations projections database; and IMF staff calculations.

**2. Labor Force Participation Rate by Region**




Sources: International Labour Organization database; and IMF staff calculations.

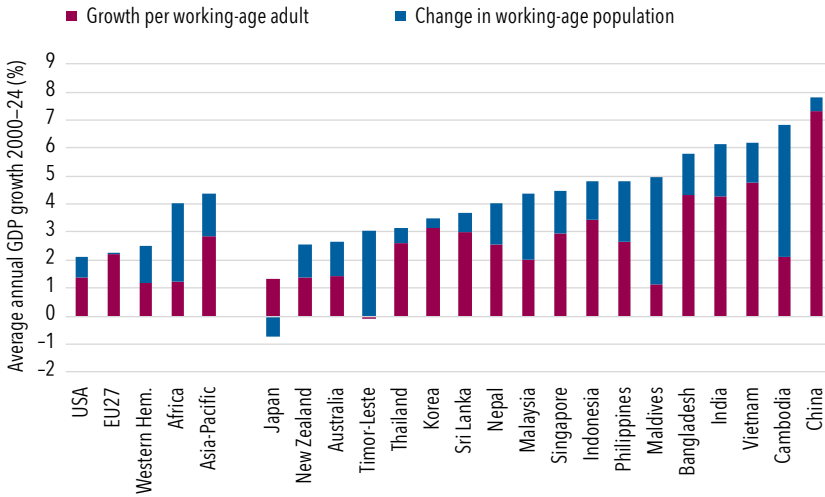
elderly workers. To illustrate their point, they show that, in Korea, the physical abilities of individuals aged 70 years in 2022 were approximately equivalent to those of a 60-year-old in 2006. These findings suggest that health improvements have been a key driver of rising longevity. Relatedly, to the extent that healthy aging enables individuals to remain in the workforce beyond the traditional retirement age of 65, and as lower fertility reduces the number of young dependents, the actual increase in the overall dependency ratio could be substantially smaller than suggested by the projected standard old-age dependency ratios defined with the traditional retirement-age threshold. Moreover, as the share of older individuals rises, demand for health care services, long-term care, and related sectors increases significantly. Higher quality and more accessible care can further improve health outcomes for older adults, helping them become even more productive as they age.

- Recent advances in artificial intelligence (AI) hold the promise for further automation to deliver medium-term productivity gains. This is especially true for those advanced economies in Asia that are among the global frontrunners in both AI innovation and adoption. As noted in Chapter 8, a large body of existing research has focused on the potential boost to productivity from AI adoption (see also Felten and others 2021, 2023, for example). AI allows for the possibility of rising growth and wage incomes for most workers (Cazzaniga and others 2024), depending on whether AI is complementary to human labor in certain occupations. In that case, the increases in economic activity and labor demand spurred by AI would offset the negative consequences of labor displacement. Chang and others (2025) study potential AI adoption scenarios in Korea and estimate the productivity gains to be between 1.1 and 3.2 percent in the first 10 years after AI adoption depending on whether AI increases demand for high-complementarity occupations.
- Examining whether aging could accelerate the adoption of new technologies, Lee and others (2020) provide evidence that labor productivity increases when older workers are employed in industries with a large share of information and communication technology in the capital stock. Most research has found that the positive effects of aging on labor productivity in information and communication technology industries are limited to high-skilled older workers (Ilmakunnas and Miyakoshi 2013), but Lee and others (2020) find positive effects on labor productivity also for less-educated older workers in Japan. As the country with the most advanced demographic transition, Japan has long illustrated how aging-induced labor shortages can encourage investment in labor-saving technology. In 1995, when Japan's working-age population began to shrink, about 700,000 industrial robots were in use worldwide, of which 500,000 were in Japan (Schneider, Hong, and Van Le 2018).

Given the heterogeneity and potentially profound macroeconomic implications of the demographic transitions in the region, it will be critical to consider alternative metrics to assess macroeconomic performance, in addition to the traditional metrics of growth of aggregate GDP or GDP per capita. As aging accelerates across the region, the demographic contribution to headline GDP observed in recent decades will diminish rapidly. GDP growth will remain an important measure, but Fernández-Villaverde, Ventura, and Yao (2025) argue that accounting for demographic change can substantially affect assessments of relative economic performance. They propose using growth in GDP per working-age adult, which corrects for population aging and is more representative for assessing the economic performance of aging countries than GDP per capita. The authors illustrate this point using Japan and the United States as an example. Japan's GDP per working-age adult grew by 1.4 percent per year between 1991 and 2019, while GDP grew by 0.83 percent and real GDP per capita by 0.76 percent. The difference reflects Japan's shrinking working-age population. Using GDP per working-age adult reduces the difference in economic performance between the United States and Japan to only 0.26 percentage point between 1991 and 2019, much lower than the gaps using either real GDP growth (1.75 percentage points) or real GDP per capita growth (0.87 percentage point).

This approach was extended to an international sample by decomposing GDP growth into the contributions of growth in the working-age population and that of output per working-age adult between 2000 and 2024. Figure 4.6 shows that

**Figure 4.6. Contributions to Average Annual GDP Growth 2000–24**




Sources: United Nations Population Prospects database 2024; and IMF staff calculations.
Note: The methodology entails decomposing logarithmic growth in real GDP (purchasing power parity in US dollars) into growth in total working-age population and growth per working-age adult. Growth was calculated as the change in logs of each variable.

aggregate growth in Asia and Pacific was similar to growth in Africa in that period, but the decomposition reveals how demographic divergences mask economic disparities. In Asia and Pacific, two-thirds of aggregate growth reflected growth per working-age adult—the second highest share after Europe. By contrast, the pattern in African economies was reversed: more than two-thirds of aggregate growth reflected the rise in the working-age population. The contribution of GDP growth per working-age adult was especially large in post-dividend countries such as China, Japan, and Korea and much smaller in early dividend countries such as Cambodia and the Philippines.

GDP per working-age adult provides a moderately different assessment of economic performance for several Asian countries than GDP per capita growth, with Japan providing the clearest example. However, as working-age population shares begin to decline substantially, the divergence between per capita and per working-age adult measures is expected to widen substantially. The nascent divergence would be the greatest in the post-dividend group, and the prospect of it increasing as aging accelerates underscores the importance of selecting appropriate metrics to assess the macroeconomic performance of aging economies.

## LABOR MARKET POLICY IMPLICATIONS

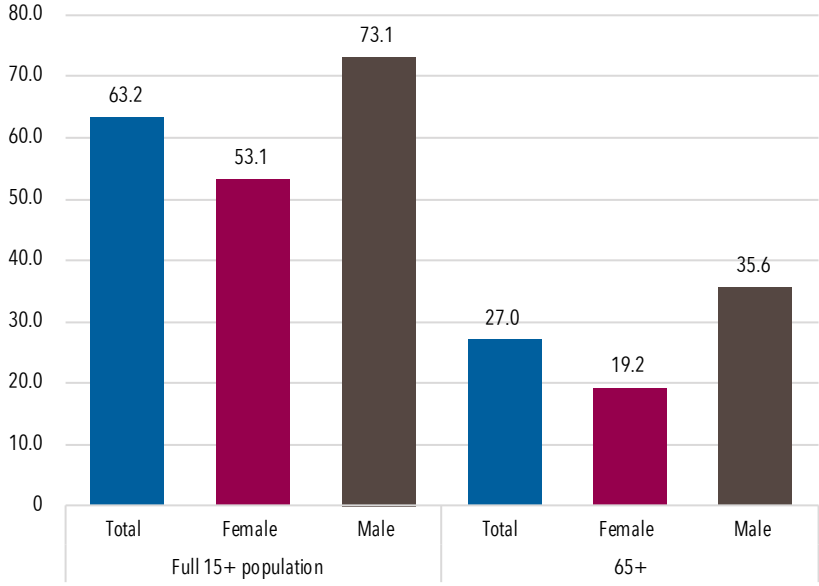
As noted above, population aging will affect the labor supply. Besides the impact on the number of available people, the impact will also depend on their participation in the labor force or labor market. Increases in the latter can, to some extent, offset a reduction in the former. Boosting the participation and employment rates of female and elderly workers or increasing the number of foreign workers can help mitigate the drag from aging on labor supply and, thus productivity and growth in the medium term (Colacelli and Corugedo 2018; IMF 2023; Xu and Chahande 2023).

### Boosting Women's Labor Force Participation

Women's labor force participation has improved across all Asian countries. The largest gains are in the post-dividend group, where participation rates reached 55 to 60 percent of the 15+ population in 2023 in China, Singapore, and Thailand, leaving room for more increases. Nevertheless, at about 53 percent, the average labor force participation rate for females is still substantially lower than that for males (73 percent) (Figure 4.7). Some countries, like Japan, have had successfully raised female participation rates to higher levels than in most G7 countries between 2012 and 2022 (Xu 2023), supported by the widespread expansion of childcare facilities and increased availability of part-time and flexible work arrangements. Other countries, including Korea, have experienced slower progress, with the participation gender gap estimated at close to 20 percentage points, which is among the largest in the OECD advanced economies (55 percent for females and 78 percent for males) (IMF 2023).

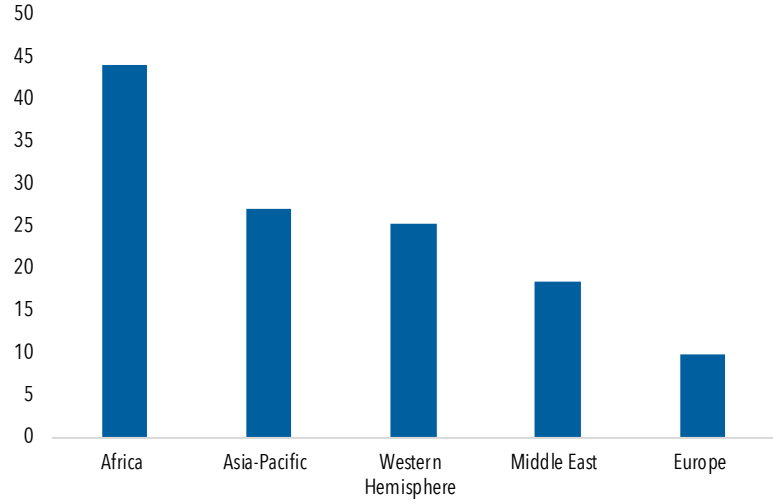
## Figure 4.7. Asia and Pacific: Labor Market Indicators

**1. Labor Force Participation Rate: Age and Gender, 2022**




Sources: International Labour Organization; and IMF staff calculations.

**2. Labor Force Participation Rate: Age 65 and Older by Region, 2022**




Sources: International Labour Organization; and IMF staff calculations.

Improving women's labor participation and employment will boost labor supply and lift growth in any economy, but the effect is especially important in late-stage aging societies. Many economies in the region have substantial scope to increase participation. Across the region, if the women's employment-to-population ratio as of 2023 (35 percent) rose to the level for men (43 percent), the workforce would increase by 112 million workers. Agarwal and others (2023) estimate that even a small increase in women's labor force participation (from 27.7 to 32 percent) could fully offset the decline in labor supply in South Asia. In Korea, increasing women's participation from 57 percent to the OECD average of 67 percent would add nearly 2.4 million workers to the labor supply.

It should also be noted that higher women's labor force participation can also improve fiscal sustainability in aging economies through several channels. First, more women in the workforce means higher employment and therefore taxable labor incomes, thereby broadening the tax base and raising revenue, which can help finance additional spending including on health care and pensions. Second, increased women's labor force participation raises social security contributions and increases the number of pension contributors to beneficiaries, thus easing pressures on the financial sustainability of pension schemes and more broadly reducing old-age dependency pressures.

Demographers expect little change in fertility rates until 2050, but uncertainty surrounding those forecasts is substantial. Research shows that both economic and noneconomic factors contribute to declining fertility rates. In high-income countries, declining fertility has been linked to economic factors such as real wage growth, higher housing costs, and increased women's labor force participation. More broadly, improved education has increased incomes (especially in emerging market and developing economies), which encouraged higher female labor force participation and is associated with lower fertility. However, Kearney and Levine (2025) show that economic factors alone do not explain declining fertility rates, suggesting that cultural change and shifting preferences away from parenthood also contribute.

## Increasing Labor Force Participation of Older Workers

Population aging has increased the age of the region's labor force. The median age is about 39 years but rises in the post-dividend group to about 43.5 years—the highest in the world. Labor force participation among older workers (age 65 and older) is estimated at about 27 percent, higher than in other regions except for Africa (Figure 4.7). In Korea, the share of older workers in the labor force nearly doubled since 2000 to reach about 24 percent in 2025, contributing to exceptionally high participation rates of 72 percent (Lagerborg and Selim 2025). Evidence also shows that older workers in Korea remain in the labor force into their late seventies (Lagerborg and Selim 2025). Participation rates have increased among both older men and women in Asia, but women's overall participation remains lower, despite rapid gains.

Healthy aging enables workers to remain productive for longer, which can increase older workers' participation in workforce and thus increase the labor supply. Gruss and others (2025) show that healthy aging increased the labor supply among older people in Korea by about 1.9 percentage points per year between 2006 and 2020.

## Raising Participation Rates of Women and Older Workers

Reforms are essential to raising labor market participation by female and elderly workers, both on the supply and demand side. Reforms in the latter are necessary to ensure that higher participation sustains employment, wages, and productivity.

Economic reforms in three areas could increase labor force participation among women and older workers in the region. Skills training can help these workers gain access to more productive jobs. Women often accept low-productivity jobs with less-favorable working conditions because they are closer to home and offer greater flexibility in terms of working hours, which helps them balance paid work with housework and childcare. In some emerging market and developing economies and low-income countries in Asia and Pacific, women typically work in agriculture, services, the informal sector, or in vulnerable employment (ILO 2018 and Figure 4.8)—jobs that often pay low wages, lack social protection, and in some cases involve unsafe working conditions (ADB and ILO 2023).[2] Women also remain underrepresented in managerial roles, including in advanced Asian economies such as Japan and Korea, which lag other G20 economies (Asao, Khera, and Vasishth 2024). In both countries, women also tend to work in jobs that pay lower wages and offer fewer opportunities for skills development and career advancement.

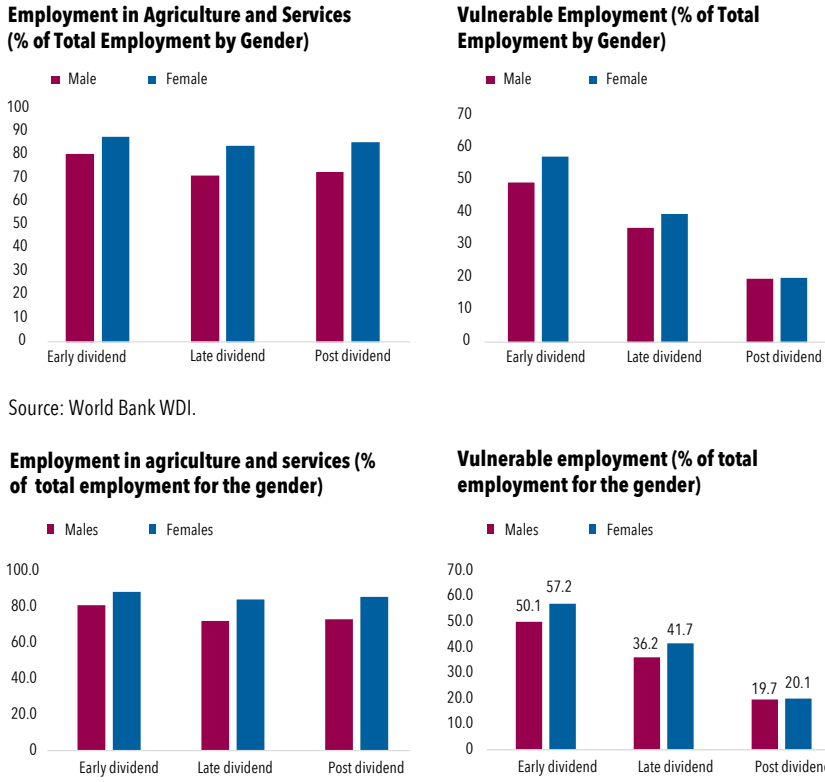
Older workers also tend to be employed in low-productivity jobs in agriculture or the informal sector. In some low-income countries and emerging market and developing economies in the region, older workers often have lower levels of education and skills (ESCAP 2023; ADB 2024). In other cases, older workers move from formal employment into informal or nonregular jobs as they retire, especially in Bangladesh, Indonesia, and Korea (ADB 2024).

To improve quality employment prospects for both women and older workers, governments could expand access to targeted skills training, provide job-matching services and employment counseling, and encourage lifelong learning. Reskilling and upskilling older workers, especially in advanced AI skills, could improve their employability relative to younger workers.

Reforming labor market institutions in three priority areas would encourage women and older workers to remain in the labor market. Eliminating regulatory barriers that discourage either group from entering or remaining in the labor

[2] The share of women employed in informal jobs in Asia is more than 30 percentage points higher than the global share, and in South Asia, the share rises to more than 85 percent (ADB and ILO 2023).

## Figure 4.8. Quality of Jobs for Women




Source: World Bank WDI.




Source: World Bank World Development Indicators.

market is one area. For example, social security thresholds can restrain second-income earners from working more (especially women) because additional work can lead to a disproportionate increase in household contributions. Laws prohibiting gender-based discrimination in hiring, dismissal, and promotion (including after childbirth), along with reforms that remove barriers to women starting and running businesses, could also create equal opportunities for career advancement and encourage labor force participation. Brussevich, Dabla-Norris, and Li (2021) estimate that reducing barriers—including barriers to hiring women in China to levels observed in advanced economies—can narrow the gender participation gap by 15 percentage points.

Less-stringent employment protection legislation, including more flexible dismissal regulations and lower severance costs, could enable hiring of more women and older workers. Agarwal and others (2023) find that easing employment protection legislation in South Asia increases women's share of employment both in absolute terms and relative to men. Such reforms could be especially

beneficial in countries where stringent legislation leads firms to hire informal or nonregular workers, who are more likely to be women and older workers. IMF (2023) finds that reducing severance payments for regular workers in Korea could increase labor force participation by 0.9 percentage point for women and 0.3 percentage point for men and reduce the overall unemployment rate by 0.5 percentage point. Expanding social protection for nonregular or informal workers could also reduce incentives for firms to hire those types of workers and improve job security.

More flexible work arrangements, supported by rapid digitalization, could also increase labor force participation among both groups. The issue remains especially relevant in advanced economies such as Japan and Korea, where work culture continues to involve long working hours, inflexible schedules, and limited use of telework, despite considerable past progress. Those conditions can lead women to choose part-time work or discourage labor force participation entirely.[3] Flexible work arrangements can also support faster returns to work among young women after childbirth and reduce setbacks to career progression. Rapidly expanding digitalization and remote work can help women overcome constraints related to distance and safety and support their entrepreneurship. Sodsriwiboon (2023) estimates that increasing digital use in South Asia to the G20 median could raise women's labor force participation rate by about 3 percentage points in three years. Similarly, more flexible work arrangements and age-inclusive working conditions would allow older workers to work at a pace better suited to their needs and delay retirement.

Some reforms can target specific groups. In Japan, affordable childcare services advanced women's careers (Asao, Smirnov, and Xu 2024b), while stronger incentives to use paternity leave can reduce gender bias in hiring decisions and increase women's participation and enable their career advancement.

Policies that support healthy aging can increase older workers' productivity and narrow the participation gap with prime-age workers. Budgets could prioritize or create space for health promotion and prevention policies that reduce the incidence of chronic disease. Targeted policies could also expand affordable access to long-term care, especially in countries where costs prevent older people from remaining active (OECD 2024). Reducing incentives for early retirement by raising retirement ages to at least 65 years would help mitigate declines in labor supply and improve the fiscal sustainability of pension systems (Chapter 5).[4] In New Zealand, a combination of a low labor tax wedge, universal access to the New Zealand Superannuation without retirement requirements, and a flat-rate pension encouraged labor force participation among older workers.

[3] As of 2024, Japan and Korea rank at the top three OECD economies in time spent in paid work.

[4] Except for Australia, Hong Kong Special Administrative Region, Japan, Singapore, and a few other countries, the retirement age in most Asian countries is below 65.

### The Role of Foreign Workers

Encouraging migration is another option to partially offset the impact of aging on labor supply, notwithstanding political, institutional, and absorptive constraints in many Asia-Pacific economies. As discussed in Chapter 7, experience in North America and Western Europe shows that migration can mitigate population decline associated with aging by reducing the old-age dependency ratio.

## CONCLUSION

Asia and Pacific is undergoing a faster demographic transition than any other region, and aging is set to reshape labor markets, fiscal positions, and growth prospects between 2025 and 2050 and beyond. Within this regional picture, the scale and timing of the transition will differ markedly across economies. Some economies are still poised to enjoy a decade or two of growth tailwinds from a demographic dividend, while others will face drags from shrinking working-age populations. In general, falling fertility rates and further increases in life expectancy imply that countries must prepare for aging to continue or even accelerate. While these trends will be associated with some economic pressures, especially on the fiscal policy side, this chapter also argues that the outcomes need not be as dire as sometimes feared, especially if countries adapt to the needs of an aging society. Nevertheless, while reforms can help countries prepare for aging, they cannot entirely eliminate aging-related pressures, especially fiscal ones.

Demographic shifts will not only change economic performance, but also the metrics needed to assess relative economic performance and to provide granular policy recommendations. These shifts call for new metrics to assess macroeconomic performance, as current metrics relying on aggregate GDP or GDP per capita growth largely abstract from the large demographic shifts in the region and therefore may only provide an incomplete assessment of economic performance. Adjusting for (exogenous) demographic structures, notably the WAP, will become increasingly necessary, particularly when it comes to future estimates of potential growth. The use of the growth of GDP per working-age adult, which corrects for population growth, should be considered.

Expanding employment opportunities for women, older workers, and foreign workers can help mitigate the drag from aging on labor supplies. Investments in healthy aging, skills development, and flexible work arrangements, meanwhile, can support longer and more productive working lives. Subsequent chapters will examine the impact of these demographic trends on other policy areas.

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CHAPTER 5

# Fiscal Implications of Aging in the Asia and Pacific Region

Nick Carroll, Michael Green, Fei Han, Paulo Medas, and Weining Xin

## INTRODUCTION

People are living longer and healthier lives, but aging populations also pose challenges to public finances. An older population can lead to higher age-related public spending on pensions, health care, and long-term care, while shrinking working-age populations can weigh on potential growth and reduce the tax base. The region's advanced economies, notably Japan and Korea, and some emerging markets are already experiencing rapidly aging populations, and most countries in Asia and the Pacific will be in that situation by 2050.

Governments in the Asia and Pacific region face challenging trade-offs as they balance maintaining existing services and transfers with the higher costs from population aging. All countries face such pressures to some extent, but their likely fiscal impact differs considerably across countries and income levels. For some countries, the main challenge is managing growing fiscal pressures largely related to rising aging costs. Social spending in developed countries in the region has been rising rapidly—fueled by age-related spending—and is above 20 percent of GDP in most cases. Others need to address structural weaknesses in pensions and health systems to reduce old-age poverty and expand health care amid elevated levels of labor informality. Governments must develop and build public support for strategies that are fiscally sustainable and balance costs and benefits across generations.

Pension and health systems in Asia and the Pacific face growing challenges associated with population aging. Age-related public spending is expected to increase by 2050, adding to budgetary pressures, while large structural gaps in pension and health systems remain. Policy options can help manage these pressures, but choices and priorities vary across income groups. Adopting reforms at an early stage of demographic transition will reduce the fiscal costs of aging.

## PENSION AND PUBLIC HEALTH SYSTEMS

The goal of public pension and health systems is to provide income in old age and deliver affordable health care. The design of pension systems reflects

country-specific policy objectives and preferences, but achieving poverty alleviation and maintaining living standards during retirement (consumption smoothing) generally are standard objectives. The design of health systems also reflects different policy objectives (for example, promoting population health and economic growth) and administrative capacity. In both cases, country-specific preferences regarding the complementary roles of government and the private sector also shape the systems. In general, governments have a lead role in health and pension systems in Asia and the Pacific, but the private sector is also significant. For example, in Australia privately funded pensions are becoming the norm, while private health spending tends to be higher in countries in the Association of Southeast Asian Nations and in Pacific island countries. Even where the private sector's role in financing or management is significant, governments often provide support to low-income households and financing for services with large public benefits (for example, preventive health).

Advanced economies and East Asia are already experiencing high public pension and health spending, both of which tend to rise as incomes rise.[1] Among advanced economies, pension and health spending already accounts for one-third or more of government spending, as in Australia, Japan, Korea, and New Zealand. However, even among developing economies and especially those experiencing rapid aging, such as China and Thailand, pension and health spending pressures are starting to rise. Annual public pension spending is highest in East Asia at about 7 percent of GDP, well above the rest of the region (Figure 5.1). Public health spending is highest in Australasia at 8 percent of GDP, more than double the spending in the rest of the region.[2]

## Pensions

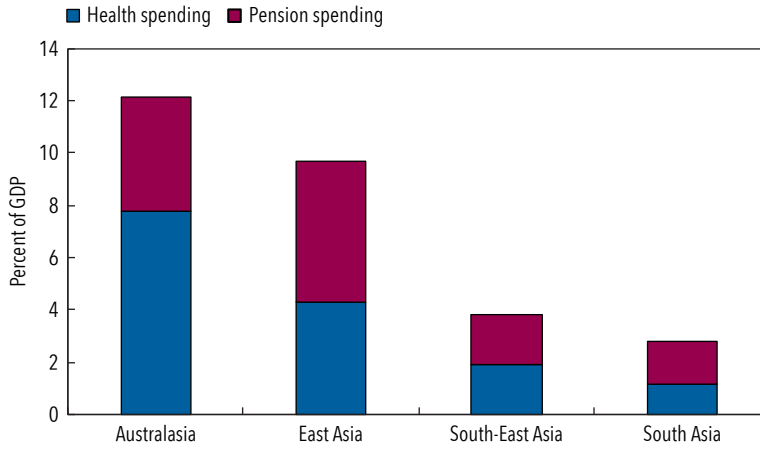
Public pension systems vary considerably across the region. Countries tend to have a mix of noncontributory pensions (mainly targeted), pay-as-you-go defined benefit systems, or funded defined contribution pensions, but the mix and importance of each tier vary across countries. Most economies in the region have mandatory, contributory, earnings-related pensions. A pay-as-you-go defined benefit system is common, integrated into the social security system and funded by contributions from employees and employers (for example, Japan, Korea, Mongolia, Thailand, and Vietnam).[3] In funded defined contribution schemes, pension contributions are paid into individual accounts or a fund, and the level of benefits depends on

[1] Public spending is defined as spending managed by central or local governments or social security agencies, whether it is financed by taxes, social security contributions, or other sources.

[2] Australasia includes Australia and New Zealand in this chapter.

[3] Defined benefit schemes pay pension benefits related to earnings and years of contributions. The schemes can be funded in advance or a pay-as-you-go system or a combination of both. Defined contribution schemes are usually based on individual private accounts that accumulate contributions during the working career. When the accounts accumulate capital in the form of financial assets, the schemes are classified as funded defined contribution. Schemes based on notional accumulated capital are known as notional defined contribution schemes.

**Figure 5.1. Public Spending on Pensions and Health Care**
*(Percent of GDP, 2022 or latest available)*




Sources: World Health Organization; World Bank; and authors' estimates based on several sources.
Note: Health spending data are for 2023 and drawn from the 2025 World Health Organization Global Health Expenditure Database.

investment returns, as in Malaysia, Singapore,[4] and Sri Lanka. In some countries, such as Australia, New Zealand, and Thailand, tax-financed (noncontributory) social pensions are prominent.[5] All those systems impose some burden on public budgets. Countries that rely heavily on social contributions might need public financing when contributions are lower than benefits. Countries that rely on private financing face fiscal risks and pressures when private pension schemes become unsustainable, or pension benefits are seen as inadequate (and thus require additional public funding).

The fiscal cost of pensions depends on demographics but also on key policy choices. Countries with higher old-age dependency ratios—the ratio of the elderly population (65 years and older) to the working-age population (ages 15 to 64)—tend to spend more than other countries (Figure 5.2). Pension spending in Australia, Japan, Korea, and New Zealand rose by 0.9 percentage point of GDP between 2000 and 2022, with especially strong growth in Japan and Korea. The wide variation in budget costs associated with public pensions in the region also reflects differences in key parameters, including coverage and retirement ages (Annex 5.1). Retirement ages have been rising in many countries in the region,

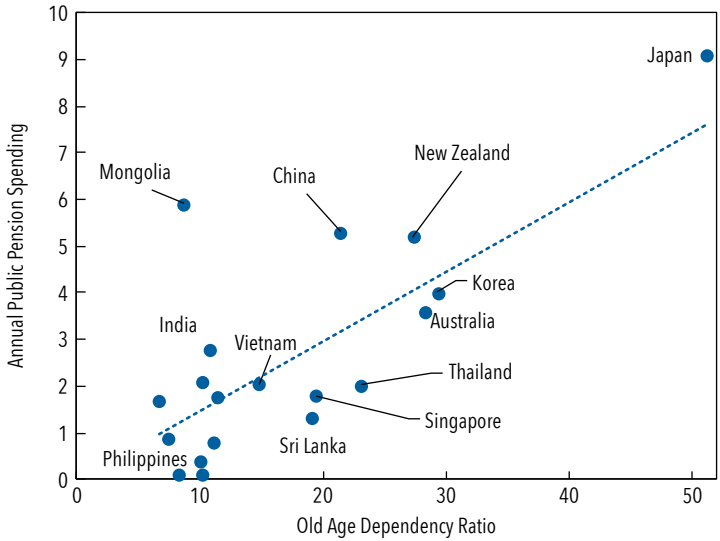
[4] Malaysia's defined contribution Employees Provident Fund largely covers private sector employees, but a defined contribution scheme for public sector employees also exists. For Singapore, a fixed, guaranteed interest rate is paid by default on all savings in the Central Provident Fund, but members can invest a portion of their savings for higher returns.

[5] Australia still spends considerable public funds on pensions as it transitions to a private-funded defined contribution system.

## Figure 5.2. Pensions in Asia and the Pacific

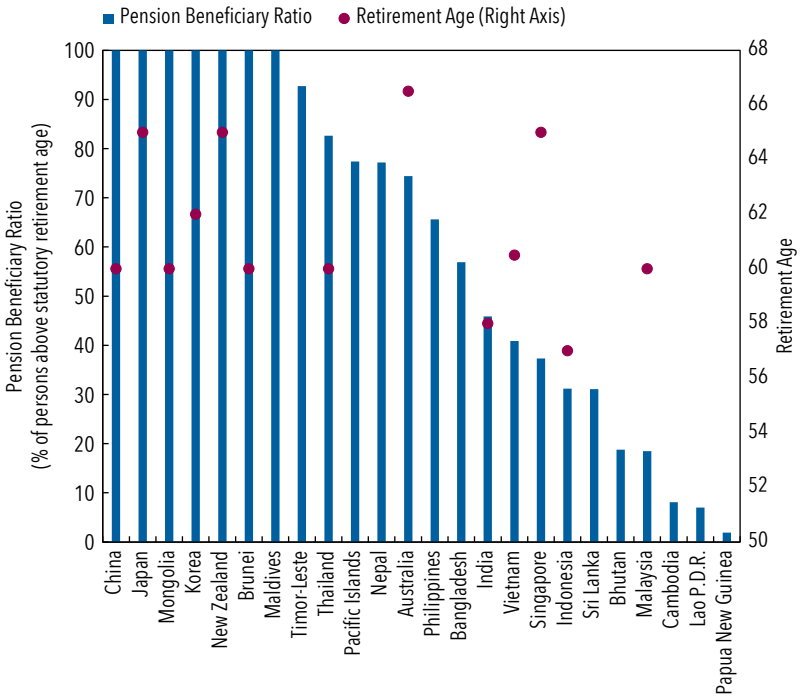
**1. Public Pensions and Demographics**

*(Percent of GDP and old-age dependency ratio)*




**2. Pension Coverage and Retirement Age**

*(Percent of population age 65 and older [lhs] and male retirement age [rhs])*




Sources: International Labor Organization; United Nations Population Prospects; World Bank; OECD; and authors' estimates based on several sources.

Note: Old-age dependency ratio is defined as population age 65 and older per 100 working-age population (ages 15–64). Pension coverage is defined as the share of age 65 and older population receiving at least one pension. Data are for 2022 or latest available. lhs = left-hand scale; rhs = right-hand scale.

but they often have not caught up with rising longevity or work capacity. In some countries, retirement ages remain well below the OECD average of 65 years. Replacement rates—the percentage of an individual's wage income that is replaced by pension income when they retire—tend to be similar to those in other regions but with a wide variety across systems within and across countries. Replacement rates in China (Box 5.2), the Philippines, and Vietnam are well above 50 percent, while other countries have rates of about 40 percent or less, such as India, Malaysia, and Sri Lanka. Countries with lower retirement ages and higher replacement rates in their publicly financed systems will face greater budgetary pressures as their populations age.[6]

Some countries collect low levels of social contributions that can lead to pension fund deficits (ADB 2024).[7] For example, some countries such as Cambodia, Indonesia, and Thailand have contribution rates of 10 percent or below as a share of wages, while only a few have rates of 20 percent or above, such as Malaysia, Singapore, Sri Lanka, and Vietnam.

Many developing economies have limited coverage of the elderly (how many receive pensions) and labor force (how many contribute to pensions, which is an indication of future coverage of retirees). This situation implies that a large share of the existing or future older-age population is at risk of not having access to pensions or only to limited noncontributory (minimum) pensions, which could lead to risks of significant political pressures for additional budgetary transfers. For example, in Indonesia, Malaysia, and Vietnam, less than half of people older than 65 receive a tax-financed pension or a contributory pension (Figure 5.2). In terms of the share of labor force that contributes to a mandatory contributory system, some countries in the region have high coverage levels (for example, Australia, Japan, the Philippines, and Singapore) but also some of the lowest (for example, Indonesia, Sri Lanka, and Vietnam), partly reflecting high levels of informality (OECD 2024a).

When designing pension systems, policymakers also need to address old-age poverty that remains elevated in the region. Old-age relative poverty rates are highest in China and Korea and lowest in Malaysia, Maldives, and New Zealand (Figure 5.3). Most countries in the region have levels above the OECD average of 14 percent (OECD 2025).[8] For some developing economies, severe, multidimensional (absolute) old-age poverty can also require different policy responses, including designing or strengthening social safety nets for the elderly. In the latest available household surveys in Papua New Guinea (in 2018) and Timor-Leste (in 2016), a majority of elderly people were in absolute poverty (UNDP and OPHI 2025). A broader response is needed in countries where population-wide poverty is even more prevalent. Policymakers also need to address other sources of

[6] A key issue with overgenerous replacement rates is that specific groups have generous pension benefits (which is common for many civil servants' schemes) and that gaps exist between rural and urban schemes.

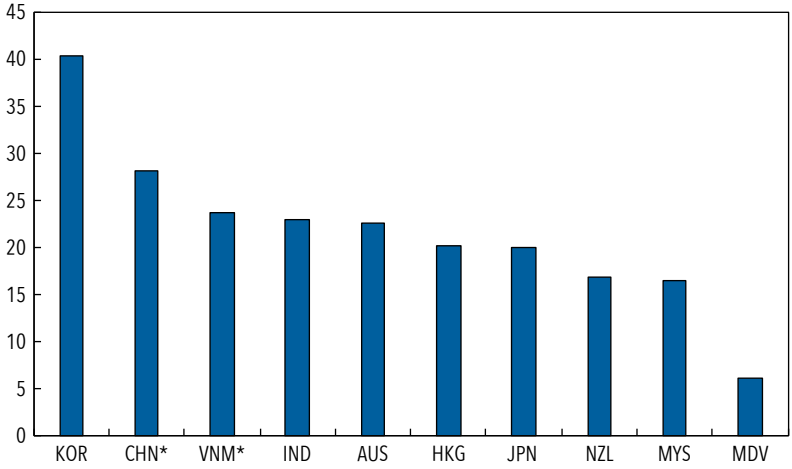
[7] For example, Mongolia has faced large pension spending, despite its relatively young population, reflecting wide coverage and generous benefits.

[8] The old-age poverty rate is a relative poverty concept based on disposable income and can overstate poverty in old age in countries with high home ownership among the older population.

## Figure 5.3. Poverty

**1. Relative Elderly Poverty Rates**

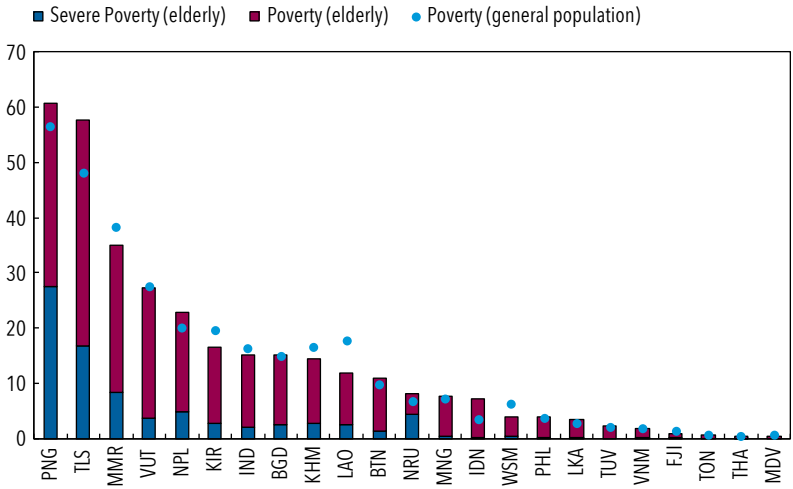
*(Percent of elderly with less than 50 percent of median household income, latest year)*




Sources: OECD; LIS; and national sources.
Note: The elderly are defined as persons aged 66 or older. Data labels use International Organization for Standardization country codes. Most recent data for China and Vietnam are for 2018 and 2013, respectively.

**2. Absolute Elderly Poverty Rates**

*(Percent of groups experiencing multidimensional poverty)*




Source: UNDP and OPHI 2025.
Note: The elderly are defined as persons aged 60 or older. Multidimensional poverty measures the population with deprivations in health, education, and living standards. Data are based on most recent country surveys during 2016–23. Data labels use International Organization for Standardization country codes.

inequalities to provide support to more vulnerable old-age populations, for example, responses targeted at rural populations or informal workers. Some countries, such as Korea, the Philippines, and Thailand, have low public pension benefits relative to GDP, which creates risks of old-age poverty and puts pressure on future public spending as countries consider plans to improve the adequacy of pension benefits. In other countries such as China, Japan, and Vietnam, high pension benefits for specific groups increase the level of pension spending and can lead to uneven benefit adequacy between groups, such as urban and rural populations (Box 5.2), and in some cases inequities.

## Health

The region's health systems vary considerably across countries, including in public participation and fiscal costs. In advanced economies, governments finance most health spending, while private financing (for example, out-of-pocket household spending or private health insurance) is more prevalent in developing economies. Health spending increased considerably in advanced economies and some emerging markets between 2000 and 2022 (Figure 5.4). The differences in rising health costs can be explained by several factors, including the level of income, aging, technology, health policies, and inefficiencies in public health spending (Soto and others 2023). Publicly financed health services tend to be more limited in developing economies. The lowest spending is in South Asia and Southeast Asia (less than 2 percent of GDP in Bangladesh, India, and Papua New Guinea).

Public health spending depends on the breadth and depth of the health benefits package and the source of financing. Health systems in the region tend to rely on government financing or voluntary private financing, including out-of-pocket spending (Figure 5.5), reflecting budgetary constraints and country-specific preferences.

High reliance on government financing is typical in the region's advanced economies, similar to other advanced economies. These countries tend to have more health professionals per capita and greater access to pharmaceuticals and medical goods. Public financing makes up about 66 percent of health spending.

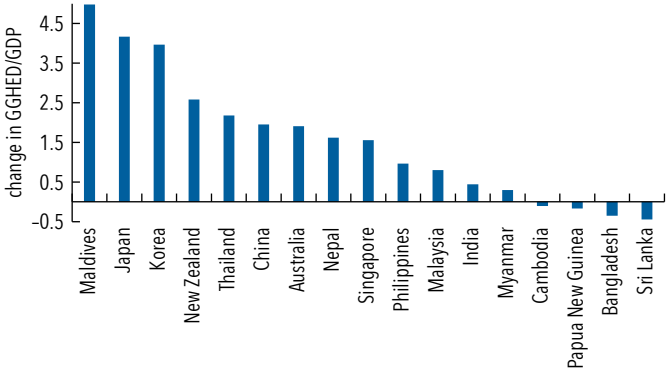
A mix of government and private financing is common in China and some countries in the Association of Southeast Asian Nations, such as Indonesia, the Philippines, Singapore, Thailand, and Vietnam. They usually have an average number of doctors and health professionals per worker but also stricter controls over government financing. Private prepaid health financing (or savings accounts in Singapore) also has a role in some cases.

High reliance on private and external financing of health services is widespread among low-income and developing countries in the region, where 54 percent of health financing is from these sources. For example, Bangladesh, India, Nepal, and Papua New Guinea, and especially the Pacific island countries rely heavily on private financing, particularly out-of-pocket spending.

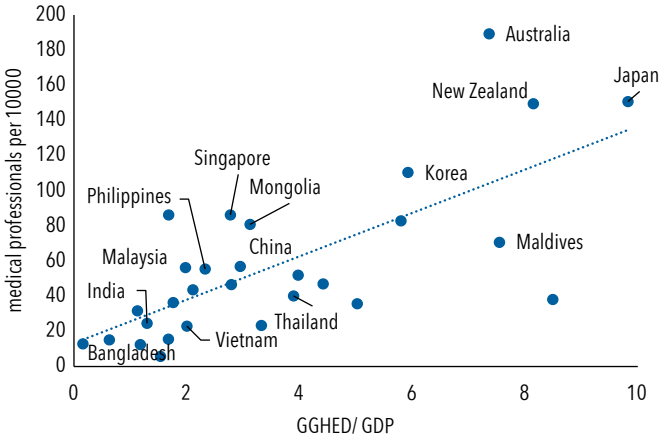
Spending on long-term residential, hospital, or home-based care and support for the elderly with daily living needs (with a social or health focus) is often significant in high-income countries. For example, data from OECD show that spending on

## Figure 5.4. Government Health Spending

**1. Change in Health Spending, 2000–22**
*(Percent of GDP)*



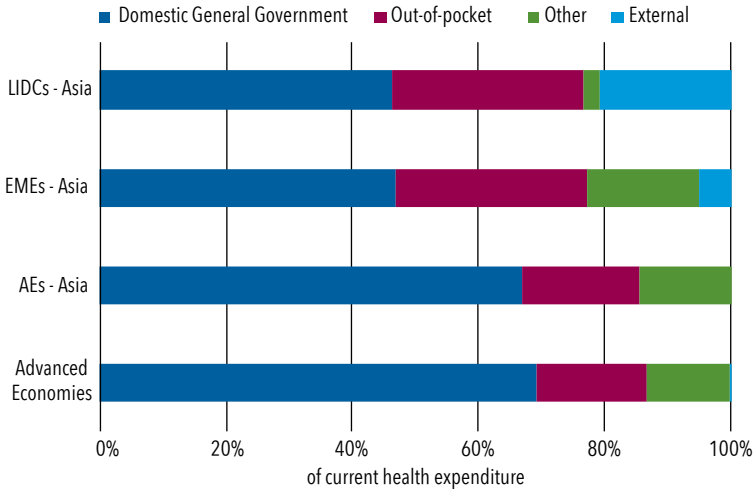

**2. Medical Professionals and Health Spending**
*(Percent of GDP)*




Sources: World Health Organization Global Health Expenditure Database; World Health Organization; and authors' calculations.
Note: "Medical professionals" refers to total (public and private) doctors and nurses per 10,000 people. GGHED = domestic general government health expenditure.

long-term care is at 3 percent of GDP or higher in several European countries that are also facing pressures from aging. By comparison, total long-term care spending in Japan is at about 2 percent of GDP and 1 to 1.5 percent in Australia, Korea, and New Zealand. Although data are scarce for other countries, indications are that spending is typically low, and the main future challenges will relate to unmet needs for care and the associated spending pressures to close those gaps.

**Figure 5.5. Current Health Expenditure by Financing Sources, 2022**
*(Percent)*




Sources: World Health Organization Global Health Expenditure Database; and authors' calculations.
Note: Excludes small emerging market economy Pacific island countries. AE = advanced economy; EME = emerging market economy; LIDC = low-income developing country.

The challenges from aging societies will increase the urgency to address weaknesses in the region's health systems and limit fiscal pressures. Widespread issues across the region include a heavy reliance on hospital-level care, system fragmentation, and lack of access to affordable essential health services (World Bank 2016; ADB 2024). In the context of a rising burden of noncommunicable diseases, systemic weaknesses in the region often lead to poor outcomes and ineffective and inefficient spending.[9] Reliance on hospital-level care rather than primary care is a common issue. For example, during 2017–20 in Vietnam, one-third of admissions could have been handled in an ambulatory care setting (Bales and others 2022). Fragmented health care delivery systems can be expensive and challenging to manage acute and episodic care.[10] A reliance on out-of-pocket expenditures leads to catastrophic health expenditures for some older households. In the Philippines and Vietnam, 10 to 20 percent of households incur out-of-pocket payments that exceed 25 percent of household expenditures. The spending required to improve health services or to close the health gap, especially in emerging markets and low-income countries, could increase fiscal costs substantially.

[9] As of 2019, the percentage of total deaths in Asia attributable to noncommunicable diseases reached 77 percent (ADB 2024).

[10] Poor older people with minor ailments in Cambodia, Indonesia, and Vietnam reported seeking care from many practitioners because of concerns about the quality and convenience of public primary health care (World Bank 2016).

## PENSION AND HEALTH SPENDING IN 2050

As the region's populations age, more people will rely on pension schemes for income and health services delivery, putting greater pressure on public finances. Spending pressures and deficits are driven by a higher ratio of pension beneficiaries to workers, and for health services, the driver is higher health spending for older people relative to younger people. How much these trends increase government spending will depend on the coverage and generosity of the programs and the size of private financing. Several other factors will also contribute to government spending pressures in the coming decades. As countries' incomes rise, their total public spending tends to rise proportionately or more. Unit labor costs also tend to rise over time in labor-intensive service sectors such as health, driven by productivity growth that is slower than the economy-wide average while wages increase in line with the rest of the economy—a phenomenon known as the Baumol cost effect. The costs of pharmaceuticals and other high-cost treatments also typically rise at a faster pace than inflation, raising their relative prices.

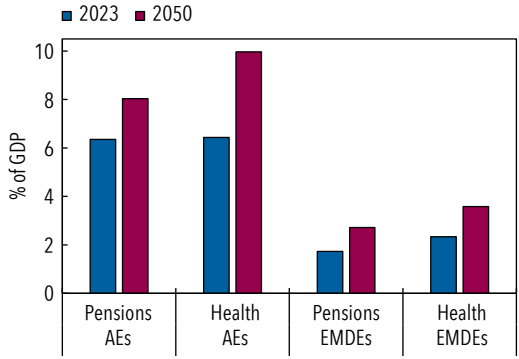
To help assess budgetary pressures, an analysis projected the effects of aging on pension and health spending pressures. The projections were conducted through 2050 for a sample of Asia and Pacific countries under a current policies scenario for health and a constant relative pension benefit amount per recipient. The projections for health spending assumed that such spending will grow relative to GDP at the rates observed in recent years.[11] The analysis considered three main elements. One, as a country's old-age dependency ratio rises, health and pension spending will increase; at the same time the contribution base for pensions will shrink as the share of the working-age population declines (the magnitude depends on policy settings). Two, health spending tends to grow faster than GDP because of the income effect, Baumol's cost effect, and other factors described above. Three, as populations age, they also become healthier at each age (healthy aging), which dampens the aging effect on health spending. The projections did not account for potential additional pension and health spending growth above historical norms to address adequacy issues (spending pressures, especially in developing economies, could be larger if pension and health system coverage or benefits are expanded). The projections also did not account for government action to limit fiscal pressures over time.

Advanced economies in the region are expected to face larger increases in spending by 2050 because of generally higher old-age dependency ratios, but spending in emerging market and developing economies will also increase.

Overall pension spending is expected to rise on average by about 2 percent of GDP in advanced economies and 1 percent in emerging market and developing economies between 2023 and 2050 (Figure 5.6). As the population ages and despite already having higher pension costs, additional large increases are expected

[11] Excess cost growth is the growth in health spending above economic growth after controlling for demographics.

**Figure 5.6. Projected Increase in Public Pension and Health Spending**
*(Percent of GDP)*




Sources: World Health Organization; World Bank; IMF; International Labor Organization; and authors' calculations.
Note: The projections are based on IMF (2023), Clements and others (2015), and Clements, Coady, and Gupta (2012), along with country-specific projections prepared by the authorities, World Bank, or other sources. AE = advanced economy; EMDE = emerging market and developing economy.

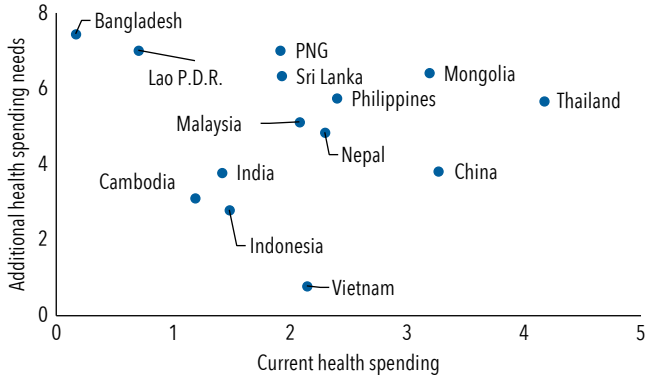
in some advanced economies, such as Korea and New Zealand. The increases in developing economies will be more contained, given their younger populations and lower coverage. However, larger increases are expected in some emerging markets, such as China, Malaysia, Thailand, and Vietnam, given rising old-age dependency ratios. Many developing countries in the region will also be facing a period of even higher old-age dependency ratios and accelerating spending pressures after 2050.

Health spending is expected to grow more than pensions. Advanced economies and a few emerging markets (for example, Australia, Japan, Korea, Maldives, New Zealand, and Thailand) are expected to face the largest increases. The higher health spending is driven mostly by excess cost growth (above demographic drivers), assuming the existing base of benefit coverage.

High uncertainty is associated with demographic changes and the risk of higher spending pressures. For example, the total fertility rate in the region has declined steeply since the 1970s from more than 5 in 1975 to less than 2 in 2025. The baseline points to stabilization at about current levels but even more decreases in fertility rates are possible to the levels in the Americas and Europe.

Countries seeking to improve existing pension systems and health services would incur even larger increases in spending. Pension systems with low coverage or high poverty rates among elderly people could improve coverage or increase pension adequacy, which are beneficial from welfare and equity perspectives but

**Figure 5.7. Health Care Spending Needs and Pressures**
*(Percent of GDP)*




Sources: World Health Organization; World Bank; International Labor Organization; and authors' calculations based on methodology outlined in Carapella and others (2023).
Note: "Additional health spending needs" refers to total health spending needs, which could be financed publicly, privately, or externally. "Current health spending" refers to public, private, and external spending.

could also increase spending significantly (Figliuoli and others 2018). Many middle- and lower-spending countries also need to balance expanding access and the quality of health services with the available fiscal space. The spending required to improve health services or close the health gap, especially in developing economies, could increase fiscal costs substantially beyond the baseline projections.[12] For example, Bangladesh, many countries in Southeast Asia, and Pacific island countries could face higher spending from pressures to expand coverage and access to services that could require an increase in annual health spending of 5 percent of GDP or more in the long term (Figure 5.7).

Rising age-related spending will likely place a significant burden on budgets, but other related factors such as tax and growth effects of aging could also have large impacts on fiscal deficits. The impact of aging can affect budgetary revenues through lower tax revenues (if it lowers productivity or reduces the labor income share of GDP) or through other aging-related tax measures. Model-based simulations for Korea and New Zealand show that aging will put significant pressure on the deficit and debt dynamics through increased spending but also through declines in tax revenues and potential economic growth (Box 5.1). Mongolia faces large and growing pension imbalances, but if the country implements recent

[12] Simulated using the methodology in Carapella and others (2023), the estimates show how much more countries at different levels of income would need to spend to have the same resourcing as benchmark countries on global health development indicators (global benchmark and frontier countries at each level of income).

## Box 5.1. Fiscal Pressures from Aging in Korea and New Zealand

Korea is one of the fastest aging countries in the world, driven by longer life expectancy that is among the highest in the world and a declining fertility rate that is among the lowest. The elderly population has grown sharply since the mid-1990s from less than 6 percent of the total population to about 20 percent in 2024. The old-age dependency ratio is estimated at 27.5 percent in 2024, nearly double that of the Asia and Pacific region.

Without reforms, age-related spending pressures are set to rise markedly over the medium to long term. Under a no-reform baseline scenario, aging is expected to result in declining potential growth over the next 25 years and put pressure on fiscal revenues. Age-related spending is expected to increase sharply, with total public spending projected to rise from less than 25 percent of GDP to 30–35 percent by 2050. In that scenario, fiscal deficits would widen from about 2 percent of GDP to 6–10 percent by 2050, while public debt is projected to reach between 90 and 130 percent of GDP.

Structural reforms to lift potential growth could reduce the debt-to-GDP ratio by about 30 percent relative to the baseline by mid-century (IMF 2025b). Fiscal reforms are also critical to maintaining fiscal sustainability. Parliament approved pension reforms in March 2025, raising the contribution rate from 9 to 13 percent and the income-replacement rate from 40 to 43 percent. These changes are expected to strengthen scheme finances—delaying the onset of National Pension Fund cash deficits from 2041 to 2048 while pushing projected asset depletion from 2057 to 2065 (NABO 2025).

New Zealand faces demographic shifts driven by low fertility rates and rising life expectancy. The old-age dependency ratio has increased markedly since the 1980s and is projected to more than double by 2100. The population includes a sizable share of working-age migrants, which helps mitigate aging pressures in the short term. New Zealand operates a two-tier pension system: the universal, flat-rate, residence-based public pension New Zealand Superannuation, funded through general taxation, and KiwiSaver, a voluntary private savings scheme incentivized by government contributions. Although the superannuation system is effective in reducing old-age poverty, population aging poses rising fiscal costs amid slower economic growth, raising concerns about fiscal sustainability and intergenerational equity.

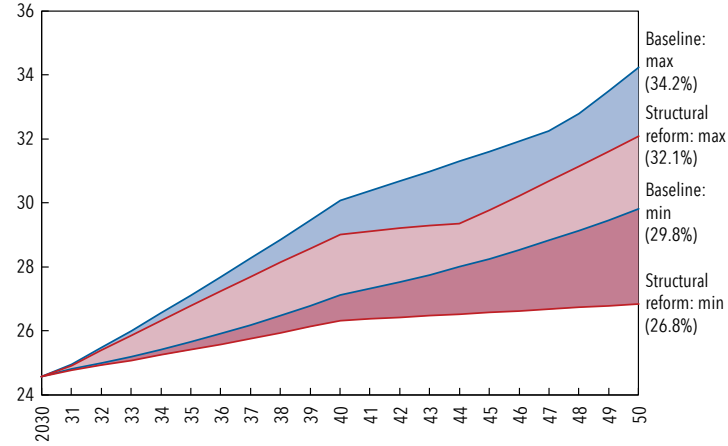
Using an overlapping generations and retirement general equilibrium macro-pension model (Baksa and Munkacsi 2016) calibrated for New Zealand's demographics, baseline projections were run under current policies and demographic trends. The results show rapidly increasing superannuation expenditures in line with the rising old-age dependency ratio, reduced consumption as households save more for longer retirements, and declining fiscal revenues amid slower economic growth. Together, these trends lead to significant deterioration of the public debt-to-GDP ratio by mid-century.

Addressing spending pressures and controlling debt dynamics will require a combination of reforms. Model simulations show the possible effects of different reforms options. These options include gradually raising the eligibility age for the superannuation to encourage longer workforce participation and slow pension spending growth; reducing the generosity of superannuation payments by effectively lowering replacement rates for new retirees, which can incentivize private savings but requires safeguarding retirement incomes; and increasing pension contributions from employees and employers to increase funding for current and

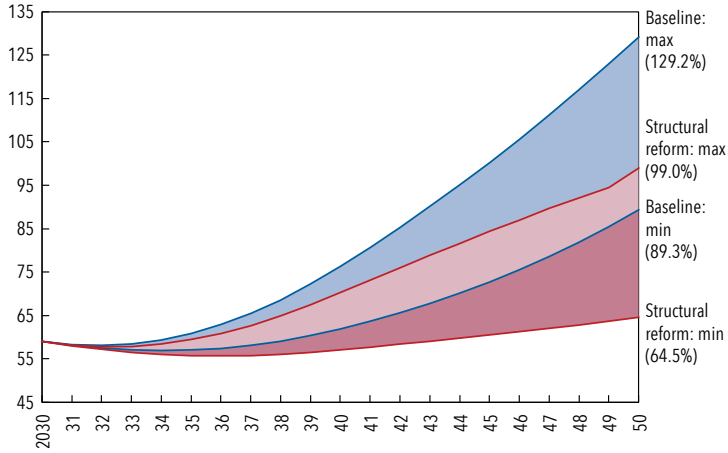
## Box 5.1. Fiscal Pressures from Aging in Korea and New Zealand *(continued)*

### Box Figure 5.1.1. Korea: Public Spending and Debt Projections (2030–50)

**Projected Total Public Spending Path**
*(In percent of GDP)*




**Projected Debt Path**
*(In percent of GDP)*

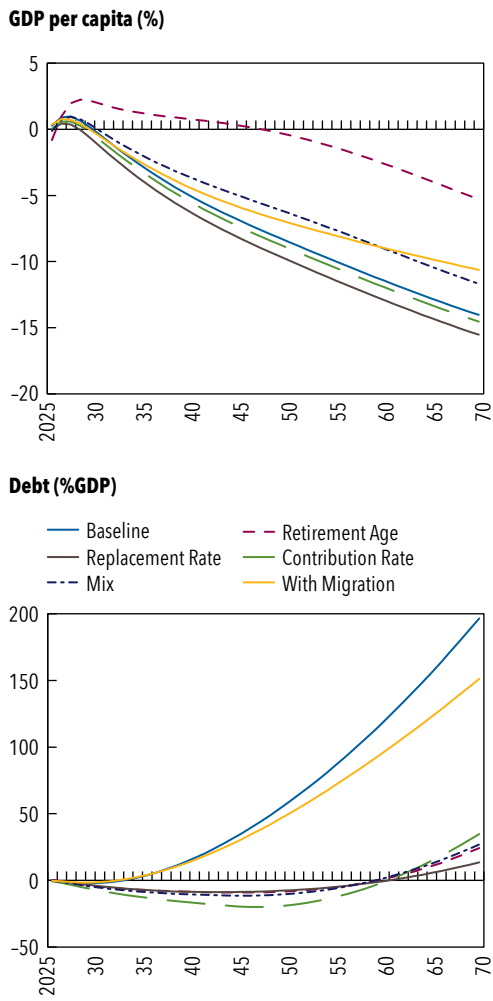



Sources: IMF staff calculations based on estimates from IMF (2025e); OECD (2023); OECD (2024b); NABO (2025); and MOHW (2024).
Note: Methodology: The social spending growth rates for 2030–50 were obtained from the sources listed, and the nominal spending path was derived using macroeconomic projections from the IMF staff's baseline macro-framework. Under the baseline scenario with no reform, real GDP growth is projected to decline to 0.9 percent by 2050, reflecting a shrinking labor force. Under the structural reform scenario outlined in the IMF (2025a, Annex X), real GDP growth was held constant at its potential of 2 percent from 2031 through 2050, because reforms fully offset the impact of labor force decline on potential growth. The Ministry of Economy and Finance's Third Long-Term Fiscal Outlook (2025–65), released in September 2025, incorporates the pension reform and reports debt-to-GDP and expenditure trajectories broadly aligned with the range of IMF staff's estimates.

## Box 5.1. Fiscal Pressures from Aging in Korea and New Zealand *(continued)*

future liabilities. Combined reforms could reduce the scale of each adjustment and distribute burdens more evenly across generations. Migration can also help slow population aging and reduce the magnitude of pension reforms needed, but migration alone is unlikely to fully neutralize the fiscal impact of aging, given the global demographic shift.

### Box Figure 5.1.2. New Zealand: Simulated Paths for GDP Per Capita and Debt (2025–70)




Source: IMF staff estimates.
Note: Projections are based on IMF (2025f, Annex V) and do not incorporate the impact of the KiwiSaver reform announced in Budget 2025 (effective in 2026).

legislative changes to reduce generous pension benefits, pension spending will likely grow more modestly throughout 2050. In Australia, public spending on pensions is expected to stabilize as a share of GDP, but tax concessions to promote private savings for old age are expected to grow and surpass the cost of public pensions (Australian Government 2023).

As concerns with rising fiscal costs spread, an increasing number of countries worldwide have been undertaking pension reforms, including in Asia and the Pacific. The number of pension reform episodes increased from about four in the 1980s to seven or more in the 2000s to 2020s as the old-age dependency ratio rose (Figure 5.8; Annex 5.2). Countries such as China, Indonesia, Japan, Mongolia, Singapore, and Vietnam have gradually raised retirement ages. The relatively low coverage or adequacy of pensions has also prompted a few countries such as Indonesia, the Solomon Islands, and Thailand to extend the coverage of mandatory and voluntary pension schemes to informal and part-time workers or increase contribution rates and replacement ratios, as in Korea. Introducing or strengthening voluntary saving schemes to complement public pensions is also a growing trend.

Countries have also begun to expand health coverage or undertake other reforms to improve health care, some of which imply significantly higher public spending. Key reforms in some countries—such as Bangladesh, Indonesia, Nepal, and Vietnam—include expanding health coverage to reach broader populations and reducing out-of-pocket spending. Lao P.D.R. and the Philippines have prioritized the expansion of primary and preventive health services, Malaysia and Singapore implemented measures to contain costs, and China and Korea redesigned provider payment systems to improve efficiency.[13] Australia also implemented several reforms to contain health costs while ensuring minimum care standards.[14]

## MANAGING CHALLENGES RELATED TO AGING: REFORM AND POLICY OPTIONS

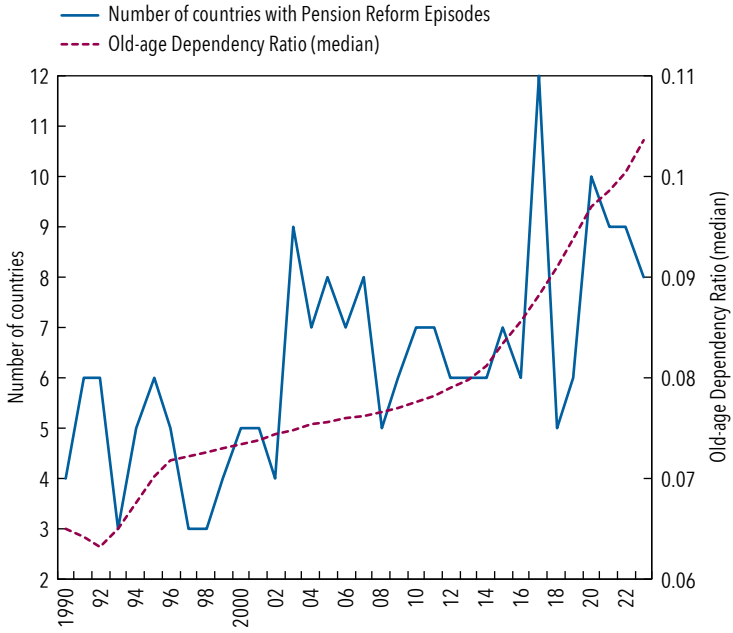
Governments will need to adjust their pension and health systems to meet the challenges posed by population aging, but many are ill-prepared for the challenges. Some have already begun to implement reforms as they face mounting spending pressures from aging and existing structural weaknesses in pension and health systems, but more efforts are needed. As Europe's experience has shown, the reform process takes time and usually involves several reforms over many years.

[13] China is transitioning from a traditional fee-for-service model to a more sustainable case-based payment system, centered on disease-based payment mechanisms.

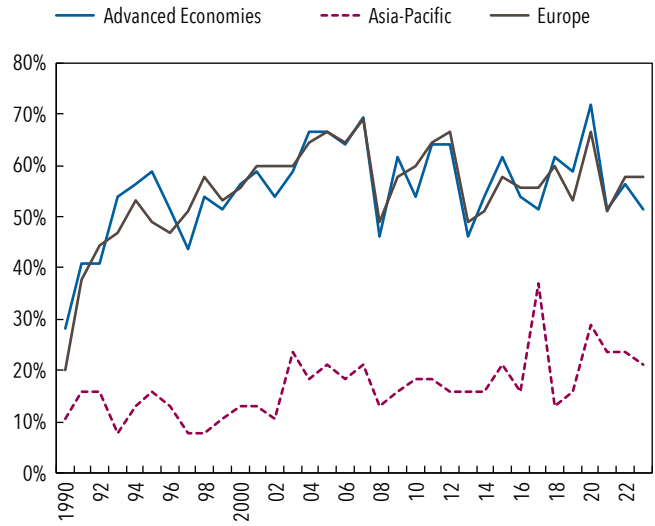
[14] Given the rapid expenditure growth in the National Disability Insurance Scheme, Australia made changes to the legislation, set targets on expenditure growth, built a new framework for eligibility including clearer definitions of what can be funded, and upgraded IT and safeguards. In addition, the 2024 Aged Care Act also made more aged care fees means-tested, and there were also reforms to incorporate minimum care standards and make it easier to have care-at-home

## Figure 5.8. Pension Reform Episodes

**1. Number of Asia and Pacific Countries with Reform Episodes versus Median Old-Age Dependency**




**2. Percentage of Countries with Reform Episodes**

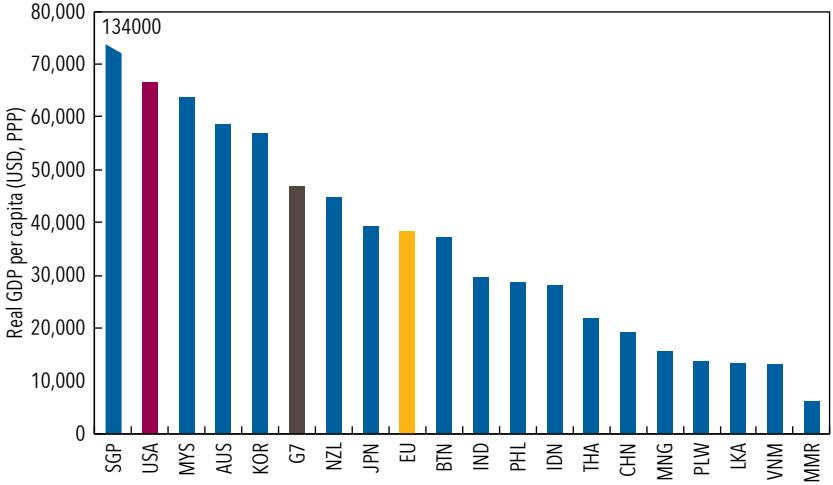



Source: IMF, Global Pension Reform Database.

Policymakers will need to determine how to balance adequate pensions and health care (especially for more vulnerable groups) with sustainable financing sources. Countries with gaps in coverage or that seek to manage fiscal pressures can promote private financing (private savings and self-insurance by households) while governments focus on social pensions and public health financing for the poorest. Another option is to rely more on changes to public systems to ensure that they are fit for purpose as societies age.[15]

Asia and Pacific countries will face daunting fiscal policy choices as their populations age at lower levels of income and with less developed pension and health systems. Advanced economies face more acute aging pressures but have had time to develop their economies and mature pension and health systems. However, many countries in the region increasingly experience aging at relatively lower levels of income (Figure 5.9). Even if the rise in age-related spending until 2050 appears more manageable, it could prove challenging because of several factors. One is the need to set priorities between addressing rising spending

**Figure 5.9. GDP per Capita 10 Years after Peak of Working-Age Share**
*(Real 2015 US dollars, purchasing power parity)*




Sources: International Labor Organization; World Bank; and authors' calculations.
Note: Forecasts assume a GDP growth rate from 2031 to 2040 equal to the 2030 rate forecast in the October 2025 *World Economic Outlook*. For example, the US peak working-age share is from 1997, and therefore the USA bar refers to GDP per capita in 2007. Data labels use International Organization for Standardization country codes. G7 = Group of Seven.

[15] Private provision can give individuals greater choice over pension and health services while reducing pressure on public financing. Public provision of pensions and health care can ensure that benefits and services reach low-income households, reduce inequalities, support risk pooling, and help address market failures.

pressures from aging, closing pension and health system gaps, and investing in development.[16] The cost to achieve the Sustainable Development Goals in emerging markets and low-income countries in Asia and the Pacific is estimated to be more than 10 percent of 2030 GDP (Flores and others 2024). Adequacy issues could also be more pronounced because future pensioners might not have been covered by contributory pension systems during working life. Another factor is rapid changes in demographics, which make adapting more challenging because spending pressures increase with rapid aging. These factors highlight the benefits for developing economies to act early to manage the impacts.

A crucial goal of the reforms should be to promote longer, healthier working lives, which would help mitigate the negative impacts on economic growth and improve the sustainability of pension systems. Longer working lives before retirement lengthen the period of contributions and shorten the period of pension benefits, both of which improve the financial position of schemes. Reforms should include raising retirement ages and creating incentives for older people to continue working for longer. Promoting healthy aging can help expand the effective labor supply among the old-age population and enhance productivity. Gruss and others (2025) show that in Korea, a decade's worth of healthy aging increased the probability of labor force participation and reduced the likelihood of retirement by nearly 20 percentage points. Labor market policies that support healthy aging, increase labor force participation among older people, and close gender gaps in the workforce can boost GDP growth (IMF 2025d).[17]

## Options for Pensions and Health Systems

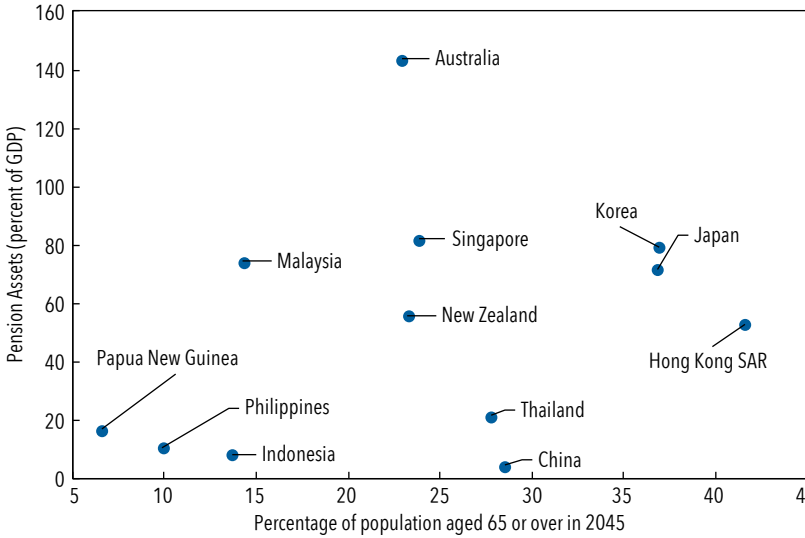
A growing elderly population rapidly increases the number of beneficiaries, and thus governments will need to monitor and adjust their pension systems. More regular monitoring of the pension system's long-term fiscal sustainability is critical,[18] along with adjusting key parameters of the systems through reforms that lift contribution rates, increase the retirement age, and address overgenerous pension benefits in some cases. Japan introduced a macro-indexing mechanism to automatically adjust benefits to demographic and economic conditions, along with a gradual increase in the retirement age (Arbatli and others 2016). Australia increased contribution rates and retirement ages over the past decade to improve

[16] Vietnam plans to increase public investment in infrastructure substantially while also facing rising spending pressures from population aging (IMF 2025c).

[17] A range of cost-effective policy interventions can also encourage later retirement and longer working lives, including reducing social contributions and income taxes for older workers. Broader demographic policies could also help. For example, migration could delay and mitigate the macroeconomic and fiscal impact of population aging, although migration alone would likely be insufficient, given the magnitude of the demographic shifts globally. Policies such as family-friendly workplaces and practices and more generous childcare benefits can also influence fertility.

[18] Monitoring should include actuarial and other modeling of the long-term costs and income of schemes.

**Figure 5.10. Pension Assets versus Current Social Protection Spending**
*(Percent of GDP)*




Sources: United Nations; OECD; and authors' calculations.
Note: "Pension assets" refers to assets earmarked for retirement at pension providers and public pension reserve funds.

resilience to an aging population. Several other countries are also gradually increasing retirement ages.

Pension reform priorities depend on existing system design challenges. Some countries rely on pay-as-you-go public social protection spending, others rely more on funded pension schemes (public or private) or a mix of both. Some countries have accumulated financial assets to manage future pension pressures to different degrees.[19] Australia has an average level of public spending among advanced economies but has substantially higher pension assets, while Japan has much higher public spending and relatively lower assets (Figure 5.10).

Countries with pay-as-you-go public social protection spending (common in Asia) will likely face the largest increases in public pension spending over time. In these cases, aging could put significant pressure on public budgets in the future, which may require raising taxes or cutting other spending to provide a similar level of pensions for future generations.

[19] Pension providers manage some of these assets, as in Australia and Singapore, but some are in public pension reserve funds and earmarked for future pension payments. In Japan and Korea, a large share of assets are in such funds.

Funded public pension schemes can smooth the rising costs over time and establish a clear link between contributions and benefits. These schemes require calibrating contribution rates to spending to ensure sustainable financing. The governance and regulatory frameworks for managing and accessing the funds need careful design and rigorous enforcement to avoid withdrawals from the fund for other than the agreed policy objectives. It requires effective asset management to ensure higher returns while containing risks.[20] Singapore's pension system is designed around individualized savings (funded defined contribution) managed by the Central Provident Fund, a mandatory social security savings scheme established and governed by Singapore law.

Funded private schemes can also help limit exposure to rising fiscal costs from an aging population. However, coverage gaps could still pressure government spending, especially on poverty alleviation. Governments also need to ensure that systems can function efficiently by developing market institutions and frameworks. Preconditions include a comprehensive pension policy, a healthy macroeconomic and fiscal outlook, financial sector readiness, and implementation capacity (Coady and Le 2020).

Each pension design has costs and benefits, but moving between schemes involves large transition costs, and governments must weigh the benefits against those costs. Transitioning from a pay-as-you-go scheme to a funded scheme, for example, requires some cohorts to contribute financing for a previous generation's pension benefits along with their own either directly or through tax financing.

Demands for greater coverage and minimum pensions to limit old-age poverty and inequality and promote consumption smoothing will likely increase. Better coordination across different pension schemes can improve both benefit adequacy and fiscal pressures. Pension systems are fragmented in some countries, and limited coordination across schemes could result in overlapping benefits, inadequate coverage, or unequal treatment across population groups, which could amplify fiscal pressures. In cases where rural-urban or interregional inequalities are significant, reallocating fiscal resources toward lower-income groups might not necessarily increase overall fiscal spending, as in China (Box 5.2). Singapore enhanced its Silver Support Scheme for low-income households by relaxing eligibility criteria relating to pension assets and annual income and increasing the payment. Thailand has made several reforms to its Old-Age Allowance social pension since 2009 and modified its coverage.

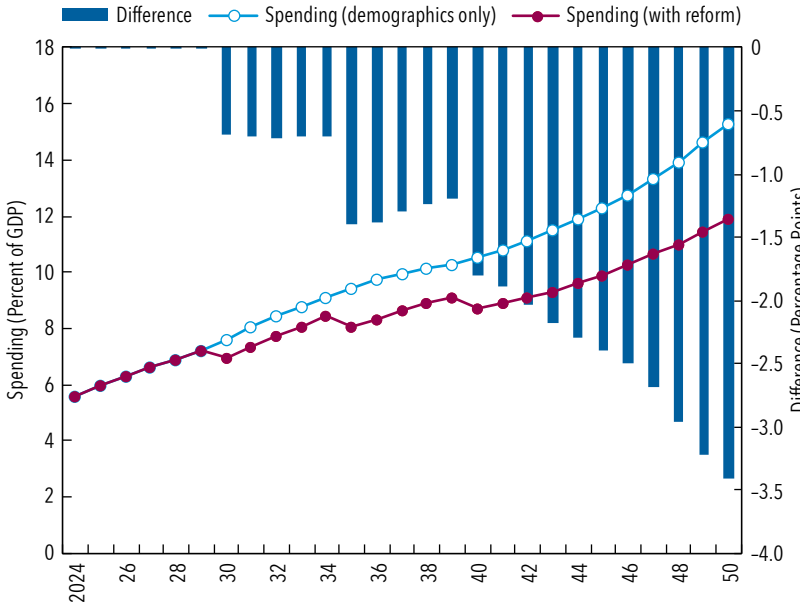
Advanced economies, where health spending is higher, will need to ensure that financing is available to accommodate future spending pressures. Health systems should rely less on costly hospital care but still ensure sufficient primary health care capacity to manage an expected increase in chronic conditions. Institutions and policies should ensure that decisions on wages and pharmaceuticals are

[20] For example, in Vietnam, the government's relatively low borrowing rates in recent years are partly because of captive demand from Vietnam Social Security, but this puts at greater risk the pension system's long-term sustainability (IMF 2024).

## Box 5.2. China's Pension Reform Options: Population Aging and Urban-Rural Inequality

China's pension system has transformed profoundly over the past several decades, evolving from a state-owned-enterprise-based model to a multitier system with near-universal coverage. Beginning with the dismantling of the so-called iron rice bowl (a system of guaranteed lifetime public sector employment) in the late 1970s, successive reforms expanded coverage to private sector, rural, and informal workers. Near-universal pension coverage was achieved when the Residents Pension Scheme and the Urban Employees' Pension Scheme were established.

### Box Figure 5.2.1. China: Impact of Population Aging and the 2024 Retirement Age Reforms on Pension Spending

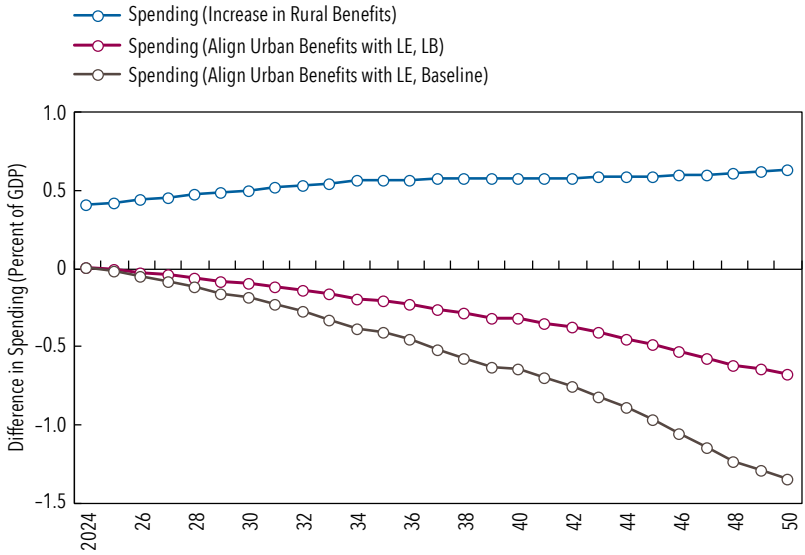



Source: Bonthuis and others (2026).
Note: This chart compares pension spending under two scenarios: legislated reform (red line) and population aging (blue line) from 2024 to 2050. Pension spending is expressed as a percentage of GDP. The blue bars (right axis) represent the difference between the two scenarios in percentage points of GDP.

However, the pension system is facing significant challenges from rapid population aging. China's old-age dependency ratio is expected to double in 17 years—a change that took 19 years to occur in Japan. Pension expenditures are projected to increase from 5.6 percent to 15.3 percent of GDP by 2050 based on an overlapping generations model calibrated for China's demographics and pension system, primarily because of the rising old-age dependency ratio (Bonthuis and others 2026). In response, the government has raised the retirement age gradually since 2024 and plans to increase the minimum contribution period to qualify for a basic pension starting in 2030. Simulations indicate that the retirement age reform could help lower pension expenditures by 3.4 percent of GDP by 2050.

## Box 5.2. China's Pension Reform Options: Population Aging and Urban-Rural Inequality *(continued)*

### Box Figure 5.2.2. China: Impact of Additional Reform Options on Pension Spending




Source: Bonthuis and others (2026).
Note: The chart shows the percent difference relative to the baseline legislated policy for pension spending under three reform scenarios: Increase in Rural Benefits, Align Urban Benefits with LE (LB), and Align Urban Benefits with LE (Baseline). The y-axis represents changes for pension spending in percentage points of GDP.

Pension benefits remain uneven between urban and rural residents. Urban residents receive relatively generous benefits, while benefits to rural residents are low. The Urban Employees' Pension Scheme, primarily serving urban workers, provides a high replacement rate, with average earners receiving 68.3 percent of their wage after retirement, surpassing the OECD average of about 50 percent. The Residents Pension Scheme provides wide coverage, reaching nearly half of the eligible population and predominantly rural residents, but the scheme offers minimal support of only 2 percent of the average wage when no contribution is made. Simulations suggest that increasing rural pension benefits while reducing urban benefits could help ease overall pension spending pressures in the long term while boosting GDP and improving the welfare of elderly rural residents. For example, doubling rural pension benefits increases total pension expenditure marginally by 0.6 percent of GDP by 2050, but it significantly improves consumption-equivalent welfare for elderly rural residents while reducing overall consumption inequality. Lowering the replacement rate for the Urban Employees' Pension Scheme by aligning the notional defined contribution component with improvements in life expectancy could potentially reduce pension expenditures by up to 1.3 percent of GDP by 2050 while raising GDP through higher labor force participation and enhanced capital accumulation.

fiscally sustainable and higher benefit interventions for some treatments and pharmaceuticals are prioritized. Japan still faces fiscal sustainability challenges but has advanced several health care reforms over the past 15 years that improve fiscal sustainability and efficiency and strengthen primary and preventive care (Sakamoto and others 2018). A 2006 reform of curative care in the Netherlands produced efficiency gains, and government efforts in Greece to unify annual tenders for hospital pharmaceuticals and medical devices reported a 10 percent overall price reduction for pharmaceuticals and a 20 percent price reduction for selected medical devices (Soto and others 2023).

Developing economies will need to focus on narrowing or closing large gaps in health service capacity and higher household burden sharing. The most immediate priority is to ensure access to essential health services such as quality maternal and child health, monitoring and management of chronic conditions at the family medicine level, and interventions that support population health, such as programs targeting alcohol and smoking rates, obesity, and accidental and premature deaths. Thailand (Box 5.3) and Korea have extended coverage to additional population groups by expanding existing systems.

Governments should also consider the boundary between private and government financing for health services. Countries have made different choices regarding the government's role in regulating, financing, and delivering health services, but private sector financing or delivery is a factor in most countries' systems. The government's role in the health system is to promote equitable access to affordable essential health services and to deliver public health. No single best model exists. Some health systems deliver more choice at a higher cost or constrain costs with waiting times for some services, but all systems have scope to consider the level of private contributions for some components (for example, targeted out-of-pocket expenditure for pharmaceuticals or prepaid private health financing).

## Challenges from Informality

High levels of informal employment in the region present major challenges for the design of equitable, sustainable pensions and public health systems (Figure 5.11). The high degree of informality among old-age workers in numerous Asia and Pacific countries implies that many lack adequate pensions or health insurance and have limited access to social safety nets.[21] More broadly, workers in informal jobs typically lack health insurance coverage and generally contribute little or nothing to pension schemes, yet they could still rely on publicly

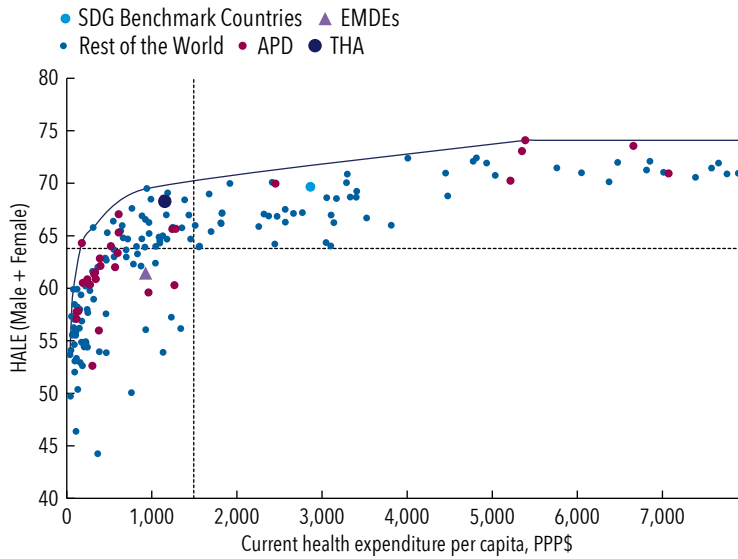
[21] The 2023 Indonesia Longitudinal Aging Survey showed that a significant proportion of older Indonesians work and expect to work into old age, and most work in the informal sector (ADB, SurveyMETER, and LD FEB UI 2023), particularly in low-paying sectors such as agriculture and low value-added services. The employment rate decreases steadily with age, from 66 percent among those in their early 60s to 30 percent among those in their late 70s, with about half of older workers working part-time. The proportion of older people who receive a regular pension income remains low (8–15 percent depending on the age cohort).

## Box 5.3. Thailand: Expanding Health System Coverage and Managing Fiscal Costs

Thailand evolved from a country with high fertility and mortality rates to one with low fertility and mortality rates. The fertility level in 2020 was below the replacement level at 1.3 children per woman, and its crude mortality rate in the same year was 7.3 per 1,000 population. This demographic shift has resulted in a rapidly aging population, with associated pressures on health care spending.

However, Thai health reforms in the 2000s are considered a model of broad health coverage with sustainable fiscal costs. Health reforms began in 2001 in the context of 30 percent of the population lacking coverage by prepaid health financing. The country historically had three major health financing instruments: a service provided by the Ministry of Public Health, a civil servants scheme, and a private sector (social health insurance) scheme. A separate levy-financed scheme covered informal workers (Tangcharoensathien and others 2019). Fiscal costs were constrained through cost efficiency throughout the system.

### Box Figure 5.3.1. Health Spending Efficiency Frontier

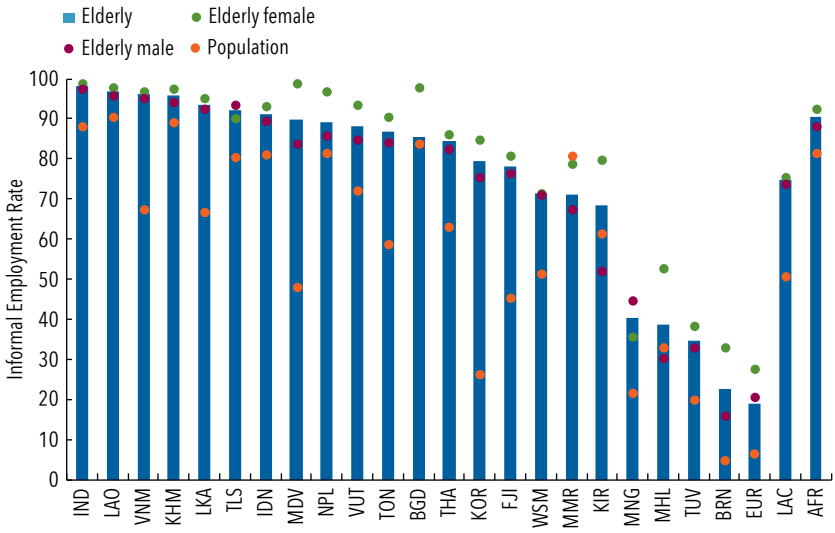



Source: IMF staff estimates based on World Health Organization Global Health Expenditure Database and Global Health Observatory data.
Note: The benefit packages of public health insurance schemes have been gradually expanded since 2002 (World Bank 2023). The spending increase improved access to health services by leveraging the capacity of the Ministry of Public Health delivery systems (Tangcharoensathien and others 2019). Although coverage expansion was large, challenges remain. Financing arrangements in the health sector are complex and fragmented. An evolving burden of noncommunicable diseases challenges a fragmented system and will require long-term care instead of episodic care. Expanded use of new drugs, procedures, and other medical technology also increase health spending (World Bank 2023; WHO 2024). EMDE = emerging market and developing economy; HALE = health-adjusted life expectancy; PPP = purchasing power parity; SDG = Sustainable Development Goal.

**Box 5.3. Thailand: Expanding Health System Coverage and Managing Fiscal Costs *(continued)***

Thailand has achieved improvements in population health outcomes in this century as a result of implementing the 2002 and 2007 reforms to embed universal health coverage. Life expectancy increased from 70.6 years in 2000 to 77.2 years in 2019. From 2000 to 2020, the infant mortality rate declined from 19 per 1,000 live births to only 7 per 1,000 live births, and the under-five mortality rate declined from 22 deaths per 1,000 live births to 9 per 1,000 live births. In addition, the maternal mortality rate decreased from 43 per 100,000 live births to 37 per 100,000 live births. Thailand is now close to the macro health spending efficiency frontier.

**Figure 5.11. Informal Employment Rates, Elderly and Total**




Source: International Labor Organization.
Note: Data labels use International Organization for Standardization country codes. AFR = Africa; EUR = Europe; LAC = Latin America and the Caribbean.

financed benefits in old age or face poverty. As a result, high informality both limits the revenue base of social protection systems and risks reinforcing inequality because formal workers (often higher-income groups) benefit more from public schemes.

A tax-financed social pension can target old-age poverty alleviation when informality is high, but the fiscal cost could be considerable. Short-term, simple solutions that reduce informality are few. In countries with concerns about

poverty and equity, a tax-financed social pension can provide a basic level of income for people in old age as part of a general safety net. Introducing mandated contributions for pensions and health requires care because they may discourage formality, but mandated contributions can be an additional source of financing for pensions and health. Incentives to formalize (such as increasing pension and health benefits associated with formality or reducing payroll contributions), combined with administrative measures, can be effective if costs are well managed.[22]

Governments must take a long-term comprehensive approach to reducing informality. They must integrate the design of contributory pensions, social pensions, and policies that facilitate other forms of elderly support and consider how all three interact (Giles, Joubert, and Tanaka 2025). Beyond general economic and development policies, successful formalization requires a balanced approach that combines labor market reforms, improved enforcement, and well-targeted social protection measures (IMF 2023). Some countries have used administrative measures to raise formality, such as strengthening rules, building administrative capacity, and increasing incentives for administrators for compliance enforcement, but those measures can be context-specific.

## The Need for a Comprehensive Reform Strategy and Timely Policies

To manage rising spending on pensions and health, governments must better integrate aging issues into the broader fiscal strategy. Managing aging will also require other reforms with large intergenerational trade-offs and broad economic effects, including broader reforms such as labor market and productivity-boosting efforts, given the size of the challenges.[23] Enhanced medium-term fiscal frameworks and planning can ensure that reforms are adequately financed, prioritized, and implemented in a sequenced manner. Quantitative assessments and projections of future spending pressures in pensions and health can help assess the urgency of reforms, guide early policy adjustments, and build public support.[24] The examples of country experiences highlight the importance of macro-fiscal conditions, quality of institutions, and public sentiment for successful pension

[22] For example, a reform in Uruguay that extended health benefits to dependent children of private sector salaried workers altered the incentive structure of holding formal jobs within the household. Bérgolo and Cruces (2012) show that informality rates fell by about 1.3 percentage points.

[23] Structural reforms to raise economy-wide productivity could help boost growth potential and improve the long-term fiscal outlook. Improvements in digitalization and AI can also have applications in reducing costs in the health sector (Bloch and Wang 2023). A healthy and educated workforce can help raise productivity and reduce informality and indirectly boost fiscal revenues.

[24] Some countries already produce long-term fiscal sustainability analyses, such as Canada and countries in Europe. Australia also prepares an intergenerational report, and New Zealand's 2025 long-term fiscal report undertakes projections beyond the medium term.

reforms. The negative impact on sentiment for stakeholders can be mitigated by undertaking reforms when economic growth is stronger, if feasible, and mitigating reforms with an increase in benefits and providing cash transfers when the pension age increases (IMF 2025d). The successful enactment of the 2009 Australian pension reform also highlights the importance of coordinated communication, policy packaging, and framing of pension age adjustments.[25] Countries with favorable demographic trends will find value in saving more now and lowering public debt to help weather the increased spending pressures when demographics are less favorable.

Strengthening revenue mobilization can be an important part of a strategy to ensure fiscal sustainability, given the region's relatively low tax revenue. Broadening the tax base (for example, value-added tax coverage and property and inheritance taxes) and enhancing tax compliance and administration can be effective. Some countries in the region have increased social contributions in the past few decades but still have room to expand such contributions to finance higher age-related spending. Although some countries also use alternative tax treatments for private savings and health insurance (including promoting higher savings), any preferential treatment needs to consider the equity implications of providing tax benefits to middle- and high-income households that tend to save more and use private health insurance more, and whether this approach is the most cost-efficient tool. Private underwriting of pension assets and liabilities can have competitive benefits, but the contingent liabilities for government budgets need to be managed.

Delaying pension and health reforms until demographic pressures fully materialize could make the adjustment more difficult and lead to significant debt sustainability risks.

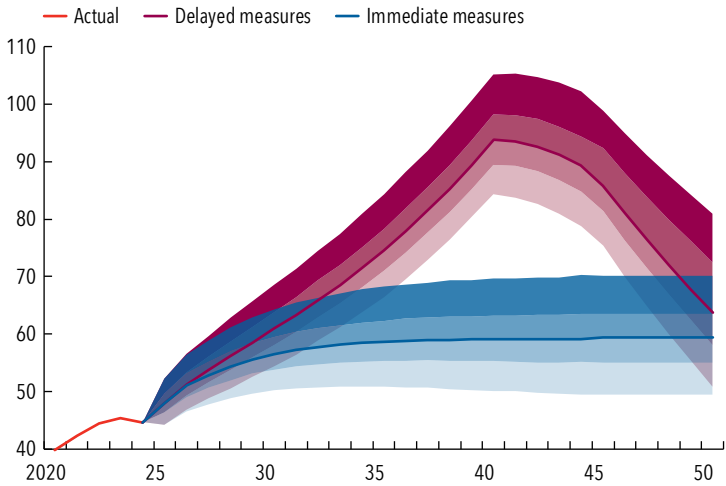
Model-based simulations show that delaying offsetting measures could significantly worsen debt dynamics. If a country is facing a large increase in public pension and health spending over the next 25 years, waiting until the last decade instead of starting reforms immediately would require larger measures later, drive debt higher, and increase the risk of debt distress if adverse shocks materialize (Figure 5.12).[26] In the example, delaying the adjustment instead of acting imme-

[25] The Australian government had strong political ownership. The pension age announcement coincided with the release of the Harmer Review, which highlighted the demographic and fiscal challenges that drove the reform. Comparisons to international reform trends in OECD countries reinforced the rationale for reform. The simultaneous enhancement of pension benefits for low-income retirees also helped gain broader support.

[26] The illustrative simulation is based on a country starting with a debt-to-GDP ratio below 40 percent and facing pension and health spending rising by 8 percentage points of GDP between 2025 and 2050. The immediate measures scenario assumes that the offsetting measures are implemented immediately, starting at 0.3 percent of GDP in 2026 and rising gradually to 8.5 percent of GDP by 2050 (for example, the tax-to-GDP ratio would need to increase by 8.5 percentage points if the adjustment were based only on taxes). The delayed measures scenario assumes that the measures start

diately would more than double it by 2050 (for example, increasing the tax-to-GDP ratio by 16.5 percentage points if based solely on tax measures). Debt would still stabilize at a higher level. A fiscal adjustment of such magnitude in a relatively short period would be difficult to implement, given social-political constraints, and would also weigh on economic growth, potentially worsening debt dynamics. These simulations underscore the importance of early and more gradual reforms to contain age-related debt pressures (Box 5.1). An earlier and more gradual adjustment would also allow the cost to be spread more evenly across generations. Kitao (2018) finds that a reform delay of 10 years can result in additional consumption taxes of about 10 percentage points at the peak. Japan eventually had to undertake major pension reforms that reduced benefit payouts through basic pension adjustments. Younger generations could be worse off in terms of lifetime consumption if adjustments are delayed.

**Figure 5.12. Debt Dynamics as Aging Intensifies**
*(Percent of GDP)*




Sources: Authors' illustrations.
Note: Pension and health spending is assumed to raise fiscal expenditures gradually, by 8 percent of GDP in 2050. The immediate measures scenario assumes that the government implements measures (through revenue increases or spending cuts) immediately, starting at 0.3 percent of GDP in 2026 and rising gradually to 8.5 percent of GDP by 2050. The delayed measures scenario assumes that the government only implements measures starting from 2041, beginning at 5 percent of GDP and increasing faster to 16.5 percent of GDP by 2050. Total sizes of measures are similar–about 100 percent of GDP–over 25 years.

in 2041 and would need to be considerably larger, beginning at 5 percent of GDP and increasing faster to 16.5 percent of GDP by 2050.

**ANNEX 5.1.**

**Selected Pension Designs and Parameters in Asia and the Pacific**

| | Noncontributory Pension Schemes | | Main Mandatory Pension Scheme | | Retirement Age[1] | |
|---|---|---|---|---|---|---|
| Country | Type of Noncontributory Scheme | Beneficiaries (percent of age 65+ population) | Contributors (percent of working-age population) | Beneficiaries (percent of age 65+ population) | Men | Women |
| Australia[2] | Targeted | 63 | 95.6 | -- | 67 | |
| China | Targeted | 20.6 | 61.8 | 68.2 | 60.25 | 55.25 / 50.5 |
| Fiji | Basic, targeted | 86.3 | 33 | 13.0 | 55 / 60 | |
| Indonesia | Targeted | 20.3 | 9 | 19.3 | 59 | |
| Japan | Targeted | 3.0 | 62.5 | 92.4 | 65 | |
| Korea | Targeted | 68.5 | 51.8 | 67.8 | 60 | |
| Maldives | Targeted | 97.2 | 26.3 | 8.8 | 65 | |
| New Zealand | Basic | 77.7 | -- | -- | 65 | |
| Philippines | Targeted | 56.0 | 19.6 | | 60 (normal) | 65 (mandatory) |
| Samoa | Basic | 94.6 | 15.8 | 1.4 | 55 | |
| Solomon Islands | No | -- | 27.7 | 26.0 | 50 | |
| Thailand | Basic | 86.2 | 32.4 | 5.6 | 60 | |
| Timor-Leste | Targeted | 1.9 | 10 | 142.4 | 65 | |
| Vietnam | Targeted | 21.0 | 23.2 | 25.0 | 61.25 | 56.75 |
| OECD average | -- | 94.6 (basic); 18.8 (targeted) | 79.2 | -- | 64.7 | |

Sources: OECD; World Bank; country authorities; and IMF staff calculations.
Note: -- means not available or not applicable.
1. Early retirement is excluded. In countries where retirement age increases incrementally, the retirement age by the end of 2025 is presented.
2. Australia's mandatory pension scheme, Superannuation, is private.

**ANNEX 5.2.**

**Recent Selected Pension Reforms in Asia and the Pacific**

| Country | Retirement Age | Noncontributory Pension Reforms | Mandatory Contributory Pension Reforms | | | | | Introducing Voluntary Pension Scheme |
|---|---|---|---|---|---|---|---|---|
| | | | Coverage and Eligibility | Qualifying Period | Contribution Rate | Benefits | Others[1] | |
| Australia | √ | √ | | | √ | | | |
| China | √ | | | √ | | | √ | √ |
| Fiji | √ | | | | | | | |
| Indonesia | √ | | √ | | | | √ | |
| Japan | √ | | √ | | | | | |
| Korea | | | | | √ | √ | | |
| Malaysia | | | √ | | √ | √ | | √ |
| Maldives | | | | | | | √ | |
| Mongolia | √ | | | √ | | √ | | |
| Nepal | √ | | | | | | | |
| New Zealand | | | √ | | | | | |
| Philippines | | √ | | | | | | √ |
| Singapore | √ | | √ | | √ | | | |
| Solomon Islands | | | √ | | √ | | | |
| Thailand | | √ | | | | | | |
| Timor-Leste | √ | | | | | | | |
| Vietnam[2] | √ | √ | | √ | | | √ | |

Sources: International Social Security Association; World Bank; country authorities; and IMF staff.
1. "Others" includes incentives for continued workforce participation, changes in conditions for early pensions, changes in administration fees, and other types of reforms.
2. Vietnam reduced the minimum contribution period for pension eligibility from 20 years to 15 years.

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CHAPTER 6

# Adapting to Change: The Influence of Demographics on Asian Banking

Haibo Li, Estelle Xue Liu, Yinqiu Lu, and Anne Oeking

The Asia and Pacific region is undergoing a pivotal demographic transition. The populations in most countries in the region are aging rapidly (though unevenly), increasing demographic pressures, while population growth is slowing and even turning negative in some economies. United Nations population projections show that the region's total population will peak in 2054, though demographic trajectories differ considerably across economies.

Aging and population decline together will shape the region's economies and financial markets in important ways. Research has examined how demographic change affects macroeconomic outcomes such as public finance and pensions, interest rates, and productivity. Still, evidence on financial systems and especially on banks remains more limited, despite the importance of banking in the region. Banks dominate the region's financial systems, especially in emerging markets, and they intermediate most savings and credit. As populations age, household financial behavior typically shifts toward lower leverage and higher liquidity, including greater deposit holdings. These shifts can influence bank funding structures, lending patterns, profitability, business models, and risk profiles. Understanding these dynamics is critical for financial stability in a bank-dominated region where financial systems might not fully mature before demographic pressures intensify. As demographic profiles diverge across economies, global and regional banks could also reallocate assets away from aging markets to younger, faster-growing markets. Such cross-border shifts could affect financial integration, competition, and stability through potential funding retrenchment or changes in the product mix that foreign subsidiaries offer.

This chapter discusses an analysis, detailed in Li and others (2026), on how demographic change—mainly aging and population growth—affects bank balance sheets, risk taking, and financial stability in the Asia and Pacific region. It uses an integrated empirical framework that traces the effects of demographic change from household behavior to bank balance sheet composition and cross-border banking activity. It combines a global bank panel with a detailed focus on the region's economies, including many emerging market economies and

one of the widest country samples used in this literature. To the best of the authors' knowledge, the analysis is the first to examine empirically how demographic change affects cross-border banking.

The analysis finds that aging is associated with lower loan-to-deposit and loan-to-asset ratios and with a shift in bank portfolios toward nonloan assets. Household behavior explains much of the shift in funding and lending, and banks' cross-border expansion is measurably stronger toward younger markets.

## DEMOGRAPHIC DEVELOPMENTS AND THEIR RELATION TO BANKING

Uneven and rapid demographic change is characteristic of the Asia and Pacific region. Currently, countries such as Bangladesh and the Philippines are classified as early-dividend countries still reaping a demographic dividend (see Chapter 4).[1] Other countries are classified as late-dividend economies (such as Malaysia and Vietnam), while China, Japan, and Korea have already moved to a post-dividend demographic state.

Many economies are growing old before becoming rich (IMF 2017a), unlike most aging economies in Europe and North America. United Nations projections show most Asian economies catching up to advanced economy levels in both the share of older people and total population growth rates by 2050. However, these shifts will occur at lower income levels and with less comprehensive social safety nets.

Two patterns stand out: sequencing and speed. Before 2010, aging was largely confined to Europe and Japan. More recently, aging has also begun in many Asian economies. Since 2010, some have transitioned from the late-dividend stage to the post-dividend stage in less than a single generation, and by 2050, most are projected to experience severe aging characterized by high old-age dependency ratios and shrinking populations. Compared to Europe's gradual demographic transition between 1990 and 2010, Asian economies have moved through the same demographic stages more quickly, leaving less time for labor markets, pension systems, and financial sectors to adapt.

Even as population age structures change rapidly, overall population growth remains positive across much of the region except for the post-dividend group. Therefore, the share of older people continues to rise even as the total population continues to grow. As a result, old-age dependency ratios are deteriorating rapidly because of composition effects, not broad population decline, with important macro-fiscal and financial intermediation implications.

[1] To determine these demographic groups, the analysis relied on the two most relevant demographic indicators: aging as measured by the old-age dependency ratio, and population growth measured by the change in total population. The changes were estimated between 2025 and 2050, based on UNDP (2024) population projections. The literature has used other classifications and indicators, but those could also be more backward-looking than forward-looking. Most classifications were found to yield broadly similar results.

These demographic pressures will increasingly shift the financial customer base toward older cohorts but with varied capacity to smooth consumption in retirement. Rapid aging in Hong Kong Special Administrative Region, Japan, and Korea, along with fast-aging emerging markets such as China, Sri Lanka, and Thailand, create region-specific challenges for banks and regulators.

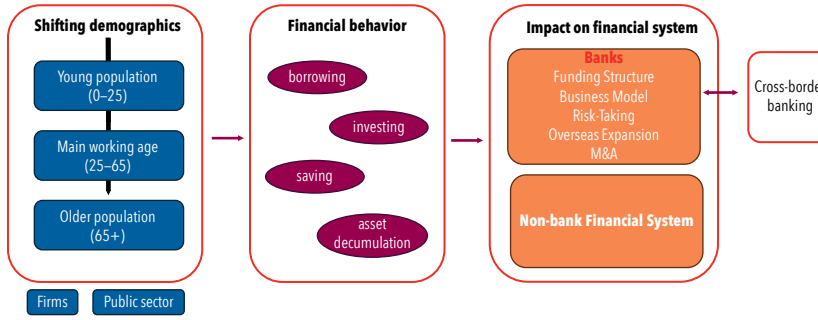
Demographic change affects the financial system most directly through shifts in household financial behavior—how people save, borrow, and invest as they age (Figure 6.1). Demographic shifts also affect firms and the public sector, both of which could also influence the banking sector. For example, aging can reduce corporate loan demand or shift sectoral borrowing, changing the composition of banks' loan portfolios. The analysis first examined household financial behavior, then the effects on the financial system—especially bank balance sheets—and finally the cross-border implications.

## LITERATURE ON DEMOGRAPHICS AND BANKING

Foundational theories linking age and the economy originated in the 1950s. Modigliani and Brumberg (1954) first developed the life cycle hypothesis to explain savings and consumption patterns across individuals' lifetimes (pp. 388–436). The hypothesis holds that individuals seek to maintain relatively stable consumption over their lifetimes by saving during peak earning years and dissaving early and late in life, when income is lower. It also implies that wealth accumulation follows a hump-shaped pattern and peaks in middle age. The permanent income hypothesis argues that consumption decisions depend on expected lifetime income, not current income (Friedman 1957, pp. 20–37). Together, these hypotheses suggest that age structure, longevity, and demographic change shape household financial behavior. Early work building on these theories focused on how demographic change affects the broader macroeconomy.

Direct discussions on interactions between aging and the financial system began in the late 1980s and 1990s, initially focusing on public pension systems

**Figure 6.1. Aging and Banking**




Source: Authors' illustration.

and the growing importance of private pensions in OECD countries (OECD 1998; Gruber and Wise 1999).[2] The literature gradually expanded to examine the effects of aging on asset prices such as housing and equities (Davis and Li 2003; Poterba 2004), saving behavior and asset accumulation (Börsch-Supan 2003), and the broader structure of the financial system, including interest rates and capital flows (Davis 2002; Davis 2006; Carvalho, Ferrero, and Nechio 2016).

More targeted research on the impact of aging on banking systems emerged in the 2010s, including case studies on Japan (for example, Imam [2013] and IMF [2017b]). Household-level studies show that aging is associated with lower risk appetite, higher deposit shares, and lower leverage after retirement (for example, Imam [2013] and Li, Huang, and Hou [2024]). Bank-level studies find that aging tends to lower credit demand and compress margins (Berlemann, Oestmann, and Thum 2014; Ramlall 2023). Recent work also identifies possible incentives for greater risk-taking in aging economies as banks search for yield (Doerr, Kabaş, and Ongena 2024; Imam and Schmieder 2026). Davis and Karim (2025) provide a broader survey of literature in this field.

Evidence for Asia is limited, especially for emerging markets, and cross-border channels remain understudied despite their importance for policymaking. Supervisory interest in the topic has increased, as recent work by policymaking institutions shows (for example, Cassino and Divekar 2025).

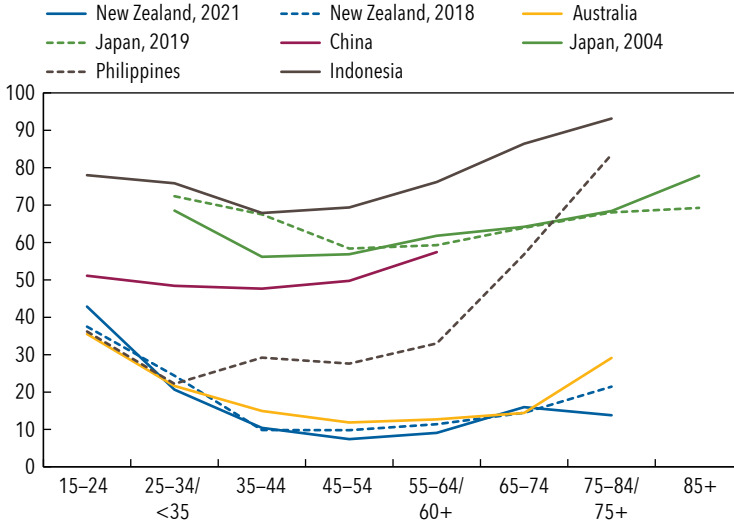
## HOUSEHOLD-LEVEL FINDINGS

Household surveys from Australia, China, Indonesia, Japan, Korea, New Zealand, and the Philippines were used to explore age group profiles of financial assets, including bank deposits, liabilities, and debt participation. Surveys from Malaysia, Nepal, and Thailand complement the analysis with partial coverage. However, the data have limitations, including sensitivity to aggregation methods and age grouping, a lack of controls for income and wealth distributions, and an inability to distinguish age, cohort, and time effects. Yet the surveys provide rare cross-country insight into how demographic change affects bank-relevant behavior in Asia and the Pacific, including wealth accumulation, debt participation, portfolio composition, and financial leverage.

Household financial assets and total assets, including real estate, peak near retirement or shortly before it. Household net worth and net financial assets also peak near retirement age. These findings are broadly consistent with the hump-shaped wealth accumulation predicted by the life cycle hypothesis and suggest some asset decumulation at older ages. Household liabilities peak slightly earlier in midlife, and the share of indebted households declines steadily with age. A significant share of elderly households still carry debt, although the data suggest that

[2] Earlier debates include the Geneva Association's Four Pillars Program launched in 1987, which emphasized long-term financing of pensions and insurance.

**Figure 6.2. Deposit-to-Financial-Asset Ratio by Age Group**
*(Percent)*




Sources: National household surveys; and authors' calculations.

continued income sources, multigenerational households, or a broad breakdown of the elderly age group explain much of this pattern.

The data also show a shift toward safer assets with age. The ratio of deposits to total financial assets follows a U-shaped pattern across age groups, falling in midlife and rising again among elderly households, thus tilting the composition toward safer assets, captured by the share of bank deposits in total assets (Figure 6.2). Bank deposit levels remain relatively stable across elderly groups, suggesting that any asset decumulation occurs mainly in other asset classes.[3]

Financial leverage and risk exposure are lower among older age groups. Both the average loan-to-asset ratio and the average loan-to-deposit ratio by age group clearly trend downward with age.

These findings are broadly consistent with the life cycle hypothesis. As people age, expected labor income declines and spending needs associated with housing and family formation fall, reducing demand for mortgages and consumer loans. Credit supply to older borrowers could also tighten because shorter remaining lifespans increase repayment risk. At the same time, aging households face

[3] However, comparing data for Japan—a post-dividend society—from 2004 and 2019, bank deposit decumulation among postretirement age groups was more pronounced in 2019 than in 2004. Because these are only two points in time, whether the tendency for older people to retain deposits is a persistent pattern over time cannot be tested, though it raises the question of whether marginal behavioral shifts have occurred as Japan's society continues to age.

longevity risk, uncertainty surrounding future medical costs, and limited opportunities to convert savings into lifelong income. As a result, they tend to hold more savings in safe and liquid assets, especially bank deposits.

Therefore, aging households have lower debt participation and hold fewer risky assets even as overall wealth declines, which could imply lower credit exposure risk but also lower profits for banks. Household loan growth and deposit growth slow in aging and shrinking economies, lowering loan-to-deposit ratios even in countries where populations continue to grow. More deposits and fewer loan drawdowns can compress bank profitability. Net interest margins would decline mostly because fewer household loans imply that larger shares of bank balance sheets are in lower-yield liquid assets.

Banks adapt to these changes by shifting toward nonloan assets and fee-based activities, potentially shifting risk toward activities that standard leverage metrics capture less effectively. The demographic impact on banks' funding and credit demand can help explain these changes.

## BANK-LEVEL ANALYSIS AND FINDINGS

The analysis constructed a panel of bank financial statements harmonized across accounting regimes and aggregated them to the country-year level to calculate key ratios, including loan-to-deposit, loan-to-asset, and deposit-to-funding ratios. The data were sourced from the Fitch Connect database, which provides global bank data spanning 1981–2023, although observations before 1989 are limited. Data cleaning procedures broadly follow Raddatz and others (2024), but the analysis used unconsolidated rather than consolidated financial statements to attribute bank balance sheets to the country of the reporting entity and better capture cross-border allocations across subsidiaries.

Country-level panel regressions were conducted to understand the impact of demographic change on bank funding structures and credit activity. The exercise examined associations between aggregated banking indicators and explanatory variables, including demographic developments and general macroeconomic controls. The banking indicators included the loan-to-deposit ratio, deposit-to-funding ratio, and loan-to-asset ratio).[4]

The baseline regressions show that aging, measured by the old-age share, is associated with material changes in bank balance sheet structure (Table 6.1). The loan-to-deposit ratio in column (1) shows that a 1 percentage point increase in the old-age share is significantly associated with a decline in the loan-to-deposit ratio of 5.2 percentage points, consistent with deleveraging as populations age. Column (2) shows that the loan-to-asset ratio declines by about 1.4 percentage points for each 1 percentage point increase in the old-age share, suggesting portfolio reallocation away from loans. Column (3) shows that the deposit-to-funding ratio rises with aging by about 1.3 percentage points for each 1 percentage point

[4] Li and others (2026) provide additional details.

**TABLE 6.1.**

Results of the Baseline Country-Level Regression

| | (1) | (2) | (3) |
|---|---|---|---|
| | Loan-to-Deposit Ratio | Loan-to-Asset Ratio | Deposit-to-Funding Ratio |
| Old-age share (65+ or total) | −5.162*** | −1.414*** | 1.296*** |
| | *(0.764)* | *(0.243)* | *(0.305)* |
| Real GDP growth rate | −0.283 | −0.004 | 0.089 |
| | *(0.195)* | *(0.067)* | *(0.056)* |
| Log real GDP per capita | 36.669*** | 3.227 | −12.596*** |
| | *(5.440)* | *(2.041)* | *(2.014)* |
| Short-term interest rate | −0.295** | −0.163*** | −0.025 |
| | *(0.142)* | *(0.047)* | *(0.068)* |
| Credit-to-GDP ratio | 54.515*** | 21.759*** | −15.921*** |
| | *(7.205)* | *(2.565)* | *(2.397)* |
| Consumer price index inflation | 0.088 | 0.003 | 0.007 |
| | *(0.127)* | *(0.048)* | *(0.062)* |
| Trade-to-GDP ratio | −4.233 | 1.427 | −1.083 |
| | *(6.402)* | *(1.806)* | *(2.365)* |
| Country fixed effects | Yes | Yes | Yes |
| Year fixed effects | Yes | Yes | Yes |
| Observations (country-year) | 1,038 | 1,038 | 1,038 |
| $R^2$ | 0.579 | 0.696 | 0.615 |

Note: Standard errors clustered by countries are shown in parentheses.
**$p < .05$; *** $p < .01$.

increase in the old-age share, consistent with stronger preferences among older households for safer and more liquid assets such as deposits, thereby boosting core deposit funding for banks.

Robustness checks using alternative demographic variables such as population growth, life expectancy, and the active-age share broadly support these findings.

The baseline results were also used to evaluate possible future banking sector dynamics and identify where demographic pressures could weigh most heavily on bank balance sheets. Regression elasticities were applied to United Nations demographic projections to illustrate future key banking indicators across economies. The exercise assumed that the same trends remain stable over time.

The exercise shows that Asia and the Pacific would experience larger declines in loan-to-deposit ratios and loan-to-asset ratios than other regions because of rapid population aging.[5] Within Asia, however, the picture differs across economies. Because they start from a high base, economies such as Australia and New Zealand would retain high loan-to-deposit and loan-to-asset ratios in the future. By contrast, other Asian economies such as China, Korea, and Thailand, which begin from a lower base and are aging rapidly, would potentially experience deleveraging and a shift in bank asset allocation away from loans, reflected in lower loan-to-asset ratios. The results point to uneven balance sheet pressures across the region.

[5] Li and others (2026) provides additional details.

## CROSS-BORDER CHANNELS: AGING AND GLOBAL ASSET ALLOCATION

Given the relationship between aging and key banking indicators, the analysis next examined whether this aging impact also relates to changes in banks' cross-country behavior. The analysis focused on cross-border asset allocation using a bank-country-level panel regression.

The bank-level panel data were also used to construct a cross-border banking allocation data set to study how banks adjust their behavior in aging economies. The Fitch Connect Ultimate Ownership data set identified the banking group or parent bank of each entity and aggregated country-specific total assets and global total assets for each banking group. Consistent with the bank-level panel data, the analysis used unconsolidated financial statements to avoid netting out exposure to foreign subsidiaries and to preserve the cross-border allocation of assets within each banking group. Banking groups operating in only one country were excluded because they do not engage in cross-border activity.

The analysis then examined the relationship between a banking group's assets in a host country and the demographic status of that host country while controlling for banking group size and various macroeconomic factors.

The findings show that banks' asset allocation is associated with aging. The estimated coefficient for the old-age share in the host country is negative and significant, suggesting that banks allocate less balance sheet capacity to aging markets, where credit demand and growth prospects are weaker (Table 6.2). The

**TABLE 6.2.**

**Results of the Cross-Border Allocation Regression**

| | Banking group's assets in a foreign country/ banking group's global total assets (percent) |
|---|---|
| Old-age share (65+ or total) | −0.851** |
| | (0.418) |
| Real GDP growth rate | 0.170*** |
| | (0.059) |
| Log real GDP per capita | −9.526*** |
| | (3.034) |
| Short-term interest rate | −0.025 |
| | (0.039) |
| Credit-to-GDP ratio | 11.623*** |
| | (3.081) |
| Consumer price index inflation | 0.018 |
| | (0.044) |
| Trade-to-GDP ratio | 2.159 |
| | (2.715) |
| Country-bank pair fixed effects | Y |
| Year fixed effects | Y |
| Home country – year fixed effects | Y |
| Observations (bank-country-year) | 2,504 |
| $R^2$ | 0.827 |

Note: Standard errors clustered by bank-country pairs are shown in parentheses.
**$p < .05$; ***$p < .01$.

results also show that the share of banks' assets in a host country increases with real GDP growth and declines with log real GDP per capita, implying that banks allocate more assets to emerging market economies with potentially lower per capita income and higher GDP growth. The asset share also rises with the country's credit-to-GDP ratio, suggesting that more developed financial markets attract capital.

## CONCLUSION

Aging is associated with significant changes in the structure of bank balance sheets and banking behavior in Asia and the Pacific region. Demographics reshape bank behavior through household channels. Household-level evidence shows that aging is associated with greater holdings of safer financial assets and lower leverage, reflected in higher bank deposits and lower loan participation. Aging also shifts bank portfolios. Panel regressions show that a rising old-age share is associated with lower loan-to-deposit and loan-to-asset ratios, indicating weaker credit demand or supply and asset reallocation away from loans. Demographics also shape cross-border banking decisions. The cross-border regression framework shows that banks allocate fewer assets to aging host markets and more assets to younger host markets, suggesting that demographic dynamics influence global expansion strategies.

Demographic change in Asia and the Pacific has altered the structure of banking. Household behavior—especially lower leverage and stronger preferences for deposits—reduces the share of loans in bank balance sheets. As populations age and in some cases shrink, banks will diversify away from lending toward nonloan and fee-generating activities, which could better match a stable deposit base. However, this shift could increase exposure to market, duration, and fee-income volatility through greater reliance on trading books and wealth or retirement services. Aging can reduce the concentration of loans in bank portfolios while reshaping risk rather than eliminating it. Although greater liquidity in the banking system would strengthen regulatory buffers, risks could reemerge in new, less regulated activities or nonbank financial institutions, making regulatory agility critical as financial systems adapt to aging economies.

Banks systematically allocate less to demographically advanced host markets and more to demographically younger host markets, adding a cross-border dimension to domestic demographic pressures. This trend can create opportunities for younger and lower-income economies seeking to attract foreign savings for domestic investment. However, poorly managed cross-border allocation can also create challenges, underscoring the need for vigilant macroprudential oversight and strong financial sector supervision. If banks reduce their presence in demographically advanced host markets, they will have less influence during business cycle swings, highlighting the importance of strengthening countercyclical policy and financial resilience.

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CHAPTER 7

# Harnessing the Demographic Dividend in Asia and Pacific: Creating Decent Jobs for Growing Populations

**Cristian Alonso, Michael Green, Raju Huidrom, and Yaroslav Hul**

## INTRODUCTION: EXPECTED DEMOGRAPHIC TRENDS BY 2050

Nine countries in the Asia and Pacific region experienced declines in their working-age population share between 2000 and 2025. Over the next 25 years, 21 of the region's economies, representing 90 percent of its population, will experience declines. The working-age population share is expected to peak in 2027. However, regional aggregates mask wide variation in country-specific demographic profiles. The region spans the full range of aging profiles, from some of the world's youngest societies to some of its oldest. This chapter focusses on the role of the former, exploring the remaining demographic promise of the region. The demographic dividend—the potential for young economies to harness their growing workforces to boost economic growth—depends on policy choices that can maximize it. Migration also shapes the demographic dividend and responds to it.

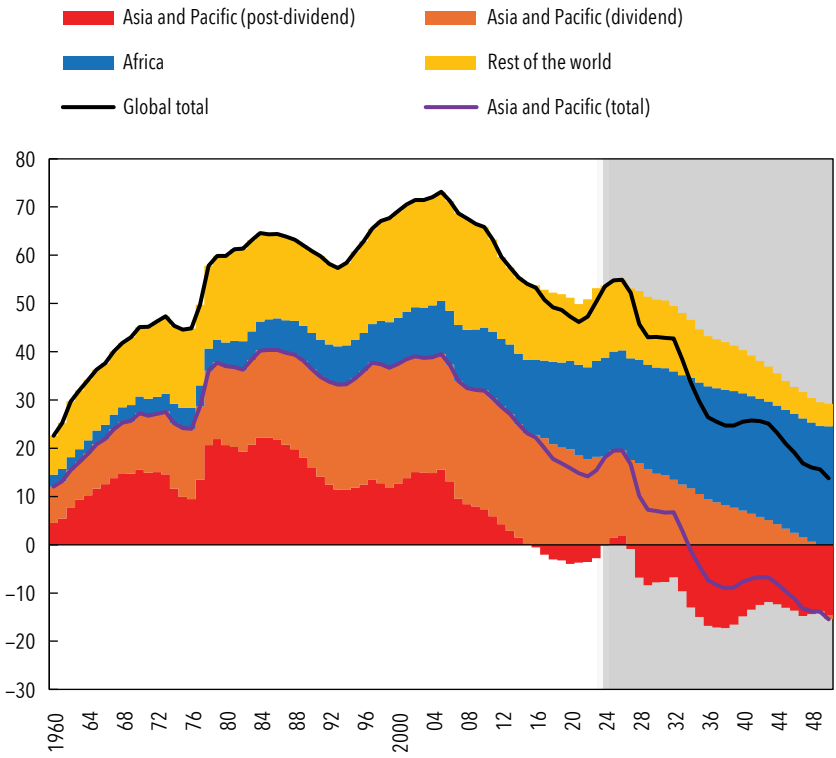
Two projected changes between 2025 and 2050 help explain Asian economies in the context of the demographic dividend: changes in the working-age share of the population and in the total working-age population. Economies are classified by their stage of the demographic transition based on these projections (Chapter 4). "Early-dividend" economies will experience growth in the working-age population faster than the total population, so that working-age population will grow in both absolute terms and as a share of the total population. Nine percent of the region's population lives in early-dividend economies, such as Nepal and the Philippines, where workforces will continue to grow during this period. "Late-dividend" economies will continue to experience working-age population growth but at a slower rate than overall population growth, leading to a decline in the working-age share.[1]

[1] The region's early-dividend economies include Bangladesh, Cambodia, Fiji, Kiribati, Micronesia, Mongolia, Lao P.D.R., Nauru, Nepal, Papua New Guinea, the Philippines, Samoa, the Solomon Islands, Timor-Leste,

Late-dividend economies account for 49 percent of the region's population. "Post-dividend" economies will experience a decline in the working-age population in both absolute terms and as a share of the total population.[2]

The next 25 years offer a window of opportunity for dividend economies to benefit from their demographic tailwind, as they continue to add workers until 2048 (Figure 7.1). This period is critical because these economies risk becoming old before becoming wealthy. Of the 27 dividend economies, only two—Australia and New Zealand—are advanced economies. For the remaining 25 economies, an analysis compared their GDP per capita 10 years after their working-age share of the population is projected to peak with the corresponding levels observed in the United States and the European Union at equivalent points in time (2017 for the

**Figure 7.1. Annual Change in Working-Age Population**
*(Millions)*




Source: UN (2024).

Tonga, Tuvalu, and Vanuatu. Late-dividend economies in the region include Australia, Bhutan, Brunei, India, Indonesia, Malaysia, Myanmar, New Zealand, Sri Lanka, and Vietnam.

[2] The region's post-dividend economies include the Chinese mainland, Hong Kong Special Administrative Region, Japan, Korea, Macao Special Administrative Region, Maldives, the Marshall Islands, Palau, Singapore, Thailand, and Taiwan Province of China.

United States and 1998 for the European Union). All 25 are projected to remain below the 2017 GDP per capita level of the United States. Four of them face greater risks, with projected GDP per capita below that of the European Union in 1998: Mongolia, Myanmar, Sri Lanka, and Vietnam.[3]

Healthy aging can extend the demographic dividend window but only modestly. As highlighted in IMF (2025b, Chapter 2), the traditional correlation between age and economic output has changed significantly in recent decades. Advances in health care, technology, and education have enabled workers to increase productivity as they age and remain economically active beyond typical retirement ages. Higher productivity at older ages and continued economic activity can support post-dividend economies, along with policies that maximize healthy aging, by for example, increasing old-age labor participation rates (Chapter 4). Similar lessons apply even for young economies because for some of them the prospective working-age share peak can differ substantially from the statutory and restrospective working-age share peak (Kotschy and Bloom 2023).[4] Still, in dividend economies, the "silver economy" is not a silver bullet because the difference between the peak years of prospective and retrospective working-age shares is small.[5]

Quantifying the demographic dividend historically provides a guide to what young economies can expect. Analysis shows that harnessing the dividend has required policy action. The right policies on education, trade, and economic regulation have enhanced the effects of the demographic dividend. Migration can also be an important lever to relieve demographic pressures in countries with a growing working-age population. Stylized facts on migration in Asia and Pacific highlight channels through which migration can benefit the region, such as reallocation of labor supply, remittances, and human capital accumulation. Offsetting the demographic challenges in post-dividend economies would require large increases in net migration, although current levels in some economies are especially low relative to peers. Migration is thus not a panacea, and policy solutions discussed in Chapters 4, 5, and 6 will also be necessary for the region to weather the demographic headwinds between now and 2050.

## THE POTENTIAL DEMOGRAPHIC DIVIDEND FOR ASIA

The demographic dividend is the acceleration of economic growth in an economy caused by an increase in the share of working-age population, especially when the share of the working-age population rises relative to that of dependents, meaning children and the elderly (Bloom, Canning, and Sevilla 2003). This dynamic

[3] This exercise assumed an unchanged GDP per capita growth rate from the 2030 IMF projection. That trend is not guaranteed to continue, and assuming so could overestimate growth potential.

[4] "Prospective working-age share" is the share of the population older than age 14 and more than 14 years away from average life expectancy.

[5] The "silver economy" is defined as the economic activity driven by the needs, consumption, and purchasing power of people aged 50 and older.

commonly occurs during the demographic transition, characterized by reductions in both the fertility and mortality rates. Initially, mortality rates fall more rapidly than fertility rates, leading to a population boom and an increase in young dependents, which typically hampers economic growth. However, once fertility rates decline and the large, young cohort enters the workforce, a significant opportunity exists for gains in per capita income through expanded labor supply, higher savings, and greater investment in human and physical capital.

The Asia and Pacific region provides compelling empirical evidence of the demographic dividend in the past. East Asia's economic boom between 1965 and 1990, when the economies grew between 3.7 and 5.5 percent, benefited from a significant demographic dividend. Bloom and Finlay (2009) estimate that growth in the overall population and the increase in the working-age share explained 10 percent of economic growth in Japan, 16 percent in the Chinese mainland, 36 percent in Korea, and 51 percent in Singapore over this period. Growth in southeast Asia's economies (such as Indonesia, Malaysia, the Philippines, Thailand, and Sri Lanka) also benefited from favorable demographic trends between 1965 and 2005. The region still has potential to harness the demographic dividend, especially in early- and late-dividend economies.

The demographic dividend can support economic growth through labor, physical and human capital, and productivity channels. A growing young population can increase creativity and innovation and improve resource allocation, leading to higher total factor productivity (TFP) (Lee and Shin 2021). The working-age population tends to save more than dependents, and those savings provide more resources for investment, allowing faster capital accumulation (Higgins 1998). After the decline in fertility rates, households have more resources to allocate to fewer children, which allows greater investment in health and education per child and raises the stock of human capital (Rosenzweig 1990; Soares 2005). Labor force participation also increases because the decline in fertility rates increases women's labor force participation (Bloom and others 2009; Soares and Falcão 2008).

Our estimate of the demographic dividend is large and significant. Following the methodology outlined in Annex 7.1, a 1 percent increase in the working-age share raises GDP per capita by 0.91 percent through its growth effect. A 1 percent higher working-age share in the previous period increases GDP per capita growth by 0.45 percent through its level effect. These results are consistent with previous findings and align with theoretical models of economic growth (Drummond, Thakoor, and Yu 2014; IMF 2015; Kotschy and Bloom 2023).[6]

[6] Additional controls in the specification have the expected signs. Lagged GDP per capita has a negative coefficient, because convergence dictates that economies closer to the technological frontier tend to grow more slowly than economies farther away from it and with more space to catch up. Capital per capita, primary education, life expectancy at birth, and quality of economic institutions, as proxied by the Economic Freedom of the World (EFW) index, have a positive and statistically significant impact on growth in GDP per capita. Standard diagnostic tests in generalized method of moments—F-statistics in the first stage, AR(1) and AR(2) serial correlation tests, and Hansen and difference-in-Hansen tests—support the validity of the instruments used. The analysis also explored whether the demographic dividend has been larger for Asia compared with other regions by interacting the two demographic variables (the change

Our estimates suggest that labor, capital, and productivity channels are all important for the demographic dividend.[7] The growth effect is positive and statistically significant for capital per capita and labor per capita, and positive but not statistically significant for TFP. The level effect is positive and statistically significant for capital per capita and TFP but small and negative for labor per capita. Together, these findings indicate that the demographic dividend contributes to economic growth by increasing the size of the labor force, boosting the accumulation of human and physical capital, and improving overall productivity throughout the economy.

The evidence points to several areas in which policy choices shape how strongly favorable demographics translate into higher growth, including education, economic institutions, trade openness, and economic structure (Table 7.1).

In education, a 1 percent higher post-primary education rate increases the growth effect of the demographic dividend by 0.25 percent, though not statistically significantly. The effect is more than twice as large and statistically significant for the post-secondary education rate, which underscores the growing importance of education in the modern economy. Post-secondary education can also proxy for the quality of primary education. Both primary and secondary education show positive level effects of the demographic dividend, but neither is statistically significant. Post-secondary education also raises GDP growth directly, in addition to its interaction with demographic variables. Previous studies reach similar conclusions (Drummond, Thakoor, and Yu 2014; Kotschy and Bloom 2023), and some authors argue that "education dividend" better describes the demographic dividend (Crespo Cuaresma, Lutz, and Sanderson 2014). However, others (Kotschy, Urtaza, and Sunde 2020) contest that unidimensional focus, which is also not apparent in the findings of significant demographic dividend impacts on TFP and capital accumulation discussed previously.

Stronger economic institutions, as measured by the overall Economic Freedom of the World (EFW) index, lead to higher GDP per capita growth.[8] However, the average index does not have a statistically significant impact on the growth and level effects for the demographic dividend. The regulation subcomponent of the index—which captures labor market, business, and credit market regulations—has a significant effect on GDP per capita growth, with business regulations especially important.

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and initial level of the working-age share of the population) with a dummy for the region. Results show that the coefficient of the initial level of the working-age share of the population was significantly larger for Asia and Pacific economies but show no significant difference in the coefficient of the change in the working-age share of the population. The result confirms that the region reaped a larger demographic dividend than other regions in the past.

[7] Estimating the specification outlined in Annex 7.1 separately for capital per capita, labor per capita (including human capital if available in the Penn World Tables), and total factor productivity allows decomposing the different channels through which the demographic dividend affects economic growth.

[8] The Economic Freedom Index and its subcomponents used in this chapter were modified to remove components that are based on the World Bank's Doing Business report as required by IMF (2023b).

**TABLE 7.1.**

Policies Shaping the Demographic Dividend

| | (1) Post-Primary Education Rate | (2) Post-Secondary Education Rate | (3) Economic Freedom Index (EFW) | (4) EFW: Regulation | (5) EFW: Freedom to Trade Internationally | (6) Manufacturing Share |
|---|---|---|---|---|---|---|
| Policy | 0.095 | 0.170* | 0.068** | 0.305** | 0.159* | 0.049 |
| | (0.063) | (0.0871) | (0.033) | (0.139) | (0.088) | (0.069) |
| Policy × change in working-age share of population | 0.246 | 0.586** | −0.123 | −0.890 | 1.073* | 0.464 |
| | (0.289) | (0.248) | (0.307) | (1.060) | (0.618) | (0.969) |
| Policy × log working-age share of population (t−1) | 0.107 | 0.183 | 0.037 | 0.373 | 0.230 | 0.179 |
| | (0.104) | (0.152) | (0.072) | (0.244) | (0.147) | (0.111) |
| Observations | 1,198 | 1,173 | 1,198 | 1,096 | 1,077 | 850 |
| Countries | 153 | 153 | 153 | 153 | 153 | 146 |

Source: Authors' estimates.
Note: Estimates were derived from a panel of countries in five-year intervals using system generalized method of moments (GMM). The GMM estimators instrument lagged income per capita, using the second lag in the differences equation and the lagged first difference for instrumentation in the level's equation. This approach uses one instrument per variable rather than per time period (collapsed instrument set). All specifications include time fixed effects. Standard errors clustered at the country level, calculated using the two-step procedure and adjusted for sample size (Windmeijer 2005). They are reported in parenthesis.
*$p < 0.10$; **$p < 0.05$.

Openness to international trade amplifies the demographic dividend. A 1 percent higher score on the EFW subindex for freedom to trade internationally increases the growth effect by 1.07 percent, and the index itself raises GDP per capita growth. More open economies can integrate into global value chains and create jobs for their young workers by contributing to global markets.

A specific economic structure is not required to take advantage of the demographic dividend. A larger share of manufacturing does not lead to a larger demographic dividend.[9] This does not diminish the role of manufacturing in several economies' past growth episodes, especially in East Asia, but indicates that manufacturing is not a necessary condition for development and that economies can pursue different development paths based on alternative economic structures (IMF 2024a).

The demographic dividend is not automatic. The preceding estimates were used to quantify how much demographic dividend economies in the region can expect to harness in a business-as-usual scenario without reforms and compare it with gains from policy action, focusing on economies in the early- and late-dividend stages and reporting results over five-year periods until 2050.[10]

Without policy action, economies can expect only minor gains from remaining favorable demographics. Gains for the region in GDP per capita would peak

[9] The manufacturing share is not a policy per se, but the analysis included it to highlight that a particular economic structure is not a necessary condition to harness the demographic dividend.

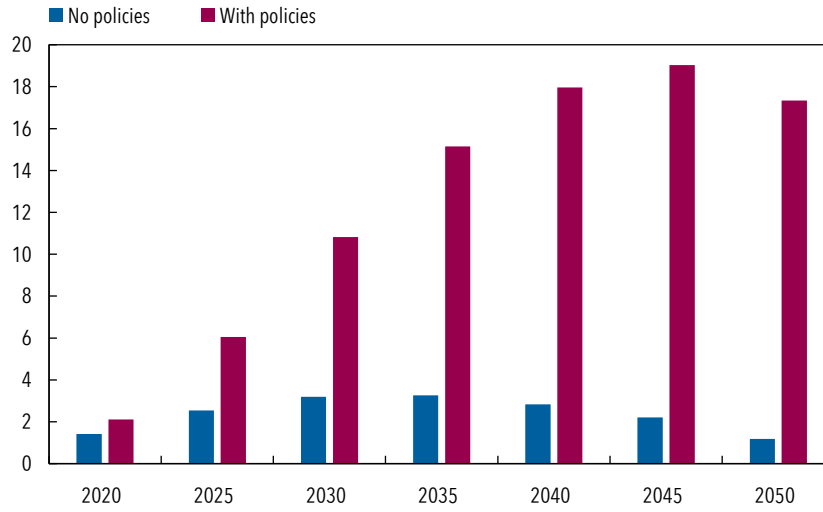
[10] Post-dividend countries were excluded from the analysis.

at just 3 percent by 2035 and then decline as demographics become a drag for some economies over the next decades.

Harnessing a sizable demographic dividend requires decisive policy action (Figure 7.2). Structural reforms that raise educational attainment and EFW indexes scores to the 75th percentile globally by 2050 would generate a much larger demographic dividend. GDP per capita would increase by 19 percent for the region by 2045 because of the combined effect of demographics and reforms. By the end of the projection period, gains would again decline as favorable demographics dissipate.

The demographic dividend under policy action is naturally larger for economies early in the demographic transition. GDP per capita in early-dividend economies would expand by 43 percent by 2050 if they enact reforms to maximize the dividend. Within this group, Bangladesh would experience a gain of 23 percent by 2050. For late-dividend economies, the dividend would peak at 16 percent by 2040 on average, with substantial differences across countries. For example, GDP per capita would be higher by 7 percent in Indonesia by 2035 and by 21 percent in India by 2045 because of the combined effect of demographics and reforms.

**Figure 7.2. Demographic Dividend: GDP Per Capita**
*(Percent relative to stable working-age share)*




Source: Authors' estimates.
Note: "No policies" compares the path of GDP per capita under the current population projections with a counterfactual that keeps the working-age share of the population constant as in 2015. "With policies" compares the GDP per capita counterfactual projections with and without demographic change after implementing the policies in Table 7.1 so that those variables increase to the 75th percentile across the world gradually by 2050. It nets out the direct effect of policies to highlight the effect mediated by demographics. Results are weighted by population. Only countries in the early and late-dividend stages are included in the calculations.

The policies outlined have historically helped harness the demographic dividend, including in many Asian economies, but such pathways could become increasingly constrained. Global trade, which once provided a robust platform for labor-intensive export-led growth, now faces headwinds amid rising protectionism and geopolitical fragmentation (Aiyar and others 2023; IMF 2023a). Rapid advances in automation and AI are reshaping labor markets, potentially reducing demand for low-skilled labor and narrowing the employment opportunities that previously absorbed large cohorts of new entrants.

Opportunities remain, however. Considerable scope exists to deepen trade integration, especially within Asia and Pacific, which could help offset external shocks and create new avenues for employment (IMF 2025a). Moreover, the number of new labor-market entrants projected over the next 25 years is significantly lower than in previous decades, which suggests that lower growth in global trade would be compatible with sufficient job creation for growing populations. The evolving nature of work also underscores the urgency of implementing active labor-market policies and building resilient education and training systems so that workers can adapt to technological change, complement emerging forms of capital, and avoid displacement (Alonso and others 2022; Cazzaniga and others 2024).

## THE ROLE OF MIGRATION IN THE CONTEXT OF DEMOGRAPHIC DIVIDEND

Migration and the demographic dividend have a two-way relationship. Migration influences the size and composition of the working-age population and thus the magnitude of the dividend. In destination economies, migrant inflows expand the labor force and can increase the potential dividend, but emigration reduces labor supply in source economies and can constrain it. Migration patterns also reflect the demographic dividend status of economies, which is one of the push and pull factors shaping flows. Labor shortages from population aging in post-dividend economies generate pull forces that attract workers from dividend economies with abundant labor but limited employment opportunities that act as a push factor. This broader demographic and economic context frames the role of migration in Asia and Pacific.

### How Migration Shapes the Demographic Dividend

Migration can reallocate working-age labor across borders by raising working-age shares in aging destinations and lowering them in origin economies, which helps smooth demographic adjustment. In destinations with shrinking cohorts, inflows of prime-age workers slow the rise in old-age dependency and sustain labor supply, moderating the demographic drag and easing policy adjustment. In origin economies with youth bulges, outward migration—especially of prime-age and skilled workers—reduces the domestic working-age share and helps relieve pressure on labor markets to create enough jobs quickly. The net macroeconomic

effect through this labor quantity channel is typically modest but meaningful for cushioning abrupt demographic shifts, especially in economies at the extremes of the age-structure distribution.

Migration is also multidimensional and redistributes skills, savings, and productivity, so the balance of gains depends on policies in both origin and destination economies. In origin economies, remittance-financed education, return migration, and diaspora links can partially offset losses in labor supply and skills. In destination economies, migration inflows increase labor input but require investment and effective integration to avoid capital dilution and brain waste and to preserve social cohesion.[11] Policies can help in both situations. Stronger financial intermediation and a better investment climate in origin economies can channel remittances into savings and productive investment. Credential recognition, language training and support, and cross-border portable benefits in destination economies could raise migrant productivity. A more predictable business environment, supported by stronger legal and regulatory frameworks in both origin and destination economies, enables knowledge diffusion and firm formation. Without these complementary policies, cross-border labor mobility can simply shift the region's aggregate growth potential instead of raising it.

Conceptually, international migration interacts with the demographic dividend through four channels: labor force participation, human capital, physical capital accumulation, and TFP. A rising working-age share can boost growth, especially with tailored policies that create jobs and increase productivity, and migration affects each link in this causal chain. Net immigration increases the effective labor force in destination economies, whereas emigration reduces it in origin economies, and migration can also influence women's labor force participation (Amuedo-Dorantes and Pozo 2006; McManus and Johnson 2020). Migration also shapes human capital by altering the skill mix in both origin and destination economies (the brain drain–brain gain dichotomy), and by influencing schooling incentives in origin economies—positively through expected earnings abroad and remittance-financed education, or negatively because of parental absence (McKenzie and Rapoport 2011; Yang 2008). Return migration and diasporas can transfer skills and technology back to origin economies (Wahba 2015). Remittances can ease financial constraints and support savings and investment in origin economies, whereas destination economies must invest enough to prevent dilution of capital-labor ratios (World Bank 2006). Several factors can affect TFP, including more diverse teams, improved matching, and knowledge flows in destination economies and by diaspora-enabled technology transfer in origin economies if integration and absorption are effective (Alesina, Harnoss, and Rapoport 2016).[12]

[11] The term "brain waste" refers to underuse of human capital. Skilled and educated immigrants are overqualified for their jobs often because of the host-country's obstacles to recognition of their foreign-earned credentials and professional experience.

[12] Migration can also affect population dynamics over longer horizons through differences in fertility behavior between migrants and natives. Migrants are typically in prime child bearing ages. Evidence

## How the Demographic Dividend Status of Economies Shapes Migration

Migration patterns reflect push and pull forces. Asymmetries in demographic dividend status across origin and destination economies form part of these forces and help explain who moves to where and what skills they take with them, linking directly to the four channels. Youth bulges, weak job creation, low real wages, limited social protection, and exposure to shocks—such as conflict, disasters, climate—in origin economies raise incentives to migrate. Aging populations and sector-specific shortages in destination economies, along with higher expected earnings, safer working conditions, and education opportunities, attract migrant workers. Policy regimes determine both the volume and composition of migration flows through visa quotas, employer sponsorship, credential recognition, benefits portability, and access to regular pathways, including seasonal and temporary programs. Recruitment costs, documentation hurdles, and uneven enforcement limit access for poorer or less connected candidates. Networks and information flows, such as diaspora ties or designated recruiters, reduce search costs and risks and reinforce persistent migrant corridors. Macroeconomic drivers, including business cycles, exchange rates, and commodity prices, also shift incentives.

## Migration Trends in Asia and Pacific

Asia and Pacific is a major source of global migration flows but a smaller destination. During 2020–24, emigration flows originating in the region were about 24 million, accounting for one-quarter of the nearly 100 million flows globally. These flows rank among the highest in the world, second only to Europe, which includes sizable war-related emigration from Ukraine. A few economies with large populations—Bangladesh, India, and the Chinese mainland—account for more than half of the total emigration flows from the region, reflecting the global scale of their diasporas. As a destination, the region received about 17 million migrants during 2020–24, less than one-fifth of global immigration flows and among the lowest regional totals, second only to Africa.

A larger share of Asian emigration flows is extraregional, whereas immigration flows mostly intraregional. Nearly 60 percent of emigration flows originating in Asia and Pacific are destined for economies outside of the region, with a smaller share remaining within the region. This pattern reflects popular destinations of many Asian emigrants—for example, Filipinos and Indians migrating to the United States, and Nepalese migrating to the Gulf Cooperation Council member countries (Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and

for OECD economies suggests that migrants also tend to initially exhibit higher fertility than the native population (Adsera and Valdivia 2023). Together, these factors temporarily support birth rates and slow population aging. However, migrant fertility often converges toward host-country norms as integration progresses, which implies that the long-term demographic impact operates mainly through migrants' age composition instead of permanently higher fertility.

the United Arab Emirates). The predominance of extraregional migration contrasts with emigration flows originating in other regions, which tend to be more intraregional. Immigration flows into Asia and Pacific differ: about 60 percent originate from Asian economies, broadly aligned with immigration flows into other regions.

Migration flows, especially emigration, have flattened in recent years. Emigration flows originating from Asia and Pacific doubled from the mid-1990s until the global financial crisis, driven mainly by extraregional flows, but they have leveled off in recent years. The pandemic might have contributed, but the retreat of globalization, widely discussed in the context of trade and financial flows, occurred at the same time (IMF 2025c). Immigration flows into the region show a similar but less pronounced flattening trend.

As share of population, emigration flows from smaller Asian economies are sizable. In absolute levels, emigration from a few large Asian economies dominates. However, as a share of the source population, emigration from smaller economies—notably Pacific island countries—is sizable and exceeds inflows. During 2020–24, net emigration flows (outflows net of inflows) from Samoa and Tonga ranged between 5 and 8 percent of their populations, whereas the equivalent for India was less than 0.25 percent. These large net outflows underscore the economic significance of emigration in the Pacific island countries. However, as a share of the destination population, net immigration flows (inflows net of outflows) into Asian economies are relatively small, averaging less than 2.7 percent (–1.1 percent net) during 2020–24.

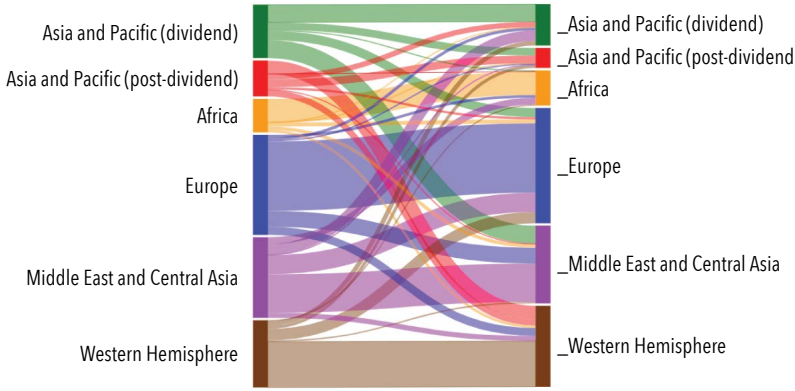
The pattern of migration flows in Asia and Pacific broadly conforms to push and pull factors associated with the demographic cycle. Net emigration flows in Asian economies, measured as share of the source population, align with demographic status: demographic dividend economies—both early- and late-dividend economies—tend to have net outflows, whereas post-dividend economies tend to have net inflows (Figure 7.3). These patterns are broadly consistent with push and pull factors. Thus, emigration flows tend to originate in economies experiencing a demographic dividend (where a large working-age population can boost economic growth) and move toward post-dividend economies (with aging populations and slower workforce growth).

Besides permanent migration, temporary emigration is increasingly relevant for many Pacific island countries. It occurs primarily through Australia and New Zealand's labor mobility arrangements (LMAs), which recruit mostly young, less-educated workers for short-term, low-skilled jobs abroad. These workers earn higher wages, send remittances home, and usually return after their visas expire. Box 7.1 describes their specific characteristics in more detail.

## Economic Implications of Migration in Asia and Pacific

Immigration flows would help ease aging-related pressures in post-dividend economies. Some emigration flows originating in Asian dividend economies are destined for post-dividend economies in the region helping ease those pressures. This

## Figure 7.3. Migration by Region




Source: United Nations Department of Economics and Social Affairs; and IMF staff calculations.
Note: Migration flows are calculated following Abel and Cohen (2019). Also see IMF (2025b).

### Box 7.1. From Departure to Development: Emigration's Past, Present, and Future Role in Pacific Island Countries

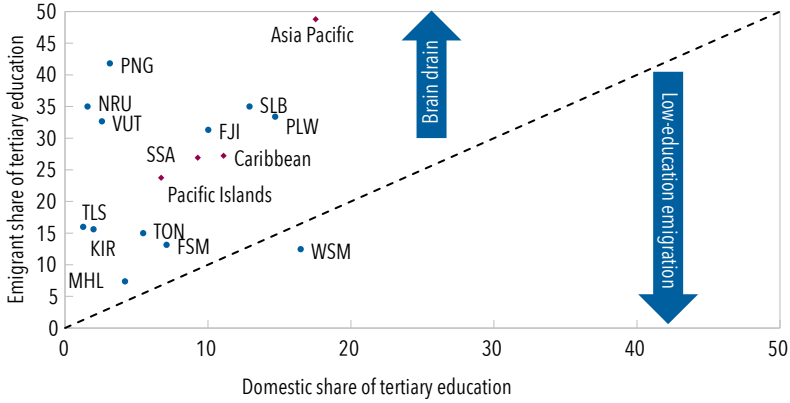
Emigration continues to shape the Pacific island countries' demographic and economic landscapes, although patterns and impacts vary significantly across the region. Some countries, such as Tonga and Samoa, have experienced substantial and sustained emigration, and their overseas populations now exceed half their domestic totals—more than double the share recorded in 1990. By contrast, Papua New Guinea and the Solomon Islands have low emigration rates, with less than 1 percent of their populations living abroad. Emigrants from most Pacific island countries relocate to Australia and New Zealand, whereas emigrants from the three Compact countries—the Marshall Islands, Micronesia, and Palau, which have special migration arrangements with the United States under the Compact of Free Association —migrate primarily to the United States, especially Guam. Intraregional migration is also common. For example, Fiji has experienced significant emigration, but its labor demand also attracts many migrants from other Pacific island countries and elsewhere.

Emigrants from Pacific island countries are typically of working age and have higher educational attainment than the domestic labor force. More than 80 percent of migrants from Pacific island countries are between ages 15 and 64, a significantly higher share than the average 64 percent in Asian countries. Although these emigrants are typically better educated than their counterparts at home, their education levels remain below the Asia and Pacific regional average because the overall educational attainment in Pacific island countries is low (Box Figure 7.1.1). In some countries, such as Palau, the Solomon Islands, and Tuvalu, the number of highly educated people living abroad exceeds the number remaining in the home country, raising serious concerns about brain drain.

**Box 7.1.** *(continued)*

### Box Figure 7.1.1. Domestic Education versus Emigrant Education, 2015

*(Percent of total working-age population)*




Sources: OECD, Database on Immigrants in OECD Countries; World Bank, World Development Indicators; and IMF staff calculations.
Note: Data labels use International Organization for Standardization country codes.
SSA refers to Sub-Saharan Africa.

The recent surge in emigration from Pacific island countries is linked closely to the expansion of the labor mobility arrangement (LMA) initiated by Australia and New Zealand.[1] Seasonal LMA workers from Pacific island countries more than doubled from about 14,000 before the pandemic to more than 36,000 in 2023, reflecting a similar increase in migration to Australia and New Zealand during the same period. In active LMA countries such as Tonga, these workers account for about 5 percent of the total population and contribute significantly to economic and financial stability. Unlike traditional emigrants, LMA participants are typically young workers with lower levels of education. They are recruited for short-term, low-skilled jobs in host countries and earn significantly higher wages than they would earn at home. Many send remittances home and return when their visas expire, reinforcing the temporary nature of these arrangements.

A mix of economic, social, and environmental factors shapes the emigration pattern from Pacific island countries. Structural constraints (including limited domestic production and employment opportunities) and the pursuit of better public services drive outward migration. Although the wage gap between Pacific island countries and destination countries has narrowed over the past decade, average wages in Australia remain more than seven times higher, reinforcing economic incentives to migrate. Climate-related risks add a new dimension to emigration, especially for low-lying atoll nations like Kiribati and Tuvalu, where rising sea levels and extreme weather events threaten habitability. Labor shortages, bilateral agreements, and immigration policies in destination countries also create emigration opportunities.[2]

**Box 7.1.** *(continued)*

Emigration brings economic benefits to Pacific island countries but also creates challenges. Decades of migration have created large, highly educated diasporas. Their remittances are vital to poverty reduction and economic development, supporting domestic output, household income, and the balance of payments in many Pacific island countries. For example, Samoa and Tonga ranked among the top five remittance-receiving countries in 2023, with remittances exceeding 25 percent of GDP (McAuliffe and Oucho 2024). However, emigration also strains domestic labor supply and human capital development. In some Pacific island countries with high emigration, such as Samoa and Tonga, population losses could exceed 25 percent compared with a no-migration scenario. Many emigrants are highly educated, which exacerbates labor shortages and concerns about brain drain.

The recent expansion of the LMA complements the traditional emigration programs in some Pacific island countries by helping address youth unemployment and easing short-term brain drain concerns. The LMA provides an alternative pathway for less-educated youth to earn income and gain experience abroad, with the option to return home. As with traditional emigration, remittances from LMA workers support economic activity and the balance of payments in several Pacific island countries. However, opportunities for skill development among LMA participants remain limited. Evidence suggests that their children are more likely to drop out of school than children in households with similar characteristics in their home countries, raising concerns about long-term human capital development.

1. The LMA refers to three programs: Australia's Seasonal Worker Program and the Pacific Labor Scheme–which were combined to form the Pacific Australia Labor Mobility program in 2022 and expanded–and New Zealand's Recognized Seasonal Employer program.

2. Bilateral agreements, including the Compacts of Free Association (between the United States and the Marshall Islands, Micronesia, and Palau) and the Australia-Tuvalu Falepili Union, also support emigration of the related countries.

pattern reflects demographic rebalancing in Asia and Pacific, in which labor surplus in demographic dividend economies helps address labor shortage in post-dividend economies. Yet the rebalancing within the region is partial because a sizable share of these flows is destined for economies outside the region and dividend economies within it. Although rebalancing is also partial in other regions, Asia and Pacific has greater scope for internal rebalancing because it includes a large share of economies with demographic dividend status.

Emigration entails a loss of human capital in dividend economies. Most emigrants are young, of working age, and better educated (Box 7.1). Thus, migration outflows reduce human capital in dividend economies, with economically significant effects, especially in the Pacific island countries. In a counterfactual without migration, estimates suggest that the working-age population share in Pacific island countries would have been higher by about 29 percentage points relative to the baseline, which incorporates migration.

However, emigration flows also present benefits in dividend economies. They help ease labor-market pressures where domestic job opportunities

remain limited and concentrated in the public sector. Emigration helps reduce constraints, especially related to youth unemployment, which is high in the Pacific island countries. Migrant workers gain employment and earn higher incomes abroad.

Both permanent and temporary migrants send remittances that can be sizable, especially for smaller Asian economies. Inward remittances exceed one-quarter of GDP in Pacific island countries such as Tonga and Samoa (Box 7.1), and in Nepal (Box 7.2). These remittances primarily support household consumption and, to a lesser extent, education and investment (Asia Foundation 2025; World Bank 2022). They are also a major source of external financing in many of these economies and are the main channel of engagement with the diaspora.

Returning temporary workers can bring skills back to their countries of origin. The LMA provides less-educated young workers with opportunities to gain work experience abroad, which they can use at home, though this channel is likely weak because opportunities for skill development among LMA participants remain limited (Box 7.1).

### Box 7.2. The Role of Remittances in Nepal

Remittances in Nepal have long been macro-critical, amounting to one-quarter of GDP in recent years and about three times the total export earnings. They provide the dominant source of foreign exchange, stabilize the balance of payments, finance the trade gap, and bolster fiscal revenues indirectly by fueling imports and domestic demand (IMF 2020). At the micro level, remittances support private consumption and help households insure against shocks and finance education (especially higher education abroad), health care, housing, and small-enterprise investment. Through deposits and asset-market purchases, remittances also channel household savings into working capital and equity for firms. These outcomes are associated strongly with poverty reduction and faster post-disaster recovery.

Despite their scale and outsize role in the economy, Nepal has struggled to translate remittances into broad-based, sustained, and inclusive growth amid persistent structural and financial friction. Vulnerabilities in the financial sector, a weak investment climate, low business confidence, shortfalls in capital expenditure execution, and limited job creation are compounded by institutional capacity gaps and regulatory hurdles. These factors raise the cost of doing business, weaken public service delivery, erode competitiveness, and dampen innovation, which blunts the growth dividend from remittance inflows. Nepal's example is characteristic of many origin economies in the region (such as Bangladesh, Cambodia, and the Pacific island countries) and demonstrates the critical nature of remittances on the macro and micro levels. However, their growth effects depend on effective intermediation, productive use, and a conducive business environment supported by enabling structural policies.[1]

1. See also Abdih and others (2008) on how an increase in remittance inflows can lead to deterioration of institutional quality.

## Quantifying the Impact of Migration

The labor quantity channel is the main mechanism through which migration reallocates the demographic dividend across economies, and a parsimonious production function framework quantifies it.[13] The method allowed inferring how projected shifts in age structure—including net migration—over the next quarter century mechanically alter effective labor input and, through it, economic output. Two baseline assumptions isolated this channel and kept scenarios comparable and focused on labor quantities rather than technology or short-term factor substitution: unchanged TFP growth (the economy's technology continues growing at its country-specific historical trend) and constant long-term capital-per-worker ratio (firms maintain a broadly stable balance between capital and labor over time). Effective labor was constructed by aggregating the number of people (across five-year age-gender cohorts) times their labor force participation and employment. Migration then alters cohort sizes. The growth impact of cross-border migration flows under these baseline assumptions was identified by comparing two scenarios—the UN medium-fertility population path "with migration" and a counterfactual "without migration"—while holding other structural features constant.[14] Both scenarios were normalized by a constant population (status quo) UN projection to remove the net effect of population change (Figure 7.4).

The analysis shows that migration smooths aging headwinds in destination economies and reshapes growth paths modestly in origin economies but on its own is generally not enough to reverse macroeconomic trends driven by underlying demographics.[15] In post-dividend economies in Asia and Pacific, adverse demographic dynamics without migration are expected to subtract about 0.5–1 percentage point from average annual real GDP growth through 2050. Sustained immigration, as incorporated into baseline projections, offsets only a fraction of this drag—about 0.1–0.25 percentage point per year in some post-dividend destinations, implying that even relatively stable inflows mitigate demographic pressures at the margin but do not fundamentally alter long-term growth trajectories.[16] A more substantial offset would require inflows to rise well above historical levels, implying a structural shift or reorientation of migration policies. For example, maintaining dependency ratios at current levels over the projection horizon would require

[13] This analysis follows the growth-accounting framework from IMF (2017) and extends the input database through 2024.

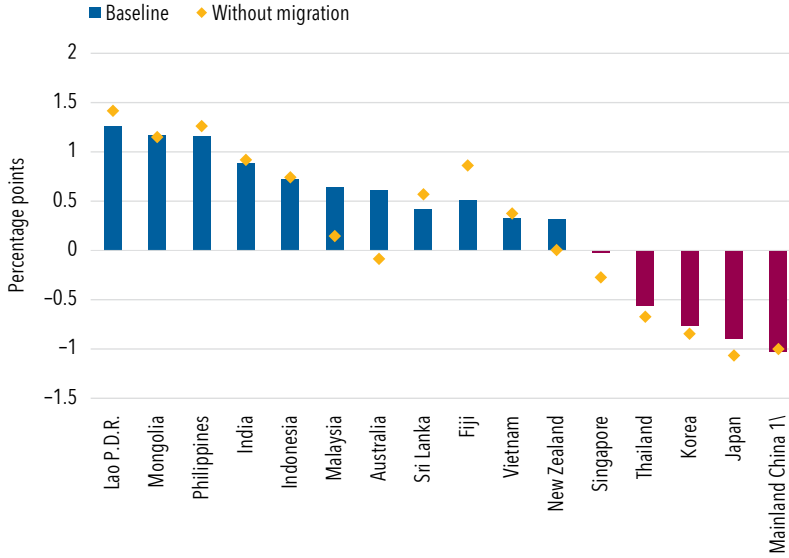
[14] The "without migration" scenario is a mechanical growth-accounting counterfactual that isolated the labor quantity effect of migration. Individuals who would have migrated under the "with migration" scenario were instead assumed to remain at the origin country and contribute to its labor force.

[15] Internal migration within an economy can also shape economy-wide macroeconomic outcomes. For example, Tombe and Zhu (2019) find that internal migration from inland provinces to coastal and urban manufacturing hubs in the Chinese mainland—facilitated by reductions in internal trade and migration costs—account for a sizable portion of the Chinese mainland's aggregate labor productivity growth during 2000–05.

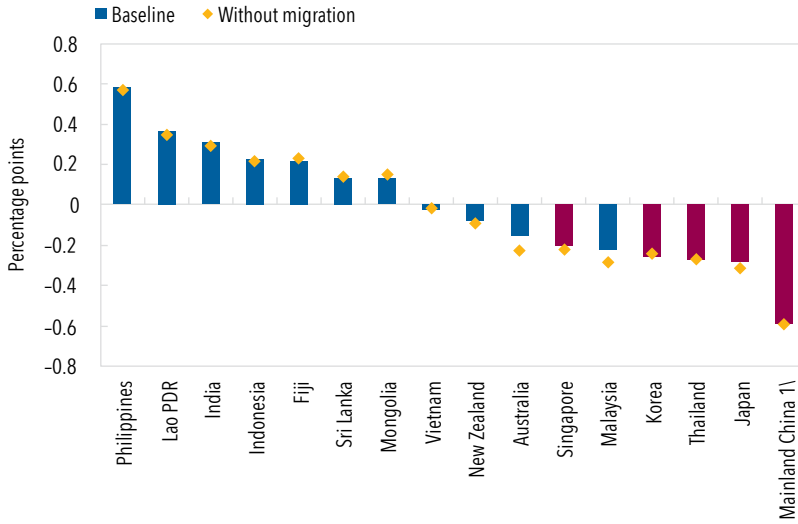
[16] "Sustained immigration" refers to a structural, policy-consistent immigration trend that is predictable and maintained over the long term versus short-lived or cyclical fluctuations.

## Figure 7.4. Impact of Migration on Growth

### 1. Impact on Real GDP Annual Growth, 2023–50




### 2. Impact on Real GDP Per Capita Annual Growth, 2023–50




Source: United Nation's World Population Prospects 2024 database: https://population.un.org/wpp/; and IMF staff calculations.
Note: World Population Prospects 2024 uses data publicly available up to the end of 2023. Red bars indicate post-dividend status for the 2025–50 projection horizon.
1/ Age-gender structure of employment data is not available.

sustained annual immigration of about 0.8 percent of the existing workforce in Korea (about 0.25 million people in 2024 terms, a three-fold increase over net inflows that year) and about 1.0 percent in Japan (about 0.7 million people, nearly five times the 2024 net inflows). In most early-dividend economies, net emigration has modest aggregate effects through the labor quantity channel, especially where migration outflows are small relative to population size (such as India and the Philippines). The impact is more pronounced in economies where temporary or permanent emigration involves a sizable share of prime-age workers, such as the Pacific island countries (Box 7.1).

## POLICY RECOMMENDATIONS

The demographic dividend is not automatic. It translates into durable growth only with policies and reforms that raise labor participation, skills, and productivity fast enough to absorb expanding cohorts. Early-dividend economies face a narrow window in which favorable age structure can lift per capita incomes. A coherent policy package must combine several levers to grow job opportunities with the workforce. It must improve the quality of labor supply, boost labor demand, and reduce friction in domestic and cross-border labor markets. It must also create an enabling macrofinancial environment and support deeper regional integration. Wealthier economies will then be better positioned to manage the higher pension and health costs associated with aging (Chapter 5).

**Invest in human capital.** Front-loading human capital investments would raise labor quality in early-dividend economies and enable the transition from a youthful population with a high dependency ratio to a more productive, higher-skill, higher-participation economy. Higher educational attainment helps harness the demographic dividend. Strengthening foundational learning, technical and vocational education, and tertiary programs that are aligned closely to labor-market needs would improve job matching and reduce skills mismatches. Continuous upskilling and digital literacy will also support this process. Stronger health and nutrition outcomes would expand healthy working years. Targeted programs for girls and women, including child and elderly care programs, would enable earlier and larger labor force participation of women. As cohorts acquire relevant skills sooner, dividend economies could reallocate labor more quickly from subsistence agriculture and low-end services toward manufacturing and modern services, raising average productivity and wages earlier in the demographic window.

Mitigating brain drain through skills partnerships and credible reintegration paths for return migrants would help preserve and even enhance human capital in the origin economies.[17] Calibrating outward mobility with expanded domestic

[17] Skills partnerships are bilateral agreements between two countries to collaborate on training and labor migration to transform brain drain into brain gain in ways that benefit both origin and destination countries. The partnerships expand training at home, align competencies with verified shortages

training in high-demand occupations, portable credentials, and time-bound return incentives (including entrepreneurship support for returnees and recognition of overseas experience) could turn migration into a skills-building loop instead of a one-way loss.

**Promote job-rich private investment.** A sustained rebalancing of policies and public resources toward job-rich, private-led investment would help translate demographic momentum into employment. Improving the quality and efficiency of public investment in sectors such as energy, transport, and logistics would crowd in private initiatives, especially in tradables and modern services.

Regulatory certainty and stronger competition would lower barriers to firm entry, scaling up, and formal hiring. Informality is widespread in the region, and harnessing the demographic dividend requires focused efforts to create formal jobs with higher wages and adequate social protection. Stronger economic institutions, especially regulatory, can support growth. Streamlining permits and inspections, strengthening contract enforcement and insolvency resolution, and simplifying compliance for small and medium enterprises (such as single-window e-registration, unified filings, and streamlined licensing) would enable private investment and support the transition of firms and jobs into the formal economy. Formal sector firms are better positioned to sign enforceable contracts, participate in public procurement and global value chains, access credit and payments infrastructure, and invest at scale. Formal workers benefit from stronger social protection, better job security, portability of benefits, and safer workplaces. At the macro level, a larger formal sector supports revenue mobilization and policy effectiveness.

Deepening financial intermediation would help channel domestic savings and remittances into firms and job creation. Broader access to credit, stronger collateral and insolvency frameworks, and wider use of digital finance could improve the conversion of household savings into productive capital formation.

**Reduce labor-market friction and harness labor mobility to match workers and jobs at home and abroad.** Greater labor-market flexibility and improved matching between firms and workers would increase the speed and quality of labor reallocation, supporting productivity gains as economies evolve. When labor demand shifts across sectors and locations, policies that ease job transitions while preserving income and employability can limit scarring and shorten unemployment periods. Focusing on protecting workers versus protecting specific jobs would reduce the cost of separations, accelerate job searching, and support timely skill acquisition, thus raising output for a specified demographic profile. Portable benefits would decouple social protection from a single employer and allow

abroad, and open safe, legal mobility channels for workers with portable credentials. Destination countries agree to provide technology and financing to train potential migrants with targeted skills in the origin countries before migration, so migrants gain the skills needed to integrate and contribute on arrival. Origin countries agree to provide the training and receive support to also train nonmigrants, thus increasing human capital instead of losing it (Clemens 2018). For more information, see the Global Skill Partnerships at https://gsp.cgdev.org/learn-more/.

continuous coverage across jobs, sectors, and regions. Data-driven public employment services would improve matching by linking vacancies, jobseekers, and training providers through one high-frequency labor-market platform. Targeted reskilling and digital literacy programs would smooth transitions into expanding occupations and shorten job search durations.

Migration can help ease demographic pressures and redistribute demographic dividends through cross-border labor reallocation, but the quantitative analysis suggests that it is not a panacea. The scale and balance of macroeconomic gains from migration depend on the complementary policies that raise employment, skills, and productivity at both origin and destination countries and at regional levels. Policies that enable the flow and integration of migrant workers could mitigate the increasing demographic pressures on labor supply in post-dividend economies (IMF 2024b). Still, migration policy remains a complementary lever that smooths demographic pressures but cannot substitute for broader reforms that expand participation, strengthen human capital, and raise productivity. In origin countries, the key trade-off lies between short-term labor supply losses and medium-term gains from remittances, skill formation, and return migration. In destination countries, the central challenge is to translate migration inflows into higher productivity and durable inclusion while aligning migration policy with broader aging strategies. Cross-border labor mobility can smooth demographic transitions at the regional level, but achieving collective efficiency requires coordinated migration policies and regulations. Temporary mobility programs illustrate how managed migration can smooth shocks and deliver mutual benefits, though effective reintegration and safeguards remain essential (Box 7.1). Well-designed structural policies amplify the gains from labor mobility, but without them, migration would mostly redistribute the region's aggregate growth potential instead of raising it.

**Scale opportunities through deeper trade and regional integration.** This approach would expand labor and product markets, accelerate knowledge diffusion, and support job creation that match the skills of growing cohorts. Greater openness to trade can help harness the demographic dividend. Lowering trade costs in goods and services through customs modernization, border agency coordination, standards and testing, and logistics would support firms participation in regional value chains. Digital connectivity and pragmatic regulatory cooperation in services would expand export niches, from IT and IT-enabled services to tourism ecosystems.

Regional initiatives that harmonize standards, support mutual recognition of education and skills, and streamline mobility for seasonal and temporary workers would better align labor supply and demand within Asia and Pacific, strengthening the region's ability to harness its demographic diversity.

**Sequence policies and reforms.** A sequenced reform and policy package that combines short-term gains in participation and matching with medium-term capacity building and institutional strengthening would maximize the demographic dividend. Early measures such as scaling well-targeted and cost-effective active labor-market policies, enhancing credential recognition, streamlining

licensing and simplifying compliance requirements for small and medium enterprises, and improving logistics corridors can deliver quick gains.[18] Sustained medium-term reforms in education quality, health and pension systems, capital market depth, and high-quality infrastructure would secure long-term dividends. Sequencing that leverages complementarities across human capital, labor-market flexibility, trade openness, and access to finance would likely yield larger and more durable gains than piecemeal efforts.

## ANNEX 7.1. ESTIMATING THE DEMOGRAPHIC DIVIDEND

A panel data approach was used to estimate the size of the demographic dividend. Following Drummond, Thakoor, and Yu (2014) and Kotschy and Bloom (2023), the following specification was considered:

$$y_{i,t} = \beta_0 \ln WAS_{i,t-1} + \beta_1 \Delta WAS_{i,t} + \Gamma X_{i,t-1} + \mu_i + \tau_t + \varepsilon_{i,t}, \qquad (7.1.1)$$

where $y_{i,t} = \ln GDPpc_{i,t} - \ln GDPpc_{i,t-1}$ is the five-year growth rate of real GDP per capita in country $i$ between periods $t$ and $t - 1$; $WAS_{i,t-1}$ is the initial working-age share of the population (at time $t - 1$) and $\Delta WAS_{i,t}$ is the change between periods $t$ and $t - 1$; $X_{i,t}$ is a set of controls, including initial GDP level, initial post-primary education rate, initial life expectancy at birth, initial level of the Economic Freedom of the World (EFW) index, and change in capital per worker between periods $t$ and $t - 1$;[19] and $\mu_i$ and $\tau_t$ are country and time fixed effects, respectively. $\mu_i$ captures all static characteristics for a given country, and $\tau_t$ captures global time trends affecting all countries in the same fashion.

Instrumental variables were used to address endogeneity concerns in the estimation. The proposed specification could be subject to endogeneity caused by reverse causality because economic growth could itself affect demographics and population characteristics through fertility and migration decisions. Thus, the analysis used the system generalized method of moments (Arellano and Bover 1995; Blundell and Bond 1998) and instrumented the change in working-age share in the population (with working-age defined as individuals age 15 to 64) with the share of individuals age 10 to 14 in the previous period, who represent a major inflow into working ages that is exogenous to contemporaneous economic conditions. The analysis also instrumented GDP per capita with its lag and lagged first difference.[20] Standard errors were calculated using the two-step procedure

[18] "Active labor market policies" refers to government programs designed to help unemployed or underemployed people find work through initiatives such as training, job search assistance, and public works (Nunn 2024).

[19] The Economic Freedom Index and its subcomponents were modified to remove components based on the World Bank *Doing Business* reports, as required by IMF (2023b).

[20] The generalized method of moments estimate addresses the bias found in dynamic panel models with fixed effects and limited time periods by using higher-order lags of the lagged dependent variable as instruments in a first-differenced and a level equation. One instrument per variable was used

and adjusted for sample size (Windmeijer 2005). The sample includes 153 economies between 1955 and 2015.[21]

The specification was later expanded to quantify to what extent policies can help harness the demographic dividend. Policy variables were incorporated into the equation to estimate as follows:

$$y_{i,t} = \beta_0 \ln WAS_{i,t-1} + \beta_1 \Delta WAS_{i,t} + \Theta_1 P_{i,t-1} + \Theta_2 \ln WAS_{i,t-1} P_{i,t-1} + \Theta_3 \Delta WAS_{i,t} P_{i,t-1} + \Gamma X_{i,t-1} + \mu_i + \tau_t + \varepsilon_{i,t}, \quad (7.1.2)$$

where $P_{i,t}$ is a set of policy variables: post-primary education rate, post-secondary education rate, EFW index to proxy for the quality of economic institutions and its subcomponents (EFW: Freedom to Trade Internationally and EFW: Regulation), and manufacturing share of GDP.

In this framework, $\Theta_2$ and $\Theta_3$ capture the effect of policy variables on harnessing the demographic dividend, whereas $\Theta_1$ more broadly captures the impact of policies on economic growth.[22]

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rather than per time period (collapsed instrument set) to minimize the problem of having many weak instruments.

[21] Sources include Wittgenstein Centre Human Capital Data, UN (2024), Fraser Institute Economic Freedom of the World Indices, Penn World Tables 10.01, and migration data compiled by IMF staff in IMF (2025b, Chapter 3) based on UN Department of Economic and Social Affairs data.

[22] This chapter focuses less on the direct impact of policies on economic growth, concentrating more on how those policies interact with demographic variables to yield a boost to growth in the form of a demographic dividend.

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CHAPTER 8

# Leveraging AI to Boost Growth and Productivity in Asia and Pacific

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AI has emerged as a transformative force to counter expected headwinds to growth in the Asia and Pacific region. For the analysis in this chapter, AI is defined as the engine of modern automation, encompassing technologies that perform complex cognitive tasks either stand-alone or as key components of modern mechanization of both manufacturing and service production. This broad definition includes large language models, robotics, and physical intelligence, and extends beyond simple software to include the capital-intensive infrastructure, such as data centers, specialized processors, and integrated systems for training and inference, required for AI deployment at industrial and national scales.

AI could be especially salient for the region, given its macroeconomic characteristics and structural challenges. Most critically, AI is a potential catalyst for augmenting human capabilities and reinvigorating productivity growth, especially when the demographic transition and the related population aging constrain labor supply growth across much of the region. The deep integration of many of the region's economies into global value chains creates scope for AI-enabled gains in logistics, process optimization, and smart manufacturing. The technology also offers pathways to accelerate the transition toward higher value-added services, potentially enabling countries to leapfrog traditional development stages.

AI adoption has been uneven, however, raising concerns about potential economic divergence. The chapter focuses on AI adoption at scale, defined as country-level adoption of AI that is reflected in capital deepening and a declining labor share—patterns historically associated with routine-biased automation. This definition is in the same spirit as the key assumption in Cazzaniga and others (2024a), building on Drozd, Taschereau-Dumouchel, and Tavares (2022) and Moll, Rachel, and Restrepo (2022). It reflects the view that adoption at scale is not simply about access to AI technology but also requires investment in AI-enabling physical infrastructure to deploy AI widely and effectively. In some cases, it entails building domestic AI production capacity,

which is particularly capital-intensive.[1] Advanced economies appear to move earlier and faster in adopting AI than emerging markets and developing economies. OECD (2025) documents wide cross-country gaps in business AI adoption, with advanced economies generally at the frontier and emerging markets and developing economies lagging. Survey-based data also point to strong momentum in enterprise uptake of AI with widening cross-country differences.[2] This uneven pace of adoption is primarily because of structural differences across advanced economies and emerging markets and developing economies and could cause economic divergence (Alonso and others 2018; Benzell and others 2021; Cerutti and others 2025; UNDP 2025).

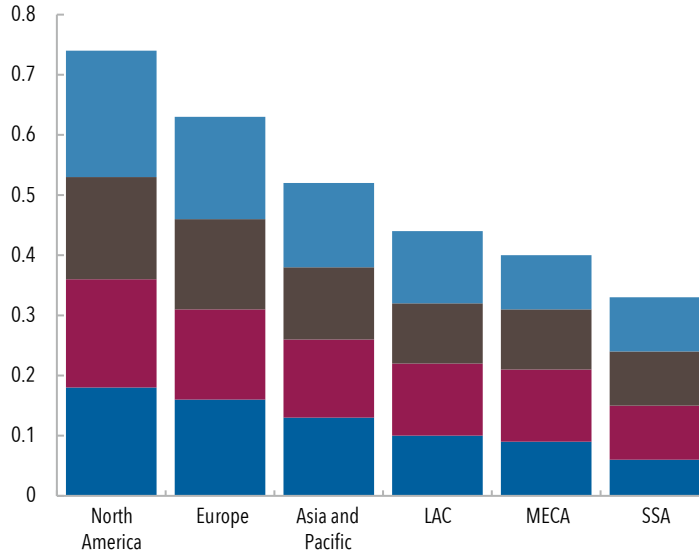
Economies with high capital intensity, a well-educated workforce, and strong technological infrastructure are better positioned to adopt and benefit from AI. Higher initial capital intensity or a lower labor share tends to make AI-related capital deepening more attractive, while a well-educated workforce and strong digital infrastructure provide the complementary inputs needed to deploy AI effectively and translate it into output gains. Advanced economies typically fit this profile, while many emerging markets and developing economies face gaps in these prerequisites that could slow adoption and limit the returns. The AI Preparedness Index (Cazzaniga and others 2024a) summarizes these structural conditions through a composite score across four key dimensions that are especially relevant for smooth AI adoption: digital infrastructure, human capital and labor market policies, innovation and economic integration, and regulation and ethics. Economies with strong technological and data infrastructure—usually associated with high capital intensity—a skilled and well-educated workforce, and adaptable and robust legal and regulatory frameworks are better positioned to adopt AI and benefit from it and therefore tend to score higher on the AI Preparedness Index.

Asia and Pacific's overall AI preparedness lies between the global frontier and most emerging markets and developing economies, though with considerable heterogeneity within the region. The average AI preparedness in the region lags Europe and North America but leads other regions such as Latin America and the Caribbean, Middle East and Central Asia, and sub-Saharan Africa (Figure 8.1). Asia and Pacific economies often perform relatively well within the same income groups, but the diverse set of countries spanning all income groups lower the regional average. Advanced Asia, which, among others, includes Japan, Korea, and Singapore, is near the frontier, whereas many emerging markets and developing economies lag. This heterogeneity creates both opportunities and risks. Some countries are well positioned to harness AI's benefits, but others could be left

[1] As AI use scales up, firms and governments may invest in upstream components—such as data centers, specialized chips and accelerators, cloud platforms, and model development and fine-tuning capacity—to secure reliable access to compute, reduce latency and data-governance constraints, and manage cost and supply-chain risks. For example, major investments are ongoing or planned in Japan and Korea in advanced chip manufacturing, high-bandwidth memory facility, and others.

[2] See, for example, the IBM Global AI Adoption Index 2023, the Microsoft Work Trend Index 2025, and the Stanford Global AI Index 2025.

**Figure 8.1. AI Preparedness Index**
*(Average within the respective region)*




Sources: Cazzaniga and others (2024); and authors' calculations.
Note: LAC = Latin America and the Caribbean; MECA = Middle East and Central Asia; SSA = sub-Saharan Africa.

behind, potentially worsening existing economic disparities within the region. The demographic divergence across the region adds another layer of complexity to AI adoption dynamics. For economies with aging populations such as Japan and Korea, AI offers crucial tools for maintaining growth despite shrinking workforces. For economies with younger populations such as India and the Philippines, the challenge lies in ensuring that AI adoption creates employment opportunities for growing labor forces instead of displacing them.

Considering AI's potential to boost output and productivity and the risk of economic divergence arising from uneven adoption, the analysis examines the potential impacts of AI adoption across Asia and Pacific's economies, from both a cross-country and within-country distributional perspective. A model-based analysis is used to investigate cross-country AI adoption timing and its macroeconomic and distributional impacts in the Asia and Pacific region. The analysis first examines the cross-country AI adoption paces and how country-specific characteristics affect them. It then evaluates the potential benefits of AI adoption and the risks it poses, including cross-country divergence from different adoption speeds and within-country inequality caused by heterogeneous AI exposure and complementarity with it across skills. Then the analysis explores policy reforms that can prepare economies for AI adoption to maximize output and productivity gains while managing potential negative distributional impacts. Besides the macro-model approach, the analysis also examines granular occupation- and firm-level exposures

to AI—including through country case studies—to understand the distributional impacts at the micro level. This work adds to the literature on the macroeconomic implications of AI and also builds on IMF research on the distributional effects of AI across and within countries (Box 8.1).

The simulation results suggest sizable but uneven AI dividends for Asia and Pacific across and within countries, underscoring the importance of well-designed policies to mitigate the divergence risks. AI is neither a panacea nor an inevitable source of harm. With timely structural reforms and carefully crafted redistributive and labor market policies, the region's economies can harness AI to support growth while containing its distributional costs.

The analysis highlights that AI adoption can lift growth significantly in countries that are structurally prepared, as noted above. In advanced economies such as Japan, Korea, and Singapore, AI adoption can raise growth immediately, mitigating the drag from shrinking workforces. In contrast, many emerging markets and developing economies in the region adopt AI only with delays, especially low-income countries, reflecting lower skill and productivity levels, and more limited capital intensity. During the pre-adoption phase, late adopters experience lower growth because of tighter global financial conditions arising from early adopters' investment pushing up global interest rates. Structural reforms that raise skills and productivity can materially change this picture, advancing AI adoption by 10 to 20 years for many late adopters and substantially magnifying the eventual growth dividend once AI is adopted.

AI adoption could also widen inequality within countries absent mitigating policies. High-skilled workers, particularly in AI-complementary occupations, benefit disproportionately from AI adoption because their wages increase much faster than low- and medium-skilled workers, thereby worsening income inequality. Micro-level evidence from occupation-, gender-, and firm-level data corroborates these macro findings. A large degree of heterogeneity in AI exposure and complementarity exists within the region, driven by different occupational compositions and the gender distribution across occupations. Firm-level productivity gains from AI appear to be skewed toward larger and more mature firms, which can create productivity and income disparities in the firm landscape. Redistributing part of the AI dividend can mitigate these distributional costs but with trade-offs. Targeted transfers can better support vulnerable groups but could entail larger growth costs than a universal program, especially in rapidly aging, high-tax economies, whereas those trade-offs are less pronounced in fiscally stronger, younger economies.

Significant uncertainty surrounds the results of the analysis. The model-based simulations are designed to provide insights into the mechanisms and direction of macroeconomic adjustments related to AI progress. Accordingly, the findings should be interpreted as scenario-based illustrations of mechanisms and policy trade-offs, rather than forecasts or policy prescriptions, especially given the large uncertainty surrounding the pace, diffusion, and ultimate capabilities of AI. This uncertainty is also reflected in the wide range of modeling approaches in the literature and the resulting dispersion in estimated macroeconomic impacts (Box 8.1), emphasizing the nascent state of understanding AI's macroeconomic impacts.

## Box 8.1. Recent Studies on the Macroeconomic and Distributional Impacts of AI

Different models have been used to analyze the macroeconomic implications of AI. Task-based models focus on productivity gains. This strand of models yields a wide range of estimates, from the more cautious findings of Acemoglu (2025)—a total boost of 0.66 percent to total factor productivity over 10 years—to the optimistic views of Aghion and Bunel (2024)—a median of 0.68 percentage point boost to annual total factor productivity growth over 10 years. Neoclassical models capture additional channels, such as changes in production function and capital flows. Among these, Alonso and others (2022) estimate that GDP per capita could be increased by as much as 40 percent over the long term in the new steady state with AI. The analytical framework in the analysis aligns more closely with the second strand of the literature: it examines AI's impact on GDP and other key macroeconomic variables by allowing AI to reshape the production function and yields estimates broadly consistent with Alonso and others (2022). The analysis uses this model-based simulation framework because of its ability to examine endogenous AI adoption and simulate dynamics of macroeconomic adjustments.

Recent research provides a growing analytical base for understanding the macroeconomic and distributional effects of AI. At the global level, Cerutti and others (2025) show that AI could raise world GDP by up to 4 percent over 10 years but would widen cross-country income gaps, with advanced economies benefiting about twice as much as low-income countries. UNDP (2025) argues that uneven AI readiness, combined with exposure to disruption, could widen inequalities across countries. Regional analyses confirm these findings: Mish and others (2025) find that productivity gains in Europe depend on human capital and innovation capacity, while Chang and others (2025) document how early AI adoption in Korea has boosted firm-level productivity but also heightened inequality risks. The evidence suggests that AI can deliver substantial efficiency gains but could widen cross-country gaps before convergence resumes as diffusion broadens.

Within countries, IMF studies have found that the effects of AI differ sharply across skills and demographics. Pizzinelli and others (2023) and Cazzaniga and others (2024b) show that advanced economies are more exposed but also better positioned to benefit through greater complementarity between technology and skilled labor and stronger occupational mobility. Rockall, Tavares, and Pizzinelli (2025) find that AI could narrow wage inequality while increasing wealth inequality through higher capital returns. Complementing these findings, Jaumotte and others (2026) emphasize that closing skill gaps through stronger foundational learning, scalable reskilling and upskilling, and policies that support job transitions is central to ensuring that AI adoption translates into broad-based employment and income gains. Country evidence from Chang and others (2025) also indicates that AI can help offset demographic pressures when supported by human capital investment and flexible labor markets.

IMF research points to fiscal, structural, and institutional readiness as crucial to spreading the gains from AI. The research consistently points to three priorities: building skills that complement technology, expanding access to digital infrastructure and finance, and using fiscal systems to share productivity gains and manage distributional pressures. Brollo and others (2024) provide a detailed analysis of fiscal instruments to broaden AI's benefits, while evidence from Khan (2024), Chang and others (2025), and Asao and others (2025) shows how coordinated innovation, labor market, and social policies can support adaptation.

Important caveats also apply to the model simulations. The small open economy assumption is more suitable for genuinely price-taking economies (for example, small advanced economies and emerging markets and developing economies) than for large economies (such as Japan or China), which influence global savings and investment, and technology development, and thus global prices. The analysis assumes a discrete binary choice on AI adoption at the country level, while in reality, adoption could be gradual, continuous, and sector-specific. The chosen framework models AI as a frontier production technology that countries can adopt and use, not as an innovation sector that countries build and export, which implies that rents accruing to AI-producing economies are not captured. The model includes only one type of capital, but AI-related investment in practice can be multidimensional and qualitatively different from traditional physical capital. AI is modeled broadly as one new technology without distinguishing among specific AI technologies. However, different AI modalities could have different adoption patterns. Robotics adoption, for example, is likely to be more capital-intensive, while systems based on large language models can be accessed from laptops and smartphones and can be more readily adopted by countries and industries with lower existing capital intensity. Different AI modalities can also have different substitution or complementarity patterns. Robotics could substitute more for specific manual tasks, while systems based on large language models could reshape cognitive and clerical tasks. The model represents within-country heterogeneity with three skill groups calibrated to education-based measures. It captures distributional impacts through skill-biased changes in factor shares and capital-skill complementarity. This stylized representation abstracts from finer-grained heterogeneity across occupations, tasks, and firms. Despite the model's ability to address both macroeconomic and distributional outcomes, it abstracts from several adjustment margins, such as evolving sectoral structure, firm entry and exit and diffusion dynamics, financial frictions, governance and regulatory constraints, and other potential channels through which AI could affect productivity. The modeling framework also captures cross-country spillovers primarily through global financial conditions while abstracting from trade linkage channels that could enable late adopters to adopt faster, such as knowledge diffusion and technology transfer.

## MACROECONOMIC IMPLICATIONS OF AI ADOPTION FOR ASIA AND PACIFIC

### AI and Cross-Country Economic Divergence

The analysis uses a small open economy general equilibrium model with overlapping generations and endogenous AI adoption decision. Following the multicountry global model of Benzell and others (2021), the country-specific model allows simulation of the macroeconomic implications of different AI adoption scenarios for individual countries (Box 8.2). In the model, firms adopt AI when it becomes profitable to substitute labor with capital, which depends on factors such as the cost of capital, the cost of labor (driven by the level of total factor

## Box 8.2. The Model and AI Adoption Decision

The analysis uses a small open economy general equilibrium model with overlapping generations in which the AI adoption decision is endogenous rather than an exogenous shock.[1] A country consists of three sectors: households that choose labor supply, consumption, and saving over the life cycle; a business sector that hires labor, rents capital, and produces a single good (which can be consumed or invested); and a government that taxes, transfers, and spends. The interest rates are taken as exogenous from the multi-region model (17 regions, each consisting of several countries) in Benzell and others (2021), in which a global capital market clears at a common rate of return determined by global demand and supply of capital. The model is run on an individual country basis, feeding in the global interest-rate path implied by the multi-region simulations, which allows simulation of country-specific macroeconomic implications of AI adoption while keeping global financial conditions consistent with a world in which AI adoption is also happening elsewhere, thereby capturing cross-country spillovers through global financial conditions.

Output is produced according to a Cobb-Douglas production function in capital and three skill-specific labor inputs, with country-specific parameters calibrated to match observed capital shares, skill distributions, and other macroeconomic targets. AI is represented as a more capital-intensive, high-skill-biased frontier technology (a production function with higher capital share and labor shares shifted toward high skill) while leaving the total factor productivity and factor-specific productivity unchanged. Although the model is calibrated based on education attainment data instead of granular occupational composition, studies suggest that higher-skill or more educated workers are more likely to be in AI-exposed jobs that AI is expected to augment tasks rather than replace them (Pizzinelli and others 2023; Cazzaniga and others 2024; Green 2024).

In each period, a country's AI adoption decision is determined by firms comparing the output produced with the frontier AI technology versus the legacy technology. AI is adopted if the frontier technology yields higher output, given current factor prices. Because the frontier AI technology is more capital-intensive and high-skill biased, adoption becomes attractive when the economy is sufficiently capital abundant in effective terms, that effective capital is large relative to effective labor inputs. In the model, countries that start with higher capital intensity are mechanically closer to this threshold, so the extra tilt toward capital embedded in the frontier AI technology is easier and makes early adoption more likely. Higher total factor productivity pushes in the same direction: the optimal capital-to-labor ratio is increasing in productivity, thus cross-country differences in total factor productivity also shift the adoption decision. High-total factor productivity countries use capital more effectively, tend to demand more capital per worker at a given global interest rate, and thus switch to the more capital-intensive AI technology sooner. High-skilled labor matters because the frontier AI technology is skill biased: because labor is not globally mobile, economies with a larger high-skilled workforce capture larger gains from shifting toward high-skill-intensive production, increasing incentives to adopt earlier. Moreover, these forces interact in general equilibrium: early adopters' higher investment demand raises the global interest rate, making capital more expensive for late adopters and delaying their adoption.

1. See Che, Xin, and Yoshida (2026) for more details on the model setup.

productivity and demographics), and the skill composition of the workforce.[3] Countries with higher capital intensity, higher total factor productivity levels, and a greater share of high-skilled labor are more likely to adopt AI earlier. As a result, early adopters are usually advanced economies. By contrast, emerging markets and developing economies would adopt AI more slowly because of structural constraints such as lower productivity and larger reliance on labor but limited high-skilled labor. In addition, global interest rates (endogenously determined by the demand for capital) can influence the timing of adoption in lagging countries. When early adopters raise global capital demand, interest rates rise, increasing borrowing costs for late adopters and potentially delaying their adoption. Government policies such as corporate income tax and labor tax also affect the decision to adopt AI.

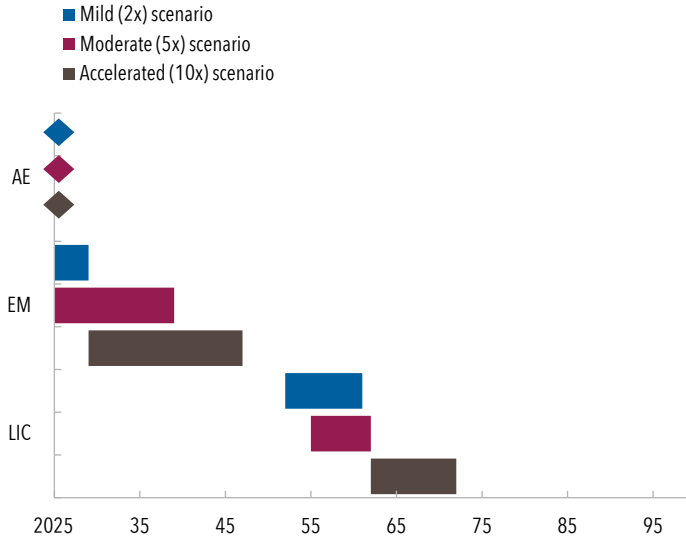
The model is used to simulate a scenario without AI and three alternative scenarios with AI and varied paces of AI progress (mild, moderate, and accelerated), reflecting the substantial uncertainty surrounding AI development. In the no-AI scenario, countries continue to rely on the traditional production technology with no AI technology present. The three AI scenarios differ in how fast the capital share increases in the frontier production—at a rate two times, five times, and 10 times the historical experience of the United States.[4] These scenarios generate differentiated impacts on global capital demand, interest rates, and countries' growth trajectories, enabling quantification of the impacts of AI adoption while acknowledging the significant uncertainty surrounding the pace of AI technological breakthroughs.

Simulation results reveal substantial heterogeneity in AI adoption timing across Asia and Pacific economies, driven by differences in initial economic conditions and structural characteristics (Figure 8.2). Advanced economies adopt immediately in all scenarios. By contrast, emerging markets and developing economies adopt with a delay, the length of which depends on the pace of AI progress. Faster AI progress, captured by a more rapid rise in the capital share, tends to be associated with longer adoption lags for emerging markets and developing economies. This is because early adopters would demand more capital under faster AI progress, which pushes up global interest rates higher and faster and makes it more difficult for late adopters to accumulate capital and switch to the more capital-intensive technology. The pattern of adoption timing reflects fundamental differences in countries' readiness for capital-intensive, skill-biased technological change. Three key factors determine which countries adopt earlier: initial capital intensity, skill composition, and overall productivity levels.

[3] The adoption decision is assumed to be binary at the country level, but adoption is likely to be a gradual process in practice. The binary adoption decision captures how much the new technology changes the production process materially on a macro-relevant scale, which does not exclude the possible existence of smaller-scale AI use among pockets of firms in countries before they are simulated to adopt AI in the model.

[4] In the United States (and in contrast to the projected capital share of 37 percent in 2050), the mild, moderate, and accelerated scenarios correspond to capital shares of 40, 48, and 62 percent in the same year, respectively.

**Figure 8.2. AI Adoption Timing**




Source: Authors' simulations.
Note: The bars show the range of adoption years within income groups, and diamonds represent a single adoption year across countries in the group. The three AI scenarios differ by the speed at which the capital share increases in the frontier production, at a rate two times (2x), five times (5x), and 10 times (10x) the historical experience of the United States. AE = advanced economy; EM = emerging market; LIC = low-income country.

Countries with higher existing capital shares are already using capital as a major production factor. When AI increases capital share even more, these countries benefit more because the output gains accrue disproportionately to their major production factor. This creates a natural advantage for capital-rich economies such as Japan, Korea, and Singapore, where capital already accounts for 40 to 50 percent of output.

The skill-biased nature of AI reinforces these advantages. As AI technology progresses, the labor share in output declines for all skill groups, but high-skilled workers retain a larger fraction of the shrinking labor share. Countries with higher proportions of skilled workers therefore capture larger productivity gains from AI. Korea and Singapore are naturally positioned to benefit more from skill-biased technological change. According to the model definition, high-skilled workers make up 10 to 15 percent of the workforce in these two countries, compared with 2 to 5 percent in countries such as Bangladesh and Cambodia.

Capital intensity and skill composition interact with total factor productivity to determine each country's initial capital-to-labor ratio. Higher levels of the latter raise the productivity of both capital and labor, but when combined with high initial capital intensity, it results in countries being more capital-abundant relative to their labor endowments. As a result, countries such as Japan and Singapore,

which have both high total factor productivity and high capital share, maintain high capital-to-labor ratios that make AI profitable from the outset.

Demographics provide an additional push toward early adoption in advanced economies. Countries experiencing population aging or slow labor force growth naturally have higher capital-to-labor ratios, other things equal, as capital deepening outpaces labor supply growth. Japan and Korea face especially acute demographic pressures, with working-age populations projected to decline by 25 to 30 percent by 2050. These shrinking labor pools make capital-augmenting technologies especially attractive because they offer a way to maintain output despite declining labor inputs. By contrast, countries such as Bangladesh and the Philippines have young, growing populations that keep labor relatively abundant and capital relatively scarce, delaying the point at which AI becomes profitable.

The three scenarios highlight the significant uncertainty surrounding the potential economic divergence arising from uneven AI adoption. Under a mild AI development path, several of the region's emerging markets and developing economies begin adopting within the next 10 to 20 years. However, under faster development, early adopters would demand substantially more capital as the capital share in production rises more quickly, pushing up global interest rates more and thereby pushing adoption for some emerging markets into the 2040s and delaying adoption for low-income countries to the 2050s and 2060s. The order of adoption can also vary across different scenarios. For example, the Philippines would adopt earlier than Vietnam under the moderate scenario but later under the accelerated scenario.[5] These results suggest that the degree of cross-country divergence driven by uneven AI preparedness is highly sensitive to the pace of global AI progress, with faster progress potentially widening and prolonging cross-country gaps without policies that accelerate diffusion and strengthen complementary factors. The rankings of the simulated adoption timing do not necessarily align with those of survey-based adoption rates. The simulations identify at which point AI induces a meaningful, economy-wide shift in the production structure, whereas survey-based adoption measures often capture partial or early-stage deployment and could be upward biased because of sampling limitations, particularly the overrepresentation of large enterprises. Several factors not incorporated into the model could attenuate or reinforce the divergence, including AI-driven total factor productivity gains, technology transfer, and learning by doing, among others.

The simulated adoption timing aligns closely with countries' degree of AI preparedness. Countries scoring higher on the AI Preparedness Index—those estimated to be better prepared for AI adoption—adopt significantly earlier in

[5] This difference may reflect the interaction between delayed adoption and demographic trends: the longer the delay, the more demographic trajectories matter. In countries with growing labor forces, such as the Philippines, the capital-to-labor ratio rises only slowly. Vietnam's faster demographic transition, by contrast, lifts its capital-to-labor ratio sooner—so when adoption is pushed far enough into the future under the accelerated scenario, Vietnam crosses the adoption threshold earlier than the Philippines.

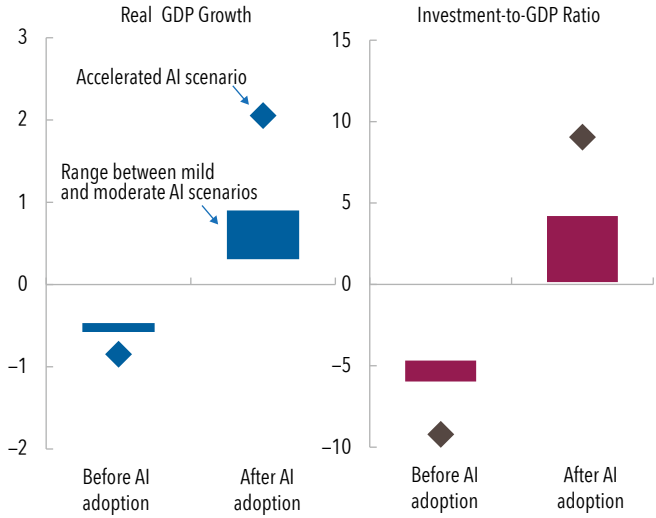
the simulations. Japan, New Zealand, and Singapore occupy the top tier of AI preparedness and adopt immediately. Middle-tier countries such as Malaysia, Thailand, and Vietnam show moderate preparedness and adopt in the 2030s and 2040s. Bangladesh, Cambodia, and Nepal are at the bottom of the preparedness distribution and face adoption delays until the 2060s or beyond. This correlation emerges from the economic logic embedded in both the AI Preparedness Index and the structural model. The AI Preparedness Index captures complementary capabilities that facilitate AI adoption, while the model-simulated adoption decision depends on factor endowments and productivity, outcomes that themselves depend on the same underlying capabilities. Digital infrastructure enables firms to deploy AI systems effectively, human capital determines the supply of skilled workers who can work along with AI technology, innovation capacity reflects the ability to adapt and improve technologies, and regulatory frameworks reduce the uncertainty and transaction costs associated with adoption. Countries with high scores on those dimensions also tend to have high capital intensity, skilled workforces, and elevated productivity that make AI adoption attractive in the model setup.

The differences in the timing of AI adoption translate directly into divergent growth trajectories across Asia and Pacific economies. Early adopters such as Japan, Korea, and Singapore experience an immediate growth increase of 0.5 to 1 percentage points annually in the mild and moderate AI scenarios. Given their high capital intensity, skilled labor forces, and aging demographics, adopting AI technology (switching to the more capital-intensive production function) would be profitable for firms in these countries immediately. This switch leads to faster accumulation of capital, with an investment-to-GDP ratio higher by 0.1 to 4 percentage points in the mild and moderate AI scenarios, which boosts growth. Immediate adoption creates a compounding advantage because the gains from AI adoption finance additional capital deepening that reinforces the countries' technological lead.

By contrast, emerging markets and developing economies do not adopt immediately. They experience lower growth by about 0.5 to 0.6 percentage points annually in the mild and moderate AI scenarios than the no-AI scenario during the pre-adoption period because they face headwinds from the changing global environment (Figure 8.3). As early adopters increase investment when switching to the more capital-intensive production, global demand for capital surges, driving up the global interest rate. Higher financing costs slow late adopters' capital accumulation, reducing investment (an investment-to-GDP ratio lower by 5 to 6 percentage points in the mild and moderate AI scenarios) and lowering growth.

The availability of AI technology and the resulting global competition for capital alters the convergence dynamics implied by the standard growth model, introducing a new channel through which economic divergence can arise. However, when late adopters eventually adopt, their growth rates recover and exceed the levels in the no-AI scenario—reflecting rapid catch-up capital accumulation under the new technology—and economic convergence dynamics reassert themselves. Over the long term, AI adoption can boost growth by 0.5 to 1 percentage point in

**Figure 8.3. Impacts on Growth and Investment of Emerging Markets and Developing Economies**
*(Percentage points)*




Source: Authors' simulations.
Note: Bars plot the range of impacts on real GDP growth and investment-to-GDP ratio between the mild and moderate AI scenarios, and diamonds plot the impacts in the accelerated AI scenario. The three AI scenarios–mild, moderate, and accelerated scenario–differ by the speed at which the capital share increases in the frontier production, at a rate two times, five times, and 10 times of the historical experience of the United States.

advanced economies and by 0.2 to 0.8 percentage points in emerging markets and developing economies in the region annually on average, depending on the pace of AI technological advance.

This asymmetry between early and late adopters highlights the risk of AI-driven divergence in the region. The technology does not simply create a one-time gap in income levels. It actively disadvantages late adopters during the transition through general equilibrium effects in global capital markets. Tighter financial conditions slow capital accumulation in late adopters, reducing their growth and pushing out their adoption further. Whether this divergence persists or eventually reverses depends critically on how quickly late adopters can overcome structural constraints and eventually adopt the new technology.

## Structural Reforms and the Pace of AI Adoption

The model simulations show that late-adopting emerging markets and developing economies in the region could face decades-long delays before AI adoption, during which they could experience slower growth. This raises a critical policy question

of whether structural reforms can accelerate adoption timing and amplify the benefits when adoption finally occurs. The analysis simulates a reform scenario for late-adopting countries that examines how improvements in economic fundamentals affect both the timing and magnitude of AI-induced growth gains.

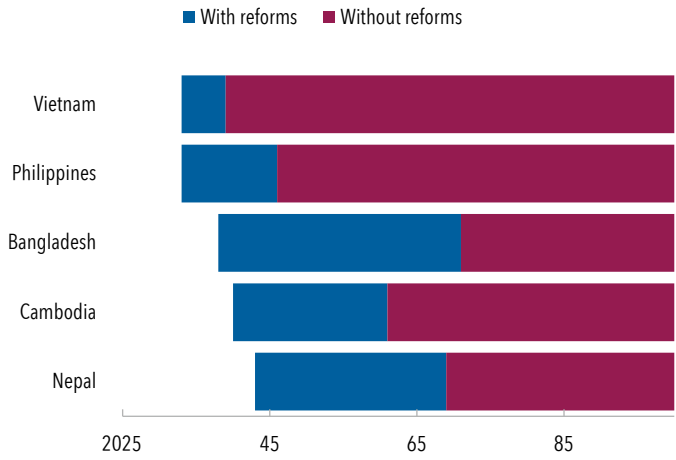
Enhanced human capital and productivity underpin the reform scenario and are calibrated to capture the cumulative effects of sustained policy efforts over a 10-year reform period (2026 to 2035). The high-skilled labor share is increased by 0.5 percentage points annually, representing a cumulative 5 percentage point increase in the proportion of workers with tertiary education or advanced technical skills by 2035. Labor-specific productivity is raised by 50 percent for high-skilled workers and 25 percent for low- and medium-skilled workers over the same period, reflecting improvements in worker effectiveness that go beyond formal education. Capital-specific productivity is raised by 50 percent, capturing enhanced efficiency in how physical and digital capital are deployed in production. These parameter changes can be mapped into real-world policy reforms across multiple domains. The increase in high-skilled labor share could be achieved through investments in human capital development, such as enhanced access to higher education, vocational training programs, and continuous skill upgrading initiatives that align workforce capabilities with the demands of advanced technologies. Improvements in labor-specific and capital-specific productivity could arise from a range of overlapping structural reforms, including institutional reforms that strengthen property rights, reduce corruption, and improve governance, thereby raising the efficiency with which both labor and capital are employed in production, financial sector reforms that deepen capital markets and improve credit allocation,[6] and infrastructure investments that raise the productivity of both physical and digital capital while also improving labor market matching and reducing transaction costs, among others.

The effects of structural reforms extend beyond simply accelerating adoption timing. Simulation results indicate a substantial acceleration in adoption for most late adopters (Figure 8.4). Countries that would otherwise adopt in the 2050s or 2060s can advance their adoption by 10 to 20 years through sustained structural reforms (see Box 8.3 for an example of AI adoption in Vietnam). This acceleration occurs because the reforms directly address some of the fundamental constraints that delay adoption: insufficient skill levels and low productivity. More importantly, the strengthened structural fundamentals amplify countries' growth gains from AI adoption, which demonstrates a double dividend from structural reforms. As found in the literature, structural reforms boost output and productivity directly (Budina and others 2023). Beyond that, in the face of AI progress, reform efforts also prepare countries better for adopting AI and therefore allow them to benefit from AI earlier and more. This underscores the critical importance of comprehensive structural reforms, especially against the

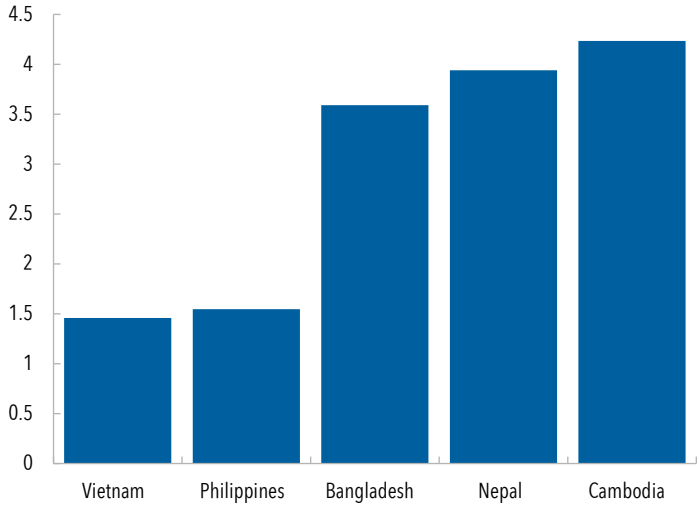
[6] IMF (2025) finds that capital misallocation has increased in the region in recent years, attributable partly to inefficiencies in financial intermediation: financial institutions favoring larger, less productive firms over smaller, more productive ones.

## Figure 8.4. Adoption Timing and Growth Gains with Structural Reforms

**1. AI Adoption Timing**
*(Adoption year under accelerated AI scenario)*




**2. Growth Gains from Earlier AI Adoption**
*(Percentage points, average per year)*




Sources: Authors' simulations.
Note: Panel 1 plots adoption years under the accelerated AI scenario (which assumes that capital share in the frontier production increases at 10 times the historical experience of the United States) with and without structural reforms. This includes increasing the high-skilled labor share by 5 percentage points, boosting labor productivity by 50 percent for high-skilled workers and 25 percent for low- and medium-skilled workers, and boosting capital productivity by 50 percent over 10 years. Panel 2 plots the average additional annual growth gains from earlier AI adoption.

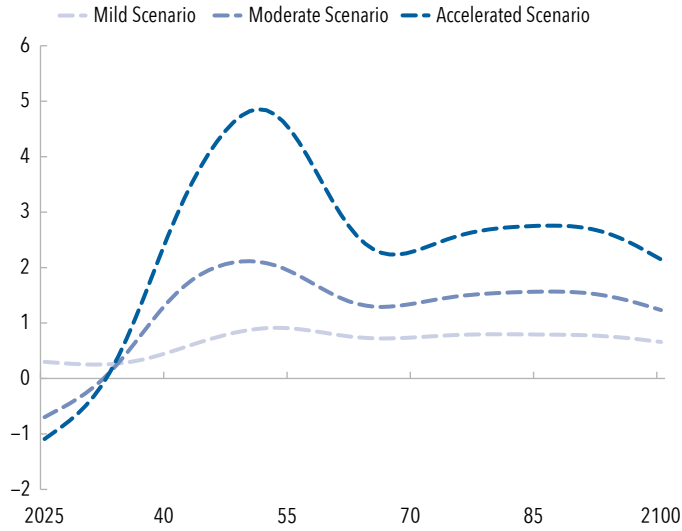
## Box 8.3. Macroeconomic Implications of AI Adoption in Vietnam

Despite initial headwinds from delayed adoption, model simulations suggest that Vietnam would catch up rapidly once AI technology is adopted extensively. During the pre-adoption phase, Vietnam would experience a temporary growth slowdown relative to the no-AI scenario—up to 1 percentage point lower under the accelerated AI scenario—driven by weaker capital accumulation amid tighter global capital markets. However, once AI is adopted extensively, growth accelerates significantly, underpinned by a structural transformation toward more capital-intensive and high-skilled production. This shift supports capital deepening, raises output per worker, and boosts GDP growth. Model simulations suggest that the initial growth boost after adoption could be as high as 1 percentage point under the moderate scenario and 2 percentage points under the accelerated scenario. As production transitions permanently to the more capital-intensive technology, the growth boost is sustained by about 0.7 percentage points (moderate scenario) and 1.5 percentage points (accelerated scenario) per year on average over the long term.

### Box Figure 8.3.1. Vietnam: Impacts of AI Adoption on Growth and Investment

**1. Real GDP Growth**

*(Percentage points, deviation from no-AI scenario)*

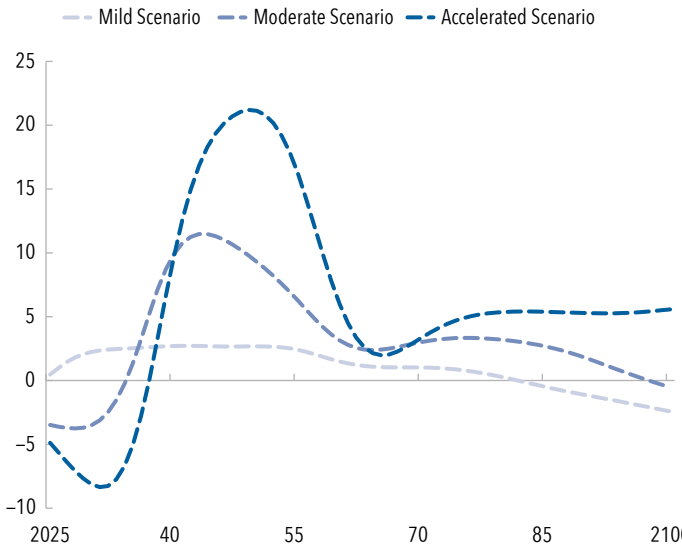



(*continued*)

**Box 8.3. Macroeconomic Implications of AI Adoption in Vietnam *(continued)***

**2. Investment-to-GDP Ratio**

*(Percentage points, deviation from no-AI scenario)*




Source: Authors' simulations.

Note: Panel 1 plots the growth impact–measured as percentage point deviation in growth rate from the no-AI scenario–for the mild, moderate, and accelerated AI scenarios, in which capital share in the frontier production increases at two, five, and 10 times the historical experience of the United States, respectively. Panel 2 plots the investment impact, measured as percentage point deviation in investment-to-GDP ratio from the no-AI scenario.

Structural reforms to boost productivity and enhance human capital could accelerate Vietnam's AI adoption significantly and magnify the growth dividend. These reforms raise Vietnam's readiness for AI, enabling adoption in the early 2030s under the accelerated AI scenario—more than five years earlier. In addition to the direct gains from structural reforms, earlier adoption mitigates the initial growth drag and raises growth by about 1.5 percentage points annually on average. These results highlight that even with remaining uncertainty about AI development and still nascent understanding of its full impact, taking a proactive approach with necessary comprehensive reforms could enhance the positive effects of AI adoption and enable better management of related risks.

backdrop of unprecedented technological progress, which requires strong economic fundamentals and capabilities to harness its benefits.

Model simulations suggest that AI is likely to be a powerful engine of growth and a source of potentially long-lasting divergence in the region unless late-adopting emerging markets and developing economies can overcome structural constraints through sustained reforms. Even in scenarios in which reforms succeed in

accelerating adoption and narrowing cross-country gaps, the transition is far from benign. The shift toward more capital-intensive, skill-biased production inevitably can reshape the income distribution and opportunities within countries. These distributional effects vary across skill groups, occupations, gender, and firms, making the right policy mix critical to ensuring that AI-driven gains are widely shared rather than concentrated.

## WITHIN-COUNTRY IMPACTS OF AI ADOPTION FOR ASIA AND PACIFIC

### AI Effects on Within-Country Inequality

AI adoption could worsen within-country inequality, despite long-term aggregate output gains. High-skilled workers are poised to gain disproportionately, especially those in AI-complementary roles, while low-skilled workers face greater displacement risk, which could widen the wage gap between skill groups (Jaumotte and others 2026). Model simulations suggest that soon after AI adoption, high-skilled workers' wages increase above their wages in the no-AI scenario, while medium- and low-skilled workers especially face lower wages for a longer period and slower wage increases than high-skilled workers, reflecting lower demand for medium- and low-skilled labor after AI adoption. All workers' wages would eventually rise above the no-AI scenario because aggregate labor productivity is lifted and aggregate output is boosted. However, wage gaps would widen across skill groups, and the faster AI advances, the larger and more persistent the gaps become. Moreover, wealth inequality could also be worsened by AI adoption and the resulting rising interest rates, especially in economies with concentrated asset ownership.[7] It is important to note, however, that the skill composition in the model is a stylized representation of heterogeneous exposure to and complementarity with AI. The simulations are therefore best interpreted as an illustrative mapping from a skill-biased, capital-deepening technology shock to relative wage dynamics, rather than as a point prediction for any specific country or occupational group. In practice, distributional outcomes are likely to be more nuanced and depend on the task content of jobs and the pace at which tasks are reorganized within occupations (as in the task-based framework of Autor and others (2003)), as well as on firm-level adoption patterns and the extent to which AI substitutes for versus complements particular tasks.[8]

Beyond the macro-level, model-simulated distributional effects, granular data on AI exposure by occupation, gender, and firm types also provide insights into the micro-level distributional consequences of AI adoption. AI exposure is slightly

[7] See Che, Xin, and Yoshida (2026).

[8] For example, some empirical studies find that lower-skilled workers experience higher gains from Gen-AI (Brynjolfsson and others 2025). Model-based studies also discuss a case where AI adoption leads to a decline in labor income inequality driven by displacement of high-skilled but low-complementarity workers (Rockall and others 2025).

lower in Asia and Pacific than in other regions because a larger share of workers are employed in low-exposure roles such as agriculture, but exposure varies widely across income groups within the region.

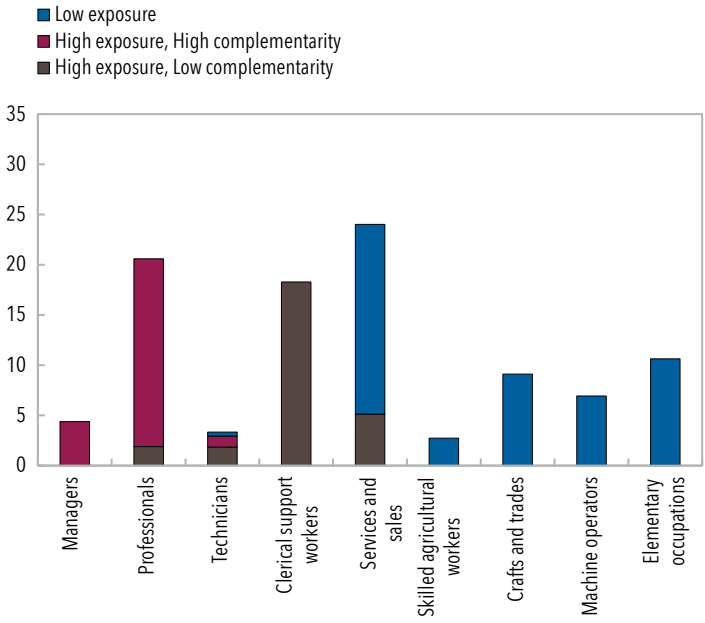
Workers in advanced economies are significantly more exposed to AI than those in emerging markets and developing economies (Figure 8.5). About half are employed in highly exposed occupations, and exposure is split relatively evenly between roles with high complementarity (AI can increase worker productivity) and roles with low complementarity (task automation and displacement risks are greater). High-skill occupations, such as managers and professionals, account for a large share of employment and tend to exhibit high levels of complementarity, suggesting considerable potential for productivity gains from AI augmentation. By contrast, clerical support workers are concentrated in highly exposed occupations with low complementarity, making them more vulnerable to automation and displacement. Even within skill groups, AI's effects vary across occupations; for example, AI is likely to be more complementary to surgeons and judges than to economists.

In emerging markets and developing economies, only about one-quarter of workers are in highly exposed occupations, reflecting a greater concentration of workers in low-skill and manual jobs with limited potential for AI integration. Employment remains heavily concentrated in agriculture, elementary occupations, and craft and related trades, all of which have relatively low AI exposure. Among the smaller share of highly exposed occupations, service and sales roles feature prominently and are often characterized by low complementarity, while managers and professionals—who tend to benefit more from AI augmentation—account for a much smaller share of employment than in advanced economies. As a result, emerging markets and developing economies face fewer near-term displacement risks from AI but are also less likely to realize substantial productivity gains unless structural transformation accelerates. These differences largely reflect variations in occupational structure and highlight the importance of education, skills development, and technological upgrading to support broader AI adoption and diffusion.

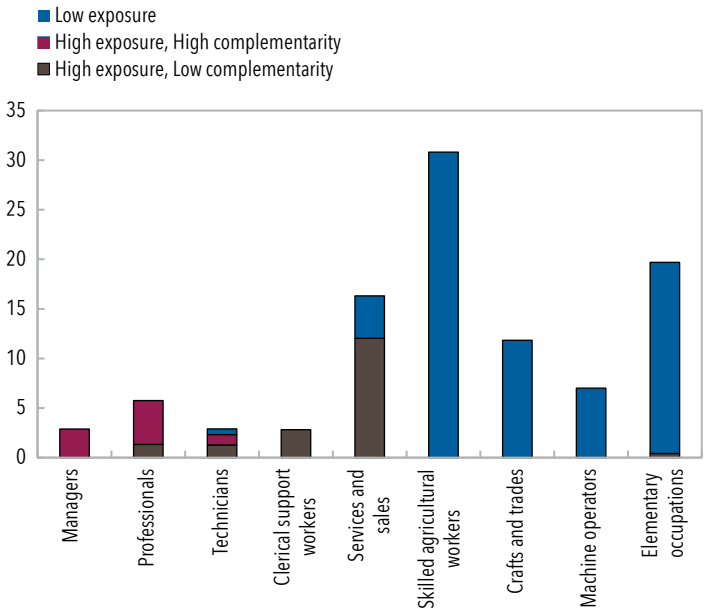
Differences in AI exposures across occupations also translate into gender disparities. Women in the region generally face higher AI exposure than men, particularly in advanced economies. Women are more concentrated in clerical support and certain service and sales occupations, roles that tend to have high AI exposure and low complementarity. As a result, women account for a larger share of employment in high-exposure, low-complementarity occupations than men. By contrast, men are more likely to work in lower-exposure occupations such as craft and related trades, plant and machine operation, and elementary occupations. At the same time, men and women are about equally represented in high-exposure, high-complementarity occupations, primarily professional and managerial roles, suggesting relatively equal access to potential productivity gains from AI augmentation. Overall, gender disparities in advanced economies stem less from differences at the top of the occupational distribution than from women's greater concentration in vulnerable, middle-skill occupations with high exposure and low complementarity. Gender differences are present but generally less pronounced in emerging markets and developing economies.

## Figure 8.5. AI Exposure and Complementarity by Occupation

**1. Advanced Economies**
*(Percent of workforce)*




**2. Emerging Markets and Developing Economies**
*(Percent of workforce)*




Sources: Cazzaniga and others (2024); and authors' calculations.

These heterogeneous AI exposures highlight the importance of skill transferability in mitigating labor market disruption, especially for workers in highly exposed occupations with limited potential for complementarity. The literature indicates that occupational mobility depends on the similarity of skills required across jobs. Using skill content data from the Occupational Information Network of Japan and drawing on Gathmann and Schönberg (2010), a measure of skill distance is constructed for Japan to explore the degree of skill transferability between occupation pairs (Asao and others 2025).

The results indicate substantial variation in skill transferability, emphasizing the need for policies that proactively support retraining and labor mobility (Table 8.1). The skill-distance analysis points to significant differences in task and knowledge requirements between occupations with high AI exposure and those with low exposure. For example, the task distance between professionals and cleaning occupations is 0.99, while the knowledge distance between sales and manufacturing workers is 0.8. These findings suggest that transitioning between these broad occupational categories would be challenging. The analysis is independent of labor market structure and holds even in a perfectly flexible labor market, so labor market rigidities would make skill transfer even more challenging. The results indicate that managerial and professional workers perform similar tasks and both are significantly exposed to AI with high complementarity. Although task requirements differ only moderately between highly exposed occupations with low and high complementarity, differences in knowledge requirements are more pronounced.

In Japan, transitions between occupations with high and low AI exposures occur relatively slowly. Less than 20 percent of workers newly hired into highly AI-exposed occupations came from low-exposure roles, and the share decreased to 10 percent for occupations with both high AI exposure and high complementarity. Conversely, about 30 percent of hires into high-exposure, high-complementarity occupations transitioned from roles with high exposure but low complementarity, consistent with earlier studies (Cazzaniga and others 2024b). Differences in task and knowledge requirements indicate that occupation-specific labor market policies could support reskilling efforts and facilitate worker transitions into high-demand roles.

In Korea, labor mobility across AI exposure and complementarity levels remains limited. Korea is among the global leaders in AI adoption, with higher exposure among women, younger people, more educated workers, and higher-income groups. College-educated workers in AI-intensive jobs tend to remain in similar occupation categories when changing jobs.

About one-third of workers leaving low-complementarity jobs move into higher-complementarity roles, suggesting a modest job ladder effect. However, mobility declines with age because both college- and noncollege-educated workers shift to low-exposure, routine jobs before retirement. Older workers also face greater difficulty returning to high-exposure roles after unemployment, with reemployment rates significantly lower than those of prime-age workers. Overall, men have higher reemployment rates, but women are more likely to transition into high-exposure, high-complementarity occupations.

**TABLE 8.1**

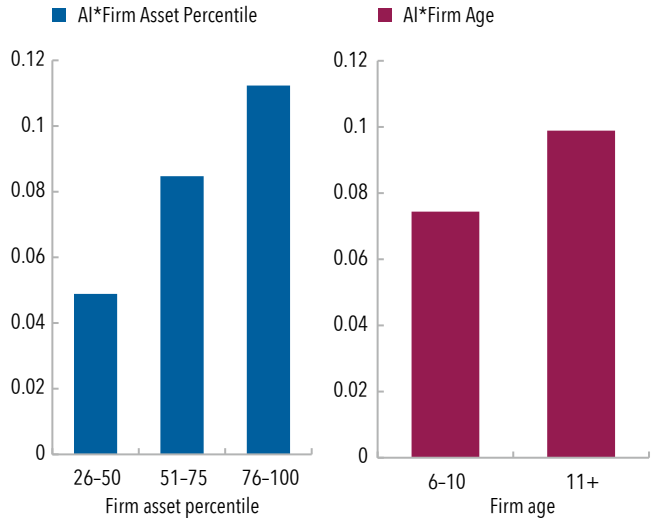
Japan: Skill Distance across Occupations

| *Knowledge* / *Tasks* | Administrative and managerial workers | Professional and engineering workers | Clerical workers | Sales workers | Service workers | Security workers | Agricultural, forestry and fishery workers | Manufacturing process workers | Transport and machine operation workers | Construction and mining workers | Carrying, cleaning, packaging, and related workers |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Administrative and managerial workers | **0.00** | 0.27 | 0.20 | 0.16 | 0.53 | 0.42 | 0.64 | 0.71 | 0.80 | 0.57 | 0.90 |
| Professional and engineering workers | 0.08 | **0.00** | 0.46 | 0.46 | 0.52 | 0.21 | 0.64 | 0.64 | 0.75 | 0.55 | 0.91 |
| Clerical workers | 0.17 | 0.18 | **0.00** | 0.23 | 0.43 | 0.38 | 0.69 | 0.80 | 0.68 | 0.68 | 0.65 |
| Sales workers | 0.22 | 0.33 | 0.09 | **0.00** | 0.33 | 0.64 | 0.58 | 0.79 | 0.73 | 0.69 | 0.72 |
| Service workers | 0.80 | 0.82 | 0.51 | 0.42 | **0.00** | 0.53 | 0.58 | 0.87 | 0.57 | 0.81 | 0.41 |
| Security workers | 0.14 | 0.07 | 0.22 | 0.39 | 0.76 | **0.00** | 0.65 | 0.65 | 0.48 | 0.54 | 0.66 |
| Agricultural, forestry and fishery workers | 0.71 | 0.74 | 0.94 | 0.85 | 0.58 | 0.71 | **0.00** | 0.38 | 0.39 | 0.42 | 0.35 |
| Manufacturing process workers | 0.64 | 0.61 | 0.90 | 0.90 | 0.76 | 0.61 | 0.12 | **0.00** | 0.36 | 0.21 | 0.38 |
| Transport and machine operation workers | 0.91 | 0.90 | 0.92 | 0.85 | 0.35 | 0.78 | 0.15 | 0.24 | **0.00** | 0.37 | 0.17 |
| Construction and mining workers | 0.43 | 0.48 | 0.78 | 0.71 | 0.81 | 0.50 | 0.17 | 0.13 | 0.41 | **0.00** | 0.49 |
| Carrying, cleaning, packaging, and related workers | 0.95 | 0.99 | 0.80 | 0.69 | 0.15 | 0.92 | 0.29 | 0.43 | 0.11 | 0.58 | **0.00** |

Source: Occupational Information Network of Japan; and authors' calculations.

Note: The skill distance index measures the degree of skill dissimilarity required by two occupations. Skill distance equals 0 if two occupations require the same skill set and equals 1 if two occupations require entirely different skill sets. Occupations in red have high AI exposure and complementarity, those in blue have high AI exposure but low complementarity, and those in black have low AI exposure.

**Figure 8.6. Korea: Productivity Gains**
*(Percent)*




Sources: Statistics Korea, Survey of Business Activities; and authors' calculations.
Note: Each bar plots the estimated coefficient of the respective interaction term: AI adoption interacting with firm asset percentile in the left panel and AI adoption interacting with firm age in the right panel.

These patterns suggest that enhancing occupational mobility, especially for older and less-educated workers, will be crucial for adapting to AI-driven structural changes and sustaining Korea's labor competitiveness.

AI adoption can also vary across firms and sectors, potentially widening productivity and income gaps. Using Korean firm-level data from the Survey of Business Activities (2017–22), an empirical analysis assesses the productivity effects of AI adoption across firms of different sizes, ages, and sectors. The estimates find that AI adoption has a significant positive effect on firm productivity and profitability, with productivity gains concentrated in larger, more mature firms, and with positive profitability effects for mature firms (Figure 8.6). Within industries, larger firms in professional, scientific, technical services, and construction sectors show productivity improvements, while mature firms in information and communication technology sector show substantial gains. Overall, large and mature firms in Korea are better positioned to benefit from AI adoption, while smaller and younger firms lag.

## Policy Options to Mitigate the Distributional Impacts of AI

Model simulations examine two alternative policy intervention programs—targeted transfers and universal transfers—to analyze whether redistribution policies can mitigate rising inequality without severely compromising growth in

three early adopters, Japan, Korea, and Singapore. The targeted transfers policy directs additional benefits to only low- and medium-skilled workers whose wages fall below a certain threshold, scaled inversely with income. This captures means-tested programs such as earned income tax credits, conditional cash transfers, and enhanced unemployment benefits. The universal transfers provide unconditional flat transfers to all households regardless of skill level or income. Unlike targeted transfers, universal transfers do not phase out with income, which avoids the implicit marginal tax rates that means-tested programs create but also provide benefits to high-income workers who do not need support. Both policy programs are financed through the endogenous tax system, including taxes on consumption and labor income, to maintain a balanced fiscal budget, with the same total cost of 5 percent of GDP annually, allowing for direct comparison of their redistribution and growth effects.[9] The simulations—including the cost at 5 percent of GDP and results—should be interpreted as scenario-based illustrations of policy trade-offs, rather than policy prescriptions, as real-world policy design involves many considerations, which depend on the objectives.

Simulations show an equity-efficiency trade-off from redistributing AI gains, the magnitude of which depends on the country and policy design. In economies such as Japan and Korea that face rapid aging, shrinking workforces, and already high tax wedges, targeted transfers that concentrate support on low- and medium-skilled workers can mitigate the rising inequality more effectively. However, they also produce labor supply disincentives where marginal hours are most valuable. Simulations show that for Japan and Korea, targeted transfers result in modestly lower GDP compared with universal transfers, despite the fact that the resulting GDP level is still much higher than that in the no-AI scenario. Targeted transfers compress the Gini coefficient more than universal transfers because the transfers direct resources only to households that experience relative wage declines, while universal transfers also channel resources to high-skilled workers who benefit from AI and have disproportionately higher wages. However, financing the targeted program would require higher taxes on high-skilled workers, which reduces high-skilled labor supply, while larger targeted transfers amplify income effects for low-skilled recipients—both of which depress aggregate hours worked. By contrast, universal transfers at similar fiscal cost achieve a more modest reduction in inequality but entails smaller distortions to labor supply, because transfers are spread more evenly and do not need to be targeted as narrowly. The negative impact on growth is amplified when existing taxes are high because the marginal distortion from raising rates is nonlinear. In this environment, a universal program financed at the same budget cost preserves marginal

[9] Maintaining this equivalent fiscal envelope is important for two reasons: first, it isolates the efficiency trade-offs of the transfer design; and second, given that the targeted low- and medium-skilled groups comprise the vast majority of the workforce (over 90 percent of total workers in most economies as calibrated in the model, and about 70-90 percent as found empirically in the AI exposure analysis), a substantial fiscal commitment would be needed to reduce inequality caused by AI even in the targeted transfer scheme.

work incentives across the distribution, yielding smaller reductions in inequality but also smaller GDP declines relative to targeted transfers.

However, where fundamentals are more favorable, the trade-off is less pronounced, and well-designed targeting can mitigate the distributional impacts with limited costs to growth. Countries such as Singapore that have lower starting tax rates, less adverse demographics, and a lower share of low-skilled labor can finance a given envelope of redistribution with smaller increases in marginal tax rates. Their broader and more dynamic tax bases allow them to raise additional revenue without sharply increasing the tax wedge on high-skilled workers, thereby limiting disincentives for the workers who are central to AI-driven productivity gains. If the share of low-skilled labor is also smaller, the fiscal burden of means-tested support falls, allowing transfers generous enough to materially reduce inequality without eroding aggregate labor supply. As simulated by the model, the impacts on Singapore's GDP under the targeted transfer program are similar to those under universal transfers, unlike in Japan and Korea.

Several additional considerations are important for practical implementation. The simulations are stylized, focusing on long-term aggregate GDP under a relatively large, fixed fiscal envelope and abstract from institutional design features, political economy constraints, and social objectives that are central to real-world policymaking. In practice, policymakers could care as much about who is protected and the social and political benefits of shielding the workers and households most exposed to AI-related displacement—dimensions that the model does not fully capture. In addition, the quantitative exercise assumes generic transfer instruments and could overstate labor supply disincentives from targeted programs in settings where program design features (such as conditionality, time limits, and robust active labor market policies) can help contain work disincentives. Overall, even though the model points to a clear equity-efficiency trade-off at the aggregate level, country preferences and institutional realities can justify a policy tilt toward more targeted support, even if this entails somewhat larger output costs than a universal transfer.

## CONCLUSIONS AND POLICY IMPLICATIONS

AI can be a powerful new engine of growth in Asia and Pacific, but the transition could be uneven and divergent, at least for a time. Structurally prepared economies—typically aging advanced economies with high capital intensity, strong human capital, and robust productivity—will adopt early and reap growth gains immediately, while many emerging markets and developing economies tend to adopt much later. The late adopters will miss out on AI's direct benefits and be squeezed by tighter global financial conditions because early adopters' investment pushes up global interest rates, which weighs on emerging markets and developing economies' investment and growth. Model simulations and granular evidence at the occupation, gender, and firm levels also show risks of inequality rising within economies: high-skilled workers and large, mature firms capture a

disproportionate share of the gains, while lower-skilled workers, women in vulnerable middle-skill roles, and smaller firms bear the burden of adjustment.

Well-designed structural reforms and redistributive policies can prevent countries from being left behind, while mitigating within-country distributional pressures. A simulated reform scenario, calibrated to a 10-year push to upgrade skills and productivity, shows that sustained improvements in human capital and in labor- and capital-specific productivity can advance AI adoption by 10 to 20 years for many late adopters and substantially magnify the eventual growth dividend, underscoring the importance of structural reforms in reaping AI's potential. The model-based comparison of targeted transfers and universal transfers for early adopters shows that no single instrument fits all circumstances. In aging, high-tax economies, well-targeted transfers compress inequality more effectively, but they also create larger labor supply distortions and somewhat lower GDP than a universal program with the same budget envelope. Such trade-offs are less pronounced in lower-tax, younger economies, where well-designed targeted transfers have limited growth costs because lower initial tax wedges and smaller shares of low-skilled workers allow for raising additional tax revenue more easily while not reducing work incentives as much.

These findings present important policy implications. Strengthening AI preparedness—especially in late-adopting emerging market and developing economies—will require broad-based structural reforms that address multiple constraints simultaneously. A comprehensive reform agenda includes scaling access to quality education from secondary (and primary where relevant, such as in countries that face a generational gap in AI adoption) through tertiary and technical and vocational training. It also includes investing in digital and physical infrastructure that raises the productivity of both workers and capital, and improving the business and institutional environment (governance, regulation, and financial intermediation) to enable firms to deploy new technologies efficiently. Priorities may differ given competing spending needs: low-income countries should first close basic education and connectivity gaps; emerging markets should scale digital infrastructure alongside reforms that crowd in private investment; and advanced economies should focus on frontier R&D, advanced skills, and regulatory frameworks.

Labor market policies should focus on mobility and skill transferability so that workers can move toward occupations and tasks that benefit from AI. Transitions across broad occupational categories are often difficult, although some pathways are more feasible, as indicated by the skill-distance analysis for Japan and by labor mobility patterns in Korea. This points to the need for active labor market policies and lifelong learning systems explicitly designed for realistic transition paths, including modular reskilling and upskilling programs aligned with employers' needs, job search and placement support that prioritizes moves along high-transferability paths, and targeted assistance for groups facing greater displacement risks, such as mid-career women and older workers. During the AI transition, expanding access to a broader range of AI applications, including those that could complement low-skilled workers and raise their productivity, can also help

mitigate displacement risks and contain inequality pressures. Efforts to reduce the high informality in the region should also be accelerated, as this would help enhance the effective implementation of social programs.

Policies that broaden firm-level adoption are essential to avoid a scenario in which AI widens productivity gaps between firms and sectors. To prevent such outcomes, governments could expand access to finance for small and medium enterprises through expanded credit guarantee programs, tailored innovation and digitalization funds, simplified collateral and insolvency frameworks, and the use of blended finance and financial technology solutions that lower the cost and risk of lending to smaller firms. At the same time, it is important to leverage AI to reduce the high informality among small and medium enterprises in the region, which would in turn help strengthen small and medium enterprises' overall access to finance and advanced technology. Improved digital infrastructure would also benefit smaller and late-adopting firms by lowering the fixed costs of AI adoption. Competition and innovation policies that promote interoperability, data portability, and fair access to foundational models can also help ensure that smaller firms can reap AI-enabled productivity gains and avoid being locked out by first movers.

Fiscal and social protection policies are essential for ensuring that the benefits of AI are shared within countries while preserving incentives to work, invest, and innovate. The overarching lesson is that redistribution can mitigate AI-related inequality without severely compromising growth but only if policy instruments are tailored to demographic structure, fiscal space, the design and capacity of existing tax and transfer systems and are anchored in credible medium-term fiscal frameworks.

Underlying all of these efforts is the need for effective governance and risk-management frameworks that sustain public trust and ensure that AI's macroeconomic benefits are durable. Key elements include robust data governance and privacy safeguards, proportionate regulation of high-risk AI applications, and strengthened cybersecurity and operational risk management, particularly in finance and critical infrastructure. As countries in Asia and Pacific develop their AI strategies, integrating these safeguards with the structural, labor market, and fiscal policies described will be essential.

AI holds the promise of helping aging economies sustain growth and younger economies create better jobs if governments act early to close preparedness gaps, support workers and firms through the transition, and design redistribution and governance frameworks that align AI adoption with both efficiency and inclusion.

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CHAPTER 9

# Digital Payments to Boost Productivity and Advance Inclusive Growth

**Paula Arias, Alexander Copestake, Corinne Deléchat, Divya Kirti, Anne Oeking, Maria Soledad Martinez Peria, Dinar Prihardini, Nur Adhi Purwanto, Ursula Wiriadinata, Martha Woldemichael, and Ying Xu**

## INTRODUCTION

Economic growth in the Asia and Pacific region has been slowing since the global financial crisis, partly because of a persistent decline in productivity growth, and the slowdown has deepened since the COVID-19 pandemic. As economies search for ways to reignite dynamism, digitalization has become a central theme. From computer technologies to e-commerce and industrial robots, technology is reshaping how businesses and households interact. Digitalization can raise total factor productivity by reducing transaction friction, improving resource allocation, and enabling innovation spillovers (Dabla-Norris and others 2023).

Within this broader transformation, digital payment systems have become foundational economic infrastructure that shape how value moves through an economy. Acting as rails for the digital economy—the underlying infrastructure that moves money between user—these systems reduce the cost and time of moving value, formalize transactions and data, enable other digital services to function efficiently, and connect domestic and cross-border markets—all of which reduce transaction friction and facilitate financial flows, supporting firm-level productivity and enabling full participation of households and smaller firms in the formal economy.

Digital (or electronic) payments are cashless transfers of value using digital channels such as the internet, mobile and smartphones, and other digital devices. These include traditional, mainly bank-based digital payment methods such as online banking, credit or debit cards, mobile wallets, and newer retail payment methods that make funds available to beneficiaries almost immediately at any time, such as mobile money and quick response (QR) codes. Digital payments also include crypto assets, such as bitcoin, stablecoins, and central bank digital currencies (Aguilar 2024).

Digital payment types can be classified by use case, payment interface, money instrument, and other characteristics. Use case distinctions include retail payments

(such as person-to-person or government-to-person) and wholesale payments (between banks, financial market infrastructures, and other financial institutions). Public and private sector providers usually operate retail payments, and central banks typically own and operate wholesale payment systems (Ehrentraud and others 2021). Payment interfaces—how the payment is initiated—include card-based payments, instant payment systems, QR codes, and in-app payments. Money instruments include bank deposits, e-money, central bank digital currencies, and stablecoins. Other classifications focus on technology, settlement type, or service channel and access point (Adrian and Mancini-Griffoli 2019; Ehrentraud and others 2021).

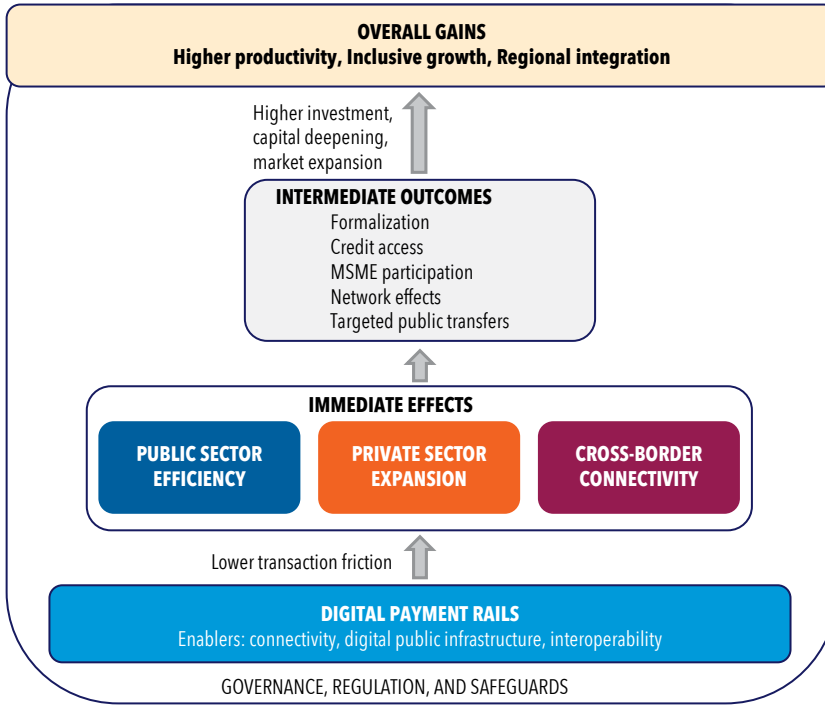
The analysis here focused on digital payment types with the greatest macroeconomic and policy relevance for the Asia and Pacific region, including domestic instant account-to-account payments, QR merchant payments (including cross-border QR), digital government payments (government-to-person and person-to-government), and real-time cross-border retail links. These payment types are scaling rapidly, exhibit strong network effects, and generate inclusion and productivity gains for households, small and medium enterprises, and governments. Wholesale central bank digital currencies frame the path toward multilateral interoperability and local currency settlement.

Alternative payment methods enabled by new technologies are mostly outside the scope of the analysis. Crypto assets (including bitcoin) have grown in market value, but their volatility limits their practicality for everyday payments. Stablecoins aim to maintain a fixed value against an asset or basket of assets and could become involved in payments over time, but adoption in the region is limited (Ehrentraud and others 2021). Large-scale implementation of central bank digital currencies remains pending across the region, but in time, they could provide the speed, efficiency, and inclusion benefits of existing digital payment instruments while offering unique advantages such as public trust and stability. Making payments faster, less costly, and more accessible delivers immediate- and longer-term gains. In the short term, digital payments enable faster government transfers and revenue collection, quicker settlement for businesses, and less costly cross-border payments (Figure 9.1). They also enable more small businesses and households to participate in the formal economy, improve access to credit, and drive network effects as adoption grows. These dynamics can boost investment, enhance transactions' efficiency, and expand markets, yielding system-wide dividends such as higher productivity, more inclusive growth, and stronger regional integration (Aguilar and others 2024; Cormier and others 2025).

Many economies and firms in the Asia and Pacific region are highly digital and at the technological frontier. But adoption remains uneven across and within countries, so that diffusion to lower-income economies, underserved households, and small and medium enterprises can provide large dividends. Without progress along these dimensions, countries risk foregoing potential revenue gains and entrenching persistent informality and fragmented payment corridors, undermining competitiveness and regional integration.

Realizing the efficiency and inclusion benefits of digital payments requires a strong enabling environment that incentivizes private sector innovation and

**Figure 9.1. How Digital Payment Rails Drive Productivity and Inclusion**




Source: Authors' illustration.
Note: MSME = micro, small, and medium enterprise.

delivery. Deliberate government policies can ensure competition, interoperability, and trust. Case studies illustrate the efficiency gains from digital payments and the policies needed to unlock their full potential, highlighting how the public sector can enable successful digital payment systems and support productivity and growth dividends.

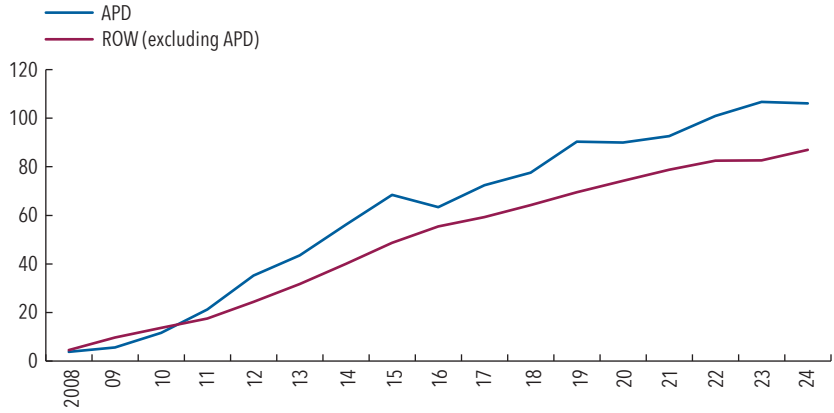
## DIGITAL DIVIDENDS AND THE DIGITAL FRONTIER

### The Digital Transformation of Asia and Pacific

The Asia and Pacific region has leapt ahead in digital connectivity, often surpassing global peers.[1] Active mobile broadband subscriptions rose sharply after 2010 and exceed one per person in most of the region's economies, placing it ahead of the rest of the world (Figure 9.2). As a result, the mobile sphere has become the backbone of digitalization in the region. Internet use, once lagging, has caught up rapidly,

[1] Usage data are based on the International Telecommunication Union's regional indicators.

**Figure 9.2. Digital Connectivity: Active Mobile Broadband Subscriptions**
*(Per 100 inhabitants, simple average)*



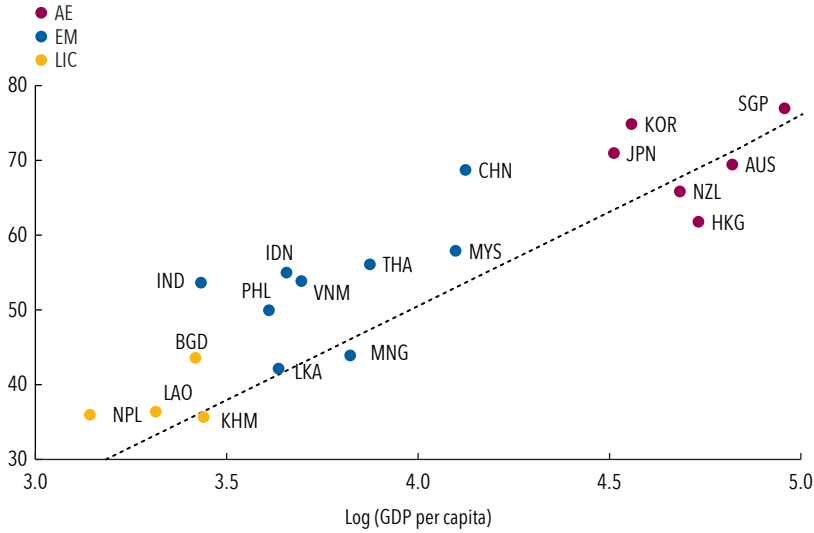

Source: International Telecommunication Union.
Note: APD = Asia and Pacific Department; ROW = rest of the world.

with about 90 percent of the region's population online as of 2018-23, nearly matching the highest penetration levels globally. The region's innovation edge has also strengthened, accounting for more than two-thirds of global patent grants in digital communications, computer technology, and information technology methods for management, including innovations in payment systems and financial technology (fintech).[2]

The region's strong digital use and innovation reflect widespread access to the internet and digital technologies. Most Asia and Pacific economies have higher levels of network readiness than global peers with similar income levels. Indicators such as the Network Readiness Index and the Mobile Connectivity Index capture multiple dimensions of digital preparedness such as infrastructure quality, affordability, skills, and governance. Both indices point to robust digital foundations across much of the region. Readiness naturally rises with income, yet most of the region's economies are above the trend line for global income readiness, suggesting that the region has a digitalization comparative advantage over the rest of the world (Figure 9.3). Although advanced economies in the region are at the frontier of network quality and coverage, several emerging markets and low-income economies lag but are catching up through continued improvements in mobile reach, affordability, and service quality. In many of these cases, similar strengths also support digital payment readiness: inclusive

[2] The IMF staff calculations used data from the World Intellectual Property Organization. The World Intellectual Property Organization defines information technology methods for management as data processing methods adapted for administrative, commercial, financial, managerial, supervisory, or forecasting purposes. This category includes patents related to payment systems, banking, insurance, and tax technologies. The organization does not publish a dedicated fintech field, but this category was used as a proxy for fintech-related innovation.

**Figure 9.3. Network Readiness Index, 2024**




Sources: Portulans Institute; and Saïd Business School, University of Oxford.
Note: The trend was calculated based on world observations. Data labels in the figure use International Organization for Standardization country codes. AE = advanced economy. EM = emerging market. LIC = low-income country.

mobile infrastructure and public-private collaboration are expanding access to transaction accounts and fast payment services.

Building on those strong foundations, digital payment adoption across Asia and Pacific has accelerated rapidly though with uneven progress. Advanced economies such as Japan, Korea, and Singapore are at the global frontier, combining near-universal account ownership (both bank or similar financial institution accounts and mobile money accounts) with highly interoperable real-time payment systems. Several emerging markets such as the Chinese mainland, India, and Thailand have also advanced to frontier levels, supported by mobile-based platforms, digital identification systems, and proactive public-private collaboration. Adoption has been slower in mostly lower-income economies and some emerging markets, constrained by gaps in identification systems, account ownership, mobile connectivity, and higher data costs. Data also show gaps within countries. For example, internet access rates for urban households are about 6 to 10 percentage points higher than rural access rates in Indonesia, Malaysia, Thailand, and Vietnam and nearly 26 percentage points higher in Bangladesh.[3] Overall, the region's experience shows that factors beyond income are important for widespread digital payment adoption.

[3] Data are from the International Telecommunication Union Global and Regional ICT database, 2023 values.

## Digital Dividends across Asia

Immediate benefits from digital payment adoption materialize through gains in public sector revenue and spending efficiency, business and income growth, economic resilience, financial inclusion and deepening, and global and regional payment connectivity. A growing body of empirical evidence across these dimensions highlights the transformative effects of digital payments.

In the public sector, digital payments strengthen revenue mobilization and spending efficiency. Online tax filing and payments reduce compliance costs and, with adequate incentives, can expand the tax base by bringing informal firms and households into the formal tax system, thereby increasing revenue collection (Deléchat and Medina 2021). Studies have shown that mandatory e-filing and e-payment can increase tax revenue collection by 4 to 5 percent of GDP (Alonso and others 2023; Nose and Mengistu 2023). Electronic verification and transfer of social safety net benefits reduce leakage and payment delays (Muralidharan and others 2016, 2025; Bird and Hanedar 2023). As revealed during the COVID-19 crisis, access to a registry or ID system allows governments to use digital payments to provide rapid support to vulnerable households and firms in the informal sector during a crisis (Deléchat and Medina 2021). Digital payment platforms can also strengthen governance through greater transparency in public service delivery (Box 9.1).

### Box 9.1. Sri Lanka's GovPay Initiative

Sri Lanka's GovPay initiative illustrates how digital payments can strengthen fiscal governance and improve public service delivery. Launched in February 2025, GovPay is a centralized platform that enables real-time digital payments for government services—including local council taxes, traffic fines, permits, and educational fees—through online banking and digital wallets. By November 2025, GovPay had processed 50,602 transactions totaling 1.4 billion Sri Lankan rupees (about $4.6 million) across more than 200 government agencies, marking an early milestone in Sri Lanka's digitalization of public finances. GovPay operates on the LankaPay Online Payment Platform, which has integrated government agencies into Sri Lanka's instant interoperable payment system, the Common Electronic Fund Transfer Switch, since 2017 (see Box 9.3).

Although still in its early stages, GovPay has potential to modernize revenue collection and service delivery significantly by reducing cash handling, limiting in-person transactions, and lowering administrative costs. It is also expected to improve bank liquidity by bringing informally held cash into the formal financial system. Its greater transparency and traceability can help reduce leakages and corruption vulnerabilities while providing faster, more reliable access to public services. Sustaining this progress will require continued efforts to expand digital financial literacy, especially in rural areas, to encourage adoption and build public trust. Strengthening regulatory compliance, ensuring cybersecurity, and investing in reliable digital infrastructure—including affordable, high-quality internet connectivity—will also be essential to unlock GovPay's full potential and support broad, equitable participation in Sri Lanka's digital transformation.

In the private sector, digital payments drive business growth and promote economic resilience. Digital payments enhance payment efficiency and shopping convenience, strengthening aggregate demand. In India, digital payment adoption has increased aggregate consumer spending and stimulated growth (Dubey and Purnanandam 2023; Agarwal and others 2024). Similarly, QR code payments in Singapore have led to significant aggregate business growth, especially for small and new merchants (Agarwal and others 2019; Agarwal, Alok, and others 2020; Agarwal, Qian, and others 2020). In the Chinese mainland, digital payments and e-commerce have reduced barriers to entry, especially for small firms in remote locations, improving aggregate welfare (Fan and others 2018).[4] Digital payments in India have enabled better risk sharing and income resilience for households and firms (Patnam and Yao 2020; Agarwal and others 2023). Outside the region, similar results have been found for Kenya and other countries in Africa (Jack and Suri 2014).

The region has room to expand government and private sector digital payments. For example, about one-third of countries in East Asia and one-fifth of countries in South Asia still deliver social security payments only in cash. In revenue administration digitalization, the region's emerging markets and low-income economies lag their peers, with only 21 percent requiring e-filing and 18 percent requiring e-payments, compared with 42 percent and 25 percent, respectively, in emerging markets outside the region.[5] In the private sector, e-commerce in the Chinese mainland has grown, with online purchases and accompanying online payments far more widespread than the region's emerging-market average of less than 40 percent of adults having made an online purchase and slightly more than 10 percent for the region's low-income economies (World Bank 2022).[6]

Digital payments expand access to financing, which is a major obstacle for firms and households in the region. A growing body of evidence shows that adopting digital payments leads to higher credit to the private sector, especially to more constrained borrowers (Bachas and others 2018, 2021; Agarwal and others 2023; Huang and others 2024; Copestake, Kirti, and Martinez Peria 2025; Copestake and others 2025). Credit growth is strongest in regions with affordable internet and where fintech lenders lead in credit expansion to new customers (Alok and others 2024). Digital payment platforms make large amounts of transaction data available and accessible at low cost (including utility bill payments and online purchase histories), thus reducing information asymmetries and adverse selection, and opening access to affordable credit. With so-called super apps' platforms, fintech lenders can process loan applications almost instantaneously with big data at their fingertips and reach households and firms with limited access to traditional banking services. Borrowers with a higher share of cashless transactions are more likely to receive loan approvals, obtain larger loan amounts, and secure lower interest rates (Ghosh and others 2026).

[4] See Copestake, Estefania-Flores, and Furceri (2024) for cross-country evidence.

[5] Data are from the International Survey on Revenue Administration database, 2024.

[6] Data are from the Global Findex Database 2025.

Digital payments are especially important in the region because the degree of financial inclusion still differs sharply within and across regional economies. Yet improvements to financial inclusion have been rapid. High-income and frontier economies have near-universal financial account ownership, but the region's emerging market and low-income economies are still catching up, with the share of adult account holders rising from 46 percent to 62 percent in the past decade. This expansion has occurred along with a sharp rise in digital payment adoption, doubling from 24 percent to 48 percent, with many individuals using an account to receive wages, remittances, or government transfers.[7] However, aggregate progress masks persistent within-country divides. Rural-urban gaps in internet use still reach 15 to 35 percentage points in several economies. Informal workers and rural households remain less likely to own an account or use digital payments regularly, and gender differences in internet access exist in some countries.[8] Closing these remaining gaps will be crucial to ensuring that expanding digital access translates into inclusive participation in financial and economic activity.

Digital payments can also promote cross-border connectivity. Several of the region's economies lead in digital cross-border payments, yet the region can still benefit from expansion because a faster, less costly, safer, and more inclusive cross-border digital payment system can lift regional trade and financial integration (Cortes and others 2024). Countries that rely heavily on tourism, migrant labor, and remittances can benefit from faster and more affordable cross-border payments. In Association of Southeast Asian Nations (ASEAN) member countries, for example, tourism accounted for 8.3 percent of the region's GDP and 12.4 percent of employment in 2023.[9] In addition, cross-border digital payment systems can benefit small and medium enterprises directly by enabling them to expand their reach beyond domestic markets. For example, using the PromptPay-PayNow link between Thailand and Singapore, a merchant in Bangkok can sell to a customer in Singapore and receive payment in local currency almost instantly and at low cost, without navigating complex international banking procedures.

Digital cross-border payments in local currency can reduce foreign exchange risks and enhance financial stability. Asia and Pacific is particularly exposed to US dollar volatility because it is the main export invoicing currency (Li and Meleshchuk 2024). In response, several Asian economies are prioritizing the local currencies in cross-border payments, leveraging digital payment technologies to strengthen financial resilience and reduce vulnerabilities to external shocks. Purwanto and Xu 2025, based on bilateral cross-border QR payment data in

[7] Data are from the Global Findex Database 2025.

[8] By contrast, mobile phone use is much more even, with small gaps in Indonesia and Vietnam and virtually none in Japan, Korea, and Thailand, pointing to the democratizing role of mobile connectivity in expanding digital participation. Data are from the International Telecommunication Union Global and Regional ICT database. Values correspond to latest available information from 2018 to 2023.

[9] About 40 percent of tourists arrive from Association of Southeast Asian Nations (ASEAN) countries and almost 80 percent arrive from the broader Asia and Pacific region.

Thailand, suggest that higher volatility between the US dollar and local currency leads to more cross-border QR payments, whereas higher bilateral local currency volatility does not. This result suggests that by avoiding fluctuations in the US dollar, QR payments help reduce foreign exchange risk and provide real-time transparency into the local currency cost of payments. By contrast, credit and debit cards transactions require several days to settle, and during that time, the payer must bear the foreign exchange risk, which increases when US dollar volatility is higher.

Recognizing these benefits, countries in the region have established multiple bilateral cross-border payment links. In ASEAN, the QR payment connectivity allows travelers to make real-time payments to a merchant in another country by scanning a QR code. Although still small in value and volume, QR cross-border payments in ASEAN countries have already surged. For example, QR payments increased by 550 percent in Malaysia and by more than 300 percent in Thailand in 2024 compared with 2023. Another payment type, fund transfer connectivity, facilitates instant remittances and fund transfers across borders between individuals using simple identifiers such as a mobile phone number or national ID.[10] Similarly, in June 2025, the People's Bank of China and the Hong Kong Monetary Authority launched Payment Connect, which enables quick cross-border transfers between Hong Kong Special Administrative Region and the Chinese mainland using only a mobile number or bank account. The Chinese mainland also introduced the Cross-boundary Interconnection Payment Gateway to allow foreign e-wallets and cards to operate seamlessly within the Chinese mainland.[11]

## ENABLERS OF DIGITAL PAYMENT BENEFITS

Unlocking the full benefits of digital payments requires both technology and an enabling policy environment. Governments must lay the foundations for adoption and trust, from supporting robust digital infrastructure to setting interoperability standards and protecting users and consumers. These policies create conditions for private innovation to flourish and adoption to spread, ensuring that digital payments drive inclusion, efficiency, and growth.

[10] These recent developments in digital cross-border payments in Asia have aligned with global and regional initiatives. The Financial Stability Board's *G20 Roadmap for Enhancing Cross-Border Payments* serves as a framework to address long-standing friction such as high costs, slow transaction speeds, limited accessibility, and a lack of transparency. Along with this global effort, the ASEAN region is at the forefront of implementing a practical, regional response through its strategic initiatives, particularly the ASEAN Regional Payment Connectivity and the ASEAN Local Currency Transaction Framework initiatives.

[11] Outside of formal corridors, global platforms such as Alipay and WeChat Pay enable quick response payments abroad through commercial partnerships, underscoring the role of private networks along with official programs.

To enable the full benefits of digital payments, ensuring connectivity is critical—universal, affordable access to reliable networks makes digital transactions possible. Governments must also invest in digital public infrastructure such as secure identity systems, instant payment rails, and data-sharing mechanisms, to provide the foundational building blocks. Policies that mandate interoperability domestically and across borders are essential to avoid fragmentation and unlock network effects. Strong governance, regulation, and safeguards covering consumer protection, cybersecurity, data privacy, and fair competition support the entire system and help ensure trust and resilience.

## Digital Connectivity

Network infrastructure and digital connectivity encompass the physical and digital systems that connect users and providers, including broadband networks, mobile data services, cloud platforms, and data centers and thus are the first building blocks for digital payments. The public sector is pivotal to ensuring competition, promoting investment in communication infrastructure, and adopting strategies to close connectivity gaps across regions (OECD 2025).

Progress in infrastructure and connectivity has been rapid in Asia and Pacific, but differences in access remain across and within countries, particularly at different income levels. Advanced economies have achieved near-universal, high-quality connectivity that supports real-time payments and digital services, whereas affordability gaps continue to limit usage in many emerging markets' and low-income economies' network coverage, particularly in rural areas. Strengthening broadband infrastructure, ensuring reliable mobile data access, and expanding cloud capacity are essential to sustaining digital payment adoption and laying the groundwork for broader digital public infrastructure.

## Digital Public Infrastructure

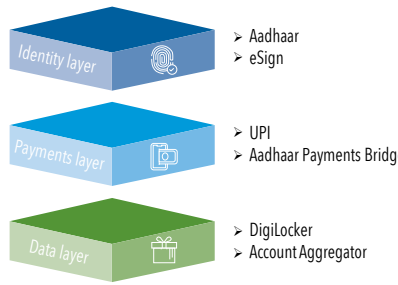
Digital public infrastructure, which makes digital payments scalable, inclusive, and interoperable, refers to a set of three shared digital building blocks powered by interoperable open standards or specifications (see Boxes 9.2 and 9.3 for an overview of India's and Sri Lanka's digital public infrastructure, respectively). This infrastructure commonly consists of digital ID systems, fast payment infrastructure, and data exchange frameworks, components that enable governments, businesses, and innovators to build their own applications on top of the common infrastructure. Well-designed digital public infrastructure can facilitate innovation and competition, expand markets, close financial inclusion gaps, boost government revenue collection, and improve public expenditure efficiency.

Digital public infrastructure acts as an enabler of digital payments adoption by easing access and expanding the benefits of digital payments. Robust digital IDs lower the cost and reduce the time involved in completing know your customer (KYC) obligations for financial institutions, and it also reduces the risk of fraud. In India, banks that use e-KYC lowered their compliance costs from $0.12 to

## Box 9.2. Lessons from India Stack

India's digital public infrastructure, India Stack, provides lessons for other countries pursuing digital transformation. A building-block approach, which unbundled complex challenges into modular, interoperable components,[1] proved critical for scalability and innovation (Box Figure 9.2.1). Each layer addressed a foundational need, whereas open standards enabled public and private sector use of shared infrastructure to develop tailored solutions across areas such as banking, education, and health.

### Box Figure 9.2.1. India's Digital Public Infrastructure Building Blocks




Source: Indian Software Product Industry Round Table.
Note: UPI = Unified Payments Interface.

India Stack has delivered significant benefits. The Indian government saved 1 percent of GDP by transferring social benefits directly to bank accounts to reduce leakage and governance reforms and using Aadhaar (digital ID) to remove duplicate beneficiaries (Government of India estimates). The infrastructure also supported a rapid pandemic response, with 87 percent of impoverished households receiving government assistance within the first few months. For some individuals, Aadhar provided access to public and private services for the first time. It also supported formalization by easing compliance, reflected in an increase in newly registered businesses for the goods and services tax. Registration allowed businesses to claim input tax credits, and goods and services tax return data can be shared with lenders to facilitate access to financing. In the private sector, start-ups used the infrastructure to launch products such as digital-only, zero commission stockbrokers and to increase competition in trade financing. In both cases, the identity and payments layers enabled seamless customer onboarding.

The government acted as a catalyst rather than the provider of each layer. Although the public sector developed Aadhar, the Unified Payments Interface is a collaboration between the Reserve Bank of India and the Indian Banks' Association, and the data exchange layer is provided privately under Reserve Bank of India regulation. Government support centered on acting as an anchor client and establishing institutions to ensure continuity in operations. Delivering social benefits through the infrastructure encouraged individuals to adopt the system and gave service providers access to a large user base, supporting innovation. This public and private collaboration was key to achieving scalability. Similarly, a public-private partnership model was instrumental in the speedy rollout of Aadhar. The government also promoted technology as a utility and created a category of public-purpose, not-for-profit companies known as National Information Utilities. This approach strikes a balance between effective service provision and limits on monopoly rents.

### Box 9.2. Lessons from India Stack *(continued)*

Supportive financial and telecommunications policies also encouraged adoption. In 2014, a government initiative to provide access to no-frills, low-cost bank accounts targeting rural women and others without access to financial services doubled the number of bank account holders. Foreign investment liberalization and the prohibition of discriminatory data tariffs helped create a competitive, open, and affordable telecommunications market. The entry of a new network operator in 2016 reduced mobile data costs by 90 percent and sharply increased data usage.

Note: See "The Value of Interoperability" section for more details.

### Box 9.3. Sri Lanka's Emerging Digital Public Infrastructure

Sri Lanka's Common Electronic Fund Transfer Switch (CEFTS), launched in 2015 and operated by LankaPay, is a unified, interoperable, instant payment infrastructure similar to India's Unified Payments Interface (UPI). Its interoperable payments layer supports multiple channels, enabling instant payments through banks, fintech payment apps, and online banking. Within the CEFTS architecture, the JustPay payment platform enables banks and fintech apps to facilitate in-app and web-based payments for utility bills, fund transfers, taxes, retail purchases, government services, and capital market transactions by linking bank accounts to third-party accounts. This open, interoperable layer has increased competition in the app market and expanded access to low-cost payment options.

LankaQR, Sri Lanka's national quick response (QR) code introduced in 2018, is another key component of the CEFTS infrastructure. LankaQR enables merchants to accept digital payments from any participating bank or fintech app through a single interoperable QR code. In 2024, it became interoperable with India's UPI, allowing tourists to make real-time QR-based UPI payments at LankaPay merchant points. LankaQR also links with Alipay+ and UnionPay China. These initiatives facilitate tourism through seamless cross-border payments and demonstrate the potential of interoperable, low-cost retail payment systems across countries. Given the similarities between UPI and CEFTS, deeper integration could facilitate trade, remittances, and broader cross-border transfers while generating economic and efficiency gains.

CEFTS also enables the modernization of government revenue collection and administration. Since 2017, major revenue-collecting agencies such as the Sri Lanka Customs and the Inland Revenue Department have been integrated into CEFTS through the LankaPay Online Payment Platform. Citizens can make payments to participating government institutions through online banking or interoperable payment apps, with funds credited instantly to government accounts. By the end of 2024, more than 40 percent of total government revenue was collected digitally through LankaPay across eight government agencies. LankaPay's real-time settlement supports accurate accounting, strengthens treasury operations, reduces administrative costs, and improves transparency and efficiency in public service delivery.

**Box 9.3. Sri Lanka's Emerging Digital Public Infrastructure *(continued)***

Although Sri Lanka's payments layer is well advanced, the identity and data layers remain under development. The National Data Exchange platform is being designed to enable secure, transparent data sharing and integration across government agencies, reducing silos and improving administrative efficiency. The Sri Lanka Unique Digital Identity Project, scheduled for rollout starting in the third quarter of 2026, is expected to provide a secure biometric and demographic identity based on the Modular Open Source Identity Platform. Once operational, the project could improve access to public services, streamline administration, and reduce fraud and duplication. Realizing these benefits will require robust data governance and privacy safeguards.

$0.06—a 99.5 percent reduction (Hariharan 2016).[12] Digital IDs are especially important for reaching vulnerable segments of the population that have traditionally lacked access to formal financial channels and other services. Adopting digital payments becomes viable for merchants, particularly micro and small enterprises, when the infrastructure is simple and inexpensive. Universal infrastructure compounds the benefits because one acceptance point can serve multiple wallets and banks (see the "The Value of Interoperability" section).

Several countries in the region have invested in digital public infrastructure to support their digital transformations. India Stack enables online, cashless, and private digital access to a variety of public and private services (Box 9.2). Similarly, Singapore developed Singpass (a digital ID), PayNow (a digital payment system owned and developed privately but with public oversight and regulation), APEX (a government data exchange), MyInfo (a consented data-sharing mechanism), and SGFinDex (a financial data exchange).[13] Thailand has two ID systems: D.DOPA, a digital ID system developed by the Department of Provincial Administration, and NDID, the National Digital Identity Platform, a public-private partnership. Thailand also has GDX, a government data exchange layer, and PromptPay, a well-developed payment system. During the pandemic, many government support programs were delivered using the Pao Tang digital wallet, which uses the PromptPay infrastructure. The scope of Pao Tang has expanded, and the public and private sectors use the platform to offer services ranging from investment products to health care. Overall, two-thirds of economies in Asia and Pacific have implemented at least one component of digital public infrastructure, though fewer than one-fifth—or 7 out of 39 economies—have all three components.[14] Yet, the extent

[12] These costs are in line with other estimates on the falling costs of opening bank accounts (see Croxson and others 2022).

[13] MyInfo, for example, has reduced the time to apply for the 2,000 public and private services accessible through Singpass by up to 80 percent, which reduces business transaction costs and improves the user experience (World Bank 2022).

[14] DPI Map. Institute for Innovation and Public Purpose, UCL. https://dpimap.org/data.

to which these blocks are utilized varies. For example, not all countries have leveraged the infrastructure rails for core public financial management functions. Globally, as of 2026, 39 countries, including Bangladesh, Bhutan, Cambodia, Indonesia, Papua New Guinea, Singapore, and Sri Lanka, have committed to designing, launching, and expanding at least one component of their digital public infrastructure securely and inclusively by 2028.[15]

Developing the infrastructure through open technology solutions can help countries accelerate adoption. An open-source approach provides greater transparency, cultivates trust, and facilitates collaboration while preventing vendor lock-in. Open-source technologies (such as software, standards, data, artificial intelligence systems, and content collections) that adhere to privacy standards, follow applicable best practices, do no harm, and are highly relevant in supporting the Sustainable Development Goals are known as digital public goods (UN 2020). They can help countries leapfrog their digital public infrastructure implementation by using proven solutions.

Countries in the region have provided and implemented open technology solutions that can support broader implementation of digital payments and its various applications, including the Modular Open Source Identity Platform from India for digital IDs, Open Source Social Protection Platform from Singapore for social protection registries, and the Unified Payments Interface (UPI) from India for payments.[16] Other open technology solutions countries are implementing include X-Road for data exchange in Bangladesh, Cambodia, and Mongolia; UPI in Nepal and Sri Lanka; and the Modular Open Source Identity Platform in Cambodia, the Philippines, Sri Lanka, and Tonga.

## The Value of Interoperability

Interoperable payment systems allow seamless transactions between users of different payment service providers, supporting digital payment growth and adding value for users in two ways: users can choose their favorite app, such as a trusted brand or one with proven reliability, and interoperability increases competition between existing and new app providers, thus encouraging existing providers to make continuing quality improvements to retain users. Overall, users benefit from higher quality and varied app choices, making them more likely to transition from cash to the system.

Copestake, Kirti, and Martinez Peria (2025) and Copestake and others (2025) developed a conceptual framework to examine the impact of interoperability and provide strong evidence of its value, focusing on India's UPI.

UPI, launched in 2016, is an instant payments platform built on the Immediate Payment Service infrastructure, India's preexisting real-time interbank

[15] These countries have joined the 50-in-5 initiative, a country-led advocacy campaign with the goal of implementing digital public infrastructure in 50 countries over five years until 2028. For more information, see https://50in5.net/.

[16] The Unified Payments Interface is not open source, but the code is made available.

electronic fund transfer service. The interface enables users to initiate bank-to-bank transfers with other users—from any bank account and through any participating app—and receive notifications through UPI apps provided by banks and third-party providers.

UPI has grown rapidly since its launch in 2016, processing about 20 billion transactions per month and dominating other electronic retail payments in India. As UPI has grown, some proxies for cash usage have declined. Interoperability has been crucial to the growth of digital payments through UPI because most transactions occur between two different apps (Figure 9.4).

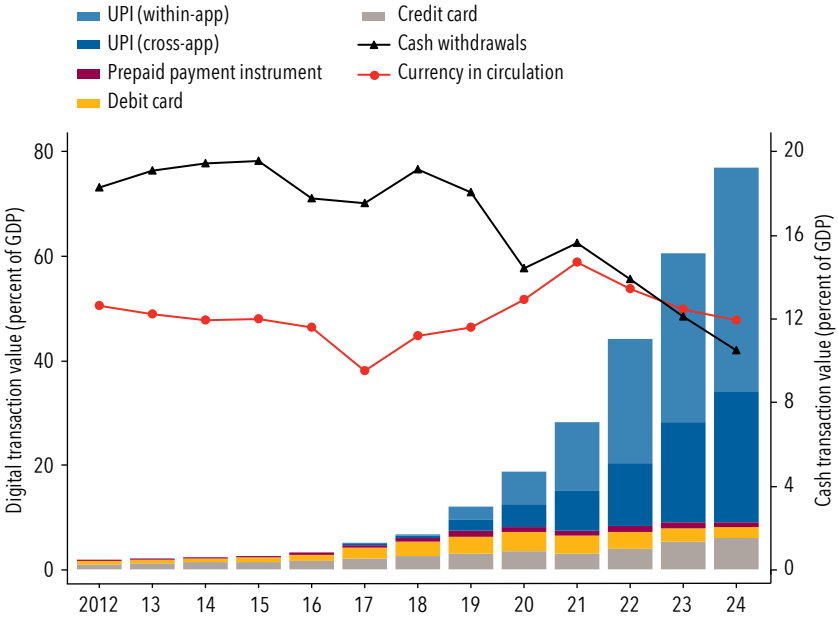
An analysis of whether greater interoperability increases overall adoption of digital payments requires identifying a relevant benchmark level of adoption—a business-as-usual outcome that would occur without interoperability—against which to measure the extent of adoption. Copestake, Kirti, and Martinez Peria (2025) and Copestake and others (2025) used an external policy change that increased interoperability and compared the responses of digital payments adoption in places more exposed to the increase with those less exposed to the increase. Specifically, after a directive by the Reserve Bank of India mandated interoperability, a major incumbent closed-loop provider joined UPI, connecting all its existing users to the system. With this integration, the interoperability of digital payments increased because two separate payment platforms were effectively combined into a single network. However, the implications of this new interoperability varied across districts. In districts where the closed-loop incumbent platform had no users before the integration, no new users were onboarded upon integration with UPI, resulting in no increase in interoperability. By contrast, in districts where the incumbent platform had many users before integration, the number of users who could send payments through UPI increased substantially, thus increasing interoperability substantially. More generally, the larger the role of the incumbent before integration (which implied a more fragmented digital payments market), the greater the increase in the number of possible connections after integration.

Copestake, Kirti, and Martinez Peria (2025) and Copestake and others (2025) find that adoption of digital payments grew substantially faster in districts experiencing a larger increase in interoperability. In Figure 9.5, the blue values plot the difference in total peer-to-merchant transaction values across both payment networks between an average district with an above-median presence of the incumbent platform before integration (and thus a larger increase in interoperability after integration) and an average district with a below-median presence of the incumbent platform before integration. During the first year after integration, the total peer-to-merchant transaction value per person increased by an average of 8 rupees per month in the above-median districts. The difference is persistent and substantial, equal to more than 80 percent of average peer-to-merchant digital payments per person in above-median districts in the month before integration. Total digital payments relative to cash withdrawals rose substantially and persistently more after integration in the districts that face greater increases in interoperability.

## Figure 9.4. Retail Digital Payments in India: Digital versus Cash Transactions

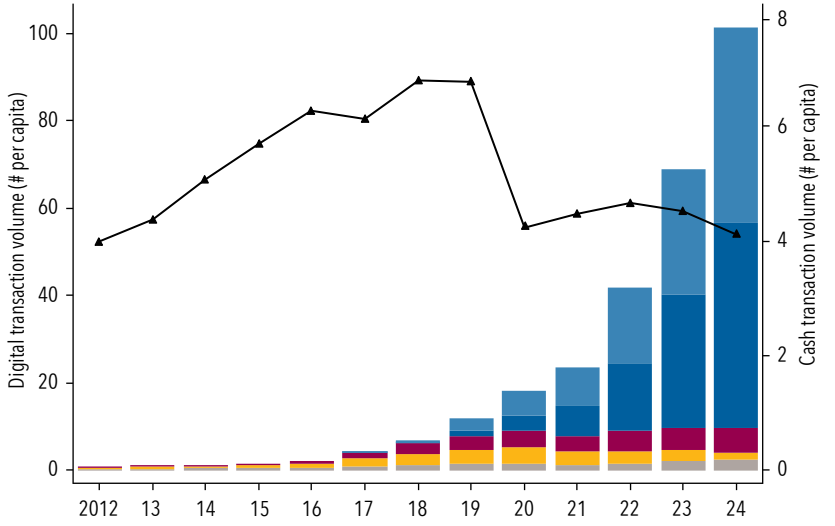
**1. Value**

*(Percent of GDP, digital [lhs] and cash [rhs])*




**2. Volume**

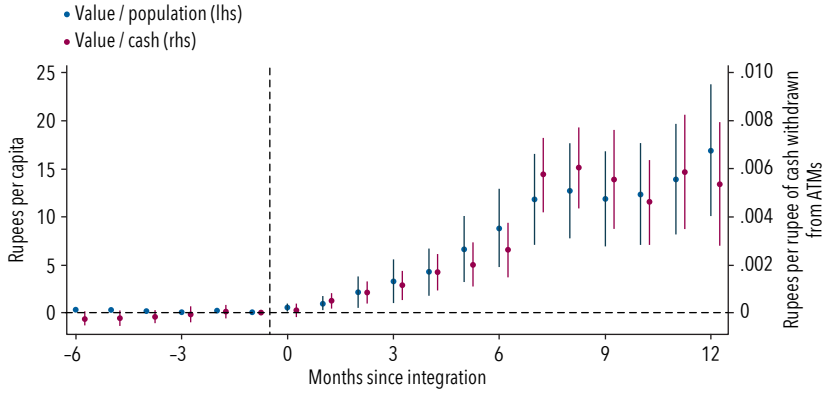
*(Transactions per capita, digital [lhs] and cash [rhs])*




Sources: Reserve Bank of India; EUROPACE AG/Haver Analytics; Bank for International Settlements; World Development Indicators; and authors' analysis.

Note: Prepaid payment instruments include smart cards and mobile wallets preloaded with value using cash, card, or other methods. The UPI includes both peer-to-merchant and peer-to-peer transactions because many retail payments to small merchants are recorded as peer-to-peer. lhs = left-hand scale; rhs = right-hand scale; UPI = Unified Payments Interface.

### Figure 9.5. Impact of Integration on Peer-to-Merchant Transaction Values Per Month




Sources: National Payments Corporation of India; and authors' analysis.
Note: The graph plots the estimated difference in the outcome variable between districts with a below- versus above-median market share of the incumbent in the month before integration–that is, between districts facing a smaller versus larger increase in actual interoperability. The outcome variable in blue is the total district-level peer-to-merchant digital payments across both digital payment platforms, measured in rupees per capita. The outcome variable in maroon is the same but normalized by ATM withdrawals–that is, measured in rupees of digital payments per rupee of cash withdrawn. The vertical bands show 95 percent confidence intervals, and standard errors are clustered at the district level. Copestake, Kirti, and Martinez Peria (2025) provide details of the estimation procedure. lhs = left-hand scale; rhs = right-hand scale.

This evidence suggests that interoperability can support adoption of digital payments and is relevant for both domestic and cross-border payments. Conceptually, the benefits of greater user choice and competition across providers are equally relevant in both cases.

## Interoperable Cross-Border Digital Payments

The many links in the current bilateral cross-border payment arrangements across Asia and Pacific have resulted in a highly fragmented system between different platforms and jurisdictions. The process of establishing a new bilateral link with each partner requires extensive and time-consuming technical, legal, and operational work from both sides, creating a costly and unsustainable model for broader regional interoperability. An alternative approach to solving interoperability issues in cross-border payments is thus needed.

Project Nexus—a collaboration among the Bank for International Settlements Innovation Hub, Bank of Thailand, Bank Negara Malaysia, Bangko Sentral ng Pilipinas, the Monetary Authority of Singapore, and the Reserve Bank of India—is a notable initiative with a scalable multilateral approach to improving cross-border payments in the region. Unlike traditional bilateral links, the project uses a standardized blueprint to connect multiple domestic fast payment systems through one central hub. This hub-and-spoke model significantly simplifies the

infrastructure required for regional and global payment connections. The project completed a successful proof of concept in 2022 and announced a comprehensive blueprint in July 2024. The incorporation of Nexus Global Payments in Singapore in 2025 was a significant transition toward operationalizing the program, with full implementation targeted for 2027.

Project mBridge is another multilateral initiative aimed at improving cross-border payments by using wholesale central bank digital currencies. This project originated in 2021 through a collaboration between the Bank for International Settlements Innovation Hub, the Bank of Thailand, the Central Bank of the United Arab Emirates, the Digital Currency Institute of the People's Bank of China, and the Hong Kong Monetary Authority. A six-week pilot conducted in 2022 involved 20 commercial banks across four jurisdictions. The pilot successfully executed more than 160 real-value transactions and demonstrated the platform's ability to facilitate real-time, peer-to-peer payments and foreign exchange transactions. mBridge reached its minimum viable product stage in mid-2024, and graduated from the BIS Innovation Hub in mid-2025. The project's Steering Committee, consisting of the four founding participant central banks and monetary authorities, will continue to guide the project's strategic direction, overseeing its enhancement, development, and operation.

Pacific island countries would especially benefit from lower cross-border payment costs. Remittance costs in those countries range from 5 to 11 percent of the transaction value, significantly higher than in Caribbean or Central American countries (Zhou and others 2024). Digital payments can help Pacific island countries overcome some of their inherent challenges. The countries' populations are small and dispersed in geographically remote areas vulnerable to climate change. Physical banking networks such as bank branches and ATMs are limited, and thus most Pacific island countries rely on cash for transactions. Loss of correspondent banking relationships exacerbates challenges with cross-border payments: six Pacific island countries have lost more than half of their correspondent banking relationships since 2011, narrowing the provision of remittance services and increasing costs (CPMI 2023, Table 11). The loss of these relationships can also affect trade, tourism, disaster relief, and humanitarian financial flows. This de-risking is driven by a reduction in banks' operational and financial risk appetites, coupled with lower returns from the Pacific region because of its small market and low level of financial activity. Given resource and capacity constraints, developing the network and digital infrastructure needed to implement digital payments effectively in Pacific island countries and rebuild correspondent banking relationships will require close collaboration with development partners.

## Governance, Regulation, and Safeguards

Robust governance and regulatory frameworks covering consumer protection, cybersecurity, data privacy, and fair competition are essential for trust and resilience across digital payments. Digital public infrastructure must balance openness and interoperability with safeguards against systemic and operational risks.

Priorities include consent-based data governance, proportional regulation to promote innovation while mitigating fraud and misuse, and clear accountability for platform operators. As digital payments are expanded, regulators also need to monitor concentration risks and ensure competitive neutrality to prevent market dominance that could undermine resilience.

Strong institutional governance is needed for digital public infrastructure to support governments' development agendas. Successful infrastructure relies on trust, and a robust data protection framework is essential to protecting citizens' privacy, preventing indiscriminate data collection, and holding companies and governments accountable for data breaches. Adequate investments in cybersecurity are required for both the public and private sector. Sound grievance redress mechanisms that rectify data mismatches, update information, and allow service denial appeals are important to build confidence in the infrastructure. These mechanisms provide a feedback loop that can strengthen the system's accuracy, which is crucial for delivering social protection benefits and preventing the exclusion of genuine beneficiaries. Similarly, efforts to close the digital divide should continue to ensure inclusive digital transformation within countries, because exclusion errors could also arise when households are unable to access and use digital tools.

Research confirms the potential gains from interoperability, but the case of India also highlights the need for payment system regulators to vigilantly protect these gains because providers have gradually introduced features designed to lock in users and recreate within-provider network effects. Many UPI providers initially offered proprietary QR codes to merchants that could be read only with the provider's app, which forced potential customers to adopt the app. The Reserve Bank of India prohibited such noninteroperable QR codes to reinforce UPI's design as a fully open and interoperable payment system (RBI 2022). Even with interoperable QR codes, some providers still framed them as readable only with a given app (RBI 2020). In addition, all major UPI app providers offer cashback program that incentivize both the sender and the recipient to use their app. Continued efforts to respond to providers' attempts to lock in users are required to ensure that users reap the gains from interoperable systems.

Digital payments at scale can raise macrofinancial risks that warrant regulatory attention. Large platforms can become systemically important because outages and cyber incidents can cause financial instability. Interoperability failures and concentration of market power can amplify shocks; while, rapid shifts in liquidity through real-time transfers could complicate monetary policy. Cross-border systems add another layer of vulnerability because disruptions can propagate across jurisdictions. High transaction volumes and instant settlement also heighten challenges for anti–money laundering and combating the financing of terrorism, requiring robust KYC, monitoring, and reporting frameworks (Box 9.4). These risks highlight the need for integrated oversight that spans payment systems, financial stability frameworks, and macroprudential policy (Adrian and Mancini-Griffoli 2019; Dobler and others 2021; IMF 2020, 2024; Huang and others 2024).

### Box 9.4. Measures to Address Loss of Correspondent Banking Relationships in Pacific Island Countries

Support from development partners is needed to sustain affordable and secure cross-border payments. An immediate but temporary solution under development by eight Pacific island countries and the World Bank is an emergency backstop that would provide correspondent banking services to eligible financial institutions if a country loses its last correspondent banking relationship in a key currency (World Bank 2024). This mechanism would ensure service continuity and prevent disruptions to cross-border flows.

A longer-term and more permanent solution is the establishment of safe payment corridors, which apply simplified due diligence to transactions assessed as posing low risks for money laundering and terrorist financing. This approach can reduce compliance costs and ease pressures from global de-risking trends. The IMF and the World Bank have developed a framework for assessing money laundering and terrorist financing risks in remittance corridors that could qualify as safe payment corridors (World Bank 2021). In Samoa, the assessment found low risks from small-value remittances and concluded that vulnerabilities in the anti–money laundering and combating the financing of terrorism framework, arising from capacity and resource constraints, can be mitigated through regional information sharing and capacity building (Markevych and Tanaka 2025). Effective safe payment corridors require an enabling legal and regulatory environment and strong support from supervisors and regulators, including those in origin countries. The IMF continues to support Pacific island countries in laying the groundwork for safe payment corridors.

The Pacific Islands Forum and the World Bank are also exploring a regional payment aggregation mechanism, the Pacific Payments Mechanism. Aggregating payments across Pacific island countries can improve the risk-return trade-offs for correspondent banks through economies of scale, helping sustain correspondent banking relationship services and potentially reduce transaction costs.

## CONCLUSION AND POLICY RECOMMENDATIONS

The Asia and Pacific region is a global leader in digital payments at the forefront of the digital transformation. Digital payments can boost productivity by reducing transaction costs, improving resource allocation, and enabling innovation. They support inclusive growth by formalizing economic activity and expanding access to financial services for households, small and medium enterprises, and other groups without access to services. The region's experience shows that technology alone is not enough to reap the benefits of digital payments. Connectivity, infrastructure, interoperability, and governance are also critical to unlocking the full benefits.

Gaps in digitalization and digital payment adoption remain both across and within the region's countries, highlighting room for continued progress. Policymakers must prioritize closing gaps in emerging market and low-income economies, enabling them to build the infrastructure and regulatory capacity needed to participate fully in the digital payment ecosystem. Digital literacy remains much lower in rural areas, among women, and in low-income

households, groups that could benefit the most from digital transformation. Targeted government and private sector initiatives are critical to enhancing digital literacy and promoting inclusive participation. Digital payment design and regulation should promote financial inclusion while preventing financial disintermediation and safeguarding financial stability.

Ensuring connectivity and providing digital public infrastructure are the foundation of inclusive, innovative, and scalable digital payment solutions. A thriving digital payments landscape depends on secure digital identification, affordable and reliable mobile data access, and integration of digital payments into government transfers and services. Well-designed and widely accessible digital public infrastructure lowers barriers for individuals and businesses to participate in the formal economy, supports efficient delivery of public services, and encourages competition and innovation.

Critical policy priorities include ensuring that payment rails are interoperable and keeping the system open and competitive, and policymakers should focus on complementary actions to reinforce those priorities. They should ensure complementary enabling investments to reduce mobile data costs and expand bank accounts and digital identity, as in India. Policymakers should also monitor the emergence of dominant private providers and if necessary take regulatory actions to ensure that providers' practices aimed at locking in users do not curtail user choice and competition.

Asia and Pacific countries should continue advancing multilateral arrangements. Bilateral cross-border arrangements common in the region are not scalable for a globally interconnected economy. As more countries join multilateral agreements, the number of necessary connections grows exponentially. A multilateral system that encourages standardization and streamlining of payment structures, infrastructure, rules, and compliance in each economy will significantly enhance efficiency and interoperability and increase local currency usage. Interoperable within-country and cross-country digital payment systems can alleviate the challenges facing small, remote Pacific island countries with limited domestic payment infrastructures and dwindling correspondent banking relationships.

Governance frameworks must evolve along with rapid innovation in digital payments. Global coordination is increasingly important for cross-border interoperability and data flows. Emerging standards on consumer protection, anti–money laundering and combating the financing of terrorism, and cybersecurity resilience will shape the next generation of digital public infrastructure. Regulators must remain agile to address new risks from AI, embedded finance, and concentration of market power to ensure that digital transformation remains inclusive, secure, and competitive.

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CHAPTER 10

# The Case for Deeper Asia and Pacific Trade Integration

**Chikako Baba, Ashique Habib, Sun Young Park, Rahul Giri, and Andrea Pescatori**

The Asia and Pacific region has experienced rapid trade growth for decades, propelled by abundant low-cost labor and falling trade costs, partly because of trade liberalization since the late 1980s. Rapid and deep integration into global value chains followed, transforming the region into a global manufacturing hub. However, the global landscape is changing. The sweeping US tariffs imposed in 2025, after earlier trade tensions and COVID-19 pandemic-related disruptions, are accelerating the reconfiguration of global supply chains. Although trade policy developments since the mid-2010s have heightened risks and pose significant challenges, they also highlight the untapped potential of deeper intra-Asia integration. The Asia and Pacific region lags in intraregional final goods trade, partly because of nontariff barriers and fragmented, shallow trade agreements. Simulation exercises point to sizable economic gains from lowering trade barriers within the region and globally, whereas complementary policies would be needed to assist workers through the transition to ensure inclusive outcomes. Policies that promote openness in trade and foreign direct investment, supported by reforms to enhance competitiveness, are essential for maintaining trade as an engine of growth.

## TRADE AS AN ENGINE OF GROWTH

Trade has long been central to the growth model of many Asia and Pacific economies. The region has leveraged exports as an engine of growth, a phenomenon documented most famously in the World Bank's 1993 report *The East Asian Miracle: Economic Growth and Public Policy.* Trade has facilitated the region's transformation into a global manufacturing hub with rising aggregate productivity and infrastructure investment, more jobs, and lower poverty (Chapter 2; Song, Storesletten, and Zilibotti 2011). However, trade tensions and global geopolitical fragmentation are challenging these achievements and could slow trade growth and reshape cross-border production networks (supply chains).

The region is highly open to trade, especially in countries in the Association of Southeast Asian Nations (ASEAN), although the degree of openness varies across economies. Asia and Pacific economies as a whole account for one-third of global

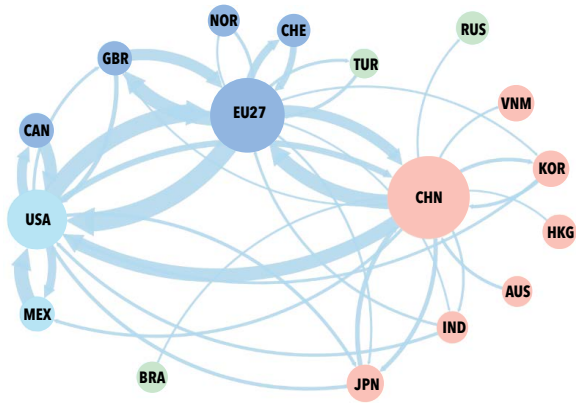
trade, in line with their share of global GDP. However, ASEAN countries stand out, with shares of global trade that exceed their shares of global GDP, even after controlling for the tendency of smaller economies to rely more on trade (IMF 2025).

Asia and Pacific economies have major footprints in global trade networks. Figure 10.1 shows the 40 largest bilateral trade corridors by value as of 2023. China is part of the largest trade networks. Along with major regional links, its exports to the European Union and the United States rank among the top 10 largest trade corridors. Larger economies in the region, such as India, Japan, and Korea, also have significant presence, especially in their exports to China, the European Union, or the United States.

China's centrality is an important feature of the region's trade network. In 2023, China was the largest export destination for eight economies in the region, representing about one-third of the region's GDP (excluding China). Many of those economies are commodity exporters, such as Australia, Indonesia, and Mongolia. China is also the largest source of imports for 16 economies that together represent 99 percent of the region's GDP (excluding China). Exceptions include some Pacific island economies that trade closely with Australia, New Zealand, and Singapore, and South Asian countries that have strong links to India, but these clusters account for only a small share of regional trade.

Oversea markets, especially Europe and the United States, are important destinations for the region's final goods exports. In 2023, the United States was the largest export destination for 10 Asia and Pacific economies, representing about 60 percent of the region's GDP (excluding China). The pattern differs in the European

**Figure 10.1. Top 40 Major Trade Corridors**
*(Based on the value of total exports, 2023)*




Sources: Asian Development Bank Multiregional Input-Output Tables; and IMF staff calculations.

Note: The size of each circle (node) represents each economy's total exports of goods and services to the world. The thickness of each arrow represents the value of bilateral export flows of goods and services between economies for the top 40 bilateral export relationships by value for the IMF members. Exports by European Union member states are aggregated, and EU internal trade is excluded from the figure. Data labels in the figure use International Organization for Standardization (ISO) country codes.

Union, another highly open trade bloc: the United States is the largest export destination for only about 15 percent of economies, reflecting Europe's extensive intraregional trade. Along with the fact that more than 60 percent of China's exports to the region are intermediate goods, these patterns underscore supply chains built around China to serve overseas markets, particularly the United States.

Since the mid-2010s, the trade landscape has been changing. Rapid shifts in global trade policy, most notably rising tariff rates and the proliferation of nontariff trade barriers, are reshaping incentives to establish cross-border production networks. Some of the region's economies benefited from the US-China trade tensions in 2018–19, during which some producers relocated production from China (IMF 2025). After the COVID-19 pandemic, the region's economies face broader headwinds from slowing growth in China and from higher US tariffs from 2025 affecting most of them. Moreover, many economies in Asia and Pacific are likely to shift from manufacturing to services (Chapter 2), implying that the contribution of manufacturing exports to growth is expected to be smaller. Together, these circumstances point to the benefits of diversifying trade relationships and expanding regional trade, especially in services, where Asia and Pacific economies have historically been more protective.

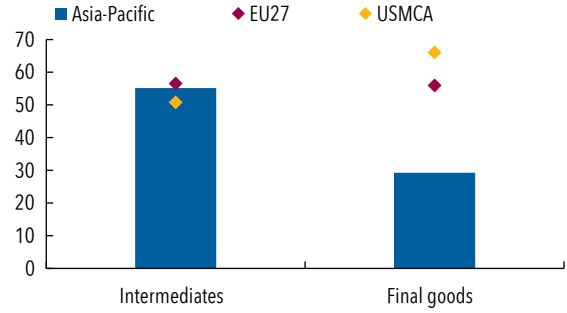
Deeper integration through lowering trade barriers could create resilience and growth opportunities. Despite the region's overall openness to trade, intraregional trade integration has room to grow, supported by stronger trade agreements. Lowering trade barriers can generate sizable long-term economic gains. The analysis presented in this chapter suggests that policies to promote greater openness in trade, supported by reforms that enhance competitiveness and macroeconomic stability (Chapter 11; IMF 2025), would benefit the region and help it navigate the evolving global trade landscape.

## REGIONAL TRADE INTEGRATION IN ASIA AND PACIFIC

Trade data show that the region is already highly integrated in intermediate goods trade, but intraregional trade in final goods has room to deepen. Figure 10.2 shows that less than 30 percent of the region's final goods exports are absorbed within the region, compared with about 60 percent in Europe and in North America. By contrast, more than half of the region's intermediate goods trade takes place within the region—a level comparable to those in other regions. This divergence partly reflects the prevalence of regional supply-chain trade that supports final goods exports to overseas markets, but it also points to trade friction within the region and the untapped potential for deeper intraregional integration.

Limited progress in intraregional integration in final goods trade may seem inconsistent with the region's efforts to promote intraregional integration through trade agreements. By 2020, the number of preferential trade agreements involving the region's economies has risen to more than 100, about half of which are among economies within the region. Most are free trade agreements intended to expand

**Figure 10.2. Intraregional Goods Exports**
*(Percent of goods exports, 2023)*




Sources: Asian Development Bank Multiregional Input-Output Tables; and IMF staff calculations.

Note: USMCA = United States–Mexico–Canada Agreement.

market access and strengthen economic integration across goods, services, and factor markets. However, the legal and institutional architecture of trade agreements reveals areas where Asia and Pacific economies could strengthen intraregional trade.

First, intraregional trade agreements tend to be fragmented in Asia and Pacific. Economies often enter into bilateral agreements with one another or with regional trade blocs such as ASEAN (IMF 2025). In contrast, regions such as Europe and North America support trade through broad, legally binding plurilateral agreements involving major regional economies. Asia and Pacific economies's reliance on a patchwork of agreements can create overlapping rules, inconsistent standards, and a lack of institutional coherence, especially for trade within Asia and Pacific (ADB 2025). Mega-regional agreements enacted around 2020 in Asia and Pacific could help mitigate these problems (for example, through harmonizing standards) if more countries sign on to them (Box 10.1).

## Box 10.1. Asia's Evolving Regional Trade Architecture

The World Trade Organization has been a cornerstone of the global trade system for the past three decades. However, the growing importance of new policy areas such as digital trade, investment, and sustainability has highlighted its limitations. Recent challenges in multilateral rulemaking and enforcement have led some countries to pursue regional trade agreements as more flexible and responsive frameworks for advancing integration and modernizing trade disciplines.

Two major regional trade agreements entered into force in Asia and Pacific around 2020. The Regional Comprehensive Economic Partnership brings Association of Southeast Asian Nations economies together with five major economies: Australia, China, Japan, Korea, and New Zealand. The agreement focuses primarily on goods trade, tariff liberalization, and stronger regional production networks. Its legal framework emphasizes inclusiveness and accommodates differ-

## Box 10.1. Asia's Evolving Regional Trade Architecture *(continued)*

ent levels of development (Petri and Plummer 2020). The Comprehensive and Progressive Agreement for Trans-Pacific Partnership, which comprises 12 countries, adopts a more comprehensive and legally binding framework, covering labor standards, environmental protection, digital trade, and investment, supported by an enforceable dispute settlement mechanism. Because of its broader scope, accession can require more politically sensitive and costly reforms, especially in areas such as agriculture, labor, environment, and regulation of state-owned enterprises (Ferrantino, Maliszewska, and Taran 2019).

Though differing in scope and legal depth, the two agreements offer complementary approaches to integration. Leveraging the strengths of both through enhanced regulatory convergence and institutional strengthening could reduce fragmentation and contribute to a more coherent, rules-based trade architecture in Asia.

**BOX TABLE 1.1.1.**

Comparison of Global and Regional Trade Framework

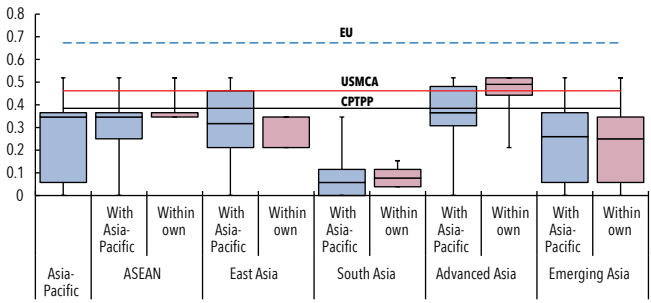
| Features | World Trade Organization (1995) | Regional Comprehensive Economic Partnership (2022) | Comprehensive and Progressive Agreement for Trans-Pacific Partnership (2018) |
|---|---|---|---|
| **Members** | 166 countries | 15 countries: the ASEAN-10 (Brunei Darussalam, Cambodia, Indonesia, Lao P.D.R., Malaysia, Myanmar, the Philippines, Singapore, Thailand, Vietnam), Australia, China, Japan, Korea, and New Zealand | 12 countries: Australia, Brunei Darussalam, Canada, Chile, Japan, Mexico, Malaysia, New Zealand, Peru, Singapore, the United Kingdom, and Vietnam |
| **Policy areas** | Trade in goods and services, customs regulations, export taxes, safeguard measures | Tariffs, services, investment, intellectual property, e-commerce, competition, and government procurement | Broad coverage, including goods, services, investment, e-commerce, state-owned enterprises, labor, environment, and digital trade |
| **Legal bindingness** | Binding, but enforcement weakened | Legally binding commitments with flexible implementation and emphasis on cooperation | Strong and enforceable commitments with clear mechanisms for compliance |
| **Enforcement** | Dispute Settlement Understanding with binding rulings but weakened functionality | Panels similar to the World Trade Organization but without appellate body; binding but more flexible with carveouts | Broad coverage with binding rulings, strong enforcement, and investor–state dispute settlement for investment disputes |

Source: IMF staff compilation.
Note: ASEAN = Association of Southeast Asian Nations.

Second, intraregional trade arrangements could be deeper. Trade literature has often used the share of legally enforceable provisions as a measure of agreement depth (for example, Hofmann, Osnago, and Ruta 2017; Dhingra, Freeman, and Huang 2023). Figure 10.3 shows the distribution of trade agreements' depth across the region's economies. Compared with agreements in Europe and North America, those in Asia and Pacific tend to be shallower, especially in areas beyond the World Trade Organization's mandate, such as competition policy, environmental protection, and e-commerce.[1] Differences across subregions are substantial: ASEAN is relatively integrated, whereas coverage in South Asia is especially shallow. Even where integration is more advanced within subgroups (such as ASEAN or members of the Comprehensive and Progressive Agreement for Trans-Pacific Partnership), agreement depth with other major regional economies is often limited.

Fragmented and shallow trade agreements can contribute to the widespread presence of nontariff barriers, some of which can hinder final goods trade in the region. Nontariff trade barriers are prevalent in South Asia and some ASEAN countries despite generally low tariff levels across Asia and Pacific (IMF 2025). Governments maintain nontariff trade barriers for various reasons, including efforts to manage the distributional impacts of trade and political economy considerations, particularly where World Trade Organization discipline constrains tariff policies (Evenett and Fritz 2015; Cadot and Gourdon 2016). Nontariff trade barriers affect both intermediate and final goods trade. However, some

**Figure 10.3. Depth Scores across Trading Partners**
*(Index, number of legally enforceable provisions out of 52)*




Sources: World Bank Deep Trade Agreements Database; and IMF staff calculations.

Note: The box plot presents the distribution of country-pair depth scores within various groups in relation to selected economies in the Asia and Pacific region (blue) or other economies in the same group. ASEAN = Association of Southeast Asian Nations; CPTPP = Comprehensive and Progressive Agreement for Trans-Pacific Partnership; EU = European Union; USMCA = United States–Mexico–Canada Agreement.

[1] The 52 areas considered include investment protection, which may influence inward foreign direct investment significantly (see Sztaierowska 2021), which may boost both imports (such as imports of final goods for the local market and intermediate goods for global value chain–linked production) and exports (for example, from export-oriented foreign direct investment).

instruments, such as standards and technical regulations, and some rationales, such as firms' efforts to maintain output prices, can especially affect final goods trade (Shapiro 2020; Fiankor, Kassa, and Lartey 2025).

The economic literature has thus highlighted the role of deeper trade agreements in reducing nontariff trade barriers and promoting trade. Studies find that deeper commitments to address nontariff trade barriers in trade agreements can strengthen global supply chains and services trade beyond the gains generated by tariff liberalization under shallower agreements (Dhingra, Freeman, and Huang 2023). For example, provisions that streamline technical barriers to trade or investment policies can boost trade by reducing regulatory hurdles and creating a more conducive environment for foreign investment.

Deeper trade agreements can also reduce barriers to services trade. Many Asia and Pacific economies maintain relatively restrictive rules in this area (see Chapter 2). Restrictions differ across countries but often include market access and regulatory barriers, such as approval requirements, local presence requirements for foreign service providers, or regulations that favor domestic companies (OECD 2025). Deep trade agreements typically include rules on competition policy or e-commerce that encourage regulatory reforms to ease those barriers. Research therefore shows that deeper trade agreements tend to provide a larger boost to services trade than to goods trade (Dhingra, Freeman, and Huang 2023).

Reducing trade barriers through deeper trade agreements can support economic growth and encourage trade even without tariff reductions.[2] For example, gains can arise from provisions that streamline technical barriers to trade,[3] reducing regulatory hurdles and helping attract more foreign investment. Interest in new trade agreements has increased globally amid heightened uncertainty about trade policy. In the Asia and Pacific region, ASEAN and China agreed in 2025 to upgrade their free trade agreement to cover new areas such as digital economy, and more countries have applied to join the Comprehensive and Progressive Agreement for Trans-Pacific Partnership. Beyond the region, many bilateral trade agreements were signed or concluded in 2025 and 2026, including agreements between India and the European Union; India and the United Kingdom; Malaysia and the European Free Trade Association; Indonesia and the European Union; and Indonesia and Canada.

## THE CASE FOR DEEPER REGIONAL INTEGRATION

Potential gains from deeper trade integration in the Asia and Pacific were assessed through a simulation exercise using a multicountry, multisector quantitative trade model, building on Cuñat and Zymek (2024). The simulations quantify the

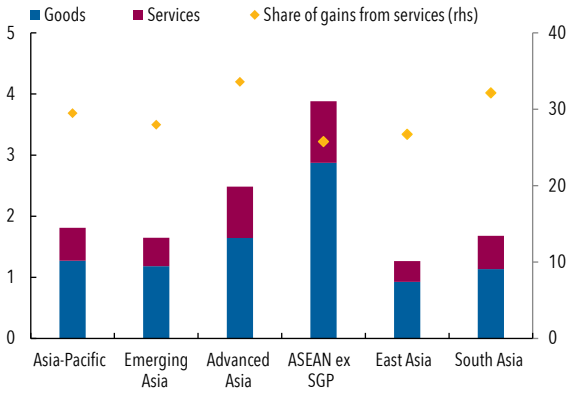
[2] Other policy measures can also promote trade integration. Chapter 11 underscores the importance of structural reforms in advancing trade integration among ASEAN economies.

[3] Technical barriers are regulations, standards, and assessment processes that could exist for legitimate purposes, such as ensuring product safety. Examples include electronics certification requirements, automotive technical regulations, agri-food labeling and testing requirements, and emerging digital and green-related standards. Cross-country differences in such procedures, combined with lack of mutual recognition, can create a barrier to trade.

long-term gains in real GDP from greater integration. The model allows capital to respond endogenously to trade liberalization, shedding light on how investment amplifies output gains through capital reallocation across countries. Following the literature (for example, Dhingra, Freeman, and Huang 2023), the extent of nontariff trade barriers between countries is measured by the number of legally enforceable provisions in their trade agreements (the depth score). Those scores are then translated into trade costs, which include the impact of nontariff trade barriers. The analysis complements the IMF work on the gains from trade integration, including from tariff reductions (for example, Rotunno and Ruta 2025).

The simulation exercise quantifies the gains for the regional economies if they moved from the current, relatively shallow trade agreements to the maximum possible depth, defined as the full coverage of legally enforceable provisions. As Figure 10.4 shows, the long-term gains would be significant: on average, real GDP across the region would rise by about 1.8 percent. A decomposition of the gains reveals that much of them arise from import liberalization, meaning from countries lowering their own nontariff trade barriers on imports from regional partners. In each regional economy, lowering import barriers allows domestic producers to use lower-cost imported intermediate inputs and consumers to purchase cheaper imported final goods, which incentivizes the reallocation of resources toward sectors in which the economy has a comparative advantage. Aggregate productivity increases as a result, which also raises returns on investment, leading to a larger capital stock and more output gains. Improved market access for exports to other Asia and Pacific economies amplifies the gains from import liberalization, with stronger regional demand reinforcing output across the region.

**Figure 10.4. Real GDP Gains by Opening Regionally**
*(Percent)*




Source: IMF staff calculations.

Note: The scenario targets maximum depth-of-trade agreements. The bars represent gains relative to the baseline without a reform. Gains from goods trade and services trade are shown separately. ASEAN = Association of Southeast Asian Nations; rhs = right-hand scale.

Several factors explain differences in gains across Asia and Pacific subregions and economy groups (the average of the economies' gains), especially an economy's initial depth of integration, the economy's size, and intensity of intermediate goods usage in produciton.

First, economies that have shallower agreements benefit more from their deepening compared with economies where agreements are already relatively deeper. This factor is a particularly important driver for the gains in South Asia, but it also contributes to the gains elsewhere in the region, including advanced economies and ASEAN economies, all of which still have some room to deepen integration with others.

Second, smaller economies tend to gain more than larger economies, even when tariffs are already relatively low, as in some ASEAN economies. Trade integration creates greater scope for efficiency-enhancing reallocation because firms in smaller economies can sell into larger markets and gain access to larger supplies of lower-cost inputs.

Third, benefits are also larger in economies where production relies more heavily on intermediate inputs, such as Korea, consistent with evidence on the cascading benefits of trade liberalization from upstream stages of supply chains (Franco-Bedoya and Frohm 2022). Lower-cost access to intermediate inputs at an upstream stage of production generates gains at each subsequent stage, amplifying the overall impact. The more intensively an economy uses intermediates, the larger those gains become.

In addition, advanced economies tend to benefit more from expanded services trade. More than one-third of their total gains are from the service sector (Figure 10.4). Emerging markets also realize meaningful gains, with services accounting for about one-quarter of total gains. Services trade responds more strongly to deeper trade agreements than goods trade, and advanced economies often have comparative advantages in service industries.

Service sector gains would be an important benefit because many Asia and Pacific economies are expected to shift from manufacturing toward services. Chapter 2 shows that this shift need not slow aggregate productivity, especially if countries move into modern, high-value tradable services. The findings reinforce that conclusion: trade liberalization can boost growth substantially through its impact on the service sector.

The region's economies could realize even larger gains by also reducing nontariff trade barriers with the rest of the world. Such benefits could arise from additional policy action or because some provisions in trade agreements naturally spill over beyond participating countries (Mattoo, Mulabdic, and Ruta 2022). Because a large share of the gains are from import liberalization, the region's economies would benefit substantially even if other parts of the world do not reciprocate. However, fully removing nontariff trade barriers could face various constraints, but even partial steps toward deeper trade integration can deliver meaningful benefits. The size of those gains will depend on how far the region can go in reducing those barriers.

Although regional economies are navigating a more fragmented global environment and balancing multiple objectives, substantial benefits—potentially for final goods—could result from pursuing trade openness, whether economies act unilaterally or plurilaterally. However, trade liberalization can affect some domestic groups adversely, including less-skilled workers (Goldberg and Pavcnik 2007), and policies must address those distributional impacts. Greater regional integration will also support external rebalancing through more diversified export destinations, complementing domestic rebalancing and stronger growth from improved investment efficiency (Chapter 3).

## CONCLUSION

Trade has been central to growth in Asia and Pacific, with China at the hub of supply chains and overseas markets providing final demand. Although some of the region's economies are highly integrated into supply chains, final goods trade continues to fall short of its potential, underscoring the importance of deeper trade agreements. Scope remains to expand trade integration within Asia and beyond. Although tariffs have declined significantly in the region, nontariff barriers remain relatively high in Asia and Pacific. A simulation exercise shows that reducing nontariff barriers with regional partners can raise the region's long-term GDP by almost 2 percent and potentially by twice as much if Asia and Pacific economies extend those reductions to global partners.

The findings highlight the importance of greater openness to trade and investment, supported by complementary policies that help workers transition to new opportunities as needed. Deepening regional trade agreements will be crucial. The economic literature shows that agreements covering a broader range of policy areas—such as competition policy, public procurement, and investment policy—tend to reduce nontariff barriers and promote trade among members. Expanding participation in newer and deeper agreements, such as the Comprehensive and Progressive Agreement for Trans-Pacific Partnership, and improving coherence across overlapping arrangements can harmonize rules and lower compliance costs. In addition, policy measures such as stronger social protection and active labor market policies to support reskilling and job transition, along with well-designed efforts to build stakeholder support, can help mitigate adverse distributional impacts during the transition.

Beyond lowering trade barriers, policies should strengthen competitiveness and improve macroeconomic stability. For example, Chapter 11 highlights the importance of structural reforms (such as improvements in logistic efficiency, human capital development, and digital government capacity) in supporting export growth. IMF (2025) also points to the benefits of liberalizing restrictions on foreign entry to take advantage of relocation by multinational firms. Such reforms would improve the region's attractiveness amid shifting global supply chains, support resilience in a more fragmented global environment, and unlock long-term growth opportunities.

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CHAPTER 11

# Association of Southeast Asian Nations Integration into a Shifting Global Landscape

Qingyu Chen, Corinne Delechat, Emmanouil Kitsios, Margaux MacDonald, Akshat Singh, Nujin Suphaphiphat, and Weining Xin

## INTRODUCTION

The Association of Southeast Asian Nations (ASEAN) stands at a critical juncture as global trade undergoes profound structural shifts. Rising geopolitical tensions, intensifying geoeconomic fragmentation, and the reconfiguration of global value chains are reshaping the international trade and investment landscape. These shifts pose new challenges to the region's long-standing development model that is anchored in openness, export-led growth, and integration into global markets. But the shifts also create opportunities for ASEAN to deepen its own regional links and strengthen resilience. As supply chains diversify and production networks become more regionalized, ASEAN's ability to deepen regional integration while remaining strongly connected to the world will be important for it to sustain growth amid persistent global uncertainty.

ASEAN leaders are again calling for deeper intraregional integration. The group of countries collectively is the world's fifth largest economy as of 2025, accounting for about 8.5 percent of the global population, and comprises fast-growing economies with expanding manufacturing capacity, rising incomes, and a growing middle class. However, intraregional trade remains limited and has stabilized at about 20–25 percent of total goods trade. Deepening intraregional integration could bring significant economic gains and help reinforce ASEAN's position as a dynamic and resilient hub within global production networks (Box 11.1).

ASEAN can harness the next phase of integration to strengthen growth amid a rapidly evolving global trade environment. ASEAN's trade and investment structure provides important context for assessing how global trade tensions associated with the imposition of higher US tariffs in 2025 are expected to affect the export performance of its member countries, making the case for enhancing regional integration. An analysis of the evolution of intraregional trade and

## Box 11.1. ASEAN Free Trade Area and Economic Integration

The ASEAN Free Trade Area (AFTA), established in 1992 under the Framework Agreement on Enhancing ASEAN Economic Cooperation, marked the first major step toward regional trade integration. Initially comprising six founding member countries (Brunei Darussalam, Indonesia, Malaysia, the Philippines, Singapore, and Thailand), the free trade area aimed to enhance regional competitiveness and attract investment by progressively reducing intraregional tariffs through the Common Effective Preferential Tariff program. Membership later expanded to include Vietnam (1995), Lao P.D.R. and Myanmar (1997), and Cambodia (1999), achieving the full participation of all 10 ASEAN members. By the mid-2000s, almost all intra-ASEAN tariffs had been eliminated among the original members, and the newer members reached near-zero rates by the early 2010s. In 2025, Timor-Leste formally became the 11th member state of ASEAN and expanded the region's economic community. These liberalization efforts culminated in the ASEAN Trade in Goods Agreement (ATIGA), which consolidated AFTA commitments into a more transparent and rules-based trade framework. Building on this foundation, on October 26, 2025, most ASEAN members signed the second protocol to amend the agreement, a comprehensive revision aimed at deepening tariff liberalization, improving mechanisms to address nontariff barriers, and promoting greater use of digital tools to facilitate cross-border trade and strengthen regional supply-chain connectivity.

Building on AFTA's success, ASEAN launched the ASEAN Economic Community in 2015 to advance beyond tariff liberalization toward deeper structural integration. The economic community envisions ASEAN as a single market and production base characterized by the free flow of goods, services, investment, and skilled labor and a more open capital regime. Policy initiatives have since broadened to cover harmonization of standards, services, and investment liberalization and improvements in logistics and customs facilitation. Key programs such as the ASEAN Single Window (which connects the National Single Windows of all ASEAN Member States, thereby facilitating the secure electronic exchange of trade-related documents) and the Master Plan on ASEAN Connectivity (a strategic framework adopted by ASEAN to enhance physical, institutional, and people-to-people linkages across the region) have helped reduce trade costs and enhance efficiency across borders. Efforts since 2020 have also expanded to digital trade, e-commerce, and sustainable investment frameworks, recognizing their growing importance for competitiveness and resilience. Although progress varies across member countries, collective action has deepened links within regional value chains and strengthened ASEAN's role in the global economy.

The next phase of ASEAN integration aims to operationalize the *ASEAN Economic Community Blueprint 2025,* shifting focus from liberalization to connectivity, inclusiveness, and sustainability. Priorities include reducing nontariff measures, improving logistics and border efficiency, promoting digital integration through interoperable data and e-commerce systems, and supporting the green transition through sustainable infrastructure and energy cooperation. Equally important is closing development gaps across members by investing in human capital, skills, and institutional capacity to enable broader participation in regional value chains. Through these reforms, ASEAN can consolidate its evolution from a tariff-based free trade area into a deeply connected and resilient economic community, capable of sustaining growth and cohesion amid increasing global trade fragmentation.

Source: Association of Southeast Asian Nations.

foreign direct investment (FDI) flows since 2005 benchmarked the status of ASEAN regional integration in 2025 relative to other major economic blocs such as the European Union and the United States–Mexico–Canada Agreement (USMCA), while also revealing structural and policy constraints to deeper intraregional integration. It also quantified the potential gains from removing policy barriers, including those related to logistics and human capital development.

Considerable untapped potential exists for deeper ASEAN integration, and structural reforms could materially expand trade in the region. Detailed trade and investment data document ASEAN's high external openness but modest intraregional trade and investment. As in the rest of the Asia and Pacific region, intermediate goods lead intra-ASEAN trade, reflecting both the structure of regional supply chains and the relatively limited size of final demand markets within the region. Export complementarity within the region is relatively modest compared with the level of complementarity with key external trading partners, while nontariff barriers and structural factors are key constraints to deepening regional integration. In addition to the potentially sizable economic gains from deeper integration and removal of nontariff barriers, empirical analysis reveals that ASEAN's intraregional export potential from reducing policy barriers can be significant (Chapter 10).

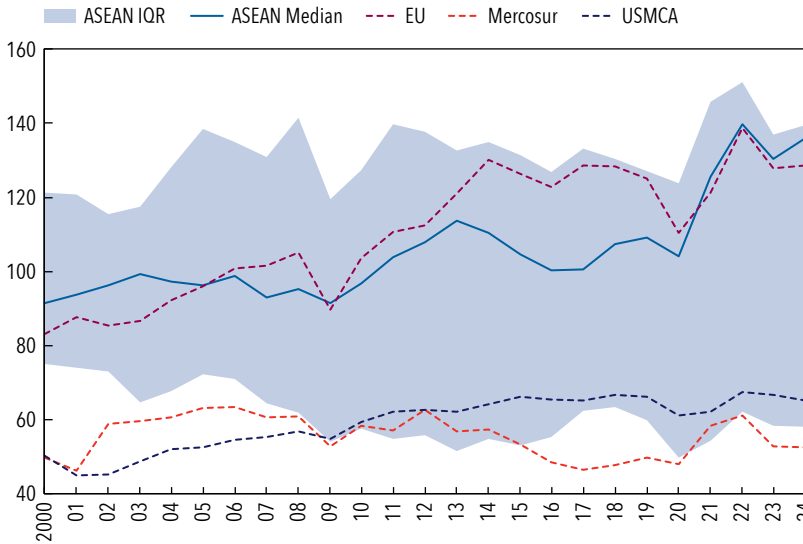
## ASEAN'S TRADE AND FDI STRUCTURE

ASEAN is among the world's most open trading regions, reflecting its members' deep integration into global value chains. ASEAN's trade openness—measured as the sum of exports and imports relative to GDP—has consistently exceeded 100 percent of GDP over the past 20 years, surpassing the levels of other major economic blocs such as USMCA, Mercosur (in South America), and, occasionally, the European Union (Figure 11.1).[1] This high degree of global openness reflects ASEAN's central role in production networks and decades of liberalization through agreements with several countries, including Australia, China, India, Japan, Korea, and New Zealand.

Within this highly open trade structure, the United States has become an increasingly important export destination. ASEAN's trade exposure to the United States has expanded significantly over the past decade, partly because of earlier trade diversion during the China–US trade tensions of the late 2010s. As a result, exports to the United States now account for a larger share of GDP in most

[1] ASEAN's 11 member states are Brunei Darussalam, Cambodia, Indonesia, Lao P.D.R., Malaysia, Myanmar, the Philippines, Singapore, Thailand, Vietnam, and Timor-Leste (which joined in October 2025). The European Union comprised 27 member states (as of 2026): Austria, Belgium, Bulgaria, Croatia, Cyprus, the Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, the Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, and Sweden. USMCA includes the United States, Mexico, and Canada. Mercosur full members include Argentina, Brazil, Paraguay, and Uruguay.

**Figure 11.1. Trade Openness across Economic Blocs, 2000–24**
*(Median for all groups; shaded area: interquartile range for ASEAN, percent of GDP)*




Sources: IMF, World Economic Outlook Database; and IMF staff calculations.
Note: ASEAN excludes Timor-Leste. ASEAN = Association of Southeast Asian Nations; IQR = interquartile range; USMCA = United States–Mexico–Canada Agreement.

ASEAN economies than in 2017. This growing reliance has heightened the region's exposure to shifts in US trade policy.

The differentiated impact of the 2018–19 trade tensions and the 2025 shift in US trade policy highlights a more challenging external environment. During escalation of US-China trade tensions in the late 2010s, several ASEAN economies benefited, though unevenly, from trade and investment diversion as production and exports relocated in response to tariff differentials. Countries such as Malaysia, Thailand, and Vietnam recorded export gains in tariff-targeted sectors, supported by increases in export values, domestic value-added, and a short-term surge in FDI (Donato and Kitsios, 2026). However, the 2025 tariff shift was broader in scope, applied across a wider range of products and trading partners, and tightened rules-of-origin provisions, which reduced the scope for additional trade and investment diversion and increased ASEAN's vulnerability to external shocks.

ASEAN's deep integration into global value chains amplified the transmission of these shocks. Many ASEAN economies are deeply embedded in global value chains, with a high share of their gross exports consisting of foreign and domestic intermediate inputs that are part of regional and global production networks (Box 11.2). Tariffs on final goods exported to the United States can thus reverberate upstream, reducing demand for intermediate goods and weakening incentives

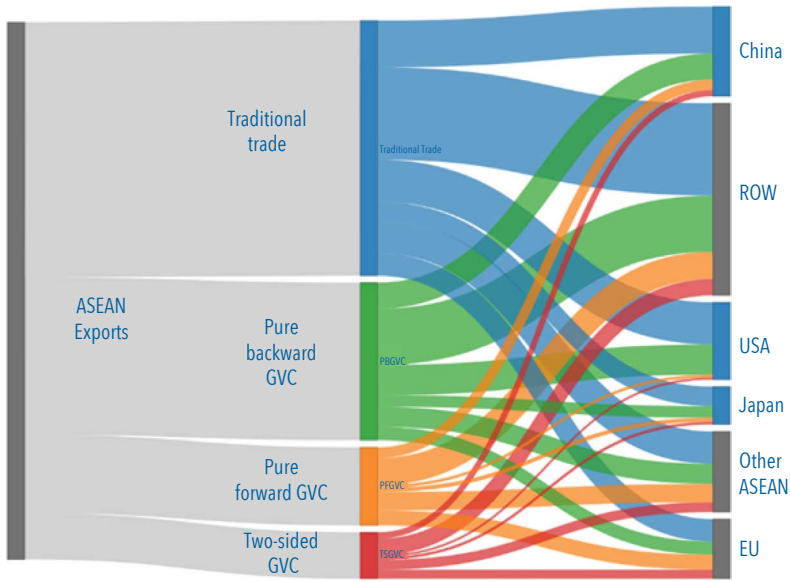
## Box 11.2. ASEAN's Role in Global Value Chains

The Association of Southeast Asian Nations (ASEAN) is deeply integrated into global value chains, and its member economies participate at multiple stages of production. A decomposition of ASEAN's exports indicates that only about half represent traditional trade—goods produced largely with domestic inputs and not part of multistage international production processes—while the remainder reflects different modes of cross-border production sharing.

Within ASEAN, traditional trade includes raw materials such as crude palm oil (Indonesia to Malaysia), natural rubber (Thailand to Vietnam), coal and liquefied natural gas (Indonesia to Thailand), and unprocessed agricultural products. Exports to China similarly consist of resource-based goods, including rubber, metal ores, and fuels. Traditional exports to final demand markets such as the European Union and the United States are dominated by garments, footwear, furniture, and processed foods.

Backward global value chain participation captures the foreign value-added embodied in a country's exports—that is, the use of imported intermediate inputs in producing goods for export, which is a key feature of ASEAN's export structure. For example, Vietnam assembles electronics using imported semiconductors from China and Korea, and Thailand imports automotive parts from Japan for vehicle assembly.

### Box Figure 11.2.1. Global Value Chain Decomposition of ASEAN's Exports, 2023




Sources: Asian Development Bank; Eora multiregion input-output table; and authors' calculations.
Note: ASEAN = Association of Southeast Asian Nations; GVC = global value chain; ROW = rest of the world.

**Box 11.2. ASEAN's Role in Global Value Chains *(continued)***

Forward global value chain participation measures the domestic value-added that leaves a country and is later embodied in another country's exports. ASEAN's upstream role is important, supplying intermediate goods such as rubber, refined fuels, and processed metals to China, Japan, and Singapore, where those inputs enter downstream production and global reexports.

Two-sided global value chain participation refers to simultaneous backward and forward links–sectors that both use imported inputs in their exports and that supply intermediates used in other countries' exports. This pattern is prominent in ASEAN's electronics and petrochemicals sectors, where economies such as Malaysia, Singapore, and Thailand import high-tech components and export higher-value intermediates to partners such as China and Korea.

ASEAN's export structure reflects its central position in global production networks, with countries acting as both upstream suppliers and downstream assemblers.

Source: Asian Development Bank

for investment in capacity expansion. ASEAN's role as a connector in China-centric supply chains means that disruptions originating from either end—US final demand or Chinese intermediate supply—can propagate across multiple economies in the region. This could amplify the impact of tariffs because even small disruptions can cascade through production networks and undermine competitiveness.

Despite ASEAN'S scale and dynamism, intra-ASEAN trade remains relatively modest (Figure 11.2). Final goods exports have stabilized at about 20–25 percent of total exports over the past 20 years—far below the levels observed in the European Union and USMCA, while services trade accounts for only about 15 percent of the region's total services exports (Figure 11.3). Trade within the region is driven predominantly by intermediate inputs and is more concentrated in upstream products, including raw materials. Capital and consumption goods, typically associated with final demand integration, account for only a small share of intraregional trade. This composition limits the scale and diversification of intra-ASEAN trade flows, reflecting both production sharing patterns and the relatively small final demand bases of many ASEAN economies.

Export complementarity within ASEAN is generally lower than with major external partners such as China and the United States.[2] For many ASEAN members—particularly the more diversified manufacturing economies such as Indonesia, Malaysia, the Philippines, Thailand, and Vietnam—export baskets

[2] The export complementarity index measures how well one country's export profile matches another country's import needs. The index is valued between 0 and 1, with 1 indicating perfect complementarity.

**Figure 11.2. ASEAN Trade Structure in 2023**




Sources: CEPII-BACI data set; and authors' calculation.
Note: ASEAN = Association of Southeast Asian Nations; BACI = Base pour l'Analyse du Commerce International; CEPII = Centre d'Études Prospectives et d'Informations Internationales.

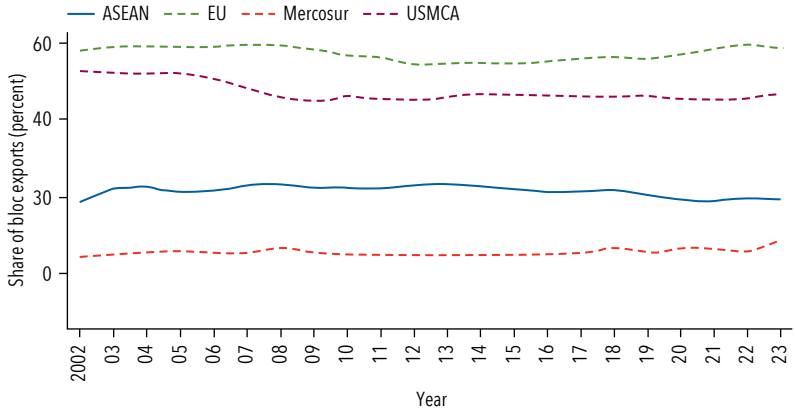
align more closely with the import needs of China and the United States than those of regional partners.

Export complementarity also varies widely across ASEAN countries, reflecting differences in industrial capacity and diversification (Figure 11.4). More advanced members tend to exhibit higher complementarity with each other, while lower-income economies such as Cambodia, Lao P.D.R., and Myanmar maintain narrow export baskets concentrated in garments, agriculture, and low-value intermediates, resulting in weaker alignment with the region's broader demand structure. Export similarity across ASEAN is also relatively low, reducing direct competition among member states, but it also limits the potential for deeper regional supply-chain integration when combined with modest complementarity and uneven industrial capability.

FDI flows within ASEAN remain limited, underscoring the scope for strengthening regional investment-trade links (Figure 11.5). Inward FDI is largely sourced from outside the bloc, with intraregional FDI shares well below those of the European Union, Mercosur, and USMCA. Most investment originates from advanced economies such as the European Union, Japan, and the United States (see Chapter 13). The analysis suggests that greater export complementarity is associated with higher trade and FDI, indicating that deeper structural and regulatory integration within ASEAN could help enhance cross-country complementarities and catalyze greater intraregional investment, supporting the development of more regionally distributed and resilient supply chains.

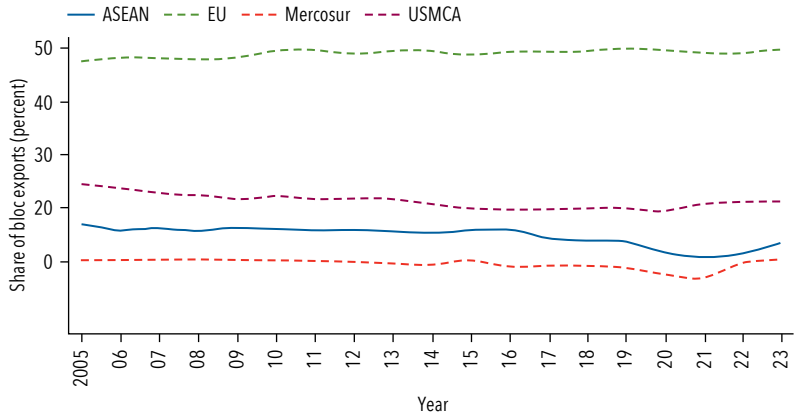
## Figure 11.3. Intraregional Exports

**1. Intraregional Goods Exports**
*(Percent of GDP)*




Sources: CEPII-BACI data set; IMF staff calculations.
Note: ASEAN = Association of Southeast Asian Nations; BACI = Base pour l'Analyse du Commerce International; CEPII = Centre d'Études Prospectives et d'Informations Internationales; USMCA = United States–Mexico–Canada Agreement.

**2. Intraregional Services Exports**
*(Percent of GDP)*

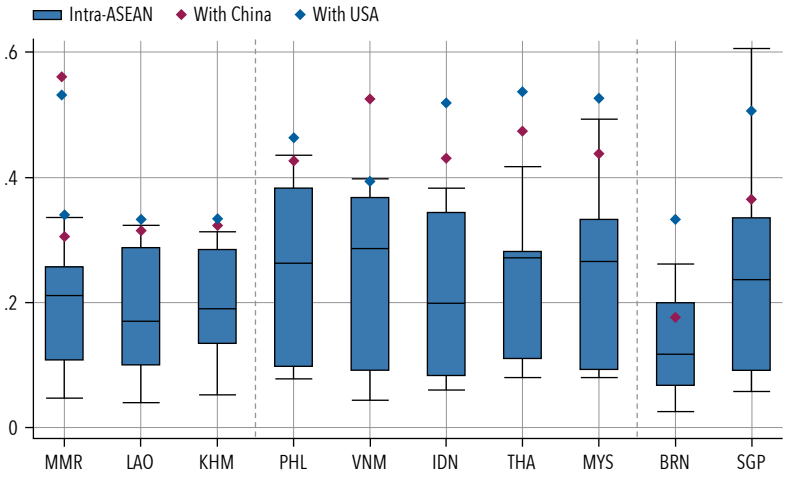



Sources: World Trade Organization–OECD BaTIS data set; and IMF staff calculations.
Note: ASEAN = Association of Southeast Asian Nations; BaTIS = Balanced Trade in Services (data set); USMCA = United States–Mexico–Canada Agreement.

Nontariff barriers remain a key constraint to deeper intra-ASEAN integration (see Chapter 10). Although tariff liberalization under the original ASEAN Trade in Goods Agreement has advanced considerably, nontariff trade barriers remain extensive and are applied unevenly. ASEAN's composite nontariff trade barrier

## Figure 11.4. Trade Complementarity in 2023

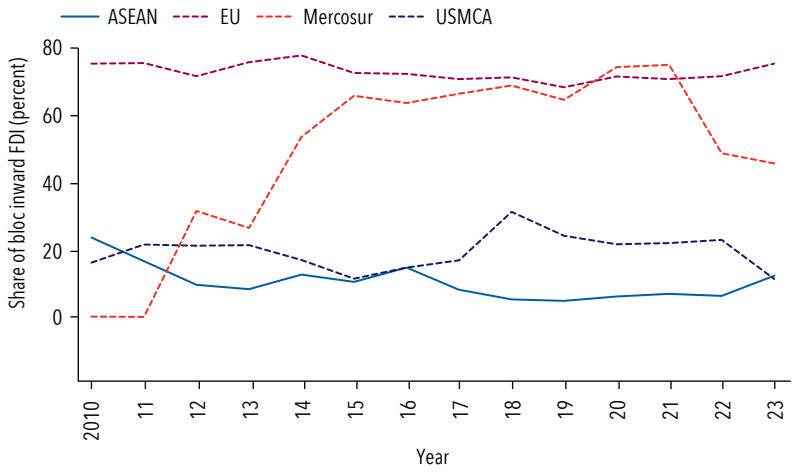
*(Index, 0–1)*




Sources: CEPII-BACI database; and authors' calculations.
Note: Data labels use International Organization for Standardization country codes. The trade complementarity index ranges from 0 (no overlap between one country's exports and another's imports) to 1 (perfect match). Box plots show the distribution of bilateral intra-ASEAN trade complementarity for each country; red and blue diamonds show complementarity with China and the United States, respectively. ASEAN = Association of Southeast Asian Nations; BACI = Base pour l'Analyse du Commerce International; CEPII = Centre d'Études Prospectives et d'Informations Internationales.

## Figure 11.5. Intra-regional Inward Foreign Direct Investment

*(Percent of total)*

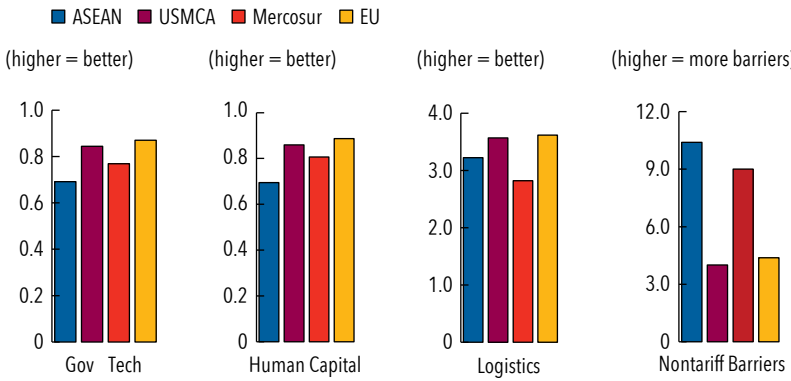



Sources: IMF Direct Investment Positions by Counterpart Economy data set; and IMF staff calculations.
Note: ASEAN = Association of Southeast Asian Nations; FDI = foreign direct investment; USMCA = United States–Mexico–Canada Agreement.

index exceeds 10, compared with about 4–5 for the European Union and USMCA (Figure 11.6). Such regulatory and administrative barriers increase trade costs, especially for smaller economies with limited institutional and technical capacity to comply with multiple regimes. The persistence of these nontariff trade barriers is closely linked to the relatively shallow institutional depth of ASEAN's trade agreements, which contain fewer provisions on regulatory convergence, mutual recognition, competition policy, and standards harmonization than those found in more integrated regional blocs.

Broader structural and institutional factors also shape the region's integration outcomes. Cross-regional benchmarking suggests that on average, ASEAN members underperform relative to advanced regional blocks such as the European Union and USMCA, across several structural aspects that underpin efficient cross-border trade (Figure 11.6). Digital government capabilities remain weaker, reflecting gaps in digital infrastructure, system interoperability, and online administrative processes. Human capital levels are similarly lower. Persistent differences in education quality, workforce skills, and vocational training limit firms' ability to move into higher-value activities. Logistics performance also trails comparator regions, pointing to constraints in port efficiency, transport networks, and customs operations. These factors influence both the cost and efficiency of cross-border trade and the capacity of member economies to participate in regional value chains.

**Figure 11.6. ASEAN: Policy Indicators**




Sources: United Nations E-Government Survey (2024); Estefania-Flores and others (2024); World Bank Logistics Performance Index (Arvis and others, 2023); and IMF staff calculations.
Note: Bars show simple averages across the member economies of each grouping. The Government Technology Index and Human Capital Index are sub-indices of the UN E-Government Development Index, each ranging from 0 to 1, with higher values indicating greater online service delivery capacity and stronger educational attainment, respectively. The Logistics Performance Index ranges from 1 to 5, with higher values indicating more efficient customs, infrastructure, and trade logistics performance, based on surveys of international logistics operators. The Nontariff barriers measure is drawn from Estefania-Flores and others (2024), with higher values indicating a greater prevalence of nontariff measures restricting trade. ASEAN = Association of Southeast Asian Nations; EU = European Union; USMCA = United States–Mexico–Canada Agreement.

## EMPIRICAL ANALYSIS: STRUCTURAL DRIVERS OF TRADE INTEGRATION

An empirical analysis on the determinants of ASEAN trade included a gravity model exercise that used a policy-augmented gravity framework to assess how structural characteristics and institutional factors (digital government, logistics performance, and human capital) shape bilateral trade in both goods and services. It also quantified the potential export gains from closing policy gaps. The model applied the Poisson Pseudo-Maximum Likelihood estimator, which is well-suited for gravity models because it accommodates zero trade flows and heteroskedasticity in the error term. This approach ensured consistent and unbiased estimates of the impact of structural variables on trade flows.[3] The analysis yielded several insights.

Digital government capability is important for both merchandise and services trade. Higher levels of digital public service delivery, proxied through the GovTech index, are associated with higher exports for both goods and services.[4] Exporter-side digital government facilitates customs procedures, licensing, and online administrative processes, while importer-side technology is especially important for services where regulatory compliance and digital documentation are critical.

Logistics performance shows a clearer relationship with goods trade than with services. Better logistics on either the exporter or importer side is associated with higher bilateral merchandise flows because they likely reduce delays, uncertainty, and transport bottlenecks that can disrupt trade flows. The absence of comparable effects for services is unsurprising, given their lower reliance on physical infrastructure. The estimates do not rule out indirect channels operating through GDP, which was controlled for separately in the gravity regression framework.

Human capital displays the strongest association with services trade. Both exporter-side and importer-side human capital correlate positively with bilateral services exports, and the magnitudes of the effects are comparatively large. A 0.1 point improvement in exporter human capital corresponds to an increase of about 6 percent in services exports, while the importer-side effect is about 2.5 percent. These patterns are in line with the skill-intensive nature of many tradable services and the importance of absorptive capacity in partner economies, though the estimates should be interpreted as average conditional correlations rather than structural elasticities.

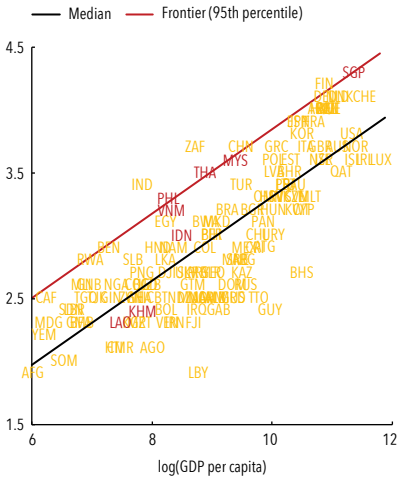
The results suggest that goods and services trade in ASEAN could respond to different structural features, with logistics more closely linked to merchandise exports and human capital more relevant for services. Digital government appears to have a cross-cutting role but with varying importance across both goods and

[3] See Annex 11.2 for a summary of the model specification and estimation approach.

[4] The GovTech index is based on the Online Services Index, a subcomponent of the UN E-Government Development Index that captures the extent to which governments leverage digital technologies for service delivery and public engagement.

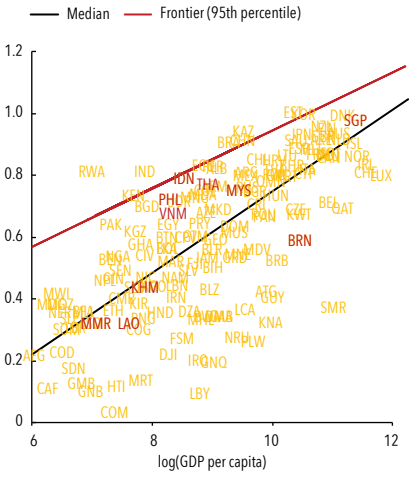
## Figure 11.7. Policy Indicators: Distance from Frontiers

**1. Logistics and GDP per capita**
*(Logistics Performance Index, 2023)*



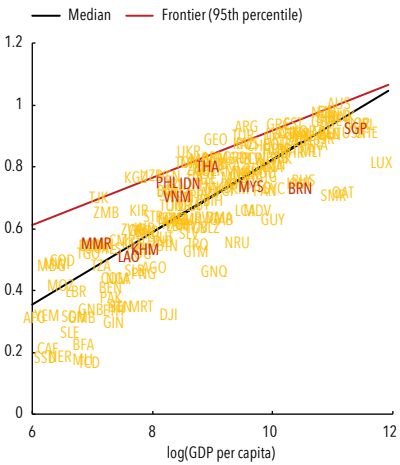

Sources: World Bank Logistics Performance Index; IMF World Economic Outlook database; and IMF staff calculations.
Note: Scores range from 1 (low) to 5 (high), with higher scores indicating better logistics performance. Data labels use International Organization for Standardization country codes.

**2. GovTech and GDP per capita**
*(Online Services Performance Index, 2023)*




Sources: UN E-Government Development Index data set; IMF World Economic Outlook database; and IMF staff calculations.
Note: Scores range from 0 (low) to 1 (high), with higher scores indicating higher quality of e-government services. Data labels use International Organization for Standardization country codes.

**3. Human Capital and GDP per capita**
*(Human Capital Index 2023)*




Sources: UN E-Government Development Index data set; IMF World Economic Outlook database; and IMF staff calculations.
Note: Scores range from 0 (low) to 1 (high), with higher scores indicating a greater level of human capital development. Data labels use International Organization for Standardization country codes.

services trade flows. The significance of importer-side characteristics in several cases also provides empirical support to the assumption that regional progress, not domestic reforms alone, can be important for deepening ASEAN trade integration.

ASEAN members differ markedly in their distance from the income-consistent frontiers for logistics, digital government, and human capital (Figure 11.7). Some remain well below attainable standards across all three dimensions. Others are closer to their respective frontiers, though all still fall short of feasible performance levels in at least one policy area. Together, these frontier distances mirror the pattern of simulated export gains, with larger potential improvements concentrated in economies that remain farthest from attainable benchmarks.

## POLICY REFORM SIMULATIONS: QUANTIFYING THE GAINS

The analysis simulated counterfactual export flows, based on the estimated gravity coefficients and each country's policy gaps, to quantify the potential impact of structural reforms on trade. The estimated gains were obtained by applying the exporter-side and importer-side statistically significant coefficients to each ASEAN member's shortfall from the policy frontier, defined as the minimum of the fitted 95th percentile benchmark and the indicator's maximum possible value. The analysis then used those adjusted gaps to simulate counterfactual bilateral goods and services export flows in 2023. Counterfactual trade gains were calculated as the observed baseline multiplied by an exponential function of the estimated semi-elasticities and the corresponding policy gaps, with aggregate gains reported as the difference from baseline intra- and extra-ASEAN trade and expressed as percent of GDP. Thus, the heterogeneity in gains mirrors the underlying gravity regression estimated elasticities, the size of each country's policy gap, and the scale of its exports relative to GDP. Overall, the findings support the view that structural trade enablers—logistics, digital government, and human capital—have distinct but complementary roles in shaping ASEAN's goods and services integration, with the highest returns in countries that are further from the policy frontier and whose exports represent a larger share of GDP.

Logistics improvements can yield large gains in goods trade for ASEAN members facing the most severe constraints (Figure 11.8.1). Lower-income economies such as Cambodia and Lao P.D.R., where logistics gaps remain substantial, would gain the most in terms of pronounced increases in goods exports under the counterfactual simulations.[5] Indonesia also would record notable gains because its distance from the logistics frontier leaves room for efficiency improvements that still yield meaningful effects, although its goods export share is lower than that of more trade-intensive ASEAN economies. By contrast, Malaysia, the Philippines Singapore, Thailand, and Vietnam exhibit smaller simulated increases. These

[5] Both Cambodia and Lao P.D.R. are taking steps to narrow these gaps, including through recent substantial investments in logistics infrastructure.

economies are clustered near the logistics frontier, and although several of them have substantial goods export shares, the limited remaining scope for logistics-driven efficiency gains constrain the overall impact.

Human capital improvements generate the largest simulated gains in services trade for ASEAN members that pair substantial skill gaps with strong service export shares (Figure 11.9.1). Singapore exhibits the largest gains, reflecting both its exceptionally high service export exposure and a meaningful distance to the human capital frontier. Cambodia and Lao P.D.R. also show sizable gains, reflecting their substantial scope for human capital improvements and the strong sensitivity of services exports to skills. Malaysia follows closely because of its remaining distance from the human capital frontier and the fact that services already account for a significant share of its GDP. The human capital levels of economies such as Indonesia, the Philippines, Thailand, and Vietnam are closer to the frontier, and those economies exhibit more moderate gains, consistent with their smaller remaining gaps and therefore more limited scope for additional improvements. By contrast, Myanmar shows somewhat larger effects because of its greater distance from the frontier.

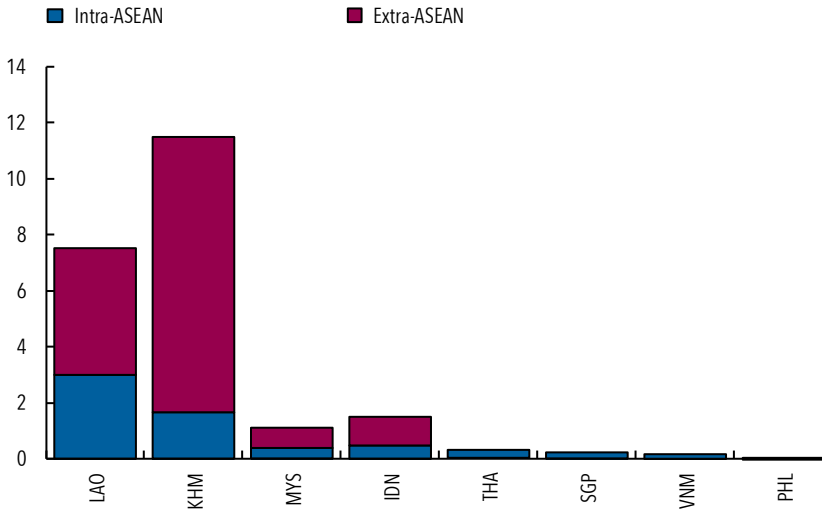
Digital government improvements yield significant gains across ASEAN, though the relative importance of such reforms differs between goods and services trade (Figures 11.8.2 and 11.9.2). For goods, the strongest gains arise in economies such as Cambodia and Lao P.D.R., where relatively large technology gaps leave considerable scope for improvements in digital administrative systems. Enhancements in these areas are likely to ease documentation and procedural steps that merchandise exporters face, which helps explain their comparatively large, simulated effects. Indonesia and Myanmar also register nonnegligible gains, consistent with the interpretation that greater distance from the e-government frontier could be associated with less streamlined trade-related administrative processes. By contrast, economies closer to the frontier—such as Malaysia, Singapore, Thailand, and Vietnam—exhibit smaller gains because the remaining scope for additional efficiency improvements in goods-related administrative procedures appears more limited. For services, the largest gains also appear notably in Cambodia and Lao P.D.R and in other economies with the widest gaps from the frontier. Brunei Darussalam and Myanmar show moderate gains, while economies closer to the frontier record comparatively smaller effects. This pattern is consistent with the gravity model estimates, which indicate that the e-government role in shaping both goods and services trade relative to logistics (for goods) and human capital (for services) is statistically significant, though more moderate.

Reducing nontariff barriers could yield substantial gains for ASEAN (Chapter 10). Recent IMF research on trade integration in Asia estimates that ASEAN could achieve gains exceeding 4 percent of GDP by reducing nontariff trade barriers, the largest potential improvement among subregions (IMF 2025). About half of those gains arise from unilateral domestic reforms such as streamlining licensing procedures, simplifying customs documentation, and improving regulatory transparency, while the rest reflect regional or multilateral efforts that lower behind-the-border barriers across members.

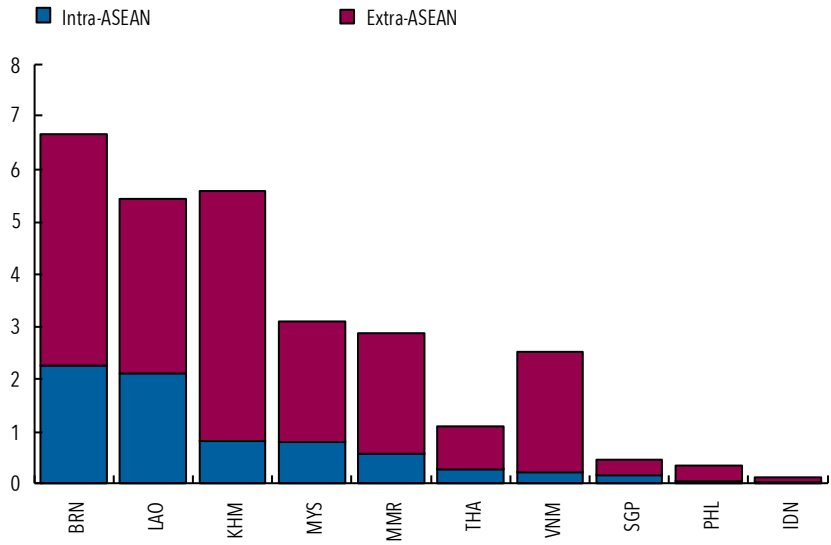
## Figure 11.8. Reform Simulation: Potential Export Gains in Goods Exports

*(Percent of GDP, 2023)*

**1. Enhancing Logistics**




**2. Enhancing Digital Government**

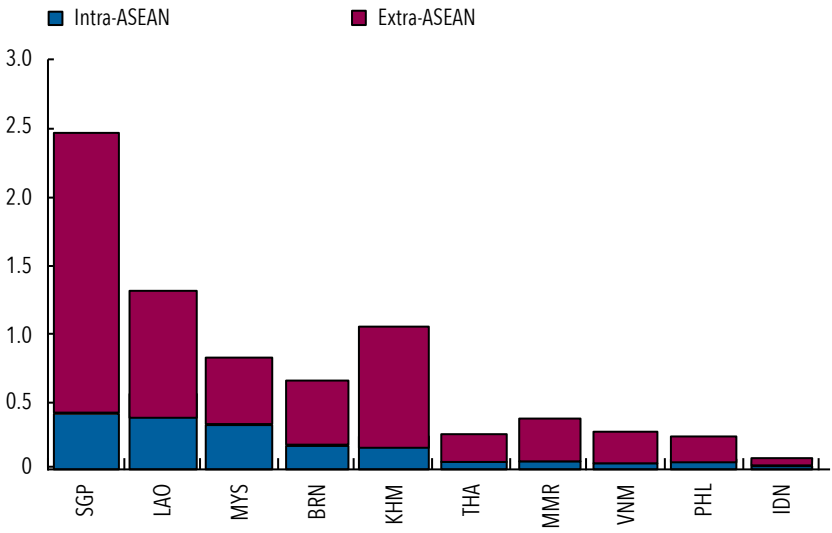



Sources: BACI-CEPII data set, UN E-Government Development Index data set; World Bank Logistics Performance Index; IMF World Economic Outlook database; and IMF staff calculations.
Note: The estimated gains were derived by applying the exporter-side and importer-side statistically significant coefficients on the respective structural indicators (Logistics Performance Index, UN Online Services and Human Capital Indices) to each ASEAN member's shortfall from income-consistent frontiers (95th percentiles) and simulating counterfactual 2023 bilateral export flows. "Extra-ASEAN" refers to ASEAN exports to non-ASEAN countries. Data labels use International Organization for Standardization country codes. ASEAN = Association of Southeast Asian Nations.

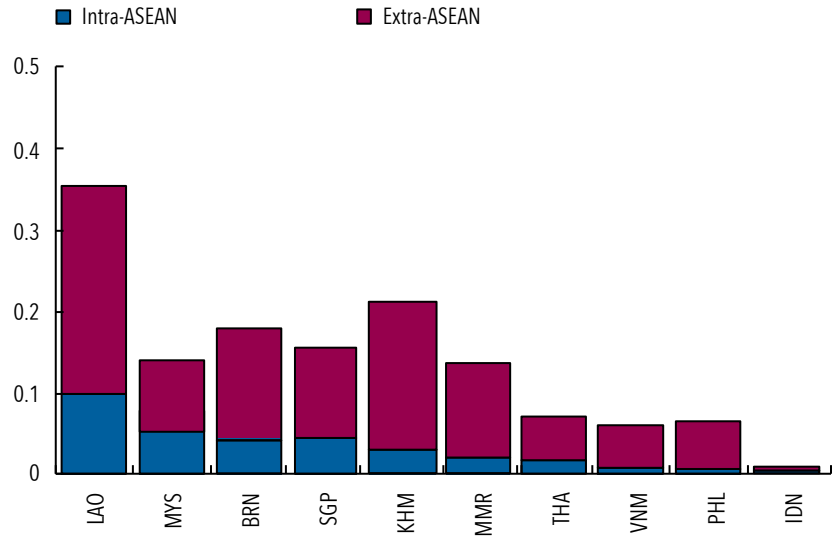
## Figure 11.9. Reform Simulation: Potential Export Gains in Services Exports

*((Percent of GDP, 2023)*

**1. Enhancing Human Capital**




**2. Enhancing Digital Government**




Sources: BACI-CEPII data set, UN E-Government Development Index data set; World Bank Logistics Performance Index; IMF World Economic Outlook database; and IMF staff calculations.

Note: The estimated gains were derived by applying the exporter-side and importer-side statistically significant coefficients on the respective structural indicators (Logistic Performance Index, UN Online Services and Human Capital Indices) to each ASEAN member's shortfall from income-consistent frontiers (95th percentiles) and simulating counterfactual 2023 bilateral export flows. "Extra-ASEAN" refers to ASEAN exports to non-ASEAN countries. Data labels use International Organization for Standardization country codes. ASEAN = Association of Southeast Asian Nations

## CONCLUSION AND POLICY IMPLICATIONS

The evolving global trading environment presents new challenges and important opportunities for ASEAN. The region has benefited greatly from a rules-based multilateral trading system and from export-driven development supported by deep integration into global value chains. However, the increasing fragmentation of the global trading system—including rising trade tensions and the announcement of broad-based US tariffs in early 2025—highlights the vulnerabilities associated with concentrated extraregional demand and globally dispersed supply chains. These shifts also underscore the opportunity for ASEAN to leverage its growing internal market to support deeper intraregional trade, investment, and production links.

The analysis of the intra-ASEAN trade structure points to scope for deepening intraregional integration. Intermediate inputs and raw materials dominate intra-ASEAN trade, with limited final demand integration. The degrees of export complementarity and similarity within the region are modest and heterogeneous across members, pointing to untapped potential for expanding regional integration. Inward FDI to ASEAN continues to be dominated by extraregional sources, while intraregional investment remains limited. Because FDI is positively correlated with export complementarity, stronger structural and regulatory alignment within ASEAN could encourage greater intraregional investment.

Potential export gains from removing policy and structural barriers to intra-ASEAN trade can be significant. The assessment focused on three key enablers, each affecting trade through different channels: logistics, digital government, and human capital. Specifically, logistics are more closely linked to merchandise exports, human capital is particularly important for services trade, and the role of digital government is cross-cutting. A country's potential export gains reflect two factors: the country's economic structure (both overall trade openness and reliance on services and merchandise exports) and distance from the frontier (defined as the calculated shortfall across the three key indicators relative to their computed benchmarks). Therefore, the reform gains vary markedly across ASEAN members. For example, Singapore is estimated to gain the most from human capital improvement, given the country's high reliance on services and its remaining gap to the frontier, while Cambodia and Lao P.D.R. are expected to gain the most from upgrading logistics.

These findings present several important policy implications.

Reducing nontariff barriers and advancing regulatory cooperation could yield substantial benefits. Even unilateral reductions in nontariff trade barriers would already deliver sizable gains, without needing reciprocal action from the rest of the world. Harmonizing product standards, simplifying customs procedures, and expanding the use of mutual recognition agreements would help reduce transaction costs and increase the predictability of intraregional trade. In this context, advancing implementation of the upgraded ASEAN Trade in Goods Agreement (ATIGA 2.0), which focuses on nontariff measures, rules-of-origin modernization, and regulatory convergence, is crucial to reducing fragmentation and supporting intraregional trade.

Strengthening digital trade facilitation through improved digital government is another important policy implication. Digital public administration has a cross-cutting role in supporting both goods and services trade. Expanding interoperable digital ID systems, enabling electronic signatures, and fully implementing the ASEAN Single Window would reduce time and costs for traders. Strengthening national and regional frameworks for data governance, cybersecurity, and digital payments would also support the expansion of digitally delivered services and reduce friction associated with cross-border regulatory compliance.

Improving logistics performance and connectivity—including investment in cross-border transport infrastructure, logistics networks, and data connectivity—remains critical to reducing bottlenecks and linking smaller, landlocked, or less-developed economies to regional production hubs. Accelerating implementation of the Master Plan on ASEAN Connectivity 2025 and enhancing public-private partnerships could help mobilize the necessary investment (see Chapter 9).

Strengthening human capital and institutional capacity would support deeper intraregional integration. Closing skills gaps and improving workforce mobility would facilitate the diffusion of technology and knowledge across borders. At the institutional level, harmonizing regulatory frameworks and strengthening digital government would enable smoother cross-border transactions and support emerging sectors such as e-commerce and digital trade.

Policies that facilitate regional sourcing, encourage supplier upgrading, and promote cross-border industrial clusters can strengthen intra-ASEAN production ties. Reducing administrative barriers for small and medium enterprises, harmonizing certification requirements, and supporting digital supply-chain platforms can also help broaden integration among member states.

Promoting intra-ASEAN investment integration is another important priority. Because FDI is strongly associated with export complementarity, deeper structural and regulatory alignment can catalyze more intra-ASEAN investment. Priorities include strengthening the ASEAN Comprehensive Investment Agreement, simplifying cross-border investment procedures, improving investor protection, and promoting investment in sectors with strong regional value chain potential such as electronics, green manufacturing, and digital services.

Greater implementation capacity and regional coordination would support the effectiveness of ASEAN agreements, which require consistent national implementation. Strengthening monitoring mechanisms, enhancing peer-learning platforms, and improving data sharing on nontariff measures and supply-chain disruptions would help sustain reform momentum.

## ANNEX 11.1. EXPORT STRUCTURE ALIGNMENT AND BILATERAL TRADE AND INVESTMENT: A GRAVITY ANALYSIS

An empirical analysis examined whether the alignment between countries' export baskets and their partners' import needs is associated with deeper cross-border economic links, including trade in goods and services and bilateral foreign direct

investment (FDI). A bilateral gravity model that augmented standard economic mass controls with measures of export complementarity and export similarity was estimated. The analysis focused on whether trading partners with more complementary export-import structures engage more intensively in both trade and new investment activity. The model was estimated for a country sample comprising 177 economies for the goods and services specifications and 170 economies for the FDI specifications

The empirical specification is given by the following:

$$y_{ijt} = \exp(\alpha + \beta_1 CI_{ij,t-1} + \beta_2 ESI_{ij,t-1} + \gamma_1 \, log\, GDP_{it-1} + \gamma_2 \, log\, GDP_{jt-1} + \gamma_3 \, log\, Pop_{it-1} + \gamma_4 \, log\, Pop_{jt-1} + \eta Tariff_{ij,t-1} + \xi_{ij} + \lambda_t) + \varepsilon_{ijt} \quad (11.1.1)$$

where $y_{ijt}$ denotes bilateral goods exports, services exports, inward FDI value, or the number of newly announced investment projects from country $i$ to $j$ in year $t$. The variable $CI_{ij,t-1}$ captures lagged export complementarity, and the variable $ESI_{ij,t-1}$ captures export similarity, both providing a direct measure of how well country $i$'s export structure matches country $j$'s import profile or similarity in their export baskets. GDP and population (lagged one year) controlled for economic scale and market size, while bilateral tariffs capture trade policy friction. Country-pair fixed effects $\xi_{ij}$ absorb all time-invariant bilateral characteristics, and $\lambda_t$ captures global shocks affecting all country pairs.

**ANNEX TABLE 11.1.1.**

Gravity Model Estimates

| | (1) | (2) | (3) | (2) |
|---|---|---|---|---|
| | | | Inward FDI | |
| | Goods | Services | Value | Projects |
| Complementarity Index | **0.727*** (0.253)** | **0.596*** (0.197)** | 0.859 (0.955) | **2.023*** (0.413)** |
| Export Similarity Index | 0.136 (0.236) | 0.371 (0.243) | 0.00164 (0.736) | **−1.239** (0.585)** |
| Estimator | PPML | PPML | PPML | PPML |
| Controls | Yes | Yes | Yes | Yes |
| Dyad + Time FE | Yes | Yes | Yes | Yes |
| Pseudo-R2 | 0.988 | 0.988 | 0.756 | 0.807 |
| Observations | 400877 | 402790 | 30461 | 30473 |

Sources: CEPII-BACI data set; World Trade Organization–OECD BaTiS data set; IMF World Economic Outlook database; Orbis Crossborder Investment database; and IMF staff calculations.
Note: The dependent variable "goods" refers to bilateral goods exports from the BACI data set, "services" refers to bilateral services exports (excluding financial services) from the BaTiS data set, and inward FDI values and project counts are sourced from the Orbis Crossborder Investment database. Control variables include lagged importer and exporter GDP, population, and bilateral applied tariffs. BACI = Base pour l'Analyse du Commerce International; CEPII = Centre d'Études Prospectives et d'Informations Internationales; FE = fixed effects; FDI = foreign direct investment; PPML = Poisson Pseudo-Maximum Likelihood.
**$p < 0.05$; ***$p < 0.01$.

The results consistently point to a positive association between bilateral complementarity and cross-border economic integration (Annex Table 11.1.1). Higher confidence interval values are strongly correlated with greater bilateral trade flows: the coefficient on lagged confidence interval is sizable and statistically significant across goods and services exports. These magnitudes imply that improvements in the alignment between a country's export structure and its partner's import needs translate into meaningfully higher trade volumes. A similar pattern emerges for FDI. Although complementarity does not significantly predict the value of inward FDI flows, possibly reflecting the lumpiness and volatility of project sizes, it is strongly and significantly associated with the number of new investment projects. This suggests that firms are more likely to establish a presence in markets where their export capabilities map closely to local demand conditions, reinforcing the role of structural alignment in shaping the extensive margin of FDI.

By contrast, the Export Similarity Index, which reflects similarity in export structures and thus potential competition, is generally unrelated to inward FDI volumes and is negatively associated with project counts. Together, these patterns imply that complementarity, not similarity, drives deeper economic ties because countries with export structures that better match each other's import needs tend to trade more and initiate more investment projects with each other.

## ANNEX 11.2. POLICY AND STRUCTURAL DETERMINANTS OF BILATERAL TRADE

To assess the structural drivers of trade integration in member countries of the Association of Southeast Asian Nations, a gravity model of bilateral trade flows was estimated for both goods and services using panel data from the CEPII-BACI and World Trade Organization–OECD BaTIS data sets.[6] The BACI data set provides harmonized trade data at the product level, while BaTIS provides bilateral trade-in-services data across a wide range of sectors.

The baseline specification is as follows:

$$\begin{aligned} Export_{ijt} = exp(&\alpha + \beta_1 \, log\ GDP_{i,t-1} + \beta_2 \, log\ GDP_{j,t-1} + \beta_3 \, log\ Pop_{i,t-1} + \beta_4 \, logP\ op_{j,t-1} \\ &+ \gamma_{11} GovTech_{i,t-1} + \gamma_{12} GovTech_{j,t-1} + \gamma_{21} Logistics_{i,t-1} + \gamma_{22} Logistics_{j,t-1} \\ &+ \gamma_{31} HC_{i,t-1} + \gamma_{32} HC_{j,t-1} + \eta \, Tariff_{ij,t-1} + \xi_{ij} + \delta_t) + \epsilon_{ij,t} \end{aligned} \quad (11.2.1)$$

where $i$ and $j$ denote exporter and importer countries, respectively, $t$ denotes year; *GDP* and *Pop* capture economic mass and market size; *GovTech* refers to the UN's Online Services Index, measuring digital government; *Logistics* is the World

[6] Base pour l'Analyse du Commerce International (BACI) is a detailed international trade database developed by the Centre d'Études Prospectives et d'Informations Internationales (CEPII), which harmonizes and reconciles bilateral trade flows that countries report to the UN Comtrade database.

**ANNEX TABLE 11.2.1.**

| Gravity Model Estimates | | |
|---|---|---|
| | (1) | (2) |
| | Goods | Services |
| GovTech (Exporter) | **0.228*** (0.024)** | **0.097*** (0.018)** |
| GovTech (Importer) | 0.036 (0.023) | **0.056*** (0.019)** |
| Logistics (Exporter) | **0.201*** (0.022)** | −0.007 (0.017) |
| Logistics (Importer) | **0.106*** (0.019 )** | −0.010 (0.016) |
| Human Capital (Exporter) | **0.153* (0.082)** | **0.600*** (0.064)** |
| Human Capital (Importer) | 0.020 (0.080) | **0.255*** (0.064)** |
| Estimator | PPML | PPML |
| Controls | Yes | Yes |
| Dyad + Time FE | Yes | Yes |
| Pseudo-R2 | 0.993 | 0.991 |
| Observations | 257935 | 259021 |

Sources: CEPII-BACI data set; World Trade Organization–OECD BaTiS data set; IMF World Economic Outlook database; and IMF staff calculations.
Note: The dependent variable "goods" refers to bilateral goods exports from the BACI data set, while "services" refers to bilateral services exports (excluding financial services) from the BaTiS data set. Control variables include lagged importer and exporter GDP, population, and bilateral applied tariffs. BACI = Base pour l'Analyse du Commerce International; CEPII = Centre d'Études Prospectives et d'Informations Internationales; FE = fixed effects; PPML = Poisson Pseudo-Maximum Likelihood.
**$p < 0.05$; ***$p < 0.01$.

Bank's Logistics Performance Index; *HC* is the World Bank Human Capital Index; and *Tariff* represents bilateral applied tariffs. All main economic variables are lagged by one year to mitigate potential endogeneity concerns and ensure that the explanatory variables reflect information available to economic agents at the time trade decisions are made. $\xi_{ij}$ captures country-pair fixed effects, controlling for time-invariant bilateral characteristics such as geographic proximity, shared language, and colonial ties, while $\delta_t$ controls for year-specific shocks that may affect trade flows uniformly across countries and over time.

The model was estimated using the Poisson Pseudo-Maximum Likelihood estimator, which is well-suited for gravity models because it accommodates zero trade flows and heteroskedasticity in the error term. This approach ensures consistent and unbiased estimates of the impact of structural variables on trade flows. Annex Table 11.2.1 shows the estimated coefficients from the bilateral gravity model estimated on the full global sample.

The results point to heterogeneity in how structural characteristics relate to goods and services trade. Exporter-side digital government is positively associated

with both goods and services exports, consistent with the idea that improvements in digital public administration may ease documentation, licensing, and compliance processes. A modest 0.1 point increase in the GovTech index is associated with about 2 percent higher goods exports and about 1 percent higher services exports. Importer-side digital government, by contrast, appears more relevant for services, which aligns with the notion that cross-border services provision often requires compliance with destination-market regulatory procedures—such as licensing, certification, data-handling rules, and online administrative processes—that depend heavily on the quality of the importing country's digital public infrastructure.

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CHAPTER 12

# The Evolving Development Landscape of Asia and Pacific: The International Economics of Strategic Interdependence

Danny Quah[1]

The Asia and Pacific region's remarkable developmental success has been built in part on mutual gains generated by cross-border trade in goods and services. However, today's geostrategic competition and the fraying of multilateralism have created new structural constraints. The emerging international economics of strategic interdependence challenges traditional thought on trade openness, economic efficiency, and the benefits of price-taking and norm-abiding behaviors. As a result, the region's small states must reconsider strategies based on their simply accepting prices and norms. The new order they build can comprise continually shifting, loosely structured, open alliances rather than be based on sweeping treaties or grand vision. Members of such alliances can cooperate whenever their interests align. Hegemonic leadership is optional, and the alliances need only to be multilateral enough.

## INTRODUCTION

The Asia and Pacific region's continued economic growth and development now face headwinds, despite a remarkable history of success. Jobs and employment, finance, digitalization, trade and integration, sustainability and the environment, industrial policy (how to implement it and how to respond to it), and demographic change are among the region's most immediate and visible challenges. However, a new challenge has also emerged: disruption of the international economic system driven by geopolitical realignment and the fraying of multilateralism. At a minimum, this challenge calls for a reconfiguration of historical global supply chains and patterns of trade. More fundamentally, and in ways that are already becoming apparent, economic security requires reassessing the importance of economic efficiency and comparative advantage in policymaking relative to trust, resilience, and strategic maneuvering. Geopolitical risk, reflecting events

[1] The chapter presents the personal views of the author and does not reflect the views of IMF staff, management, or its Executive Board.

or actions leading to economic disruption, instability, or losses, is becoming an increasingly important force shaping the international landscape.

Some of the risks facing the region and the world arise from what is often described as a rivalry between China and the United States or even a new Cold War, but that framing is misleading and obscures more than it clarifies.

In traditional international relations thought, such as Strassler (1996, Book Five, para. 5.88) or Waltz (1979, 72), small states are price takers, and their capacity to affect the global environment is generally not studied. However, the disruption of multilateralism gives small states a strong incentive to influence cross-national outcomes, and if not globally, then at least in ways that advance their own interests. They can do so either regionally or plurilaterally while seeking to shield themselves as much as possible from geopolitical disruptions elsewhere. To navigate this newly disrupted international system, countries in the region can pursue several strategies, including alignment, acquiescence, adaptation, and mitigation. China and India, as economic giants, are not representative of the region's agency and influence in this regard.

## ASIA AND PACIFIC: REMARKABLE AND UNEXPECTED SUCCESS

The Asia and Pacific region's striking rise in the global economy occurred during a period when economic policymaking and geopolitical analysis aligned in advancing ideas of economic efficiency, comparative advantage, and political convergence (Armstrong and Quah 2026; Quah 2026a). Although consensus on those ideas was nearly global, their application was far from universal. The region's success was therefore both remarkable and surprising.

Table 12.1 shows that in 1990, the region's contribution to global economic growth was one-quarter, and by 2023, that share had more than doubled to

**TABLE 12.1.**

Asia and Pacific Successes

| | 1990 | 2023 |
|---|---|---|
| Contribution to global growth *(percent)* | 1/4 | 2/3 |
| GDP per capita *(thousands of US dollars)* | 1.6 | 9.1 |
| GDP *(trillions of US dollars)* | 5.5 | 40 |
| Total population *(billions)* | 3.2 | 4.4 |
| Working-age population *(billions)* | 2.0 | 2.8 |
| Employment *(billions)* | 1.6 | 2.2 |
| Extreme poverty *(percent)* | 52 | 6.5 |
| Life expectancy *(years)* | 66 | 75 |
| Adult literacy *(percent)* | 71 | 91 |

Sources: IMF; International Labour Organization; United Nations Educational, Scientific, and Cultural Organization; World Bank, World Development Indicators; and author's calculations.

two-thirds. Over the same period, inflation-adjusted per capita GDP increased more than fivefold, and total GDP grew nearly eightfold. Extreme poverty—the share of the population living on less than 1 international dollar a day—fell from more than one-half to one-sixteenth of the population. Life expectancy at birth increased by a full decade, and adult literacy rose by 20 percentage points.

## Systemic Diversity and Uneven Success in Asia and Pacific

Even at this aggregate level, some emerging risks are already apparent. The working-age population increased from 2 billion to 2.8 billion, but employment grew more slowly, from 1.6 billion to 2.2 billion, and thus growth and development failed to generate a corresponding increase in employment opportunities for the region's working-age population.

This mismatch between employment and demographic trends becomes even more pronounced at the country level. Across the region, the dependency ratio—the ratio of the population not of working age (younger than age 16 or older than age 65) to the working-age population—has been constant at approximately 36–37 percent over the past three decades. Yet this apparent stability masks substantial variation across countries. Some of the region's smallest countries and those in its northeastern area are undergoing sharply rising dependency ratios and declining fertility rates.[2] In Singapore, the share of the population ages 65 and older increased from 13 percent of the total population in 2015 to 21 percent in 2025 and is projected to reach 25 percent by 2030. In Korea, the share reached 20 percent in 2025 and is projected to rise to 34 percent by 2040 and 40 percent by 2070. In Japan, 28 percent of the population was ages 65 and older in 2025, when the median age was 50. In China, those 65 and older were only 7 percent of the total population in 2000 but that share had risen to 16 percent by 2025.

The region's overall dependency ratio has remained constant over the past three decades despite these demographic shifts, reflecting the offsetting effect of younger and growing populations elsewhere in Asia and Pacific, particularly in India, presenting a different challenge. India has experienced a demographic bulge of approximately 500 million in its working-age population since 1990, but employment growth has not kept pace. Over the same period, total employment grew by only 286 million, equivalent to just under 60 percent of the increase in the working-age population.

This regional mismatch between employment and demography compounds the development challenge. Even if productivity were unaffected and populations could move frictionlessly across national borders, the region's long-term demographic trends, calibrated to current population dynamics, would imply a steady-state decline of 7.5 percent relative to trend.[3] More realistically, productivity

[2] Data are from the United Nations Educational, Scientific, and Cultural Organization, UN World Population publications, and the International Labour Organization.

[3] Analysis in Annex 12.1 derived this in a model of McKendrick-von Foerster distribution dynamics. Two assumptions were critical: the aging population does not influence productivity trends adversely, and reallocation across the region's labor markets is frictionless.

trends will slow, and populations will be unable to move freely across the region. Therefore, the 7.5 percent decline should be viewed as a lower bound, with the actual impact likely to be larger.

## Asia and Pacific: Large but also Small

Table 12.2 shows the substantial variation in population across the region's countries. Considerable policy attention appropriately focuses on the economic transformations of China and India, but countries with populations in the billions are exceptional. In 2023, the median country in the region had a population of only 30 million, meaning that half of the region's countries were roughly the size of Malaysia (about 33 million) or smaller. Nearly 80 percent of the region's population is concentrated in the top 10 percent of most populous countries, so the distribution has a long left tail, with a mean-median ratio—a standard measure of inequality—of 3.5. In 2023, the mean-median ratio had already declined from 4.0 in 1990. The global distribution is even more skewed toward small states (Table 12.2). Compared with Asia and Pacific, the world's median country is smaller, and population inequality is greater, while a similarly large share of the global population is concentrated in the top 10 percent of most populous countries.

Population size and economic performance (GDP per capita measured at market exchange rates) are correlated both across the region and across the world. Historically, smaller states in the region tended to have higher per capita incomes than larger economies. The same relationship held globally but was more pronounced in Asia and Pacific. In 1990, large economies such as Bangladesh, China, India, Indonesia, and Pakistan had relatively lower incomes, whereas income in smaller states such as Brunei, New Zealand, Singapore, Switzerland, Sweden, and the Gulf states was higher. Over time, however, per capita incomes in the region's larger economies—China and India—increased, weakening the relationship between size and economic performance both within the region and globally. Although the negative correlation persists in Asia and Pacific, even at half its original magnitude, it has nearly disappeared at the global level.

Across the world, smaller economies also historically tended to be more successful, though the United States has long been an outlier because it has a large

**TABLE 12.2.**

Asia and Pacific Population versus World Population

| | Asia and Pacific | | | World | | |
|---|---|---|---|---|---|---|
| Year | Median Population (millions) | Mean-Median Ratio | Top 10% (percent) | Median Population (millions) | Mean-Median Ratio | Top 10% (percent) |
| 1990 | 19 | 4.2 | 71 | 4.5 | 6.0 | 75 |
| 2023 | 30 | 3.5 | 77 | 9.5 | 4.3 | 73 |

Sources: UN World Population Prospects 2024; and author's calculations.
Note: Population figures are across countries. The mean-median ratio is a standard measure of inequality. "Top 10%" denotes population share of the top 10 percent of the cross-section of countries.

population and high per capita income. As in Asia and Pacific, however, the rise of larger economies has caused the historical negative correlation between size and economic performance to nearly disappear.

If this dynamic continues, the region's smaller economies could start lagging the larger states in both growth and incomes. In economic reasoning, many factors predict the success of large economies, including increasing returns to scale, specialization and diversification, industrial complexity, and the ability to undertake research and development at scale and thus advance technology.

By contrast, small states have limited populations and thus smaller workforces. These states will invariably produce too much of what their people can and not enough of what their people need. In such circumstances, cross-border trade becomes even more important. Large economies can contemplate economic isolation if international circumstances compel them to do so, but they will be poorer at the margin as a result. Conversely, small states succeed economically and geopolitically when they can navigate the international environment successfully.

## Asia and Pacific and World Order: Rules-Based Multilateralism

The dynamics in Tables 12.2 and 12.3 can be summarized succinctly. In the modern era, all countries in Asia and Pacific were small states. Their markets were small because incomes were low, populations were small, or both. Those economies that eventually grew did so by learning to trade internationally, which overcame the limitations imposed by population size. Larger economies can leverage economies of scale, specialization and diversification, and technology, but small states remain structurally dependent on the international system that prevailed through most of post–World War II period: an open global economy governed by clear rules that neither favored nor disadvantaged the region's economies or smaller states. Because small states have more difficulty self-insuring and cannot project coercive power, their prosperity and stability depend on the predictability of the rules-based international order.

That international system has become less hospitable both to cross-border trade and small states. Understanding the drivers of that change can help the region's smaller states navigate the resulting disruptions.

**TABLE 12.3.**

Asia and Pacific and the World: The Dynamic Effects of Size

| Correlation | 1990 | 2023 |
|---|---|---|
| Asia-Pacific | −0.45 | −0.20 |
| World | −0.30 | −0.07 |

Source: Author's calculations from *World Economic Outlook*, World Bank, and UN World Population Prospects.
Note: The correlations are between population and GDP per capita measured at market exchange rates.

## THE THREE-SHOCK CHALLENGE

The international system that prevailed through most of the 80 years after World War II—and under which Asia and Pacific achieved its development success—had several defining features. It was a rules-based order organized around multilateralism, centered on US unipolarity and thus hegemony, and it promoted liberal norms and values.

The system was never perfect. The United States acted as benevolent hegemon but sometimes broke the rules without seeking either prior approval or subsequent forgiveness. Some countries believed the system disadvantaged them while favoring others, but the system never promised to right historical wrongs or deliver equality. Even when major powers claimed to provide global public goods, the benefits they received were often at least as significant as their contributions. Overall, postwar circumstances generally aligned forward-looking self-interest with global citizenship and broader public-spiritedness. The result was an international system that proved effective, eventually giving rise to globalization, an international monetary and financial system, and global institutions that served as arbiters and repositories of rules.

That system has evolved from one characterized by benevolent hegemony to one that provides fewer global public goods, increasingly favors the exercise of power over a rules-based order, and is marked by more unilateral action and less multilateral consensus. Trade has become a contest for control over strategic choke points and balance of payments positions rather than a competition based on productivity. Although more than 70 percent of world trade might still take place under World Trade Organization multilateral rules, trade disruptions imposed by individual countries have more than tripled since 2019 (Georgieva 2023). The international monetary system's single-platform model is no longer viewed solely as a pathway to achieving greater efficiency and economies of scale in the provision of global insurance; it has also become viewed as an instrument of geopolitical coercion through sanctions.[4]

Three features of the previous order are especially important for Asia and Pacific: a level playing field for engagement and exchange, peaceful dispute resolution, and cooperation in addressing shared challenges. These three pillars form the core of multilateralism. Their erosion poses the greatest risk to the region's smaller states. For these states to respond effectively requires understanding the shocks that are disrupting these three pillars.

Three distinct but interacting shocks have been important proximate causes of this disruption: the China shock, the US shock, and the multilateralism shock.

[4] Different descriptions of these changes are available, especially in the international relations and policy scholarship literature: Ikenberry (2026) provides a valuable summary of the various descriptions of these changes. Recent accounts in economics include, among others, Armstrong and Quah (2026); Clayton, Maggiori, and Schreger (2026c); Gopinath and others (2025); Mattoo, Ruta, and Staiger (2024); Posen (2025); and Quah (2026b, 2026c, 2026d).

## The China Shock: Scale, Speed, and Competition

Cross-border trade has long been recognized as generating mutual gains at the national level through the logic of comparative advantage. However, relative winners and losers can emerge within any economy. International win-win outcomes do not guarantee domestic win-win outcomes, and much of the analysis of trade and inequality is built on this insight. For cross-border trade to have any effect, it must alter the vector of relative prices. But any change in relative prices means that some factors of production will face adverse price movements: relative prices cannot increase for every commodity.

When production factors and workers can move frictionlessly across sectors and occupations, from those where relative prices have fallen to those where they have risen, the gains from cross-border trade are magnified. When such adjustment cannot occur, unemployment and economic stagnation emerge in adversely affected sectors, even as sectors that benefit experience stronger economic performance. How widespread these negative effects emerge as equilibrium outcomes and how long they persist depend on local geography, the pace of domestic adjustment, and the scale and speed of productivity improvements in competing imports.

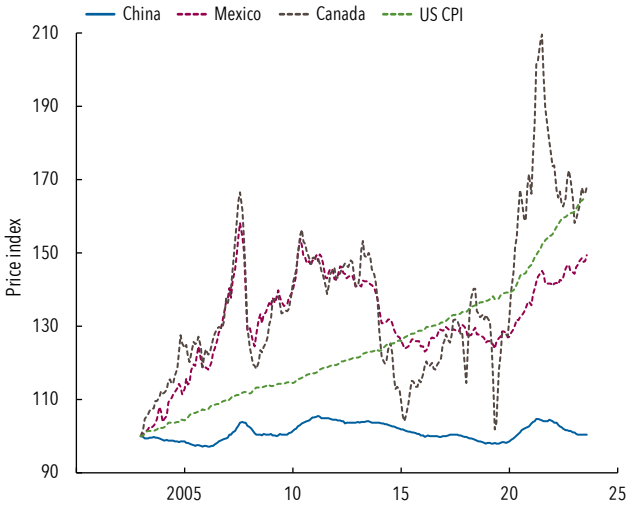
When these dynamics were observed in the United States, they became known as the China shock (Autor, Dorn, and Hanson 2013; Kennedy and Mazzocco 2022). Nothing in this narrative is unique to the United States, and policymakers and analysts elsewhere have adopted the same terminology.

Reasserting the benefits of comparative advantage does not negate reactions to the China shock. The problem lies in the combination of the effectiveness of the exporting country (China) and the economic adjustment mechanisms operating in the importing economy (the European Union, Indonesia, the United States, and other jurisdictions where the China shock narrative has emerged). The China shock is therefore consistent with economists' understanding of comparative advantage.

Empirical studies of the China shock typically use the quantity of imports from China as the causal variable and, in effect, treat the trade balance as a key indicator. This reasoning overlaps with international macroeconomic analyses, in which current account surpluses and deficits are caused directly by aggregate saving and consumption decisions, not productivity and competition (Gourinchas and others 2024). However, the description of the China shock does not refer to trade balance accounts at all. Its underlying driver is a change in relative prices. Figure 12.1 provides strong evidence of that change. The reasoning does not depend on how much changes in relative prices also affect the current account balance.

From the perspective of other nations, the China shock is a supply shock, and the most effective responses will recognize it as such. Beyond suggesting that policy should focus on prices rather than quantities, and that the current account balance could therefore be a distraction, this reasoning carries another key policy

**Figure 12.1. US Import and Domestic Prices**




Source: Quah 2026a.

Note: The graph shows import prices of imports into the United States from Canada, China, and Mexico, along with the US CPI. Over the 20 years shown, import prices from China remained flat, ending the period just 0.5 percent higher than at the beginning. Import prices from Canada at times recorded higher inflation than the US CPI and ended the period with prices 68 percent higher than at the start. Imports from Mexico followed a similar though less extreme pattern, with prices ending 49 percent higher than at the start of the period. The US CPI inflated by 65 percent over this sample. CPI = consumer price index.

implication. When confronted with a China shock, measures intended to contain the exporting country or weaken its economic performance cannot be unambiguously beneficial or harmful to domestic well-being. Policymakers must instead weigh the distributional trade-off between lower import prices, which benefit consumers and downstream domestic industries, and the harm those same prices inflict on firms and workers in directly competing sectors.

## The US Shock: National Security and Hegemonic Power

The China shock can be understood as the inadvertent consequence of an exporter nation's pursuit of increased productivity and higher profits. In terms of international relations, that interpretation casts the China shock as similar to a large nation's strong use of economic aggression to advance its position. By contrast, the US shock is an exercise in economic statecraft or geoeconomics.[5] Unlike

[5] Contrary to common belief, "geoeconomics" does not refer to the economic counterpart of the broader and more encompassing term "geopolitics." In referring to the US shock and in the scholarly

shocks such as tsunamis, earthquakes, or floods, which arise from natural events, the US shock involves intention.[6]

At the end of World War II, the United States began building what Ikenberry (2026) calls the liberal international order. This order was rules based, founded on international law and multilateral principles, coordinated through international institutions, and led in a unipolar but benevolent manner by the world's most powerful nation, the United States.

In broad terms, this system conforms to the principles of hegemonic stability theory (Kindleberger 1996), which holds that the world is more stable when a benevolent hegemon oversees it. In practice, the hegemon acted as consumer and banker of last resort, provider of global security, issuer of the world's reserve currency, and supporter of the multilateral international architecture. It warned the rest of the world about revisionist nations that sought to undermine the rules-based multilateral order and took action against them. It provided global public goods and leadership in activities that the economics of increasing returns to scale suggests are most efficiently concentrated in a single agent.

However, as the rules-based multilateral order succeeded and created economic opportunities for perimeter nations, the income distribution across nations became more equal. In the 1950s, the United States accounted for 40 percent of the global economy, but that share has since fallen to 25 percent. The United States is no longer as dominant, and unipolarity has given way to multipolarity. Prosperity increased overall and became more broadly shared, which improved humanity's well-being, but it also meant that the cost–benefit ratio for the hegemon turned unfavorable (Gaspar, Hagan, and Obstfeld 2018; Quah 2026c).[7]

Although the direction of change is consistent with predictions, the elasticity of the hegemon's response to this reversal has been surprising. Rather than simply retreating from its historical role in supporting rules-based multilateralism, the hegemon actively sought to rewire the international system to

literature more generally (for example, Baldwin 1985; Blackwill and Harris 2016), geoeconomics focuses on a hegemon's decision making and means "the use of economic instruments to accomplish geopolitical objectives" (Blackwill and Harris 2016). Thus, geoeconomics is the study of how a hegemon pursues and optimally achieves its geopolitical goals using economic tools ("a means of working the US's geopolitical will in the world" [Blackwill and Harris 2016, 99]) and is therefore also called economic statecraft. In this sense, geoeconomics is subordinate to geopolitics, not its economic counterpart. See also footnote 7.

[6] Modern technical versions of geoeconomics (for example, Clayton, Maggiori, and Schreger 2026a, 2026b) correctly maintain that distinction when studying how a hegemon shapes the decisions of foreign entities and thereby exercises and advances its hegemonic power. By contrast, an international economics of strategic interdependence also encompasses how smaller nations organize themselves in response to hegemonic power. Of these two sets of actions, only the first falls within geoeconomics, while both, together with the general equilibrium outcomes of such interactions, make up an international economics of strategic interdependence. See also footnote 7.

[7] Related, Obstfeld (2024) documents how specific economic developments such as falling competitiveness, disruptive capital movements, and growing financial pressures on currency can undermine support for hegemonic presence generally, if not multilateralism specifically.

its own advantage. This momentum manifested in a range of US actions, including tariffs, sanctions, extraterritorial interventions, and withdrawals from long-standing agreements.

Unlike conventional narratives on the geopolitical actions of the United States, this interpretation focuses on the hegemon's place in the world rather than on US–China geostrategic rivalry, although that latter is certainly consequential. But the hegemon's tariff and trade actions have targeted many countries that the US administration judges to be acting against American interests, not just China. The US shock therefore arises from the relationship between the United States and the rest of the world, not simply from its relationship with China. A Cold War analogy therefore likely obscures more than it clarifies. For Asia and Pacific countries not on the front line of the tensions between the two countries, many other challenges should take priority.

Where US–China rivalry matters is not its substance but in its effect on behavior. In the United States, geostrategic competition heightens attention to national security. Such a political environment makes it easier to pursue more economic statecraft, whether economically justified or not. Under national security imperatives, economic statecraft no longer targets traditional domains of economic competition. Major powers shift their attention from improving productivity relative to their rivals to securing choke points against them. In doing so, they transform competition that can potentially improve overall well-being into contestation that produces lose–lose prisoner's dilemma or epic fail outcomes (Armstrong and Quah 2026).

Policy implications for the region's smaller states follow from the analysis. In the narrow sense that tariffs and sanctions restrict markets, the US shock can be understood as a demand shock, just as the China shock is a supply shock. Arguably, the more critical conclusion is that the US shock is driven by long-term trends, not notions of whoever leads the current administration. The region's countries need to recognize that this demand shock differs from those hypothesized in the study of business cycles. The US shock is instead a long-lived demand shock.

## The Multilateralism Shock: Cost-Benefit Calculation for Global Leadership

The cost–benefit calculations for the United States also apply to other major advanced economies and to countries that consider themselves middle powers.

In the early years of rules-based multilateralism, the United States occupied an overwhelming unipolar position that made it easy to support the system (implied that a cost–benefit ratio allows it) or, as Ikenberry (2005) wrote, to be "a producer of world order" (p. 133).

When other states ascend to become the power that produces world order, whether their calculations will differ is an open question. Optimistic observers suggest that middle powers will enter the role with less ideological baggage and more vivid memories of what the rules-based multilateral order can deliver.

However, the basic economic calculation will still hold. Like falling dominoes, each replacing the one before it, the same cost–benefit logic will apply. As a result, no rules-based multilateral order that requires hegemonic resources can be sustained permanently in the long term.

Such partial equilibrium actions of major powers weaken the supply-side provision of public goods. At the same time, the multilateral system is experiencing heightened demand for solutions. The world's most urgent problems have become increasingly global in nature, and climate change, pandemics, and financial instability are among the most pressing. These challenges cannot be addressed through unilateral national action but instead require multilateral cooperation. The growing gap between supply and demand in the multilateral system creates additional stress for small states.

## NAVIGATING THE INTERNATIONAL ECONOMICS OF STRATEGIC INTERDEPENDENCE

Asia and Pacific small states face the challenge of responding to these emerging disruptions. Historically, successful strategies have drawn on traditional approaches to build economic security: because small states are internationally price takers and norm abiders, they respond best to economic disruption through domestic reforms that raise productivity, become more essential and indispensable in international engagement, and improve economic efficiency through the comparative advantage mechanism in open trade.

However, in a world dominated by China, the United States, and multilateralism shocks, traditional strategies can have adverse unintended consequences. Trade openness can be associated with increased foreign supply and reduced foreign demand and can therefore raise unemployment and heighten domestic dissent. Price taking and norm abiding can leave small states vulnerable to pressure by major powers and with no way to respond. If the small state tries to adopt a resilience-building strategy of raising domestic productivity or making itself more indispensable to the international trading system, that might just further increase incentives for foreign action against it and thus paradoxically expose its vulnerability. Whether pursuing the opposite of traditional strategies will produce better outcomes is unclear. A small state might conclude that if it does not accept prices as given, then it should apply retaliatory tariffs. But that leads to a downward spiral in which tariffs rise without generating positive effects on demand for domestic production. Similarly, bluntly rejecting the norms of a potentially antagonistic major power without first laying the necessary groundwork can invite punitive economic statecraft. A small state might also conclude that the alternative to trade openness is autarky (economic self-sufficiency and closed borders). But it will quickly discover that it cannot diversify sufficiently like a large economy can and will soon realize that autarkic restrictions excessively constrain its own economic prospects.

## A Basic Toolkit for Economic Diplomacy: Align, Acquiesce, and Adapt and Mitigate

The international economics of strategic interdependence can be understood as the study of how small states navigate this landscape and the outcome their actions produce.[8]

To move beyond the stark binaries implied by traditional strategies, small states can consider a mix of alignment, acquiescence, and adaptation and mitigation. Alignment involves taking sides or, more generally, accepting the bundles of attributes—such as trade and technology—offered by a great power. Acquiescence involves taking actions to placate major powers in an effort to stabilize the international space available to smaller states. Adaptation seeks to minimize the symptoms of disruption caused by major powers and mitigation addresses their causes.

Alignment alone is potentially optimal in a zero-sum world. For Southeast Asia, Khong and Liow (2025) analyze changes in the alignment of individual small states in the context of US–China rivalry. What matters is not the substance of the rivalry itself but the behaviors it produces.

Outside a zero-sum environment, however, small states should reject the binary choice of alignment and instead develop a portfolio of relationships. The decision-making horizons of great powers and small states do not necessarily coincide. Great powers might seek long-term alignment because it allows them to organize and consolidate spheres of influence. However, small states might align only opportunistically to address a specific challenge and then move on to address the next problem. In some issue areas, such as green and renewable energy development, one partner country could offer stronger opportunity. In others, such as governance standards for artificial intelligence and digital technologies, a different partner country could be important. The region's countries can also deepen ties with regional bodies such as the Association of Southeast Asian Nations to build a diversified dynamic network of relationships without global hegemony or, more generally, in plurilateral groups of countries without a hegemon (Buzan 2011, 2023; Herz and Ho 2026; Quah 2026c, 2026d). This idea can be compactly

[8] In the context of this chapter, the international economics of strategic interdependences, not geoeconomics, is the counterpart to geopolitics. Strategic interdependence refers to general equilibrium and systemic observable consequences for the international economy that extend beyond the defensive or offensive single-agent decision-making strategies emphasized in geoeconomics. Such potential complications include the operating characteristics of a leaderless world order, the responses available to lesser powers when a hegemon shifts from providing public goods to extracting economic surplus, and the conditions under which smaller states should form flexible coalitions for economic security against natural shocks and the intentional actions of major powers. Clayton, Maggiori, and Schreger (2026c) seek to generate equilibrium systemic predictions and therefore go beyond geoeconomics by incorporating the responses of smaller nations to the hegemon's economic statecraft. However, those smaller nations act noncooperatively, in contrast to some of the multilateral strategies considered in what follows. See also footnotes 4 and 5.

described as a "G-minus" world, an international order where great power leadership is optional.

As the multilateralism shock has unfolded, universal multilateralism has stalled. However, smaller groups can work toward restoring at least a rules-based three-pillar multilateral order. The new order need not be fully multilateral; it need only be multilateral enough. Nor does such an order require a hegemonic leader as long as the order is not large and sprawling. Instead of a world order built around China and the United States or the Group of Seven, the new system can comprise G-minus groupings.

The region's small states can collaborate through plurilateral networks on specific areas such as sustainable finance, digital governance, and regional supply chains. They can also leverage specialized expertise to help shape standards in areas such as financial clearing and settlement systems (for example, Project mBridge and Project Nexus). For such expertise and technological innovations, small states are just as capable as large ones.

Managing strategic interdependence across economies requires a sophisticated layered approach to economic diplomacy, understood as economic statecraft for small states. The first layer might well be alignment, qualified in the manner described previously. The second layer could be acquiescence: the temporary acceptance of necessary constraints to create space for longer-term recalibration. The third layer of adaptation and mitigation draws inspiration from the global climate crisis. All states are small when facing environmental change, yet they can adapt for the global climate crisis. They can build resilience by treating the symptoms of environmental change (relocate populations away from vulnerable areas and change agricultural practices, for example), and they can mitigate by addressing its causes (such as reducing carbon emissions and shifting energy use from hydrocarbons to renewable sources). The same logic applies for managing strategic interdependence. Small states should make their domestic economic structures more resilient to the actions of great powers while reducing their exposure to those powers through appropriate regional or plurilateral groups. Economic relationships can be rewired to incorporate redundancy and resilience, diversifying supply chains away from known choke points and single points of failure, even if doing so entails loss of economic efficiency.

Asia and Pacific small states should build incentive-compatible coalitions rather than pursue universalist arrangements. Durable cooperation does not require sweeping treaties or grand institutional redesign that might be sufficient but not necessary for rules-based multilateral order. Flexible and overlapping arrangements based on only inadvertent cooperation can prove more resilient than formal structures that great powers abandon once their interests diverge.

Asia and Pacific small states should move toward a flexible topology of networks that are incentive compatible, G-minus, and multilateral enough (Quah 2026c, 2026d), not because such an order is itself aspirational, but because it reflects the reality that the original rules-based multilateral order is not incentive

compatible and is no longer available. Unless the old rules of multilateralism are rewritten, major powers will have no incentive to continue supporting it over the long term.

## CONCLUSION

The Asia and Pacific region's growth and development over the past 80 years have been remarkable and, in many respects, unexpected. This overarching trajectory of success comprised differentiated arcs traced out by states of different sizes. Yet the progress of small states, which has underpinned a large part of the region's progress, is now at risk as the world order shifts from rules-based multilateral openness to geoeconomics and strategic interdependence.

The region's small states must continue to strengthen domestic resilience and leverage supply chains, but they can no longer rely solely on the passive strategies of price taking and norm abiding. Instead, they need to adopt a proactive approach to economic diplomacy, drawing on strategies of alignment, acquiescence, and adaptation and mitigation. By working together in incentive-compatible, G-minus, multilateral-enough diversified groups, Asia and Pacific small states will be better positioned to navigate the emerging international landscape of strategic interdependence.

## ANNEX 12.1. MCKENDRICK-VON FOERSTER DISTRIBUTION DYNAMICS FOR ASIA AND PACIFIC DEMOGRAPHY

A simple calibration shows a 7.5 percent decline in steady-state levels of per capita GDP across the Asia and Pacific region if current demographic trends continue.

An economy's demography is depicted by Annex Figure 12.1, with the horizontal axis denoting age. Two threshold values $Y$ and $W$ on this axis are significant: with $0 < Y < W$, the value $Y$ is the upper age limit of young individuals before they commence work, and the value $W$ is the upper age limit of the working population before they retire. Denote the populations of each of the three age categories separated by $Y$ and $W$ as $N_Y$, $N_W$, and $N_Z$, respectively.

**ANNEX TABLE 12.1.1.**

Summary of Demographic Transition and McKendrick-von Foerster Distribution Dynamics

| | Age $x$ | Number | Net Flows |
|---|---|---|---|
| **Young** | $x \leq Y$ | $N_Y$ | $bN_W - \alpha N_Y$ |
| **Working** | $Y < x \leq W$ | $N_W$ | $\alpha N_Y - \beta N_W$ |
| **Aged** | $W < x$ | $N_Z$ | $\beta N_W - dN_Z$ |

Let positive constants $b$ and $d$ denote birth and death rates based on working-age $N_W$ and aged $N_Z$ populations, respectively. The input flow into $N_Y$ is then $b \times N_W$ while the output flow out of $N_Z$ is $d \times N_Z$.

At the same time, however, there are output flows out of each of $N_Y$ and $N_W$ into the next higher age category at a constant rate: everyone ages 1 year every 12 months. Define the uniform densities $\alpha \stackrel{\text{def}}{=} (Y - 0)^{-1} = Y^{-1}$ and $\beta \stackrel{\text{def}}{=} (W - Y)^{-1}$ and write these output flows as $\alpha N_Y$ and $\beta N_W$, respectively. These are, simultaneously, input into and output out of working age. Collecting these results, net flow for the young $N_Y$ is $bN_W - \alpha N_Y$; for the working age, $\alpha N_Y - \beta N_W$; and for the aged, $\beta N_W - dN_Z$.

Annex Table 12.1.1 summarizes the demographic transition as McKendrick-von Foerster distribution dynamics. Next, define working-age weighted versions:

$$r \stackrel{\text{def}}{=} (N_Y + N_Z)/N_W$$

$$y \stackrel{\text{def}}{=} N_Y/N_W, \qquad z \stackrel{\text{def}}{=} N_Z/N_W,$$

with $r$ recognized as the dependency ratio. Differentiate with respect to time to obtain the dynamics of working-age weighted ratios:

$$\dot{r} = \dot{y} + \dot{z}$$

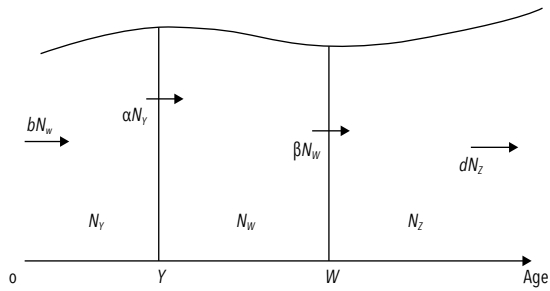
$$\dot{y} = b - (\alpha - \beta)y - \alpha y^2$$

$$\dot{z} = \beta - (d - \beta)z - \alpha yz.$$

In steady state $\dot{r} = \dot{y} = \dot{z} = 0$, so that

$$y^* = \frac{\beta - \alpha \pm \sqrt{(\alpha - \beta)^2 + 4\alpha\beta}}{2\alpha}.$$

## Annex Figure 12.1. Demographic Transition and McKendrick-von Foerster Distribution Dynamics



and

$$z^* = \frac{\beta}{d - \beta + \alpha y^*},$$

with the implied $r^* = y^* + z^*$.

If $\theta$ is underlying average productivity, then measured GDP per capita is

$$\frac{\theta N_W}{N_Y + N_W + N_Z} = (1 + r)^{-1}\theta.$$

This expression shows that even if productivity $\theta$ remains unaffected, higher dependency ratios will dilute per capita GDP from the combination of fewer working-age workers and more dependents.

In steady state, net flows are zero for both young and aged, so that $y^* = b/\alpha$ and $z^* = \beta/d$, and therefore

$$r^* = y^* + z^* = \frac{b}{\alpha} + \frac{\beta}{d} \text{ in steady state}$$

(that is, what the previous more intricate expressions collapse to upon imposing that the working population must also be in steady state, given steady state for the young and the aged).

The region's demography is not yet in steady state. But the $r^*$ equation nonetheless correctly flags the forces at work. A low birth rate $b$ lowers the immediate youth dependency ratio. However, over time, coupled with falling old-age mortality $d$, a lower birthrate does increase the overall dependency ratio $r^*$.

Next, draw on data from OECD, World Bank, and the United Nations to calibrate the model. Measured total fertility rates across Asia and the Pacific are about 1.5–1.7, suggesting an approximate birth rate $b$ = 0.018. Old-age mortality is approximately 10 percent. The standard IMF and World Bank convention gives $\alpha$ = 0.067 and $\beta$ = 0.02. These imply steady-state values $y^*$ = 0.27 and $z^*$ = 0.20, and therefore a steady-state dependency ratio $r^*$ = 0.47. Comparing this with the current dependency ratio of 0.36 means that at a fixed average productivity, per capita GDP in steady state will be 1.36/1.47 = 0.925 times the current value, that is, lower by 7.5 percent.

In developed economies, a typical business cycle might change GDP by 3–4 percent over two years, so a permanent level change of twice that is considerable. As with economic fluctuations, some parts of the cross-section will feel this effect more strongly than others. The 7.5 percent decline is what would occur if the demographic decline were adjusted by having people reallocate frictionlessly across the different nations of the region. In reality, labor mobility is much less, so that value of 7.5 percent should be viewed as the lower bound for those economies most affected by an aging population.

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CHAPTER 13

# The Changing Landscape of Financial Integration in Asia and Pacific

Cristian Alonso, Tristan Hennig, Henry Hoyle, Haibo Li, Monica Petrescu, Ying Xu, and Yizhi Xu

The Asia and Pacific region has undergone a profound transformation since 2000, increasing its share of global trade and GDP. In this chapter, a systematic analysis of cross-border financial data explores whether the region's role in global finance has expanded commensurately and whether it has become more integrated.[1] The analysis combined descriptive analyses of historical trends in financial positions with network analysis to examine how interregional and intraregional financial links have evolved across the region's economies. A gravity model was used to study the interaction between trade and financial integration and identify the macroeconomic and policy variables driving financial positions.

A key finding is that the region's role in global finance lags its role in global trade. The region's overall financial integration is low, as measured by the ratio of external assets and liabilities relative to GDP (Table 13.1). However, substantial heterogeneity exists across the region. Financial integration is high for the region's advanced economies but low for the region's emerging market economies.

Intraregional financial integration is also low but varies significantly by instrument. This pattern points to substantial potential for improvement, which could provide benefits from larger and more competitive markets for financial services and more regional diversification and external risk sharing (ADB 2025). The analysis measured intraregional financial integration by intraregionally held

[1] The analysis used data from several sources. The IMF Balance of Payments and International Investment Position Statistics provided cross-border stocks and flows data aggregated by country. Bilateral data on foreign direct investment (FDI) positions were from the IMF Coordinated Direct Investment Survey (recently renamed Direct Investment Positions by Counterpart Economy), available since 2009. Bilateral data on foreign portfolio investment positions were from the IMF Coordinated Portfolio Investment Survey (recently renamed Portfolio Investment Positions by Counterpart Economy), available since 2001 but with some gaps in the early 2000s for China and India. Bilateral trade data on imports and exports were from Direction of Trade Statistics. The Bank for International Settlements Locational Banking Statistics provided bilateral data on cross-border banking exposures for selected country pairs since 2005. Macroeconomic variables were from the World Economic Outlook Database. For more details on the methodology, see (Alonso and others, 2026).

**TABLE 13.1.**

**Asia and Pacific: Share in Global Total of Selected Indicators**
*(Percent)*

| | 2001 | 2009 | 2019 | 2023 |
|---|---|---|---|---|
| *Trade* | 24 | 29 | 34 | 33 |
| *GDP* | 24 | 26 | 34 | 33 |
| **External assets and liabilities** | | | | |
| *Liabilities* | 13 | 11 | 16 | 16 |
| *Assets* | 17 | 15 | 21 | 21 |
| **Instrument** | | | | |
| *Foreign direct investment* | *n.a.* | 12 | 16 | 20 |
| *Foreign portfolio investment* | 12 | 12 | 14 | 14 |
| *Cross-border bank lending* | 17 | 11 | 21 | 19 |

Sources: IMF, World Economic Outlook Database; IMF, Coordinated Direct Investment Survey data set; IMF, Coordinated Portfolio Investment Survey data set; Bank for International Settlements (BIS), Locational Banking Statistics Database; and IMF staff calculations.
Note: Asia's share of global trade and cross-border financial positions are measured in terms of aggregates of imports and exports and assets and liabilities, respectively. External assets and liabilities exclude foreign exchange reserves. Foreign direct investment, foreign portfolio investment, and bank lending exclude positions between the Chinese mainland and Hong Kong Special Administrative Region.

cross-border assets and liabilities relative to regional GDP and network centrality scores. Intraregional integration is increasing in foreign direct investment (FDI) and from low levels in cross-border banking but remains limited in foreign portfolio investment (FPI).[2] A few advanced economies act as primary recipients and sources of intraregional FPI, whereas other economies have only a minor role. By contrast, intraregional dynamics in FDI and cross-border banking are richer, with more economies engaged in intraregional flows. Multiple economies have emerged as intraregional intermediaries (regional hubs), including international financial centers.[3]

Trade and direct investment integration appear to reinforce each other. Gravity model analysis indicates a significant association between trade and FDI but not between trade and FPI or cross-border banking liabilities. Therefore, stronger regional trade integration (which Chapters 10, 11, and 12 advocate) would likely increase cross-border direct investment flows but not necessarily other flows, such as FPI.

As economies reach higher income levels and deepen domestic financial markets, cross-border financial integration typically increases. Stronger financial integration can provide benefits by reducing the cost of capital, improving risk

[2] Network analysis was used to highlight the richness of intraregional financial integration and how it evolves. Centrality measures from network analysis help clarify how much intraregional positions are concentrated in a small number of key economies or if they are more evenly distributed with stronger participation and broader exposure of multiple economies. Annex 13.2 describes key network concepts in more detail.

[3] Annex 13.1 outlines the challenges in interpreting financial data for international financial centers.

management, increasing efficiency in capital markets and other tradable services, and harnessing economies of scale in regional financial hubs. Increasing FDI links would also create opportunities to improve productivity through technology transfer and knowledge diffusion. However, stronger financial integration can also raise risks by heightening exposure to global shocks and the global financial cycle. Countries should seek to capture the benefits of greater financial integration in the region while mitigating the risks.

## ASIA IS NOT TO GLOBAL FINANCE WHAT IT IS TO GLOBAL TRADE

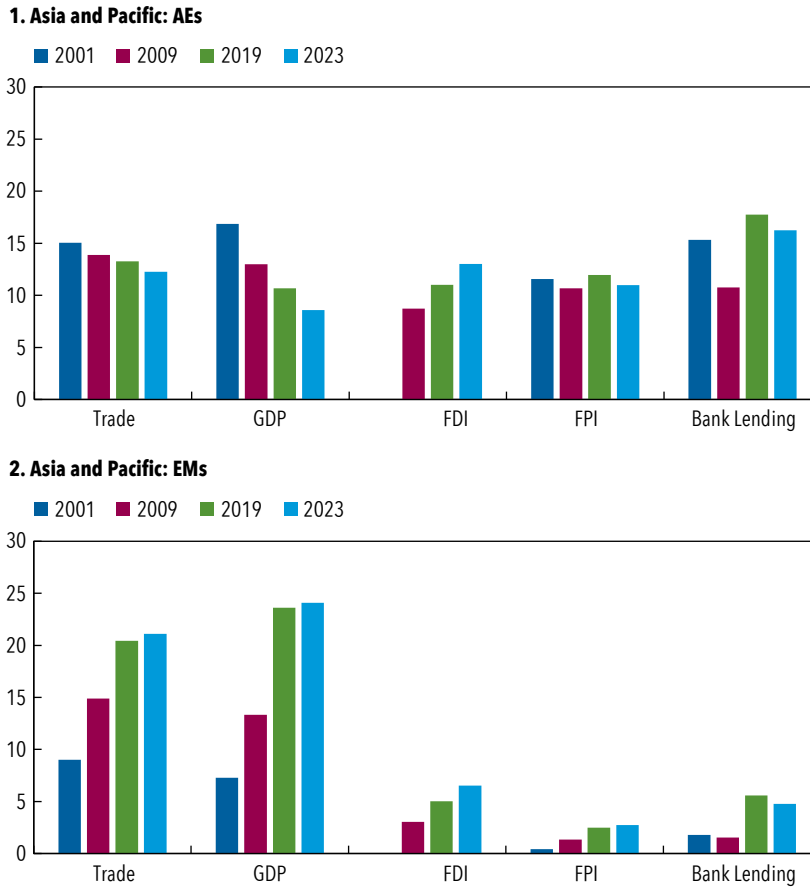
The Asia and Pacific region's share of global GDP and trade increased from less than one-quarter in 2001 to one-third by 2023, but financial integration increased more slowly. Based on IMF International Investment Position data, the region's holdings of external assets accounted for only 21 percent of the global total in 2023, and liabilities accounted for only 16 percent of the global total in the same year, both only modestly higher than in 2001 (Table 13.1). The region's share of global FDI positions increased gradually from 12 percent in 2009 to 20 percent in 2023 but remained below its share of global trade. Its share of global FPI positions remained broadly unchanged at about 14 percent after 2010. The region's share of cross-border banking claims doubled from the 2009 level to one-fifth in 2023, although it remained well below its share of global GDP and trade. Compared with other regions, the Asia and Pacific region is relatively less integrated into global finance, a finding documented in earlier work (Garcia-Herrero, Yang, and Wooldridge 2008; Borensztein and Loungani 2010; Ananchotikul, Piao, and Zoli 2015; Llovet Montanes and Schmukler 2018). Unlike those studies, the analysis also excluded positions between China and Hong Kong Special Administrative Region where possible, making the findings of low financial integration even more pronounced, especially for FDI. Intra-China positions account for 30 percent of the region's FDI claims and 24 percent of its FDI liabilities.[4]

The evolution of the region's financial integration varies across income groups. Advanced Asia's share of global GDP and trade has decreased since the early 2000s, whereas its share of FDI, FPI, and cross-border banking claims has either stayed stable or grown, maintaining levels similar to those observed in GDP and trade (Figure 13.1). By contrast, emerging Asia's share of global finance has not matched its rapid growth in global GDP and trade.

The higher degree of financial integration in the region's advanced economies than in its emerging market economies is also reflected in the cross-border asset and liability flows. Although the stock perspective provides a more comprehensive view of financial integration, recent flow dynamics confirm that the stock measures do not simply reflect advanced economies' longer history of financial

[4] Because these positions are large relative to other positions within the region, removing these links also provides a more accurate picture of financial integration within the region. These positions are used for network analysis but would not alter bottom line results significantly.

**Figure 13.1. Asia's Share of Selected Global Indicators by Income Status**
*(Percent of global total)*




Sources: IMF, World Economic Outlook Database; IMF, Coordinated Direct Investment Survey data set; IMF, Coordinated Portfolio Investment Survey data set; Bank for International Settlements (BIS), Locational Banking Statistics Database; and IMF staff calculations.
Note: Asia's share of global trade and instrument-level positions were measured in terms of aggregates of imports and exports and assets and liabilities, respectively. External assets and liabilities exclude foreign exchange reserves. Foreign direct investment, foreign portfolio investment, and bank lending exclude positions between the Chinese mainland and Hong Kong Special Administrative Region. AE = advanced economy; EM = emerging market; FDI = foreign direct investment; FPI = foreign portfolio investment.

openness. Between 2013 and 2023, advanced economies' average annual increase in gross cross-border financial claims (the sum of cross-border assets and liabilities for FDI, FPI, and other investment) was 22.5 percent of GDP, higher than that of advanced economies in any other region during the same period. Emerging markets' average annual increase in financial claims was only 6 percent of GDP, the lowest among emerging market economies in any other region. Advanced

economies are also disproportionately responsible for the region's position as a major net exporter of capital. Measured by region-wide growth in FDI, FPI, and other investment claims less liabilities, net capital exports from the region were comparable to Europe's in dollar terms between 2013 and 2023 and were about four times its net accumulation of foreign exchange reserves. The region's advanced economies accounted for two-thirds of the region's net capital exports by this measure.[5]

Asian economies' capital has been geographically concentrated and largely flowed to elsewhere in Asia and the Americas. As of 2023, 42 percent of Asian economies' FDI claims were on other Asian economies, whereas another 40 percent was roughly split between the Americas (mostly the US) and offshore financial centers in the Caribbean. Asian economies' cross-border foreign portfolio holdings are vis-a-vis the Americas (45 percent), offshore financial centers (18 percent), and other Asian economies (17 percent). Europe and other EMs have absorbed a relatively smaller share.

## FOREIGN DIRECT INVESTMENT: LOW OVERALL FDI, GROWING INTRAREGIONAL LINKS

Aggregate FDI in the Asia and Pacific region, measured as the sum of each economy's inbound and outbound positions, remains comparatively low relative to regional GDP. Aggregate FDI positions in the region were just 45 percent of GDP, compared with more than 100 percent of GDP in Africa, Europe, and the Middle East and North Africa (Table 13.2, panel 1).

Aggregate FDI in the region still grew faster than in most regions between 2013 and 2023 (Table 13.2). Although global FDI slowed broadly after the COVID-19 pandemic, the region was less affected (UNCTAD 2024). Consequently, the region's share of global FDI positions expanded (Table 13.1). The resilience of FDI in the region reflects rapid economic growth and increasing participation in global trade, which is closely associated with FDI.

FDI positions among economies in the region are similarly smaller than in other regions but are expanding. The intraregional FDI stock amounted to about 8 percent of regional GDP in 2023 (Table 13.2, panel 1). Although far below the level in Europe (which is highly integrated by this measure), intraregional FDI stock was slightly lower than in Africa and the Americas and slightly higher than in the Middle East and Central Asia. However, intraregional FDI has continued to grow in the region even as it contracted relative to GDP in the rest of the world. In dollar terms, growth in intraregional FDI during the past decade exceeded that in every region except for Africa.

[5] Data are based on balance of payment figures that prohibit excluding flows between the Chinese mainland and Hong Kong Special Administrative Region. However, the net capital exports of advanced economies remain strong even if excluding Hong Kong Special Administrative Region because it accounts for only 15 percent of advanced economies' by this measure of net capital exports for 2014–23.

**TABLE 13.2.**

**Foreign Direct Investment and Foreign Portfolio Investment: Selected Measures**

| | Foreign Direct Investment (FDI) | | | | Foreign Portfolio Investment (FPI) | | | |
|---|---|---|---|---|---|---|---|---|
| | In percent of regional GDP, 2023 (percent) | | Average growth rate, 2013–2023 (percent) | | In percent of regional GDP, 2023 (percent) | | Average growth rate, 2013–2023 (percent) | |
| | Inbound and Outbound | *Intra-Regional* | Inbound and Outbound | *Intra-Regional* | Inbound and Outbound | *Intra-Regional* | Inbound and Outbound | *Intra-Regional* |
| **Asia-Pacific** | **45** | **8** | **7** | **7** | **59** | **5** | **6** | **7** |
| Europe | 157 | 50 | 1 | 0 | 279 | 86 | 4 | 2 |
| Middle East and Central Asia | 103 | 7 | 7 | 4 | 80 | 6 | 8 | 6 |
| Africa | 156 | 12 | 3 | 7 | 108 | 2 | 4 | 10 |
| Western Hemisphere | 63 | 9 | 5 | 5 | 152 | 33 | 9 | 12 |

Sources: IMF, World Economic Outlook Database; IMF, Coordinated Direct Investment Survey data set; IMF, Coordinated Portfolio Investment Survey data set; and IMF staff calculations.
Note: Heat map colors indicate the percentile value within a given column: red represents the lower percentiles, and green represents higher percentiles. Asia and Pacific values exclude foreign direct investment and foreign portfolio investment positions between the Chinese mainland and Hong Kong Special Administrative Region. Growth rates are calculated based on US dollar values. Country groupings are based on IMF area department coverage areas.

Although FDI exposures are small compared with those in other regions, network analysis revealed widespread FDI interconnections among regional economies without reliance on one central hub, much like trade connections. This finding is consistent with findings of stronger links between trade and FDI than between trade and other financial instruments. Some economies act as regional hubs for FDI, such as Singapore, but other economies have also developed significant intraregional positions, including Association of Southeast Asian Nations (ASEAN) member countries and India (Figure 13.2).[6] Many smaller economies maintain bilateral connections with both regional hubs and many regional counterparts. As a result, centralization in the FDI and trade networks is relatively low compared with that in networks based on other financial instruments. Within this decentralized FDI network, the Chinese mainland remains the most central node, partly reflecting rapid growth in investments throughout the region.[7] However, its importance as a source or destination in the FDI interregional network has declined slightly over the past 15 years, likely because FDI into China has fallen.[8]

[6] Singapore and other advanced economies that are not international financial centers have been the dominant source of intraregional FDI growth. Singapore accounted for 22 percent of the increase between 2014 and 2023, and the other advanced economies that are not international financial centers accounted for 42 percent.

[7] See IMF (2024b, pp. 40–51) for more information on China's outbound FDI into the region.

[8] Regional investment into the Chinese mainland and Hong Kong Special Administrative Region dropped sharply between 2014 and 2023, as did the share of investment into Asian advanced economies.

## FOREIGN PORTFOLIO INVESTMENT: LIMITED CROSS-BORDER INTEGRATION

The Asia and Pacific region stands out globally for its relatively small stock of foreign portfolio investment relative to GDP. Gross FPI claims in Asia, measured as the sum of each country's inbound and outbound positions, amounted to 59 percent of GDP in 2023, whereas the corresponding figures in other regions ranged from 80 percent to 280 percent of GDP (Table 13.2, panel 2).[9]

Equity investment is more prevalent than debt in the region's portfolio positions, both as assets and as liabilities. Portfolio investments, both globally and within the region, increased more in equity than in debt, especially after the global financial crisis, when total holdings of equity investments surpassed debt investments.[10] For liabilities, global investors have historically held more Asian equities than Asian debt securities (Shirai and Sugandi 2018). By 2023, total global holdings of the region's equity investments were almost twice the total holdings of the region's debt investments. The share of nonresidents participating in domestic debt markets in the region has been declining, partly because interest rate differentials have narrowed in recent years. However, the persistently low level of portfolio integration suggests that structural factors are also involved.

FPI into the region originates largely from advanced economies and financial centers and is distributed unevenly. The region's advanced economies remain the main destinations for portfolio investment from outside the region, although their share declined from about 80 percent to 60 percent in the past 20 years. China and India became significant destinations for portfolio investments during the past decades, but China's share has declined slightly since the pandemic. ASEAN countries excluding Singapore increased their share of portfolio investment until the early 2010s, reaching a peak in 2012 before declining.

Intraregional portfolio positions remain modest at just 5 percent of regional GDP in 2023 and have shown limited growth (Llovet Montanes and Schmukler 2018). By comparison, the share is six times higher in the Americas and more than 16 times higher in Europe, and it has increased rapidly in both regions since the 2000s.

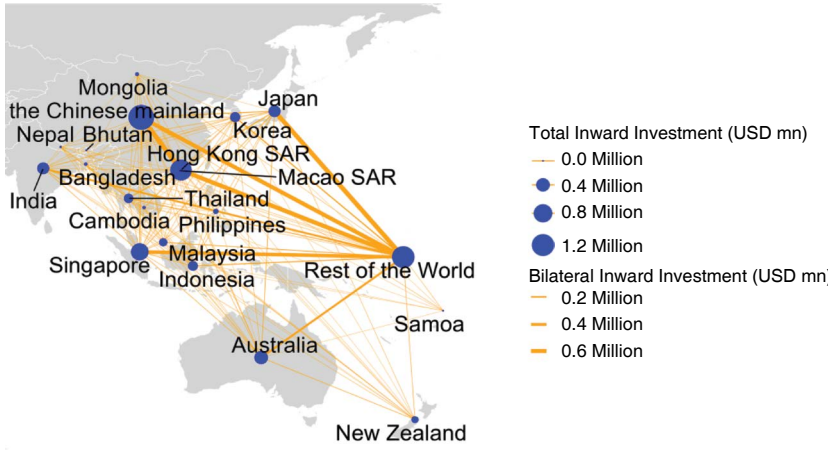
Network analysis revealed minimal intraregional FPI integration. Portfolio investment exposures within the region are concentrated in advanced economies, making the network highly centralized (Figure 13.3). Japan remains by far the

[9] The limited scope of foreign portfolio investment integration in the region has coincided with dynamic growth in markets for capital raising. Equity initial public offering activity has been booming in the region, with total values reaching nearly $400 billion during 2021–23, accounting for more than 40 percent of global initial public offering proceeds in the same period—up from only 14 percent in the 1990s. Domestic investors are likely to have absorbed most of those issuances because the region's foreign portfolio investment integration is limited.

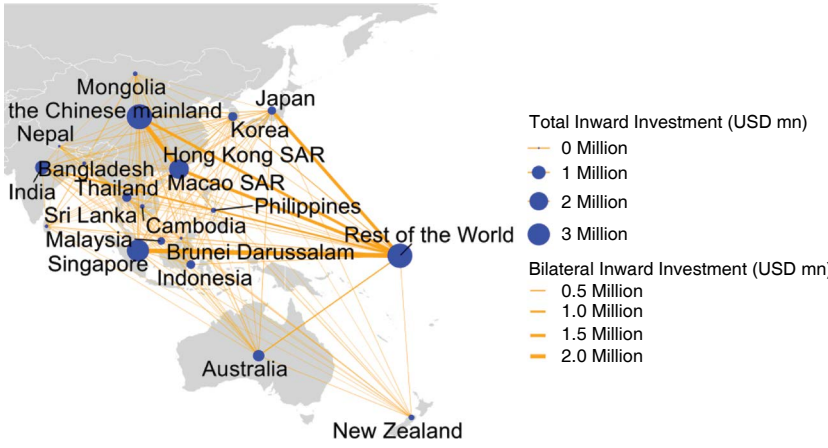
[10] Asia and Pacific's share in foreign holdings of US Treasury securities decreased from a high of more than 60 percent in 2005 to less than 40 percent in 2024, illustrating the region's declining outbound debt security investment.

## Figure 13.2. FDI Position in Asia, 2010 vs. 2023

**1. Network Graph of Cross-Border Inward Direct Investment in Asia (2010)**




**2. Network Graph of Cross-Border Inward Direct Investment in Asia (2023)**

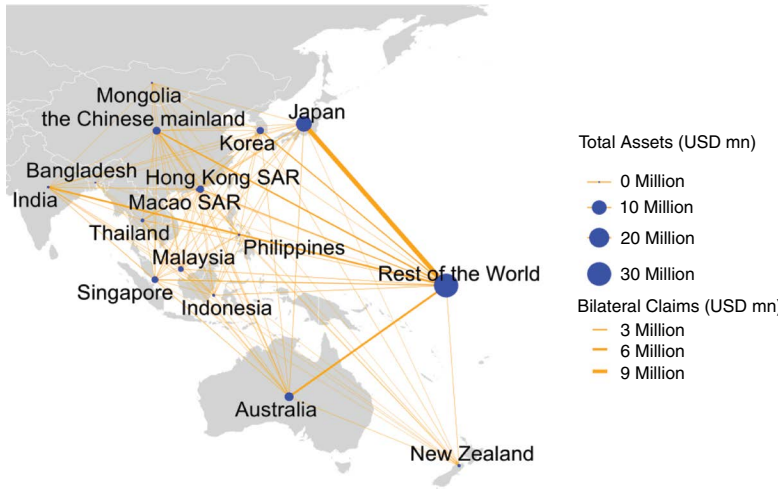



Sources: IMF, Coordinated Direct Investment Survey data set and IMF staff calculations.

largest recipient and source in the network, even though its centrality has declined slightly during the past two decades as other economies' share in regional FPI grew. Australia is the second largest economy in intraregional exposures, and its importance has increased gradually. The Chinese mainland and India are emerging participants but remain relatively small, and the Chinese mainland's importance has stagnated since 2019.

Asia and Pacific's bond markets are showing slightly more intraregional integration from a low base. Countries' portfolio debt claims are still mostly in

**Figure 13.3. Network Graph of Cross-Border Portfolio Investment in Asia (2023:H2)**




Sources: IMF, Coordinated Portfolio Investment Survey data set and IMF staff calculations.

countries outside the region, but the share of claims on other countries in the region has grown from 13 percent in 2012 to 20 percent in 2023. This increase reflects a notable slowdown in the accumulation of portfolio debt claims on countries outside the region, especially after 2018, led by a drawdown in European bond positions (Figure 13.3). Similarly, the share of the region's cross-border debt liabilities held by nonregional investors dropped slightly from 2013 to 2023, most notably in advanced economies and international financial centers.[11]

## BANKING: STRONG GLOBAL LINKS FOR ADVANCED ECONOMIES, GROWING REGIONAL INTEGRATION FOR EMERGING MARKETS

Some advanced economies in the region have become major global centers of cross-border banking, playing important roles inside and outside the region. Cross-border lending remains more limited for emerging markets.

The region's advanced economies became more integrated into global cross-border banking between 2005 and 2024. Growth in their banking liabilities to

[11] These figures possibly underestimate the shift because claims from Western Hemisphere international financial centers (for example, the Cayman Islands) are often shell companies with Asia and Pacific owners, and their share of total portfolio holdings of the region's debt grew by 7 percentage points. International financial centers in the region also experienced the region's fastest overall growth in portfolio debt liabilities, rising to 25 percent of the regional total.

economies outside the region has far outpaced economic growth, rising from less than 15 percent of their GDP in 2005 to more than 30 percent by 2024.[12] In nominal terms, extraregional liabilities nearly tripled in the same period.[13] Advanced economies have also become major sources of cross-border bank lending to other regions, primarily to advanced economies rather than emerging market economies (Llovet Montanes and Schmukler 2018). Assets (claims on economies outside the region) increased from less than 15 percent of lender GDP in 2005 to nearly 30 percent by 2024. Among the advanced economies, Hong Kong Special Administrative Region and Japan are the largest lenders and borrowers in interregional cross-border banking.

Emerging market economies are much less integrated into global banking networks. Their banking liabilities to economies outside the region are low as a share of borrower GDP—especially compared with those of advanced economies—and have not kept pace with their rapid GDP growth. By 2024, their banking liabilities to economies outside the region were less than 3 percent of GDP. The Chinese mainland is the largest recipient of interregional cross-border lending among emerging market economies. Europe and the United Kingdom remain key sources of bank lending to the region.

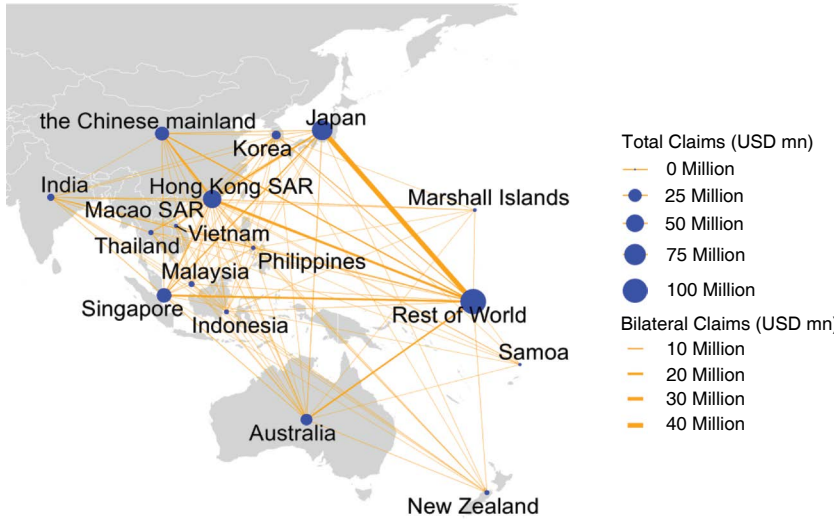
Intraregional cross-border banking is becoming increasingly important (ADB 2025). For advanced economies, banking liabilities to economies within the region nearly tripled as a percent of GDP between 2005 and 2024, although they remain slightly lower than banking liabilities to the rest of the world. These liabilities mostly reflect those among Hong Kong Special Administrative Region, Japan, and Singapore, while the Chinese mainland is also emerging as a primary lender to advanced economies. For emerging markets, intraregional banking liabilities also grew rapidly in nominal terms during the same period, broadly keeping pace with GDP growth. However, intraregional banking liabilities remain low overall at 2 percent of borrower GDP. Hong Kong Special Administrative Region and Japan are the main regional lenders for the region's emerging markets.

The intraregional cross-border banking network is becoming less centralized. A few economies once dominated cross-border banking within the region, but additional regional hubs have emerged in the past two decades and reduced their role (Figure 13.4). Japan's relative importance within the region has declined because of those new regional hubs, though it remains a major source of cross-border bank lending and maintains strong global connections. Singapore has continued in its role as a major banking hub, whereas Australia's growing importance partly reflects

[12] All metrics expressed as a share of GDP for a subgroup of countries refer to the total assets or liabilities of the subgroup as a share of the total GDP of the subgroup.

[13] Data include cross-border bank lending to all sectors (banks and nonbanks). The sample of countries serving as the source and destination of cross-border lending is limited because of data availability. Data encompass most major banking hubs inside and outside the region. For years for which reported claims were unavailable; estimations used counterparty-reported liabilities. Singapore's role after 2021 could not be estimated accurately because of data gaps.

**Figure 13.4. Cross-Border Bank Lending in Asia (2024)**




Sources: BIS, Locational Banking Statistics Database and IMF staff calculations.

connections with Pacific economies.[14] Hong Kong Special Administrative Region is overtaking Japan and Singapore as the primary source and destination of regional bank lending. The Chinese mainland has also become more significant within the regional banking network. As other economies have grown and challenged Japan's dominance in the regional banking network (and Singapore's to a lesser extent), the network has become less centralized and less dependent on a small number of countries. However, the regional banking network does not exhibit the same level of integration as the trade and FDI networks, which include many more substantial bilateral connections among smaller economies.

## UNDERSTANDING FINANCIAL POSITIONS: A GRAVITY MODEL APPROACH

A gravity model framework was used to analyze bilateral financial positions, understand how trade and financial links interact across borders, and identify the economic and institutional factors that shape those relationships. Inspired by Newton's Law of Universal Gravitation, the gravity model is a standard tool for analyzing bilateral economic flows, initially in trade and more recently in investment and finance. The model's appeal lies in its intuitive structure: economic interactions between two countries tend to increase with economic size and decrease with geographic or economic distance. Although originally developed for

[14] Singapore's role in the regional cross-border banking network could be understated because of data limitations in recent years.

trade, the model has proved effective in explaining cross-border financial flows, including FDI, portfolio holdings, and bank lending (Portes and Rey 2005; Yotov and others 2016).

The analysis first examined whether trade flows are associated with bilateral financial positions. Given the region's deepening trade integration, the analysis tested whether countries that trade more also invest more in each other. A panel gravity model was estimated with trade flows as explanatory variables for different types of bilateral financial positions: outward FDI, inward and outward portfolio investment (both total and disaggregated into debt and equity), and outward bank lending.[15] To isolate the effect of trade, the analysis controlled for a rich set of confounding factors using multiple fixed effects. First, country-time fixed effects for both origin and destination countries absorbed all country-specific demand and supply shocks (which include domestic macroeconomic conditions such as GDP, interest rates, and policy stance) and global shocks (such as global financial and economic conditions), provided those global shocks do not affect financial relationships with different counterparties differently. Second, country-pair fixed effects captured time-invariant bilateral characteristics such as distance, common language, and historical ties between the origin and destination of financial positions. This framework allowed credible estimation of the correlation between trade and financial positions after accounting for those other factors.

The results show a robust link between trade and FDI but not between trade and other financial instruments. A 1 percent increase in trade flows between origin and destination countries is associated with a 0.14 percent increase in the stock of outward FDI. Trade flows are not significantly associated with all other financial positions. A 1 percent increase in trade is associated with a 0.05 percent increase in cross-border bank liabilities, but the result is not statistically significant. The coefficients for portfolio investment are smaller and not statistically significant. Therefore, stronger trade and FDI links are mutually reinforcing, but the relationship between trade and portfolio investment or bank lending is unclear. Trade integration could facilitate direct investment (or vice versa), which does not necessarily translate into broader financial integration.[16]

[15] Annex 13.3 describes the gravity model in more detail. The analysis also estimated the reverse specification with trade as the dependent variable and financial positions as the independent variable. The signs and magnitudes of size, distance, contiguity, common language, and colonial history are consistent with those in the literature (Yotov and others 2016), validating the data set and estimator. The robust link between trade and FDI but not between trade and other financial positions also holds in this reverse specification.

[16] A particular direction of causality between trade and FDI is not implied from the results. FDI can cause trade if a foreign firm sets up its own retail and servicing operations in a destination country ahead of increasing its exports. Trade can alternatively cause FDI if growing exports to a destination market warrant shifting part of the production process to that market. Either way, FDI and trade appear strongly and positively interlinked.

The analysis then examined macroeconomic and institutional drivers of financial positions. Estimated fixed effects from the first stage regressions were regressed on macroeconomic and institutional variables. Country-time fixed effects were regressed on indicators such as GDP, inflation, interest rates, lagged current account balances, the financial development index (Sahay and others 2015), and institutional metrics proxied by some components of the Fraser Institute's Economic Freedom data set (Gwartney and others 2025) and capital flow management measures in the IMF's Annual Report on Exchange Arrangements and Exchange Restrictions (AREAER) database (IMF 2024a).

The results are consistent with the literature, suggesting that financial integration between economies reflects their combined economic size and other push-pull factors. Higher GDP leads to higher financial positions for both the origin and destination economies. Relatively high inflation in the origin economy incentivizes outward financial flows as investors seek protection against domestic price pressures. Inflation in the destination economy increases overseas bank borrowing and portfolio investment but is detrimental to direct investment. Larger current account surpluses in the past typically correlate with larger financial positions in source countries and smaller financial positions in recipient countries, consistent with their balancing role in which changes in the current account reflect changes in net foreign assets. Stronger financial development in the source country leads to larger cross-border financial positions.

Institutional reforms can help destination economies attract financial positions. Smaller governments and stronger legal systems, as measured by the relevant Economic Freedom subindexes, are conducive to larger financial inflows. Weaknesses in some of those indicators in the origin country lead to larger outflows as investors seek stronger institutional frameworks abroad. Greater freedom to trade is especially important for attracting bank lending and portfolio flows, though it appears less relevant for attracting FDI, likely because FDI could help foreign firms circumvent trade restrictions by establishing subsidiaries in relevant export markets.

Capital flow management remains relatively restrictive in the region. A composite index of capital flows management measures shows that although the region has eased those measures since 2000, it remains one of the most restrictive regions, second only to sub-Saharan Africa (IMF 2024a). Significant controls remain on liquidation of direct investment, real estate transactions, and personal capital transactions. The impact of capital flow management measures on the gravity model estimates is mixed, because even though looser measures can attract additional financial inflows, they can also allow outflows that previous restrictions prevented.

## POLICY RECOMMENDATIONS: THE WAY FORWARD

The Asia and Pacific region's role in global finance remains smaller than its role in international trade. Although the region's share of global financial flows and foreign direct investment is growing, especially within the region, participation in

portfolio investment and cross-border bank lending is still comparatively limited. This finding is driven largely by the more limited financial integration of the region's emerging markets, whereas advanced economies are strongly integrated into global financial markets. Although the analysis did not assess the factors behind this trend directly, strong evidence shows that the region's trade-oriented growth models have been more conducive to FDI integration than to integration through portfolio investment or bank flows. As the region continues to grow in economic size and its financial markets develop, stronger cross-border financial positions will be expected to accumulate, as has already occurred in the region's advanced economies.

Stronger financial integration can benefit the region's economies, especially emerging market economies, but policymakers must balance those benefits against the associated vulnerabilities.

Financial integration can deliver substantial economic gains. It expands access to financing and lowers the cost of capital, which can stimulate growth and job creation (Obstfeld 1994). For emerging markets, it can support a transition from export-led growth to a model centered on domestic investment and consumption. Integration can also encourage domestic financial market development, reducing overreliance on banks (Levine 1997), enhancing macroeconomic and financial resilience (Nazar Van Doornik and others 2024; IMF 2025), and strengthening capital market discipline and governance (Rajan and Zingales 2003; Kose and others 2006). From a risk-management perspective, it enables investors to diversify portfolios and helps local borrowers preserve access to financing during localized economic shocks (Colacito and Croce 2010). The development of regional financial hubs can augment access to liquidity and generate economies of scale. FDI can facilitate the diffusion of knowledge, including technological and managerial expertise and on-the-job training (Borensztein, De Gregorio, and Lee 1998; Berthélemy and Demurger 2000).

However, deeper financial integration also brings vulnerabilities. It increases domestic economies' and financial systems' exposure to regional and global shocks and to shifts in global investor sentiment. As integration deepens, the resilience of financial networks becomes increasingly important (Gai and Kapadia 2010). In this context, multiple hubs (such as financial centers Hong Kong Special Administrative Region and Singapore) can reduce exposure to localized shocks in a systemic economy more effectively than a single hub. Policymakers must also navigate complex cross-country regulatory differences that can hinder coordination and oversight. Data gaps and the opaque routing of financial flows through multiple financial centers often constrain effective risk analysis.

Rising protectionism and geoeconomic fragmentation globally threaten cross-border financial integration in the Asia and Pacific region. Barriers to trade and FDI can dampen cross-border capital flows. Geopolitical tensions have been found to affect portfolio investment allocations (Catalán, Fendoglu, and Tsuruga 2024; Gopinath and others 2025). Greater integration within the region can help economies navigate increasingly complex global dynamics.

To harness the benefits of financial integration while mitigating its risks, policymakers should prioritize a balanced and forward-looking strategy. Priorities include maintaining up-to-date financial sector regulations, implementing well-calibrated macroprudential policies, and strengthening supervisory frameworks to preserve systemic stability. Addressing data gaps, especially those related to nonbank financial institutions and the ultimate sources and destinations of investment, is essential for effective risk monitoring. Policymakers could pursue gradual liberalization of capital transactions along with continued progress in cross-border payment infrastructure to facilitate smoother financial flows. Regional cooperation can be strengthened by expanding intraregional swap lines to bolster the financial safety net and aligning regulatory standards in areas such as investor protection, dispute resolution, and bankruptcy procedures. Capital flow management measures should be used judiciously to mitigate volatility, consistent with the IMF Integrated Policy Framework and the Institutional View on capital flows.

## ANNEX 13.1. INTERNATIONAL FINANCIAL CENTERS: METHODOLOGICAL CHALLENGES

Hong Kong Special Administrative Region and Singapore are international financial centers that have become increasingly important within the Asia and Pacific region. Coppola and others (2021) and Daamgard, Elkjaer, and Johannesen (2019) note that international financial centers can pose challenges for capital flow surveillance and assessments of financial integration. Specifically, their financial intermediation function can obscure the true source and destination of capital flows in standard data, overstate links to real economic activity (for example, foreign direct investment [FDI] into shell companies), and in some cases, the nature of the financial claim (such as when an international company issues bonds in an international financial center but repatriates the funds through an FDI transaction). Although the chapter uncovers evidence of the salience of these data issues, it leaves deeper exploration for future analysis.

The increasing prevalence of international financial centers in the region and the ubiquity of holding companies and funds in that network point to growing potential for data gaps that complicate assessments of financial integration and links between financial and real activity. Most FDI investments into these centers represent financial investments in holding companies or funds, which might not be linked directly to real economic activity in that region (for example, if it redirects Malaysian capital to Europe through a private equity structure) or at all (for example, if it establishes ownership links between networks of special purpose entities; Daamgard, Elkjaer, and Johannesen 2019). In Hong Kong Special Administrative Region, more than two-thirds of outbound and inbound FDI positions are investment and holding companies. In Singapore, investment and holding companies make up 50 and 62 percent of outbound

and inbound FDI positions, respectively, and have grown rapidly, with most of the net growth in FDI from 2020 to 2023 flowing into investment and holding companies.

## ANNEX 13.2. A PRIMER ON NETWORK ANALYSIS

Network analysis provides a powerful lens through which to examine how economies are structurally connected through global trade and finance. Unlike other approaches, it captures both direct and indirect links, revealing a broader web of interdependencies.

Eigenvector centrality captures the importance of an individual economy (node) in the network of trade or financial flows, reflecting both the strength and number of direct links and the extent of connectivity with other key players in the network.[17] This measure of centrality captures influence and systemic relevance of economies, making it useful in financial networks in which indirect exposures and contagion channels are important.[18] For example, a financial center with fewer direct links can still be highly central if it connects other major hubs, indicating its potential role in amplifying or absorbing shocks. The centralization of the network captures whether transactions in goods or financial instruments are routed primarily through a few key hubs (high centralization) or are more evenly spread among economies (low centralization).[19]

An analysis applied this framework to the trade network as a comparator for financial networks covered in detail in this chapter. Strong links within the region were found but with changing roles for key players. Most notably, China's centrality in the network expanded sharply in the late 1990s and 2000s, consistent with its rapid integration into the global economy. China's rise coincided with a decline in Japan's centrality, which has become a distant second in relevance in the intraregional trade network. India and countries in the Association of Southeast Asian Nations are generally well integrated, with Vietnam's centrality rising, especially in the 2010s. Hong Kong Special Administrative Region and Singapore are much smaller players in this network than in the financial networks. The trade network does not rely on key hubs. Most economies connect bilaterally with regional counterparts, and centralization is low.

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[17] The centrality measure used is weighted eigenvector centrality. For mathematical details, see *External Sector Report: Global Imbalances in a Shifting World,* Online Annex 2.2.

[18] See Glasserman and Young (2016) for a literature review of the applications of network analysis to the modern financial system and the links to systemic fragility.

[19] The relationship between network structure and financial stability is complex, but generally, although more centralized networks can benefit from increased liquidity and economies of scale in large hubs, heavy reliance on a small number of hubs can increase exposure to systemic shocks (see, for example, Allen and Gale [2000], Elliott, Golub, and Jackson [2014], and Nier and others [2007]).

## ANNEX 13.3. GRAVITY MODEL OF FINANCIAL POSITIONS

An analysis estimated the following specification:

$$X_{ij,t} = exp\left[\sum_{\kappa=0}^{2} \beta_k log(Trade_{ij,t-k}) + \mu_{i,t} + \delta_{j,t} + \varphi_{i,j}\right] + \varepsilon_{ij,t} \qquad (13.3.1)$$

where, $X_{ij,t}$ are the levels of different bilateral financial positions of origin country $i$ on destination country $j$, including foreign direct investment, foreign portfolio investment in debt and equity, and cross-border bank lending. $\sum_{\kappa=0}^{2} log(Trade)_{ij,t-k}$ denotes the sum of the contemporaneous, one-period lagged, and two-period lagged log trade, where trade is measured as the sum of imports and exports between countries $i$ and $j$. Therefore, $\sum_{\kappa=0}^{2} \beta_k$ measures the correlation between trade flows and financial positions. As Yotov and others (2016) recommend, country-time fixed effects of both origin country $i$ and destination country $j$ ($\mu_{i,t}$ and $\delta_{j,t}$) were included to absorb all the domestic factors specific to each country at each point in time and common across its counterparts, such as economic growth, interest rates, policy mix, and country risk. Asymmetric country-pair fixed effects ($\varphi_{i,j}$) were also included to capture all time-invariant bilateral factors between the two countries—such as distance, common language, and colonial ties—while allowing for asymmetries in how these factors affect financial positions. Thus, the fully saturated specification captures the effect of trade on financial positions after accounting for all other confounding factors. Equation 13.3.1 was estimated using the Poisson Pseudo-Maximum Likelihood estimator, and standard errors clustered at the country-pair level.

In the second step, the analysis regressed the fixed effects estimated in equation 13.3.1 on macroeconomic and policy variables to understand the role of other factors on financial positions. Specifically, the country-time fixed effects of origin country ($\mu_{i,t}$) and destination country ($\delta_{j,t}$) were regressed on their domestic macroeconomic conditions and institutional metrics:

$$\mu_{i,t} = \mathrm{B}_o\, Macro_ori_{i,t} + \Gamma_o\, Institutional_ori_{i,t} + \zeta_i + \kappa_t + \upsilon_{i,t} \qquad (13.3.2)$$

$$\delta_{j,t} = \mathrm{B}_d\, Macro_des_{j,t} + \Gamma_d\, Institutional_des_{j,t} + \eta_j + \kappa_t + \tau_{j,t} \qquad (13.3.3)$$

In equation 13.3.2, $Macro_ori_{i,t}$ represents a series of domestic macroeconomic variables in the origin country, including log real GDP, consumer price index inflation, short-term interest rate, current account balances, and financial market development; and $\mathrm{B}_o$ represents their coefficient vector. $Institutional_ori_{i,t}$ denotes a series of policy variables in the origin country, proxied by the indices from the Fraser Institute's Economic Freedom data set (Gwartney and others 2025) and capital flow management measures in the IMF's AREAER database (IMF 2024a); and $\Gamma_o$ represents their coefficient vector.

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CHAPTER 14

# Industrial Policy in Asia and Pacific: Could the Wave Lead to Structural Transformation?

**Paula Arias, Vanya Georgieva, Rahul Giri, Maria Gonzalez-Miranda, Ashique Habib, Anne-Charlotte Paret, Tatjana Schulze, Arthur Xie, Weining Xin, Yichen Xu, and Dilan Yang**

The use of industrial policy has intensified in the Asia and Pacific region since the late 2010s, in line with a well-documented global surge (Juhász and others 2023; IMF 2024b; Evenett and others 2025; Juhász and others 2025). Economies implement these policies for multiple and sometimes overlapping reasons. Stated motives in policy announcements show that some economies use industrial policy to promote structural transformation, both for traditional development objectives and for trying to position themselves favorably amid demographic pressures, climate change, and rapidly evolving automation and digitalization technologies (Evenett and others 2024; Evenett and others 2025). High-profile examples of industrial policy in some economies, sometimes followed by rapid growth in targeted industries, such as solar panels and electric vehicles in China, could lead other economies to reconsider industrial policy as an alternative. Geoeconomic fragmentation and national security considerations have also prompted economies to use industrial policy to support de-risking strategies. Many economies in the region use industrial policy to take advantage of shifting trade, investment, and financial integration patterns or to respond defensively to the policies of major economies and rising protectionism.

By how much this regional deployment of industrial policy could drive sound economic structural transformation that can support long-term growth is uncertain. This chapter embarks on an in-depth analysis of the recent industrial policy wave in Asia and Pacific. It assesses the objectives of industrial policy and how effective it may have been in delivering durable and targeted changes that could signal strong shifts in economic structures. The analysis relies on a data set of industrial policy interventions covering 2009–24 and extracted from policy announcements recorded in the Global Trade Alert. Arias and others (2026) provide additional technical details and discussion of the results.

## INDUSTRIAL POLICY DEFINITION AND DATA

The analysis defines "industrial policy" as targeted government interventions aimed at supporting specific domestic firms, industries, or narrowly defined economic activities to achieve specific national economic or noneconomic objectives (IMF 2024a; Baquie and others 2025).[1] Although this definition is widely accepted, operationalizing it inevitably requires judgment. Much of the recent literature has focused on a narrower set of policies selected according to stated intentions in policy announcements or policies targeting a specific list of products and sectors closely associated with industrial policy (Juhász and others 2025 Evenett and others, forthcoming). However, stated motives are often incomplete or opaque, serve multiple purposes, or are framed in broad terms that are difficult to map to specific mechanisms or testable predictions.

Accordingly, the analysis relies on the broader industrial policy definition stated, under which all vertical policies that affect a specific sector or product are considered industrial policy. The definition does not filter interventions based on predetermined objectives or motives and does not limit coverage to only preselected strategic sectors ascribed to industrial policy. This broad approach expands the coverage of economies in Asia and Pacific in the analysis and, more importantly, allows the data to reveal the possible strategic intent of the interventions through novel analytical techniques. The industrial policy data set consists of observations that record whether an industrial policy intervention existed in an economy, the policy tool used, the products and sectors targeted, and the year of implementation.[2]

## THE LANDSCAPE OF INDUSTRIAL POLICY IN ASIA AND PACIFIC

Industrial policy in Asia and Pacific has received considerable attention, but no systematic assessment has determined who uses it and when, what it targets, and how it is implemented. The analysis seeks to fill that gap.

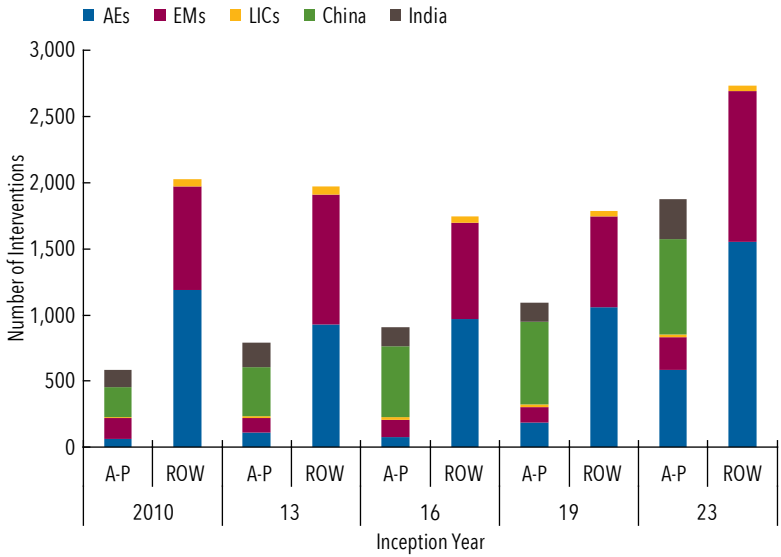
The region has participated in the global surge in industrial policy deployment, and its emerging market economies have used industrial policy more intensively since 2009. By contrast, advanced economies have accelerated industrial policy implementation only since the COVID-19 pandemic (Figure 14.1). Several advanced economies in the region had deployed industrial policy as part of previous rapid development efforts in the mid- to late 20th century, which is not captured in this data set.

- Industrial policy interventions focus heavily on industry and manufacturing, with about 80 percent of the interventions in Asia and Pacific targeting these sectors. The share is similar in the rest of the world and has remained

[1] This economic definition of IP, based on identifying preferential policy support for some sectors or firms, may differ from legal interpretations (e.g., based on WTO frameworks).

[2] Industrial policy takes the value of 1 when applied and zero otherwise. This measurement does not incorporate any information on the size or intensity of the interventions, and thus all measures are counted equally, even if their economic relevance differs.

**Figure 14.1. Industrial Policy Interventions: Asia and Pacific versus the Rest of the World**

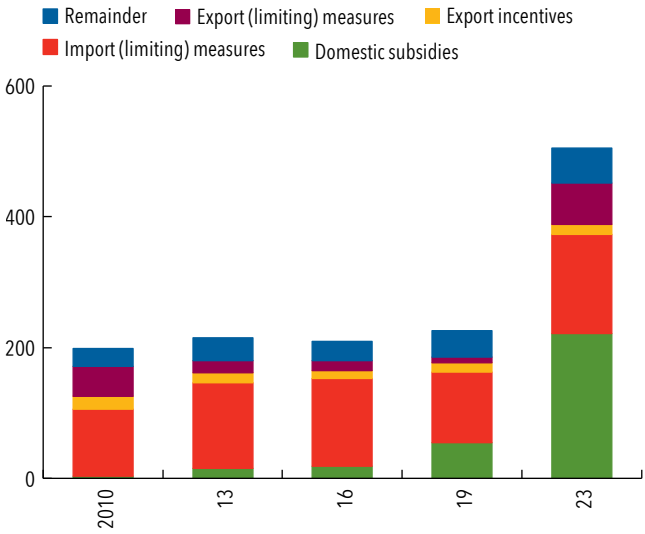



Sources: Global Trade Alert; CEPII-BACI data set; and IMF staff calculations
Note: Industrial policy interventions reflect the number of sectoral and vertical policies implemented per their inception date. The reported intervention counts do not include subnational level jurisdictions. AE = advanced economy; A-P = Asia and Pacific; BACI = Base pour l'Analyse du Commerce International; CEPII = Centre d'Études Prospectives et d'Informations Internationales; EM = emerging market; LIC = low-income country; ROW = rest of the world.

broadly stable over time. The sustained focus of industrial policy on manufacturing contrasts with the rising role of modern services, such as IT, which have become an important driver of structural transformation in some economies. Most industrial policy interventions in the region also target products for which economies have no revealed comparative advantage (RCA). Products with long-standing comparative advantage, or legacy RCA, form the second most targeted group. Interventions rarely target products that have recently gained or lost RCA. These findings suggest that although industrial policy might have been used to sustain export competitiveness, its motivations likely extend beyond competitiveness in global markets.

- Domestic subsidies and import-limiting measures are the most common industrial policy tools in the region. The broad use of subsidies in Asia and Pacific surged during the pandemic (Figure 14.2), mirroring a trend documented in the rest of the world.
- Asia and Pacific economies stand out for layering multiple tools simultaneously on the same products. In recent years, about 40 percent of actively targeted products have been subject to three or more types of policy tools in the region, compared with less than 20 percent in the rest of the world. Such

**Figure 14.2. Industrial Policy Tools in Asia and Pacific, excluding China**




Sources: Global Trade Alert; CEPII-BACI; and IMF staff calculations
Note: Industrial policy interventions reflect the number of sectoral and vertical policies implemented per their inception date, and classified as discriminatory. The reported intervention counts do not include subnational level jurisdictions. BACI = Base pour l'Analyse du Commerce International; CEPII = Centre d'Études Prospectives et d'Informations Internationales.

layering could reflect an intricate policy design strategy. However, it could also reduce transparency and effectiveness if multiple tools work at cross-purposes and lack coordination in implementation.

## INDUSTRIAL POLICY TARGETING CHOICES AND POTENTIAL IMPLICATIONS

The scale of industrial policy, its intensification since the pandemic, and its increasingly multi-instrument nature raise an important question of whether the interventions exhibit any patterns that could indicate their potential to promote structural transformation shifts. The relatively short period covered by the available industrial policy data—about 15 years—limited the scope for direct assessment of the impact on high-level structural transformation, defined as the long-term transition of economies from agriculture to industry and later to services. A more modest approach assessed whether industrial policy appears oriented toward strategies designed to achieve such transformation. The analysis draws on the existing literature to identify industrial policy strategies that could deliver gains and the characteristics that distinguish them. The analysis then investigated whether industrial policy interventions in the region align with those characteristics.

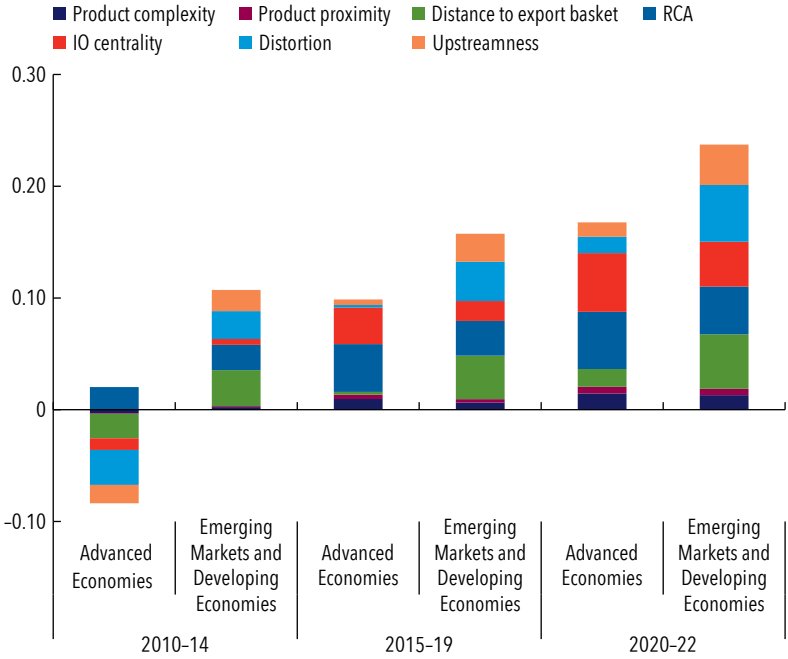
The literature identifies several strategies that could support structural transformation. Some focus on promoting technological upgrading and diversification in the product space, for example, by targeting technologically complex products (Hausmann and Hidalgo 2011) or moving toward new products that are highly related to those already present in the economy's product or export basket (Fortunato and others 2015). As discussed by Hidalgo and others (2007) and Reed (2024), economies pursue such strategies with different levels of risk. A less risky strategy might seek to scale up a product in which an economy already specializes or a product closely related to existing specializations (safe bets). By contrast, targeting attractive, highly complex products that are farther from current areas of specialization could represent a high-risk, high-reward strategy (moonshots) (Mazzucato 2021). The economic literature generally justified such strategies as responses to market imperfections or externalities. For example, underexploited economies of scale could justify efforts to expand production of safe-bet products, whereas some moonshots could involve the provision of public goods, such as vaccines or green technologies. A different, though not mutually exclusive, approach focuses on addressing specific sectoral market failures and friction that reduce efficiency. Examples include strengthening sectors that occupy key positions in value chains or domestic production networks, thus supporting growth and development more broadly through positive spillovers and cascading effects (Liu 2019; Georgieva 2025).

These three strategies point to specific indicators that can help identify them in practice. The indicators fall into three broad categories: product characteristics (such as product complexity and relatedness to other products); the relationship of a product to an economy's existing specialization (such as RCA and the distance between an economy's current export basket and the product); and measures of frictions and distortions in value chains and production networks (such as a product's position in the value chain, input-output centrality in the domestic production network, and markup-based sectoral distortion centrality).[3] Metrics from other complementary industrial policy databases, particularly the New Industrial Policy Observatory by Evenett and others (2024), provide additional insights into highly strategic products, including those associated with resilience, security, or climate mitigation motives.

The analysis first uses a machine learning approach to identify the factors that best predict the likelihood of industrial policy to target a product or sector (Figure 14.3). The findings present an overall mixed picture, with no single factor dominating the prediction of industrial policy incidence. The results also show that the relationships between the likelihood of being targeted and its drivers are mostly nonlinear. For example, industrial policy tended to target downstream products overall, but the likelihood of targeting also increased for highly upstream products, such as metals and fuel. Despite this complexity, several patterns emerged.

[3] These indicators of market frictions allow the analysis to identify industrial policy targeting such frictions in specific sectors. However, the analysis lacked sufficient information to determine whether the industrial policy in the strategies aligned with safe bets and moonshots also intended to address a market failure or externality and whether industrial policy could be justified on those grounds.

**Figure 14.3. Time Evolution of Determinants Contributing to Industrial Policy Incidence Probabilities**




Sources: Global Trade Alert; CEPII-BACI data set; and IMF staff calculations.
Note: Industrial policy interventions reflect the number of sectoral and vertical policies implemented per their inception date. Does not include subnational level jurisdictions. BACI = Base pour l'Analyse du Commerce International; CEPII = Centre d'Études Prospectives et d'Informations Internationales; IO = input-output; RCA = revealed comparative advantage.

- Some patterns in Asia and Pacific economies were consistent with efforts to build on existing export or production baskets and to target highly connected sectors with potential to generate spillovers. In general, products that were closer to an economy's existing export basket were more likely to be targeted. Similarly, products that occupied more central positions in production networks were more likely to have higher targeting rates. These patterns are consistent with strategies intended to promote technological upgrading or generate positive intersectoral spillovers.
- Nonlinearities suggest that other factors also influenced targeting decisions. For certain products far from an economy's existing export basket, the likelihood of targeting increased with distance from the export basket. Similarly, some products with less central positions in the production network were more likely to be targeted, which is consistent with the presence of moonshot strategies in the region.

To better understand whether the characteristics of targeted products or sectors align with potential industrial policy strategies that could promote structural

transformation, the analysis next uses a clustering exercise to assess how closely industrial policy targets aligned with theories of structural transformation before implementation. The econometric approach applies an atheoretical algorithm that allowed all industrial policy interventions in the data set—for both Asia and Pacific and the rest of the world—to cluster flexibly around groups of indicators associated with products that interventions might target. The approach does not assume that the resulting clusters would align clearly with rationales. The analysis then examines whether such alignment emerged. About three-quarters of the total industrial policy interventions in the data set align well with the industrial policy strategies in the literature, and three clusters stand out.

- A first cluster, accounting for more than one-quarter of industrial policy interventions in the region, is consistent with promoting products that can be considered safe bets. Products in this cluster tend to be attractive: relatively complex and about half regarded as strategic. They also tend to be products for which the economy already had RCA or were close to the economy's existing export basket. Therefore, these products are within reach in the domestic production matrix, and thus public support directed through industrial policy toward their expansion appears to be a relatively low-risk approach, including from a technological standpoint.
- Another cluster also focuses on attractive products, characterized by higher complexity and some degree of product-relatedness. About half of those products are regarded as strategic as defined in the New Industrial Policy Observatory database. The products are considered moonshots because they were far from the economy's existing areas of specialization and the economy had little or no RCA in their production. Therefore, supporting production through industrial policy interventions reflects a relatively high-risk, ambitious approach entailing leapfrogging existing capabilities. In Asia and Pacific, interventions in this cluster account for about 30 percent of total industrial policy interventions.
- A third cluster of interventions formed around market imperfections and alleviating friction at key points in production chains or networks. This cluster is linked to sectors with highly central positions in domestic production input-output networks or exhibit potential market distortions that needed to be addressed, proxied by the presence of markups. Targeting such sectors suggests a strategy aimed at reducing frictions, addressing market failures, and easing bottlenecks. Products associated with these interventions are relatively less strategic in nature. Interventions in this cluster account for slightly less than 20 percent of interventions in the region.
- Mapping a strategy for a fourth cluster, the remaining group of interventions, was difficult. This cluster comprises unattractive products with low complexity, unrelated to the economy's areas of specialization (low RCA), and in sectors that were not particularly central to production networks or distorted. Policymakers could have had other rationales not captured in the analysis, such as employment protection, that would explain the targeting,

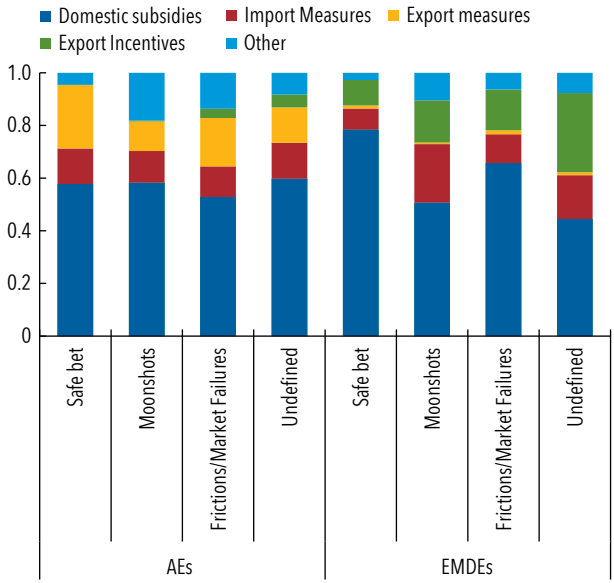
but the available data did not support linking these interventions to any specific structural transformation ambition.

Comparison of strategies between income groups in Asia and the Pacific as well as the rest of the world revealed advanced economies in the region targeting moonshots more intensely than advanced economies elsewhere. It is interesting that the targeting by emerging markets in the region is broadly balanced across the three strategies, whereas those in the rest of the world appear much more intensely focused on moonshots. This pattern is partly influenced by the relatively larger propensity of China to focus on "safe bets". Excluding this economy, the region's emerging market focus on moonshots is at similar rates to those observed in the rest of the world. Importantly, products that are "safe bets" are economy-specific in this analysis, depending on how close a product is to each economy's current specialization level; whenever an economy has developed a relatively broad export basket and diversified production structure, more products are close to its existing areas of specialization and therefore deemed "safe" from its perspective.

Although the clusters align with several industrial policy strategies, the policy tools used across clusters suggest imprecise targeting that could limit effectiveness (Figure 14.4). As expected, domestic subsidies were the most common tool across the estimated clusters. Emerging market economies in Asia and Pacific use some export incentives, which can suggest efforts—perhaps inspired by the export-oriented strategies of the Asian Miracle economies (Stiglitz 1996)—to use external markets to boost safe bets and moonshots and achieve diversification and technological upgrading. However, import-limiting measures are also a widely used tool. Because such measures often remain in place for long periods and cover a large set of products, substantive import-substitution efforts appear to factor into all strategies.

The evidence at the implementation stage presents a mixed picture of the potential for the region's industrial policy to trigger structural transformation. The characteristics of targeted products and sectors generally align with strategies that could promote structural transformation. A large share of targeted products either build on existing specialization (safe bets), aggressively expand into more ambitious areas (moonshots), or focus on sectors with the potential to address market failures and friction. However, a meaningful share of targets falls within a decidedly unattractive cluster, and the policy tools used across all strategies suggest a lean toward import substitution. Even when observable product or sector characteristics and policy tool choices indicate potential for structural transformation (for example, export-promoting efforts for attractive products), important details of policy design remain unobservable and will be critical for favorable outcomes. For example, the available data do not confirm that industrial policy strategies targeting safe bets and moonshots are based on well-identified market failures or externalities. In addition, information is lacking about safeguards to prevent political capture and distortions that could cause misallocation, or the fiscal and other economic costs of the implemented industrial policy interventions (Garcia-Macia and Sollaci 2025). These caveats remain

**Figure 14.4. Implicit Strategies: Choice of Instruments in Asia and Pacific**
*(Share of interventions)*




Source: IMF staff calculation.
Note: Safe bets: targeting attractive products that build on an economy's existing specialization or on one related closely to it. Moonshots: targeting attractive products that are far from the current specialization and could be a high-risk, high-reward bet. Frictions and market failures: targeting to alleviate frictions and at key points in production chains. AE = advanced economy; EMDE = emerging market and developing economy.

important, although they are difficult to address in a systemic cross-economy analysis.

## THE EFFECTS OF INDUSTRIAL POLICY

Significant shifts in an economy's broader structure can take decades to coalesce. The relatively short span of available industrial policy data used in the analysis—nearly 15 years, from 2009 to 2024—allows for only a tentative assessment. The focus is on identifying robust signals of durable economic shifts caused by industrial policy. Two considerations were essential: outcomes should improve significantly for targeted products or firms compared with untargeted counterparts, and any improvement must be durable if it is to trigger structural transformation. Targeted support can raise an industry's output, but for that increase to qualify as successful in promoting structural transformation, the near-term improvement

should unlock a long-lasting improvement, indicating, for example, that the support allowed firms in the industry to catch up with new technologies or that a hurdle or market failure was overcome.

Two complementary approaches identified whether industrial policy interventions produced broad-based, systematic, and durable effects consistent with structural transformation. The first approach focused on external outcomes and relied on trade-based data. The analysis examined whether industrial policy measures in Asia and Pacific generated statistically significant and durable changes in trade flows, RCA, or both, at the product level, following Baquie and others (2025) and IMF (2025). The second approach gauged the domestic effects of industrial policy using firm-level data and assessed whether industrial policy contributed to structural transformation by shifting productive resources toward targeted industries. Firm-level data from the Orbis database were used to test the hypothesis that interventions should boost the targeted sectors through increases in productivity or factors of production (labor and capital).

The results provide mixed evidence of the effectiveness of industrial policy in Asia and Pacific. A few cases show significant and durable effects that could contribute to structural transformation. However, industrial policy does not appear to be associated with a general or systematic impact on economic indicators. Some policy tools could have transitory effects on trade and competitiveness, whereas the firm-level analysis shows no systematic improvement in productivity, employment, or capital accumulation in the targeted sectors. Some evidence points to gains for advanced economies in the region, notably the east Asian ones, in both trade and firm-level variables, possibly indicating a role for stronger institutions and historical memory of effective industrial policy, which is beyond the scope of the analysis.

## Industrial Policy Impact on Trade and Competitiveness: A Product-Level Analysis

A product-level analysis first examined trade and competitiveness because improved external performance can indicate that industrial policy successfully exploited economies of scale and overcame frictions. Greater participation in competitive global markets can also encourage firms to raise productivity and impose market discipline on firms benefiting from industrial policy. Indirectly, if export growth is large and sustained, it could support labor reallocation and the sectoral composition of value added, thus promoting structural transformation. The empirical approach of the analysis, detailed in Arias and others (2026), followed a local projections difference-in-differences methodology that used the performance of nontargeted products as controls over various time horizons (see Ruta and Rotunno 2024; Baquie and others 2025; IMF 2025). The assessment covered the most widely used industrial policy instruments: domestic subsidies, export incentives, and import-limiting measures.

The findings for product-level trade suggest that industrial policy achieves export promotion or import substitution (Krueger 1985) in some cases, but the gains do not appear systematic or durable (Table 14.1).

- Although domestic subsidies appear to support exports in the region, the impact is not durable. A more granular view by income level suggests that the short-term boost observed in the region is driven largely by a temporary increase in exports in the region's emerging markets (excluding the large ones). Among advanced economies, East Asian ones show some positive but significantly lagged effects.
- Export incentives can increase exports of targeted products in the region, but the effect is only temporary. Advanced economies, specifically the East Asian ones, explain most of the gains.
- Domestic subsidies appear to reduce imports temporarily, suggesting import substitution. Imports of subsidized products in Asia and Pacific advanced economies decline about two to three years after subsidies were established. Import-limiting measures, such as import quotas or bans, show some near-term effectiveness in parts of the region. A significant but small and short-lived reduction in imports occurs at implementation in both East Asian advanced economies and emerging markets.

Analyzing the impact on competitiveness, measured by RCA, also shows no systematic gains. A higher RCA indicates that an economy produces a product more

**TABLE 14.1.**

**Heat Map of Industrial Policy Effects on Exports and Imports**

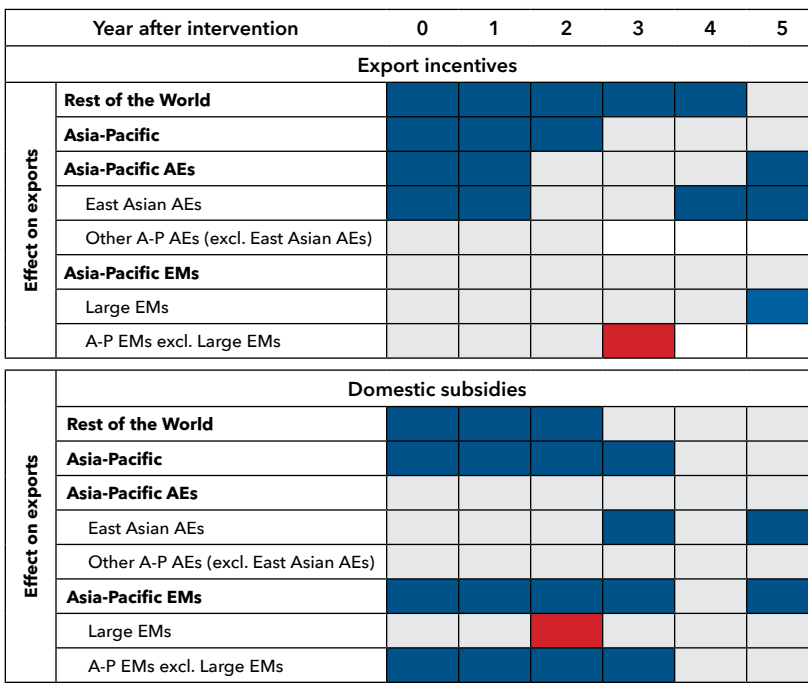


**TABLE 14.1. (*Continued*)**

**Heat Map of Industrial Policy Effects on Exports and Imports**

| Effect on imports | Year after intervention | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|
| | Import-limiting measures | | | | | | |
| | **Rest of the World** | | | | | | |
| | **Asia-Pacific** | | | | | | |
| | **Asia-Pacific AEs** | | | | | | |
| | East Asian AEs | | | | | | |
| | Other A-P AEs (excl. East Asian AEs) | | | | | | |
| | **Asia-Pacific EMs** | | | | | | |
| | Large EMs | | | | | | |
| | A-P EMs excl. Large EMs | | | | | | |

| Effect on imports | Domestic subsidies | | | | | | |
|---|---|---|---|---|---|---|---|
| | **Rest of the World** | | | | | | |
| | **Asia-Pacific** | | | | | | |
| | **Asia-Pacific AEs** | | | | | | |
| | East Asian AEs | | | | | | |
| | Other A-P AEs (excl. East Asian AEs) | | | | | | |
| | **Asia-Pacific EMs** | | | | | | |
| | Large EMs | | | | | | |
| | A-P EMs excl. Large EMs | | | | | | |

Source: IMF staff estimates.
Note: The table shows the effects of industrial policy intervention on exports (left-hand-side) and imports (right-hand-side) each year after the product-level intervention. Blue cells indicate statistically significant positive coefficients at the 5 percent level, and red cells indicate statistically significant negative coefficients. Gray cells indicate coefficients that are not statistically significant. Results are shown for each subgroup breakdown by income category and breakdown within each income category. Blank cells indicate an insufficient number of observations to estimate the effect. "Large emerging markets" refers to China and India. AE = advanced economy; EM = emerging market.

competitively, as reflected in its relative specialization in the product in export markets. The analysis focused on the two industrial policy tools most relevant for export competitiveness: domestic subsidies and export incentives.

- The results show that both domestic subsidies and export incentives raise RCA in Asia and Pacific but only for products that are already competitive. Even then, the effect is short-lived. In both advanced economies and emerging markets, after an initial increase, the positive effect achieved decreases over time, suggesting that the gains in competitiveness could result from the immediate cost support provided by industrial policy, not from durable, gradual improvements in structural productivity. The findings also show that domestic subsidies increase the RCA immediately in both advanced economies and emerging markets in Asia and Pacific while export incentives in emerging markets have a delayed impact, yet neither effect appears dura-

ble. In advanced economies, export incentives produce some temporary positive impact on the RCA, especially for products that are not yet competitive, but those gains dissipate over the medium term.

Overall, the short duration of industrial policy's positive effects on trade flows suggests that such policies generally have limited effectiveness and are unlikely to lead to structural transformation. Industrial policy tools can affect trade flows and competitiveness in Asia and Pacific economies by boosting exports, triggering import substitution, or increasing competitiveness where they already exist. However, the effects are generally short-lived, suggesting that the underlying mechanisms do not generate fundamental structural changes to the productive and trade structure of the targeted sectors. For example, fiscal support to exporters through subsidies or export incentives could increase export levels, allowing a one-time, temporary increase in exports by lowering costs and improved pricing. However, the evidence does not indicate that this gain systematically translates into a structural gain, such as more efficient cost structures. Similarly, the temporary effectiveness of import substitution raises questions about whether the protected domestic industries can sustain gains over time, become competitive, and replace imports cost-effectively. If industrial policy does not improve the cost structure, consumers ultimately bear the costs through lower quality or the need for prolonged financial support. Exploring these issues in greater depth falls outside the scope of the analysis. What drives the stronger effects of new industrial policy observed in the East Asian advanced economies also warrants further examination. Given these economies' long experience with industrial policy, they likely learned about policy design over time. The findings thus suggest that using industrial policy could, in principle, be feasible under policy designs that include appropriate incentives.

## Industrial Policy Impact on Domestic Variables

Beyond exports, industrial policy could lead to domestic production shifts that provide another indication of structural transformation. Industrial policy globally appears to have become increasingly inward-looking, with a growing emphasis on import substitution. In addition, about two-thirds of industrial policy interventions target products for which economies have no comparative advantage and therefore little or no presence in global markets. The effects of industrial policy in the region could thus be reflected most clearly in domestic variables, especially within the industry and manufacturing sector, which implements the largest share of industrial policy interventions. The analysis therefore examined whether industrial policy interventions generate significant and long-lasting firm-level changes in targeted sectors, using Orbis firm-level data and a group of Asia and Pacific economies with sufficient available data.[4]

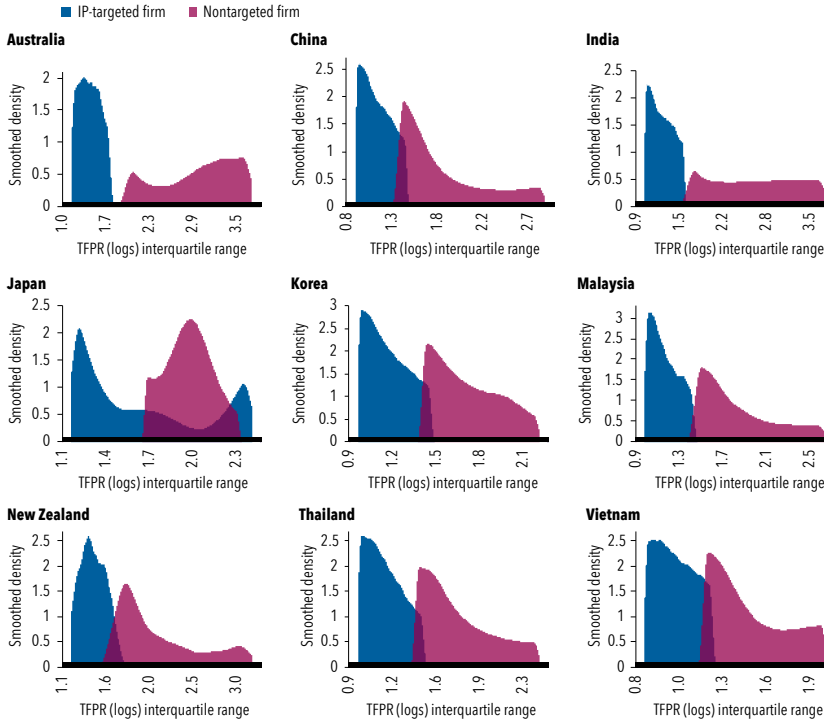
[4] Data were available for Australia, China, India, Japan, Korea, Malaysia, New Zealand, Thailand, and Vietnam.

The association between industrial policy and firm productivity does not provide much evidence that industrial policy has improved domestic firm outcomes or driven structural transformation through productivity changes.

- A first and striking empirical regularity is that industrial policy in the region is systematically associated with lower-productivity industries. The distribution of firms' total factor productivity (TFP) is lower in targeted industries than in untargeted industries in every economy in the sample except for Japan, where the distribution for targeted firms is bimodal. However, the broader pattern does not indicate whether industrial policy systematically targets less productive firms or sectors (for example, to correct a market failure); whether industrial policy is targeted imprecisely from the outset; or if firms become less productive after an intervention (for example, because of capital, credit misallocation, or other distortions).
- Most economies in the sample experienced a decline in the median productivity among firms in the targeted sector after an industrial policy intervention. Comparing the distribution of firms in the two years before the first intervention with the distribution afterward suggests a broad shift of the mass of firms toward the left, implying lower average productivity (Figure 14.5). The median firm TFP of targeted firms declines before, during, and after the intervention, with effects lasting for up to three years after implementation. The declines are most pronounced in Australia, India, Japan, Malaysia, and Vietnam. Although the pattern does not establish causality, it provides no evidence of systematic productivity catch-up associated with industrial policy at the aggregate level across economies.
- When productivity changes occur in industrial policy-targeted firms, technological improvement generally does not drive them. A decomposition analysis assessed the contributions of incumbent firms improving their own technology (the within firm component), incumbent firms gaining market share (the reallocation component), and firm market entry and exit. Some economies show substantial contributions from technology improvement within incumbent firms, including India, Japan, and Thailand. Other economies show the opposite impact, including Australia, Malaysia, and New Zealand. However, TFP changes in firms not targeted by industrial policy appear to be larger. In some cases, aggregate TFP gains among nontargeted firms exceed those among targeted firms and are often driven partly by technological improvements, though with some exceptions, such as New Zealand and Vietnam. Overall, the findings indicate that industrial policy's role in promoting technological upgrading is varied and unclear.

These findings also raise the possibility that industrial policy could be targeted imprecisely from the outset and systematically reach less productive firms. Targeting errors could arise from measurement inaccuracies, misperceptions about the relative productivity of industries, or motivations other than structural transformation.

**Figure 14.5. Asia and Pacific: Distribution of Firms' Total Factor Productivity**
*(Smoothed probability density)*




Sources: Global Trade Alert; Orbis database; and IMF staff estimates.
Note: Industrial policy interventions are mapped at the Nomenclature of Economic Activities (NACE) four-digit level to Orbis for 2009–22. The total factor productivity revenue estimation follows Ackerberg, Caves, and Frazer (2015) and Díez, Fan, and Villegas-Sánchez (2021).

An investigation into the impact on capital and labor inputs and TFP also finds no systematic gains. Improvements in any of those dimensions would be important because, regardless of the objective of industrial policy, success should result in greater production in the targeted industries. The analysis focused on the three main policy tools used in the region—domestic subsidies, import-restricting measures, and export incentives—and assessed their effects on three outcome variables: TFP, and labor and capital (the two main factors of production). Several findings emerged.

- For the Asia and Pacific sample, the impacts of industrial policy on domestic firm outcomes (productivity, capital, and labor) are mixed and generally insignificant. Across policy tools and outcome variables, most results are either statistically insignificant or inconsistent with objectives commonly

associated with industrial policy, such as raising productivity or increasing production inputs and output. Regression results show that TFP declines for the average targeted firm when export incentives are introduced, and the effect is significant only temporarily. Domestic subsidies and import-restricting measures have slightly positive though mostly insignificant effects on TFP. However, domestic subsidies have a negative and significant effect on employment, whereas export incentives and import restrictions have no statistically significant effects. Capital also shows no significant response to most policy tools except for a negative effect associated with domestic subsidies.

- Results for advanced economies in the region show that domestic subsidies produce some positive though temporary effects. Domestic subsidies increase both employment and capital in firms operating in targeted sectors, including in East Asian advanced economies. However, the effects are not durable, and the increase takes at least two years to emerge and loses significance by the fifth year. Other industrial policy tools show no significant effects on TFP or production inputs.
- Firms in the region's emerging markets tend to experience significant negative effects from industrial policy. Unlike the temporary increases in capital and labor associated with domestic subsidies in advanced economies, the same policy tool appears to reduce employment and capital in the average targeted firm in emerging markets. In addition, import-restricting measures have a long-lasting negative effect on average TFP, suggesting that import-substitution policies could be especially detrimental to productivity in emerging markets, without the possible offset of increasing the factors or scale of production.
- Interventions targeting moonshot and safe-bet products through export incentives or domestic subsidies are durably effective in increasing capital. Both strategies are associated with significant and durable expansions in capital. However, the effects on employment are mixed across policy tools, which could reflect increased capital intensity resulting from industrial policy. The effects on TFP remain insignificant, possibly because the productivity gains associated with producing more complex products take longer to materialize than the period the analysis covered. Another possibility is that industrial policy supports production expansion through greater use of inputs, particularly capital, without generating observable technological or productivity improvements during the estimation period.

## CONCLUSION

Careful analysis of the increasingly intensive use of industrial policy in Asia and Pacific between 2009 and 2024 yields several insights.

Industrial policy interventions in the region vary widely in both objectives and implementation. Economies target different types of products through a range of policy tools, primarily in manufacturing and industry, often focusing on products

for which they have no RCA. The region also layers industrial policy interventions much more heavily than the rest of the world, which can reduce transparency and effectiveness if different measures work at cross-purposes.

Industrial policy interventions in Asia and Pacific economies show alignment with structural transformation strategies, including efforts to pursue technological upgrading and diversification and to address market failures in critical sectors. However, the choice of policy instruments raised questions about effectiveness.

The evidence suggests that industrial policy is not systematically generating durable effects that could support structural transformation in Asia and Pacific. Industrial policy in the region appears associated with positive effects in some cases, for example, increasing export values or competitiveness in targeted products or shifting at least some production inputs toward firms in targeted sectors. However, the effects are not systematic, are limited to specific cases, and generally lack durability, raising questions about the mechanisms underlying effectiveness. Even when policies designed to promote exports or competitiveness generate positive effects, their temporary nature suggests that they could reflect financial support that lowers costs and improves pricing temporarily for specific products, not structural changes in production or market structures that increase productivity or competitiveness. Notably, some of the few durable effects identified in the analysis at both the macro and firm-levels occurred in advanced economies, particularly East Asian advanced economies.[5] A detailed study of those economies falls outside the scope of the analysis, but their strong institutional frameworks and long experience with industrial policy—including participation in the first wave of industrial policy in Asia—suggest that some aspects of their industrial policy design warrant closer examination.

The findings underscore the need for caution in the use of industrial policy and for moving ahead with necessary horizontal structural reforms. Consistent with other studies and the mixed evidence on underlying policy motives, the lack of systematic empirical evidence of positive effects raises important questions about the appropriate scale, pace, and design of industrial policy. Estimating the costs of industrial policy interventions is outside the scope of the analysis, but the modest gross benefits identified suggest that many of the interventions could result in net costs. The findings strengthen the case for careful and restrained use of industrial policy, grounded in thoughtful design, clearly identified market failures, and rigorous cost-benefit analysis. Complementing industrial policy with broad horizontal structural reforms can improve its effectiveness (Baquie and others 2025; IMF 2025) and, in some cases, eliminate the need for it altogether.

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CHAPTER 15

# The Macro Implications of the Energy Transition in the Asia and Pacific Region

**Sandra Baquie, Margaux MacDonald, John Spray, Sneha Thube, and Alice Zhang**

The energy transition poses fundamental yet diverse challenges to the growth models of countries in the Asia and Pacific region. The region contributes to two-thirds of global growth and accounts for more than half of global greenhouse gas emissions. Shocks related to the energy transition threaten economic growth amid limited fiscal space, existing production and value chain structures, and geoeconomic and trade fragmentation, which make the transition a macro-critical challenge.

Policymakers at the 28th Conference of the Parties of the United Nations Framework Convention on Climate Change called for countries to transition away from fossil fuels in energy systems. Falling costs of renewable energy sources and long-term climate targets and policies could result in a contraction in the global demand for fossil fuels. Renewable and low-carbon energy sources have become more competitive. The share of renewables (solar and wind) in total electricity generation increased from 1.78 percent to 15 percent globally between 2010 and 2024 and accounted for nearly three-quarters of the overall increase in power generation in 2024 (IEA 2025). During the same period, the installed electricity capacity of all renewables surpassed the installed capacity of both coal and natural gas in 2024.

The energy transition will have important implications for the region's commodity exporters, especially Australia, China, and Indonesia. Many commodity exporters rely on extractive industries to support growth, provide employment, boost fiscal revenues, and maintain sustainable external balances, though uncertainty related to the net impact and timing of the transition is high. For fossil fuel importers, the green transition is expected to have a net positive effect on energy security if investments align to address new challenges posed by an increased reliance on renewables (Kim, Patnon, and Schwerhoff 2025). An analysis of Asia and Pacific countries' potential to pivot toward greener sectors for production and export identifies options that can provide relief from costs of the transition in terms of growth, employment, and external balance, while supporting lower emissions. Firms can also upgrade technologies to reduce emissions intensity, potentially lowering aggregate emissions and boosting

productivity growth. Domestic policies that improve the ease of doing business, institutional quality, and trade and capital account openness will be important to attract foreign direct investment (FDI) to support the green transition, especially in fiscally constrained emerging market and developing economies.

## THE ECONOMIC IMPLICATIONS OF ENERGY TRANSITION IN ASIA AND PACIFIC

The ongoing energy transition will affect fossil fuel demand and supply, but the speed and composition of the changes remain highly uncertain. Figure 15.1 shows the projected global production between 2022 and 2050 for coal (panel 1), crude oil (panel 2), and natural gas (panel 3) for a scenario in which, under current policies, the global temperature increase is limited to 2.7°C by 2100 relative to preindustrial levels. These projections represent a middle-of-the-road pathway in socioeconomic development and are consistent with the equilibrium average global temperature by the end of the century under the Shared Socioeconomic Pathway (SSP) scenario SSP2-4.5, from the United Nations' Intergovernmental Panel on Climate Change Sixth Assessment Report, *Climate Change 2023: Synthesis Report,* which also underpins the IMF's baseline scenario.[1]

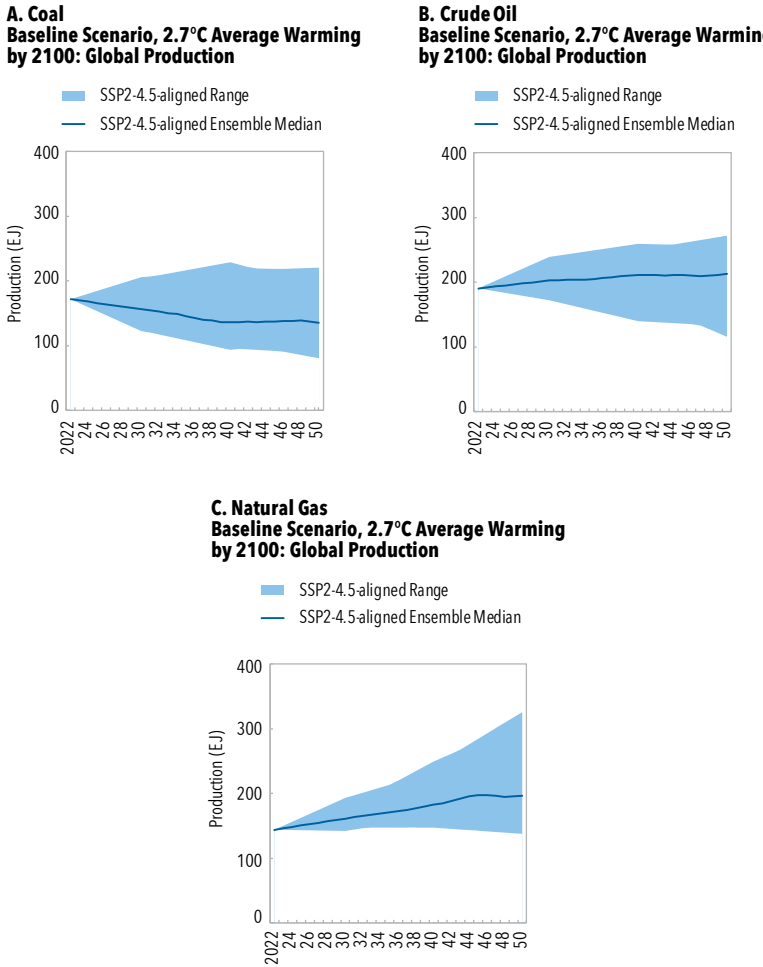
Demand for coal is broadly projected to decline. However, the relatively lower carbon content of natural gas and oil, coupled with residual demand in the transportation sector, might sustain the demand for these fuels if they serve as bridge fuels until the technologies needed for system-wide, low-carbon electricity generation; transport; and carbon capture, use, and storage are developed and made economically viable. Therefore, the Sixth Assessment Report model ensemble projects a median increase in natural gas demand of 37 percent between 2022 and 2050, with a range of 126 percent increase to 4 percent decrease. Similarly, median crude oil demand is expected to increase by 12 percent from 2022 levels, with a range of 43 percent increase to 39 percent decrease.

The Asia and Pacific region is central to the global energy coal transition. It produces nearly 80 percent of the world's coal, 8 percent of global oil, and 15 percent of global gas. Australia and Indonesia lead coal exports, accounting for 34 percent and 19 percent, respectively, of global coal briquette export value in 2023. The region also accounted for 76 percent of global coal imports, with almost 50 percent going to China, India, and Japan in 2022 (IEA 2025).

Despite the clear importance of the energy transition, the implications for the region's countries will be heterogeneous. Fossil fuel exporters will face complex and interlinked challenges of adjusting to economic transformation while their revenues decline and financial sector risks increase from stranded assets (Mesa Puyo and others 2024). Key factors that determine the country-level impact of the

[1] Fuel demand projections under alternate scenario SSP1-2.6 (which represents the optimistic scenario with an average warming of 1.8°C by 2100), and SSP3-7.0 (which represents the pessimistic scenario with an average warming of 3.6°C) are available in Spray, Thube, and Zhang (2025).

**Figure 15.1. Fossil Fuel Production Pathways**
*(Baseline scenario, 2.7°C average warming by 2100: global production)*




Sources: IMF staff calculations based on data from International Energy Agency; World Energy Balances database, 2024; and Intergovernmental Panel on Climate Change AR6 Scenario Explorer and Database (2022).

Note: SSP = Shared Socioeconomic Pathway.

energy transition include the type of fossil fuel a country produces and exports, with coal exporters expected to experience strong demand declines in the medium term; the cost of extraction, with lower-cost producers expected to gain market share while high-cost producers lose market share or wind down production; and country characteristics, with stronger economic headwinds for countries with less diversified economies. However, the quantitative magnitude of these effects and how they will differentially influence countries in the region remain unclear.

## Modeling the Transition Away from Coal

An analysis used the IMF-ENV model to examine the impacts of the global transition away from fossil fuels, especially coal because of its importance in the region (Chateau and others 2025).[2] Scenarios differ in two aspects: the level of global mitigation ambition and the specific fossil fuel targeted for reduction. Each of the scenarios implies a progressively lower level of coal demand on the global scale. The regional composition of the demand and supply is endogenous in the model. No deployment of carbon capture, use, and storage was assumed and thus no abated fossil fuel power generation by 2035.

The two initial scenarios focused on a coal phasedown approach, and the third concentrated on a broader decarbonization strategy. Regional emissions are endogenous in the model except for the Below 2°C scenario, in which regional emission trajectories were exogenously defined using projections from phase IV of the Network for Greening the Financial System to ensure that global temperature rise is limited to 2 °C by the end of the century.

The first scenario, G7, relates to the Group of Seven (G7). In this scenario, coal demand for power generation falls to zero in the G7 economies by 2035. The phasedown focuses on unabated coal. The analysis assumed no significant level of coal power generation and emissions reduced through carbon capture, use, and storage over the next 10 years.

The second scenario is G7 plus. Along with the G7, coal demand for power generation in China and India declines in line with their 2030 nationally determined contribution targets (a moderate reduction in coal demand for both nations, not an elimination).

In the third scenario, Below 2°C, all countries are constrained by a regional emissions budget aligned with a global 2°C temperature target by the end of the century.

### *Asia and Pacific Region*

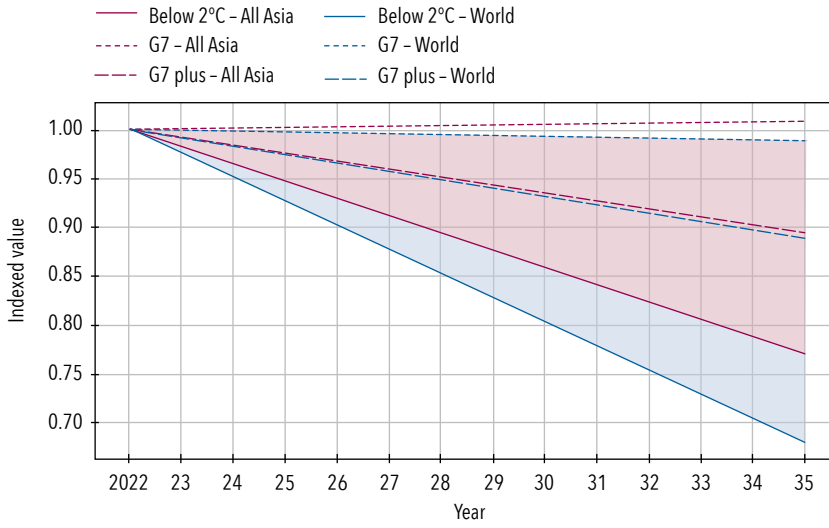
Under the model baseline, future global coal investments remain primarily located in Asia and Pacific, and a smaller proportion of natural gas investments occurs there. By 2035, 82 percent of global capital stock in coal is concentrated in the region, with China holding the largest portion at 31 percent. The region's proportion of global investments in natural gas is approximately 15 percent by 2035, with Australia and "other developing Asia" each representing about 5 percent.

Total investments in different sectors vary according to the specific design of each policy scenario. Compared with 2022, total capital stock in coal at the global scale falls marginally under the G7 scenario but decreases by 32 percent under the Below 2°C scenario (Figure 15.2, panel 1). At the regional scale, total capital stock

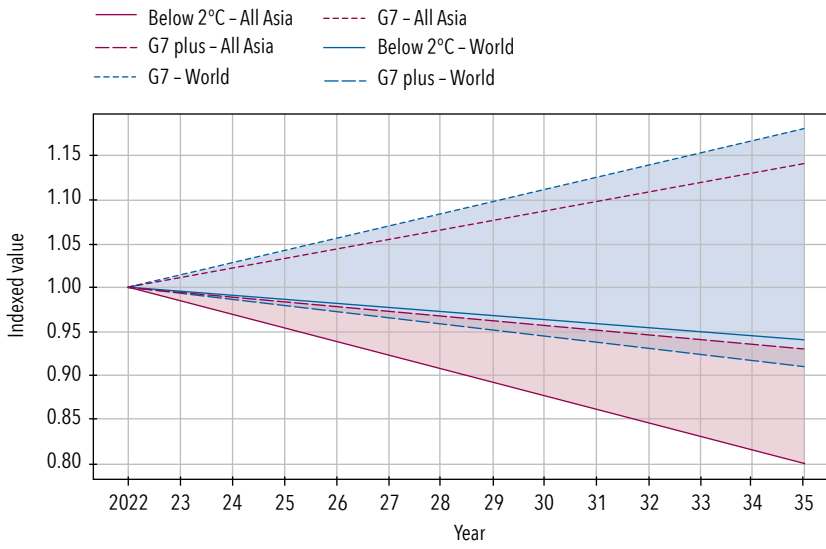
[2] IMF-ENV is a global, dynamic, computable general equilibrium framework built on input-output tables, covering 160 countries and 76 commodities. The model is designed to assess large structural shifts resulting from decarbonization efforts. It provides high sectoral and regional granularity, supports a wide range of policy instruments, and captures both domestic and cross-border general equilibrium effects through detailed modeling of trade flows.

## Figure 15.2. Impacts of Global Energy Transition in Asia and Rest of World

**1. Capital Stock in Coal Mining**
*(Percent relative to baseline, 2035)*




**2. Capital Stock in Gas Extraction**
*(Percent relative to baseline, 2035)*




Source: IMF staff calculations.

in coal in Asia and Pacific rises marginally under the G7 scenario but decreases by 23 percent under the Below 2°C scenario (Figure 15.2, panel 1). By contrast, total capital stock in natural gas at the global scale could increase by 18 percent when policies target only coal phasedown, while the capital stock would fall by 5

percent when policies are fuel agnostic. For Asia and Pacific, policies that target only coal phasedown lead to a 16 percent increase in capital stock in natural gas but a decrease of 20 percent when policies are fuel agnostic.

A notable shift occurs in the geographical allocation of coal investments. In the G7 scenario, the share of capital stock in coal mining increases in Asia and Pacific to 84 percent but declines in the rest of the world. Under the G7 plus scenario, coal investments decline in Australia and Indonesia as China and India reduce their demand for coal. Coal mining capital in the rest of the region could increase slightly in these scenarios because of general equilibrium price movements in coal. By contrast, under the Below 2°C scenario with decreased demand for coal worldwide, the overall share of capital in the region rises to nearly 87 percent, and investment tends to become more concentrated in Australia and Indonesia. Imports from those two countries are preferred because their costs of extraction and transport are comparatively lower. In this scenario, domestic coal investments in China and India also decline because of domestic decarbonization efforts. The change in the origin of coal imports is also reflected in the shifts in market shares. In an increasingly decarbonizing world, when global coal demand falls, the market shares of low-cost producers such as Australia and Indonesia increase.

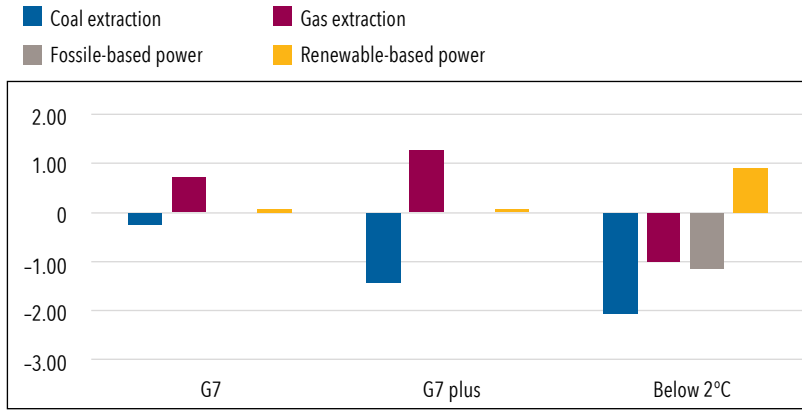
The varying impacts on capital stock highlight the challenges investors face when deciding between short-term and long-term investments in fossil fuels (Figure 15.2). Capital stock in the coal extraction sector declines across the scenarios because new investment in the sector decreases and existing capital assets depreciate. However, the model results show that the speed of transition, the countries leading the transition, and the policies they implement can affect the magnitude and location of coal production significantly. Depending on market and policy factors in a region, investing in natural gas extraction can be rational if decarbonization targets coal specifically, but this incentive disappears when implementing a fossil fuel–agnostic energy transition plan. Considerable uncertainties surround international commitments and collaboration on decarbonization policies, resulting in a complex investment landscape that increases the potential for both overinvestment and underinvestment in fossil fuels.

### *Case Study: Australia*

Australia is a large global exporter of minerals and has ambitious domestic energy transition targets. Using the same set of simulations for an analysis of Australia, three main narratives emerge. Coal and gas output are set to fall in the baseline, generating substantial adjustment costs (Figure 15.3). Different global scenarios lead to vastly different economic outcomes, which could shape the structure of the Australian economy, labor demand, and investment. Also, Australia's low extraction costs relative to other fossil fuel exporters generate resilience to the global transition.

Policies can help minimize costs and maximize benefits during the transition (IMF 2024). Australia's diversified economy and range of trading partners enable it to address short-term economic fluctuations and shift production to different

**Figure 15.3. Impact on Labor Demand in Australia**
*(Thousands relative to baseline, 2035)*




Source: IMF staff calculations.

sectors as needed, a result of sustained macroeconomic stability and efforts to develop a supportive business environment. Revenues from fossil fuels have helped the country maintain low debt and low borrowing costs. Australia is building capacity in critical minerals and green metals and is developing its renewables sector, which provides a natural hedge to declining fossil fuel demand.

## DIVERSIFYING THE ECONOMY TOWARD LOW-CARBON SECTORS

The global energy transition presents risks to exporting countries, but it also implies the development of low-carbon value chains to meet a substantial rise in demand for low-carbon products, which could be an important opportunity for exporting countries and others. Ongoing global trade policy uncertainty has left many countries in the region—where trade has been a major driver of growth—with a need to build economic resilience, enhance diversification, address supply-chain vulnerabilities, and expand trade arrangements to reduce dependence on traditional trading partners.[3] Thus, integrating into low-carbon value chains could help achieve these goals. For Asia and Pacific countries to contribute to the development of low-carbon value chains, they must acquire the expertise to produce the associated low-carbon products. If the composition and production steps of low-carbon products are similar to products already in production, seizing the related opportunity should be relatively easy. Identifying

[3] For a detailed discussion of resilience building, see Aligishiev, Bellon, and Massetti (2022), Dabla-Norris and others (2021), and Lim and others (2024).

such opportunities is therefore critical, including quantifying expertise similarity across products.

The Harvard Growth Lab defines "green products" as products that contribute to 10 low-carbon technology value chains: batteries, critical minerals, electric grid, electric vehicles, green hydrogen, heat pumps, hydroelectric power, nuclear power, solar power, and wind power. Using this definition to identify green products, the analysis considered three dimensions to assess whether a product represents an opportunity for export diversification within the green transition: market size, ease of production and export, and learning opportunity.

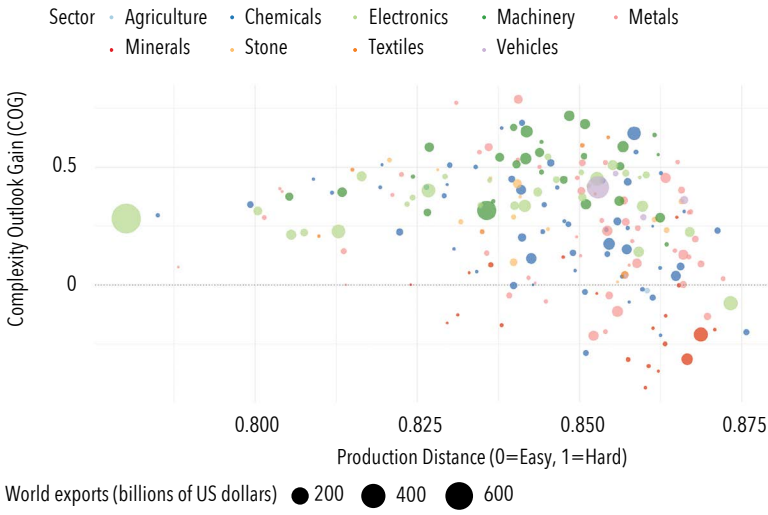
First, policymakers might want to develop products for which the global market is relatively important, and the analysis measured this market size in nominal terms (billions of US dollars). Second, ease of export production ("know-how distance") captured low-carbon products that can be easy to produce using the country's existing export production capabilities, defined on a scale of 0 to 1 (Hausmann and others 2014). The know-how distance is close to 0 if the country exports products similar to a product that policymakers are considering and close to 1 if it exports very few products similar to the one under consideration. The mean know-how distance in Asia and Pacific countries is 0.87 with a standard deviation of 0.11. Third, policymakers could be interested in developing a low-carbon product that represents a learning opportunity, quantified with the Complexity Outlook Gain, defined as the change in the average complexity of products in the vicinity of the production frontier[4] (at a small know-how distance) if the country exported the considered product (Hausman and others 2014). If this dimension is positive, exporting the product would enable the country to gain expertise (know-how), thereby opening the door to more complex products. If it is negative, exporting the product would help expand the country's production frontier toward simpler products. For Asia and Pacific countries, the Complexity Outlook Gain ranges from –0.82 to 1.7, with a mean of 0.2 and a standard deviation of 0.41.

Based on these dimensions, export diversification opportunities are balanced across green and nongreen products in Asia and Pacific. For green products only, Figure 15.4 plots how difficult it would be for a country to produce this product for export (average know-how distance on the *x*-axis) and how much of a learning opportunity the product represents for countries not exporting it (average Complexity Outlook Gain on the *y*-axis). In advanced economies, Complexity Outlook Gain and know-how distance are negatively correlated, which suggests that nonexported products with greater know-how distance are simpler to produce than exported products. Thus, advanced economies in Asia and Pacific have little to learn from exporting products with distant expertise, likely because their existing exports are complex. However, in emerging market and developing economies, complexity outlook gain and know-how distance are positively correlated,

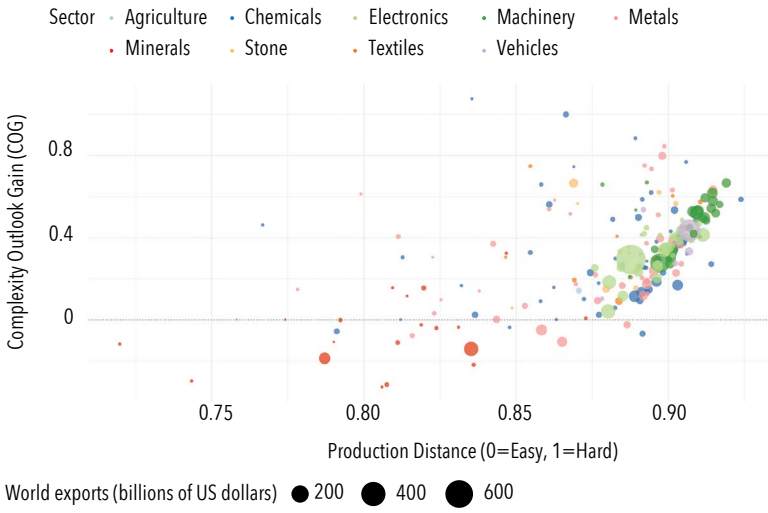
[4] The production frontier corresponds to the maximum outputs an economy can produce given its resources and technology.

## Figure 15.4. Export Diversification Opportunities in Green Products, by Sector, in 2023

**1. Asia and Pacific Advanced Economies**




**2. Asia and Pacific Emerging Market and Developing Economies**




Sources: The Growth Lab at Harvard University (2025) "Country Trade by Product, Unilateral trade, HS92 4-digit" [Data Set] and IMF staff calculations leveraging the latest available year at the time of writing, 2023.

which implies the possibility of acquiring skills by exporting products that are more complex to produce.

The analysis identified promising sectors for advanced economies to boost exports as sectors that combine a large global market with a short know-how distance (high feasibility), represented by large dots on the left of Figure 15.4, panel 1.[5] This suggests that opportunities exist at the sector level for these countries to expand their production in the electronics, machinery, and chemicals sectors to benefit from the energy transition. At the product level, the five most promising green products for the region's advanced economies (and their uses) are electronic integrated and printed circuits, semiconductors, telephones, and instruments for measuring electricity. Such products are important in the value chains for electric vehicles, wind and solar power, and heat pumps, among other green product lines.

However, variation exists across countries. For example, in New Zealand, the production of aluminum shows potential for expanded production and is an important input to solar, wind, and nuclear power; electric vehicles; electric grid; critical minerals; and heat pumps. In Australia, which faces possibly sharp declines in demand for coal and gas, mining minerals critical to the energy transition represents an opportunity, including developing production of iron ores for heat pumps and electric grid value chains. Ramping up its production of copper could also enable Australia to seize opportunities in the electric grid, critical minerals, and solar and nuclear power value chains. Australia could also benefit from the likely increase in demand for manganese, used in the production of green hydrogen, nuclear power, and batteries.

Unlike the region's advanced economies, its typical emerging market and developing economy could find it difficult to acquire the necessary expertise to seize opportunities in the electronic and machinery sectors. Still, the analysis highlighted other promising opportunities to diversify exports in other sectors that combine large global markets and feasibility (large dots on the left of Figure 15.4, panel 2), such as minerals, chemicals, and metals. Many emerging market and developing economies could strengthen their economic resilience by developing the extraction, cultivation, and production of raw materials critical to low-carbon technology value chains, such as copper, uranium, iron, zirconium, and rubber.

Although mineral products offer promising development opportunities, on average, expanding production in these sectors might not be a stepping stone for emerging market and developing economies to diversify exports, as shown by the negative complexity outlook gain associated with minerals (red dots below the *x*-axis in Figure 15.4, panel 2). When learning potential is given equal weight to market size and feasibility (large dots in the upper-left quadrant of Figure 15.4, panel 2), the most promising opportunities for the average Asia and Pacific

[5] Index to highlight promising products that have a large market and are easy to produce is defined as: $Promising\ index_p = Std\ (-distance_p) + Std\ (\log(exports_p))$. This formula does not include learning opportunity because Figure 15.4, panel 1 suggests that exports are already complex in advanced economies.

emerging market and developing economy emerge in the chemical, stone, and electronic sectors. As with advanced economies, variation exists across countries, and the energy transition opportunities for emerging market and developing economies vary by country resources and production capabilities. China, for example, is well positioned to benefit from the energy transition in the minerals, electronics, machinery, and chemical sectors, aligning it with those of advanced economies. India's accessible growth opportunities lie in expanding its iron production, boosting textiles for electric vehicles, and increasing output in chemicals and metals such as iron derivatives and aluminum. Some low-income economies, such as Bhutan and Cambodia, could also benefit from opportunities in the vehicles and metals sectors, including producing parts for motor vehicles or developing exports of iron and steel to contribute to the electric vehicle value chain.

## FIRM-LEVEL ANALYSIS OF TECHNOLOGY AND STRUCTURAL TRANSFORMATION TO ENHANCE ECONOMIC RESILIENCE AND REDUCE EMISSIONS INTENSITY

The region leads in clean technology development and drives an affordable green transition through enhanced energy efficiency and increased investment. Deploying existing technologies can deliver more than 80 percent of the global emissions reductions needed by 2030, according to the International Energy Agency (Hasna and others 2023). The region can also leverage trade and green FDI and scale research and development to advance decarbonization while boosting productivity and environmental performance.

The global energy transition will involve a shift away from fossil fuels toward the low-carbon value chain, but the transition period could be lengthy. During that period, oil and natural gas will still be important inputs into production and consumption in most countries, and thus reducing emissions intensity—defined here as firm-level carbon dioxide emissions over revenues—will also be an important part of meeting climate targets. Targeting only emissions-intensive sectors could be too broad an approach because significant heterogeneity exists in firm-level emissions intensity even within sectors. Firms with relatively high emissions intensity account for a considerable share of aggregate emissions (Lyubich, Shapiro, and Walker 2018), in both emerging market and developing economies and advanced economies. If firms with higher emissions intensity than their peers drive a large share of aggregate emissions, policies that support physical capital upgrades with new technologies (which are often green-biased) and investment in productivity-enhancing intangible assets can reduce firm-level emissions intensity and help lower aggregate emissions (Capelle and others 2023). Such policies can also raise aggregate economy-wide productivity.

Despite those benefits, many firms continue to operate with older, more emissions-intensive technologies, possibly because increased exposure to elevated

temperatures stemming from climate change reduces firms' revenues and operating income (Pankratz, Bauer, and Derwal 2023). Evidence suggests that firms with greater climate exposure face increased costs of capital, reduced investment activity, and lower investment levels (Agoraki and others 2024). Climate policies that help firms replace older, more emissions-intensive capital can therefore reduce emissions intensity and potentially lower overall emissions.[6] Carbon pricing is one example. It creates stronger incentives for firms with higher emissions intensity to upgrade capital because they face greater tax burdens.[7] Subsidies, which Asia and Pacific countries commonly use, can also be highly effective though distortive and more costly. When targeted to purchases of the newest vintage of physical capital and to investment in intangible capital, subsidies will be costly up front because they imply replacing capital stock. However, they can also imply substantially greater declines in emissions in the long term (Capelle and others 2023), boost productivity, and lift potential growth.[8]

The analysis used firm-level data on emissions, balance sheets, and income statements for more than 1,600 firms across Asia and Pacific from 2010 to 2022. "Emissions intensity" is defined as the residuals extracted from regressing the log of $CO_2$ emissions divided by revenues (megatons per millions of US-dollar revenue) on country-industry fixed effects (where industry is defined at the four-digit Standard Industrial Classification code) fixed effects.[9,10] The data reveal significant within-industry heterogeneity in the environmental performance of the region's emerging market and developing economies. This heterogeneity is greater than that between firms in emerging market and developing economies in other regions and than the heterogeneity between firms in Asia and Pacific advanced economies (Figure 15.5). Firms in the region's emerging market and developing economies and advanced economies have higher median emissions intensity in most sectors. Thus, improving the emissions intensity of lagging firms could reduce total emissions.

The data also show a positive correlation between firms with older capital, fewer intangible assets, lower productivity, and higher emissions intensity. The correlations hold for firms in both emerging market and developing economies and advanced economies. This pattern suggests that the main findings of Cappell

[6] However, policies that increase access to capital could raise total emissions because they raise GDP (Capelle, Kirti, and Villegas-Bauser 2025).

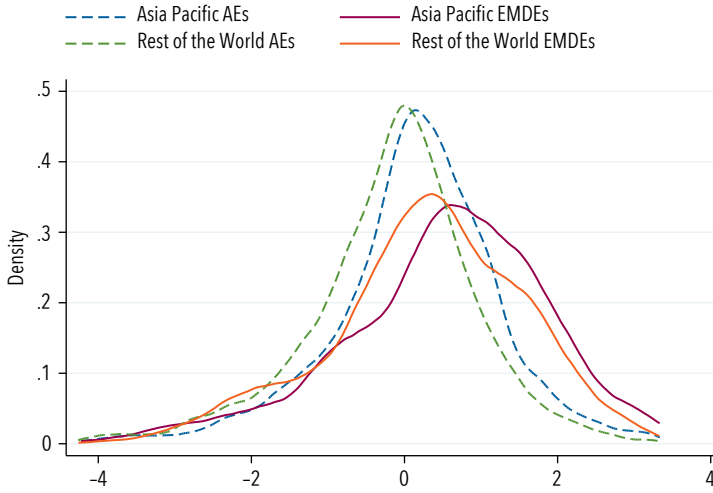
[7] A carbon tax also has important sectoral implications. Sectors that depend highly on energy for production are most affected and thus would see the largest decline in energy usage.

[8] Subsidies distort the efficient allocation of inputs. A subsidy for new vintages of capital will increase demand for capital and for labor in capital-producing sectors, increase real wages, and crowd out labor from other sectors. The capital goods sector will expand to produce new capital goods, which will generate more emissions and offset some of the gains from sectors upgrading to new capital.

[9] $CO_2$ emissions are defined as $CO_2$ equivalent scope 1 (direct) and scope 2 (indirect emissions from purchased energy) emissions, based on self-reported emissions following the Greenhouse Gas Protocol.

[10] Emissions data are from ICE Data Services. Balance sheet and income statement data are from S&P Compustat Global. The data cover corporations that issue publicly traded securities. Data on energy consumption are from DataStream. Capelle and others (2023) combined and cleaned the data.

**Figure 15.5. Heterogeneity in Emissions Intensity**




Sources: Capelle and others 2023; and IMF staff estimates.

Note: The figure plots the kernel densities of the log of emissions intensity (measured as emissions over revenues) after controlling for industry multiplied by year fixed effects. Because the density of emissions intensities is plotted, the median value of emissions intensity in each region is represented by the peak of the corresponding line in the figure. Four-digit Standard Industrial Classification is used. AE = advanced economy; EMDE = emerging market and developing

and others (2023)—that investment in newer capital embodying advanced technologies and in intangible assets can reduce firms' emissions intensity—likely also apply to the region.

Counterfactual exercises show that improvements in the technology and production processes of lagging firms can reduce emissions even with existing technologies in each industry.[11] The first exercise calculated how total emissions would change if all the region's firms with emissions intensity above the median in each industry-year improved their emissions intensity to the within-industry median value. In this case, emissions would fall by more than 50 percent. Additional counterfactual exercises showed that if every firm in emerging market and developing economies had capital no older than that of the median firm in its industry-year sample, or intangible-asset shares at least as high as those of the median firm, emissions could fall by about 4 to 6 percent. These simple counterfactual exercise do not account for several factors. Replacing older capital with newer capital could raise productivity, increase production, and therefore increase emissions, offsetting the reduction from lower emissions intensity. However, lower emissions intensity would achieve stronger productivity and growth, a desirable goal.

[11] This result, which is not proved causally in the counterfactual exercises, is shown more rigorously in the modeling exercise of Capelle and others (2023).

## FOREIGN DIRECT INVESTMENT AND THE GREEN TRANSITION

The region's energy transition will require large and sustained climate finance. Emerging market and developing economies face an annual climate finance gap of at least $800 billion (Lim and others 2024). The private sector must cover a large share of investment needs, given fiscal constraints in many countries, yet many emerging market and developing economies face systemic barriers. These barriers include gaps in the climate information architecture that hamper accurate assessment and tracking risks; resource constraints, inadequate coordination, and lack of institutional oversight at the national level; and a shift toward protectionist industrial policies and geoeconomic fragmentation, which threaten to imperil collective and cooperative action on climate change (Lim and others 2024).

Financing needs for the climate transition are substantial, including support for displaced workers and industries, expansion into new opportunities in low-carbon value chains, and reducing emissions intensity. Meeting countries' announced climate commitments by 2030 would require annual energy investment to double relative to 2021–24 levels and reach $130 billion (IEA 2024). Although investing in novel and greener technologies incurs short-term costs, it is associated with higher productivity and output in the medium term and can reduce emissions intensity.[12]

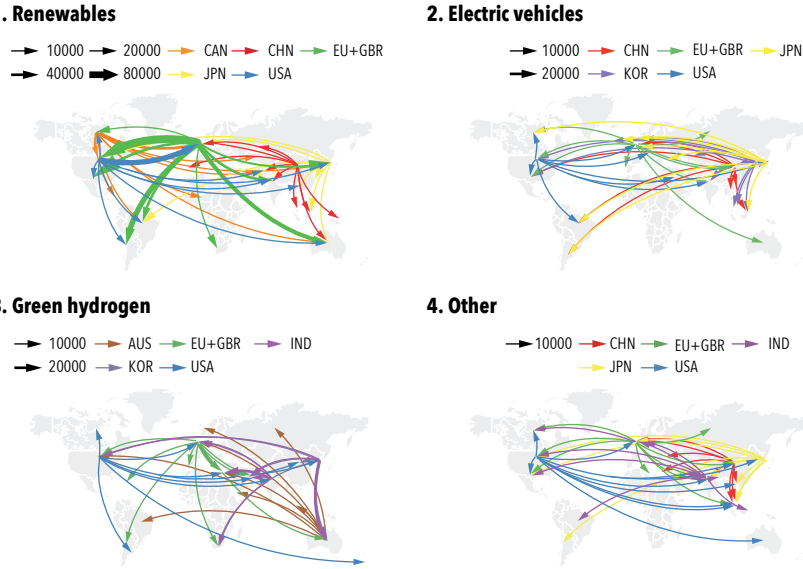
Policies that encourage investment across firms and sectors could support the transition. Horizontal policies, such as improving access to credit, can encourage green investment (Budina and others 2023). When market failures prevent necessary investment without public support, disciplined and well-targeted industrial policies could help address the distortion if their expected benefits exceed their costs. Such policies could support technology upgrades among firms that have the highest levels of emissions intensity.[13] Baquie and others (2025) find that industrial policies targeting green products have a stronger impact on competitiveness and innovation than policies targeting nongreen products. Capelle and others (2023) also find that carbon taxes can be beneficial while creating fewer distortions and requiring lower up-front costs.

FDI will also be essential, especially in emerging market and developing economies with limited fiscal space. FDI can help bridge the renewable energy investment gap, fund green projects, and facilitate access to new technologies (Jaumotte and others 2024; Soto 2024). Figure 15.6 shows that some countries in the region are already major sources and destinations of green FDI (Jaumotte and others 2024). Technology diffusion and innovation are also crucial for expanding the adoption and production of low-carbon technologies, facilitating substitution with cleaner energy sources, and seizing green transition opportunities (IFC 2025). These factors

[12] Higher productivity and output could limit the reduction in absolute emissions or even increase it. Still, it would help lower emissions intensity while supporting higher growth, which the region's emerging market and developing economies need. It is also consistent with the principle of common but differentiated responsibilities under the Paris Agreement.

[13] See Baquie and others (2025) for a comprehensive discussion of industrial policies and their use.

**Figure 15.6. Green Foreign Direct Investment Flows by Technology**




Source: Jaumotte and others 2024, figure 3.

Note: The figure shows total green foreign direct investment in millions of US dollars between 2003 and 2023 for the largest five source countries and the 10 largest destinations in each category. Data labels use International Organization for Standardization country codes.

are particularly important for the region's emerging market and developing economies, where export potential depends heavily on skill acquisition.

Strong structural fundamentals (governance, education, and openness to trade and capital) can help attract FDI in low-carbon technologies and renewable energy (Jaumotte and others 2024). Improvements in the business environment can encourage FDI in low-carbon sectors and firms, facilitate higher growth, and create fiscal space for additional domestic investments. Ambitious structural reform packages that strengthen governance, improve business regulation, and increase external sector openness (both trade and capital accounts) can have significant effects. In emerging market and developing economies that begin with weak performance in those areas, such reforms can increase output by up to 8 percent within four years (Budina and others 2023). The reforms can also reduce emissions and energy intensity. Countries that expand their climate policy portfolios also show higher numbers of climate change–mitigation patent filings and increased low-carbon technology trade flows (Hasna and others 2023).

Domestic climate policies can attract green FDI inflows into the renewable energy sector, especially in countries with solar potential and low dependence on fossil fuels. Evidence suggests that closing the climate policy gap between the average emerging market and developing economy and the average advanced economy could triple the ratio of green FDI to GDP in emerging market and developing economies and close 40 percent of the private renewable energy investment gap in

emerging market and developing economies, excluding China (Jaumotte and others 2024). Regional cooperation could help expand the wind and solar energy sectors even more. Four countries in the region are among the top 10 in the world in operating solar capacity: China (which accounts for 48.4 percent of the global total), India (7.8 percent), Japan (3.4 percent), and Vietnam (1.4 percent).[14] China also leads in wind power farms, accounting for 31.5 percent of the global total.[15]

Country case studies show that climate policies are instrumental in attracting FDI, but these measures entail fiscal costs and can produce unintended consequences, such as resource misallocation, rent-seeking behavior, and cross-border spillovers from nontariff interventions. Indonesia introduced electric vehicle regulations in 2019 that established domestic content rules that mandated automakers to produce a minimum threshold of local components used in the production of vehicles and provided fiscal and financial incentives for investment. The incentives attracted $18 billion in announced FDI for electric vehicles and batteries (IMF 2023; Jaumotte and others 2023). Thailand adopted electric vehicle regulations in 2020 that support domestic electric vehicle adoption and provide subsidies to investors. FDI announcements related to electric vehicles nearly doubled to $907 million during 2020–22, with 78 percent originating from China (Jaumotte and others 2024). India's National Green Hydrogen Mission, launched in 2023, is a comprehensive strategy to develop the production, export, and use of green hydrogen (MNRE 2023). It includes several policies that could attract FDI, including financial incentives for production and pilot projects designed to build domestic demand.

## CONCLUSIONS

The Asia and Pacific region is a powerhouse for global growth and critical to the global green transition. The energy transition is equally important for the region's long-term growth and economic resilience, especially amid ongoing trade tensions and the search for a new equilibrium in the global trading system. Underinvestment in energy resilience threatens environmental and macroeconomic objectives. Coherent, well-designed policies adopted across the region will shape both its own energy transition and the global transition.

However, considerable uncertainty surrounds future fossil fuel demand in Asia and Pacific. Scenario analysis using the IMF-ENV global model suggests that the energy transition will have significant macroeconomic effects on the region's economies by 2035, and the effects on various fuels will depend largely on policy choices. The range of possible macro outcomes complicates long-term investment decisions and increases the risk of stranded assets and carbon lock-in. Country

[14] Data are from Global Energy Monitor Global Solar Power Tracker, Top 20 Countries/Areas by Operating Solar Farm Capacity in Megawatts (MWac) database, February 2025, accessed August 21, 2025.

[15] Data from Global Energy Monitor, "Top 20 Countries/Areas."

characteristics will determine the effects of the transition. Coal exporters with high average extraction costs will face the largest initial impacts, and demand will shrink in countries with low average extraction costs, even as their market share increases. For natural gas exporters, revenues could increase under some scenarios.

Critical minerals producers will likely benefit from the transition and would be an important source of resilience. However, initial projections suggest that gains from critical minerals will not offset losses from fossil fuel exports. Deployment of technologies such as carbon capture, use, and storage could extend the use of coal, while technologies such as green hydrogen could replace coal in the hard-to-abate sectors, though existing policies do not point to large-scale adoption of those technologies.

The findings of the analysis show the importance of building diversified and resilient economies that can absorb shocks from the energy transition and adapt to a world in which higher tariffs challenge the long-standing growth model for the region. At the same time, rising demand for low-carbon products and the development of low-carbon value chains to meet the demand can create important opportunities. Advanced economies have opportunities in electronic-related sectors of low-carbon value chains that combine large global export markets with capabilities close to existing expertise. For emerging market and developing economies, critical minerals industries have the closest expertise and largest export market size, but opportunities also exist in sectors with higher value added, such as chemicals and electronics, especially when countries prioritize learning potential. Firm-level investment in technology upgrades, even in carbon-intensive sectors, can support growth while progressing toward emissions targets. Given the need to rebuild fiscal buffers in several countries, policies that attract FDI can also help ease pressure on fiscal budgets for supporting the transition. However, not all sectors and workers will benefit from the green transition.

High youth unemployment and jobless growth have become major policy challenges in several countries in the region. Higher tariffs in many labor-intensive sectors have amplified those challenges and contributed to social tensions. The region should also leverage opportunities from Green jobs and the future of work to create opportunities to increase the job content of growth and strengthen employment multipliers (Alexander and others 2024). Where job displacement occurs, potentially compounded by AI, active labor market policies, training, and other forms of household support can help mitigate job losses. Well-designed, narrowly targeted green industrial policies can help countries capture emerging opportunities, but rapid technological change will require policymakers to remain flexible. Financial sector risks will also require close monitoring, particularly exposure to stranded assets and the associated implications for labor markets.

Several important areas warrant additional research. Diversification of energy trading partners has been shown to underpin energy security dynamics within and across countries, and understanding those links in the context of emerging low-carbon value chains is important (Kim, Panton, and Schwerhoff 2024). Generative AI will also become increasingly important in helping firms and sectors adopt new technologies and integrate into new value chains. Anticipated

reductions in AI use costs and future technological advancements could increase productivity and consumption while reducing emissions substantially, but significant challenges remain, including that AI itself generates considerable emissions (IMF 2025, Chapter 1; Stern and others 2025).

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